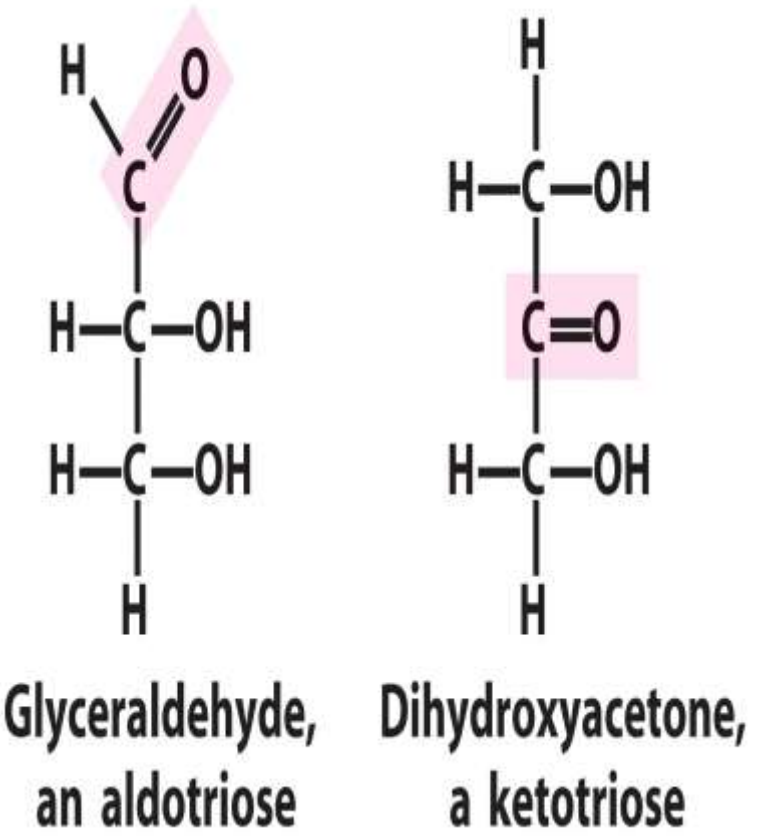
Carbohydrates

Carbohydrates

- Carbohydrates are polyhydroxy aldehydes or ketones, or substances that yield such compounds on hydrolysis
- Many, but not all, carbohydrates have the empirical formula ***$(CH_2O)n$***; some also contain nitrogen, phosphorus or sulfur



Carbohydrates

Classification

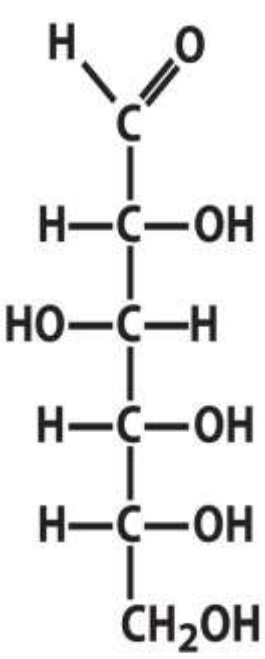
- Carbohydrates are classified into **four** major groups:
  - Monosaccharides
  - Oligosaccharides
- Disaccharides
  - Polysaccharides

Carbohydrates

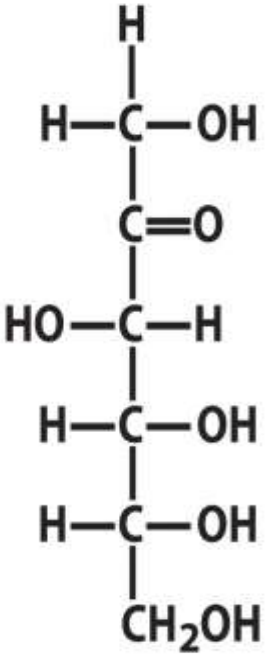
- **Monosaccharides** (simple sugars) are those which cannot be hydrolyzed further into simpler forms
- The backbones of common monosaccharides are *unbranched* carbon chains in which all the carbon atoms are linked by single bonds

Carbohydrates

- In **the open-chain form,** one of the carbon atoms is double-bonded to an oxygen atom to form a ***carbonyl group***
- Each of the other carbon atoms has a hydroxyl group



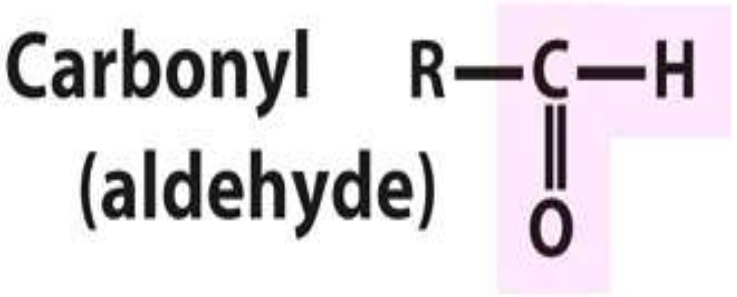
D-Glucose,
an aldohexose



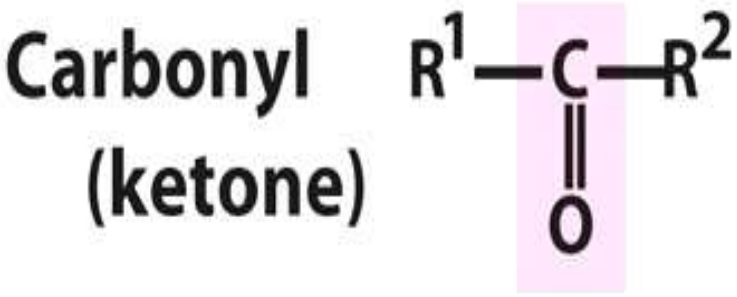
D-Fructose,
a ketohexose

Carbohydrates

- If the carbonyl group is at an end of the carbon chain (that is, in an aldehyde group) the monosaccharide is an ***aldose***

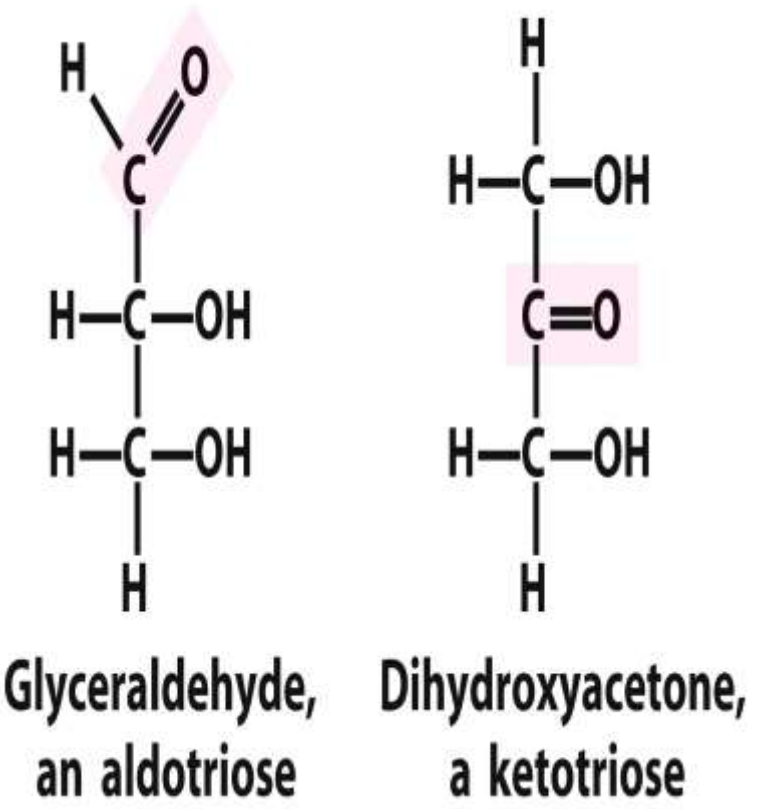


- If the carbonyl group is at any other position (in a ketone group) the monosaccharide is a ***ketose***



Carbohydrates

- **The simplest** monosaccharides are the two three-carbon trioses:
  - Glyceraldehyde,an aldotriose
  - Dihydroxyacetone, a ketotriose



Carbohydrates

- The carbons of a sugar are numbered beginning at the end of the chain nearest the carbonyl group

- All common monosaccharides and disaccharides have names ending with the suffix “-ose.”

Carbohydrates

- Ketoses are designated by inserting "ul" into the name of a corresponding aldose
- For example, ribulose is the ketopentose corresponding to the aldopentose ribose
- There are a few exceptions however, for example, fructose and dihydroxyacetone

Carbohydrates

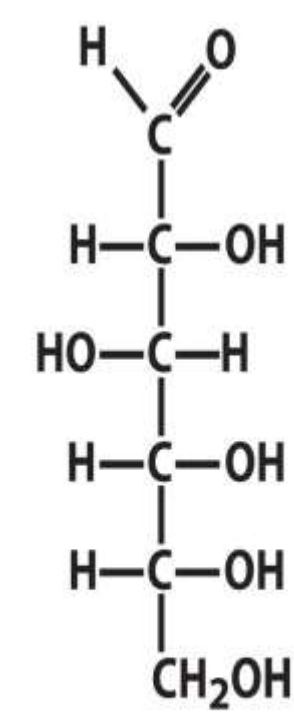
- Monosaccharides can be subdivided further depending upon:
  - Number of carbon atoms
  - Whether aldehyde or ketone groups are present as *aldoses* or *ketoses*

- General formula Aldosugars Ketosugars

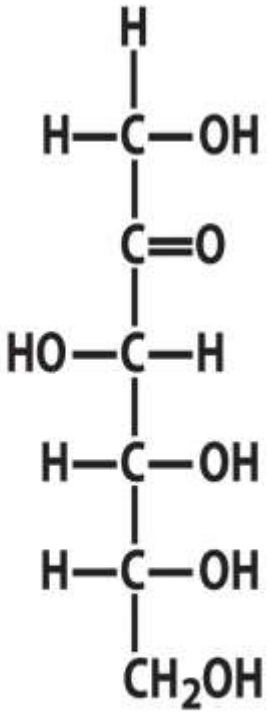
Trioses $C_3H_6O_3$	Glyceraldehyde	Dihydroxyacetone
Tetroses $C_4H_8O_4$	Erythrose	Erythrulose
Pentoses $C_5H_{10}O_5$	Ribose	Ribulose
Hexoses $C_6H_{12}O_6$	Glucose	Fructose

Carbohydrates

- The **hexoses** which include the aldohexose D-glucose and the ketohexose D-fructose, are the most common monosaccharides in nature



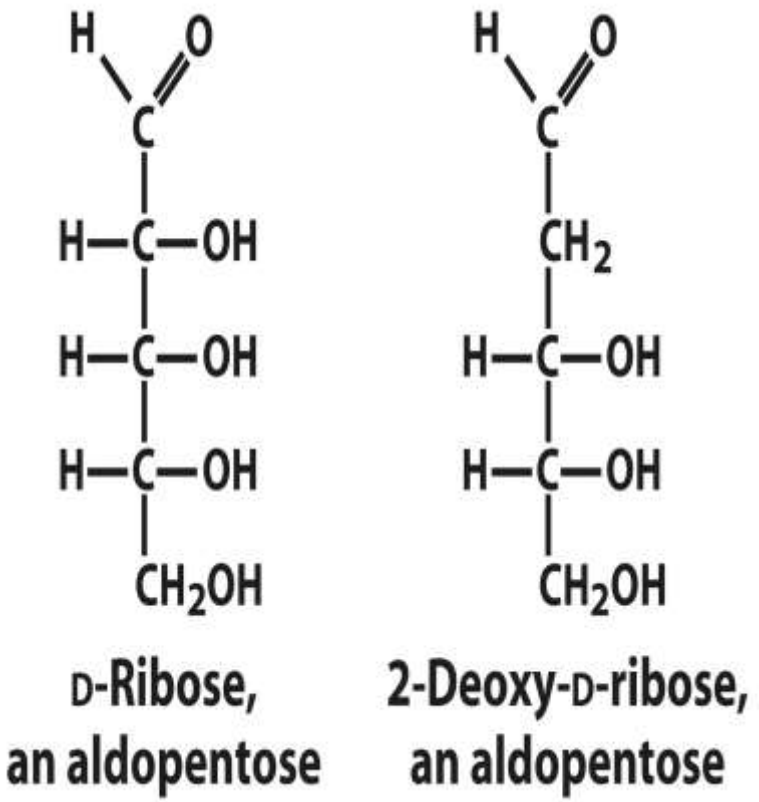
D-Glucose, an aldohexose



D-Fructose, a ketohexose

Carbohydrates

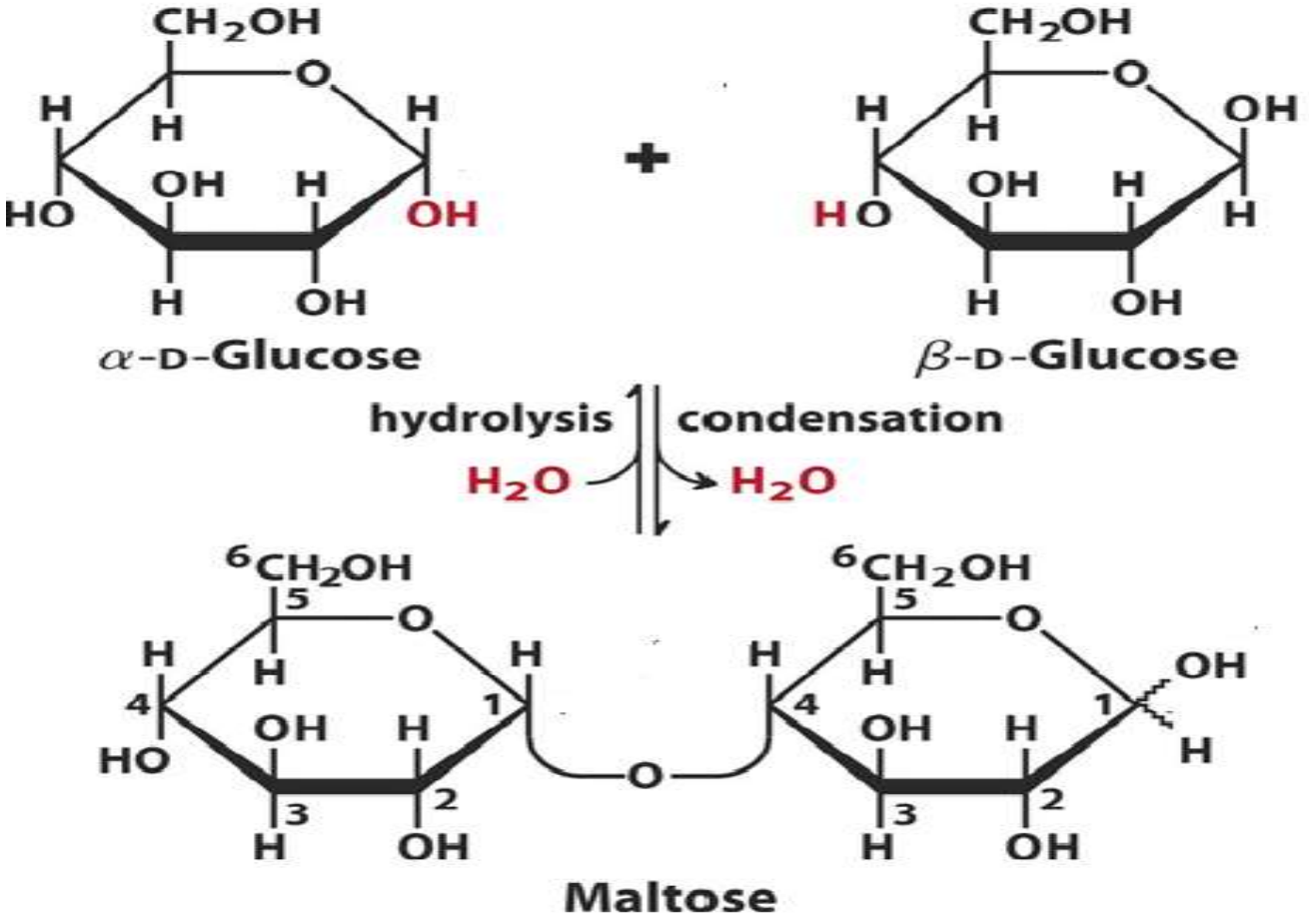
- The **aldopentoses** D-ribose and 2-deoxy-D-ribose are components of nucleotides and nucleic acids



Carbohydrates

- **Disaccharides** are those sugars which yield two molecules of the same or different monosaccharides on hydrolysis
  - Maltose - 2 molecules of glucose on hydrolysis

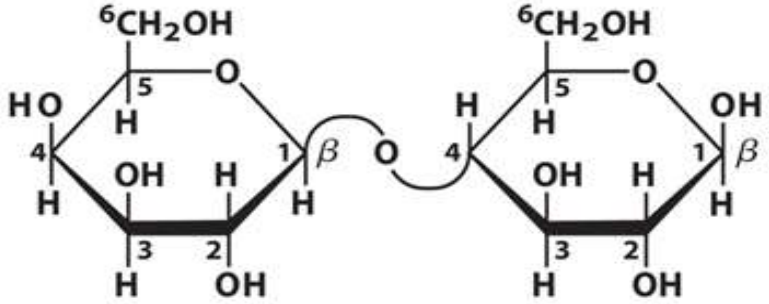
Carbohydrates



Carbohydrates

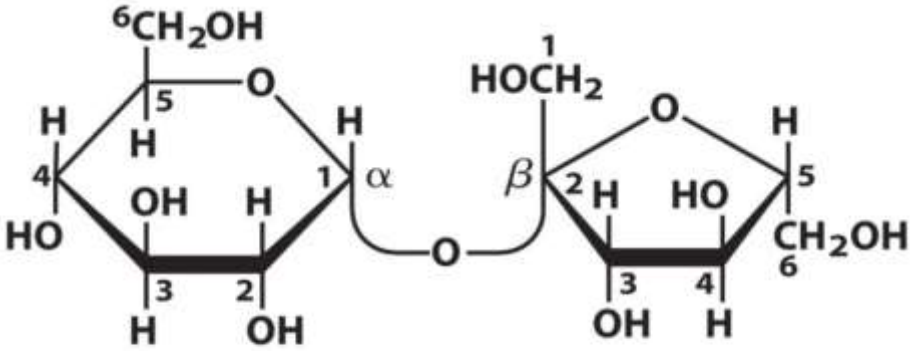
- Lactose – 1 glucose & 1 galactose on hydrolysis
- Sucrose – 1 glucose & 1 fructose on hydrolysis

Carbohydrates



Lactose

Galactose Glucose



Sucrose

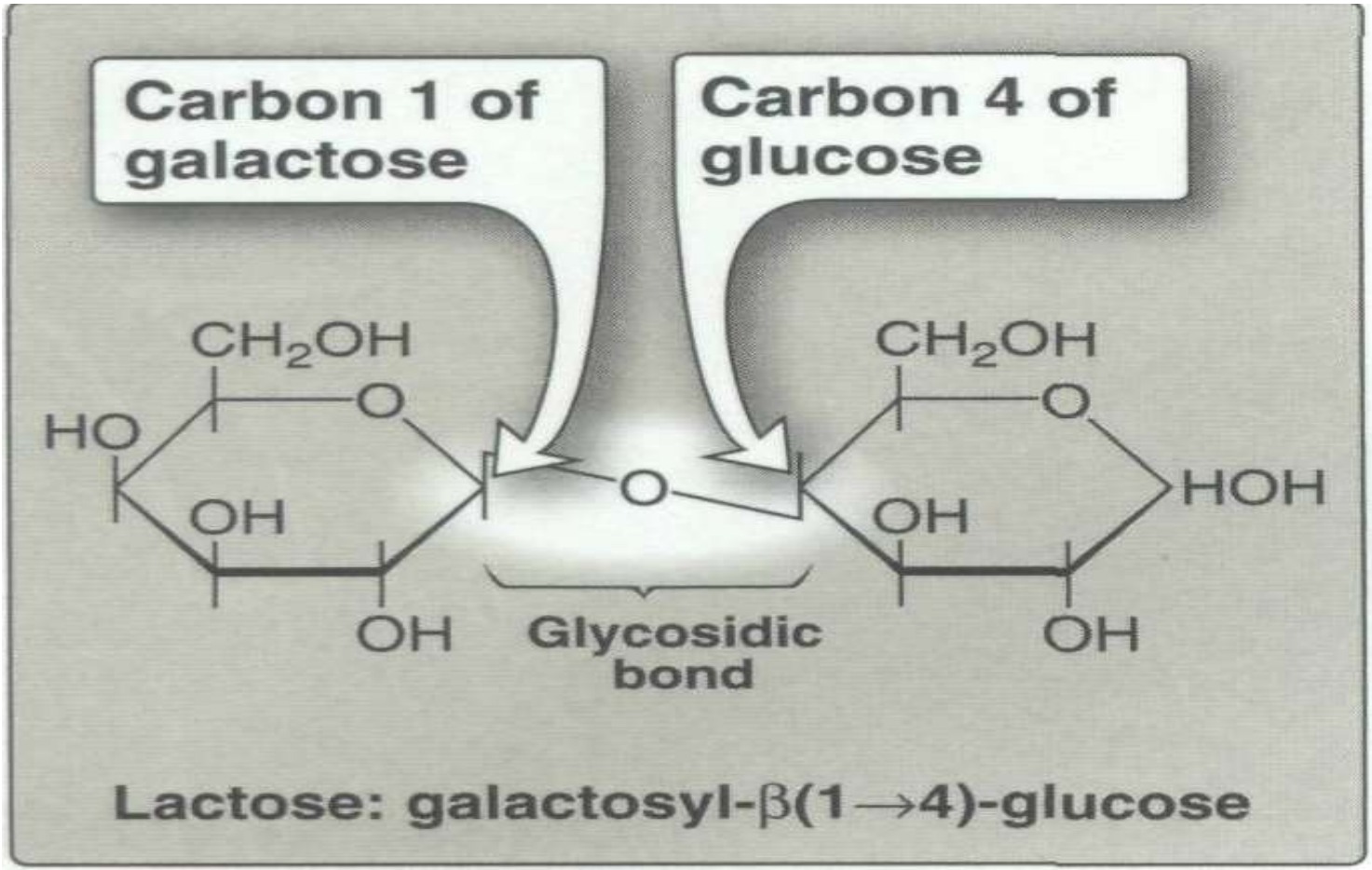
α-D-glucopyranosyl β-D-fructofuranoside

Glc(α1↔2β)Fru

Carbohydrates

- **Oligosaccharides** are short chains of monosaccharide units or residues (3-10) joined by characteristic linkages called glycosidic linkage

Carbohydrates

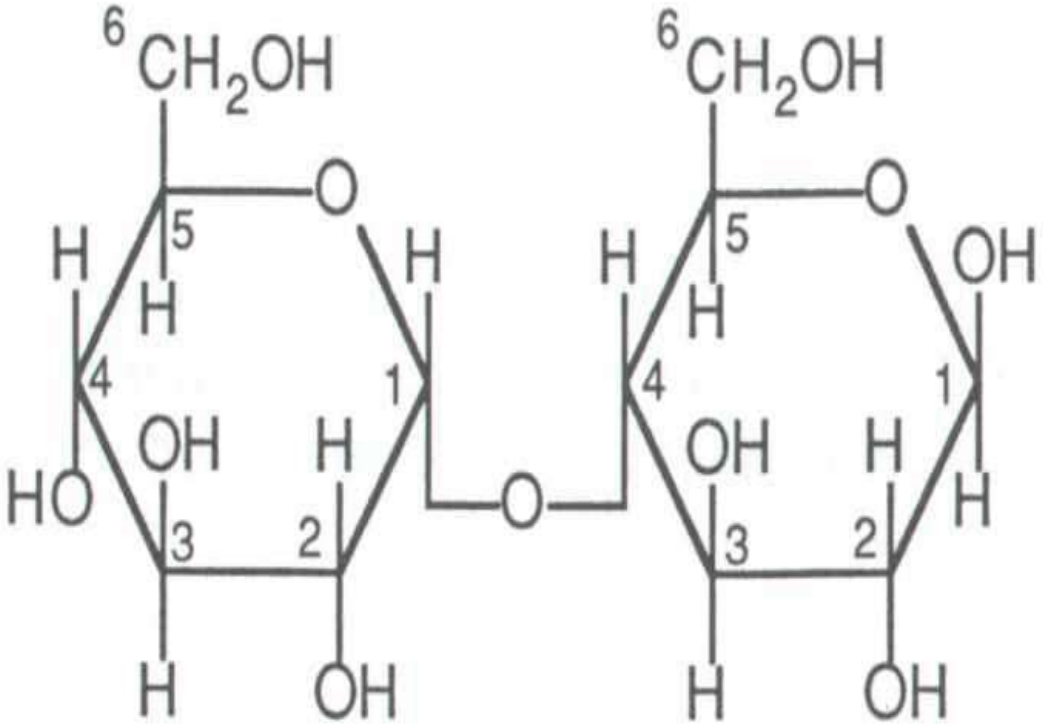


Carbohydrates

Glycosidic bond

of maltose is formed between the OH of carbon 1 and carbon 4 of 2nd glucose monomers. Thereforer, it forms an alpha (1– 4) glycosidic bond.

Carbohydrates



Maltose

Carbohydrates

- **Polysaccharides** (Glycans) are those which yield more than 10 molecules of monosaccharides on hydrolysis
- Some have hundreds or thousands of units
- Some polysaccharides, such as ***cellulose***, are linear chains; others, such as ***glycogen***, are branched

Carbohydrates

- Both glycogen and cellulose consist of recurring units of D-glucose, but they differ in the type of glycosidic linkage and consequently have strikingly different properties and biological roles

Carbohydrates

- Polysaccharides are further divided into two groups:

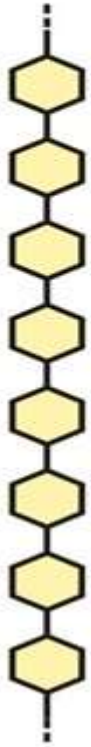
1. Homopolysaccharides (homoglycans): Polymers of same monosaccharide units e.g. starch, glycogen, inulin, cellulose, dextrins, dextrans

2. Heteropolysaccharides (heteroglycans): Polymer of different monosaccharide units or their derivatives e.g. Mucopolysaccharides (glycosaminoglycans)

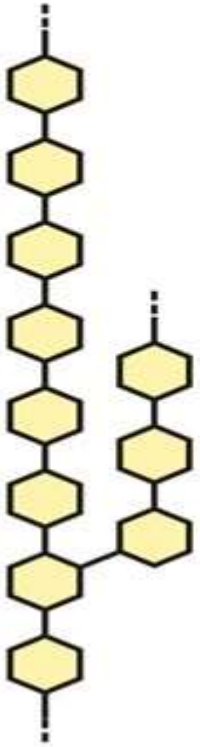
Carbohydrates

Homopolysaccharides

Unbranched

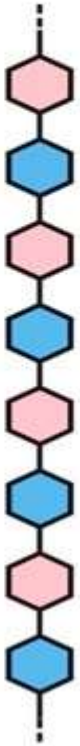


Branched

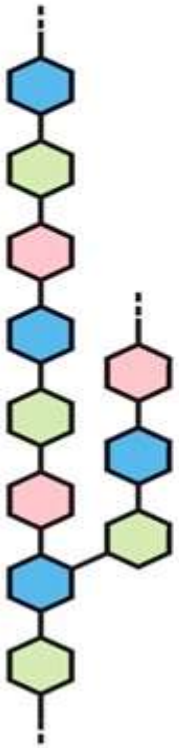


Heteropolysaccharides

Two monomer types, unbranched



Multiple monomer types, branched



Carbohydrates

Biomedical Importance

- Chief source of energy
- Constituents of compound lipids and conjugated proteins
- Lactose , the principal sugar of milk
- Degradation products utilized for synthesis of other substances such as fatty acids, amino acids, cholesterol, etc

Carbohydrates

- Constituents of mucopolysaccharides which form the ground substance of mesenchymal tissues
- Certain carbohydrate derivatives are used as drugs like **cardiac glycosides.**

Carbohydrates

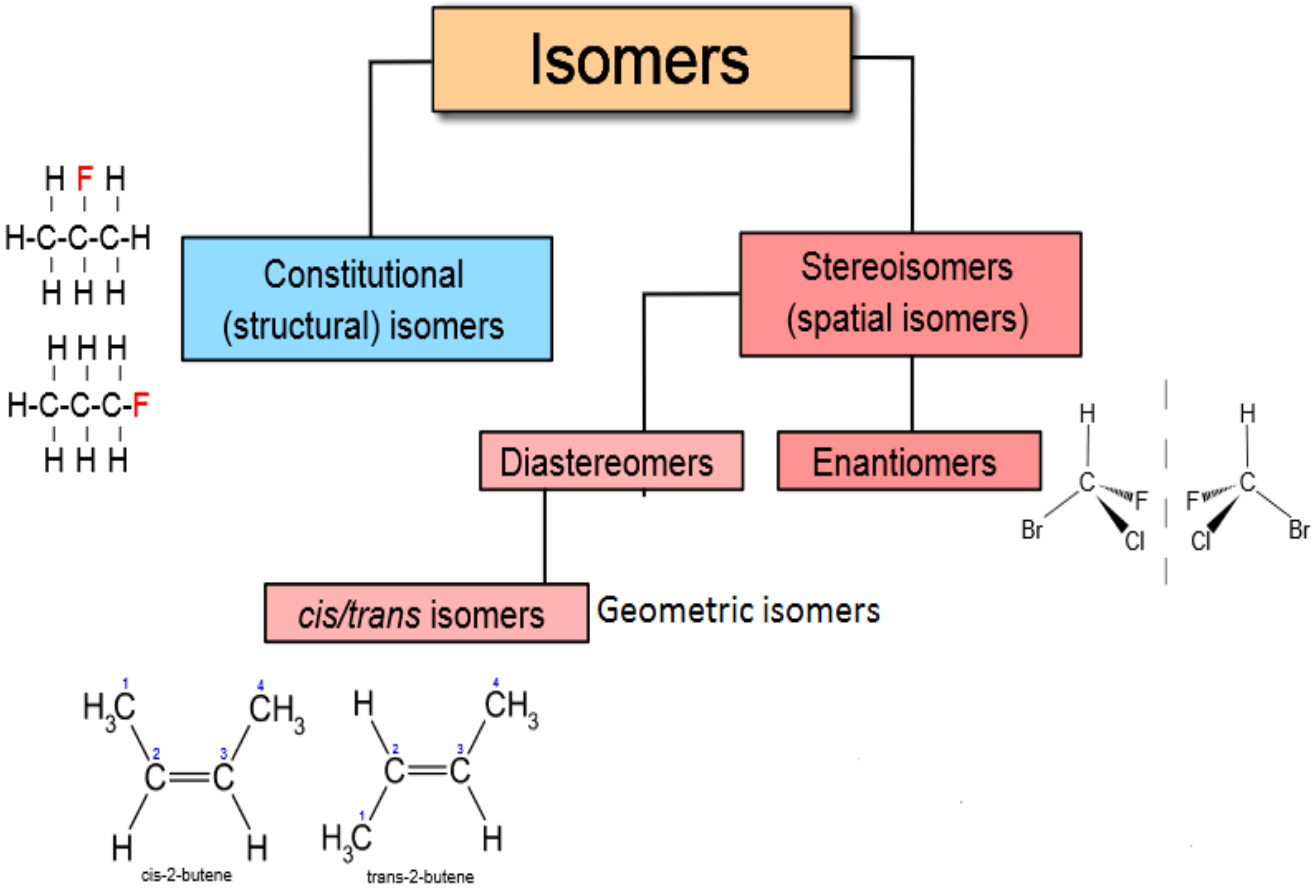
- Inherited deficiency of certain enzymes in metabolic pathways of different carbohydrates can cause diseases e.g. **galactosemia, glycogen storage diseases, lactose intolerance,** etc
- Derangement of glucose metabolism is seen in **Diabetes mellitus**

Carbohydrates

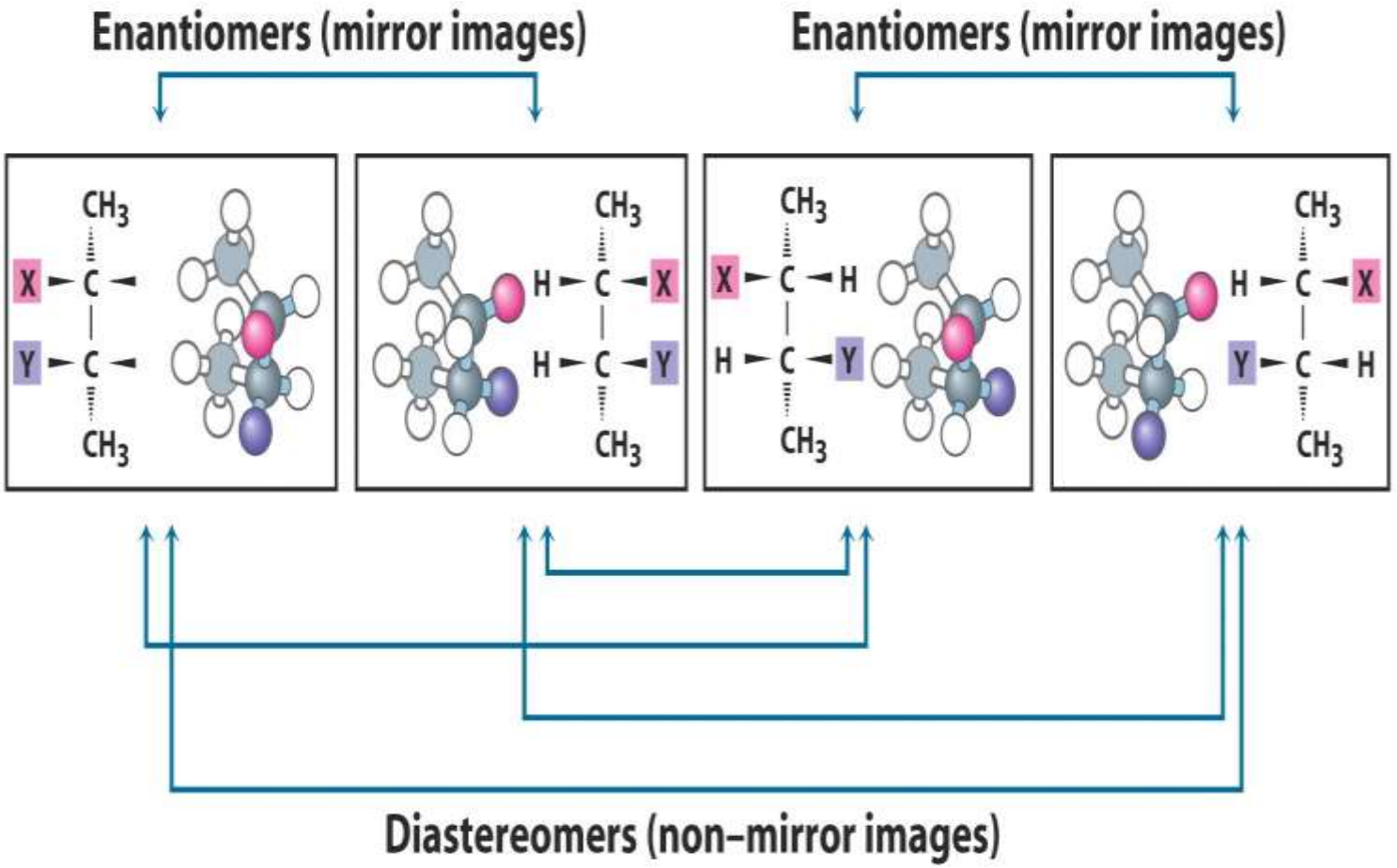
Isomerism

- The existence of two or more compounds having the same molecular formula but a different arrangement of atoms within the molecule

Carbohydrates



Carbohydrates



Carbohydrates

Isomerism

- Compounds which are identical in composition and differ only in configuration (the fixed spatial arrangement of atoms) are called ***stereoisomers***

Carbohydrates

- Stereoisomers that are mirror images of each other are called ***enantiomers***
- Pairs of stereoisomers that are not mirror images of each other are called ***diastereomers***

Carbohydrates

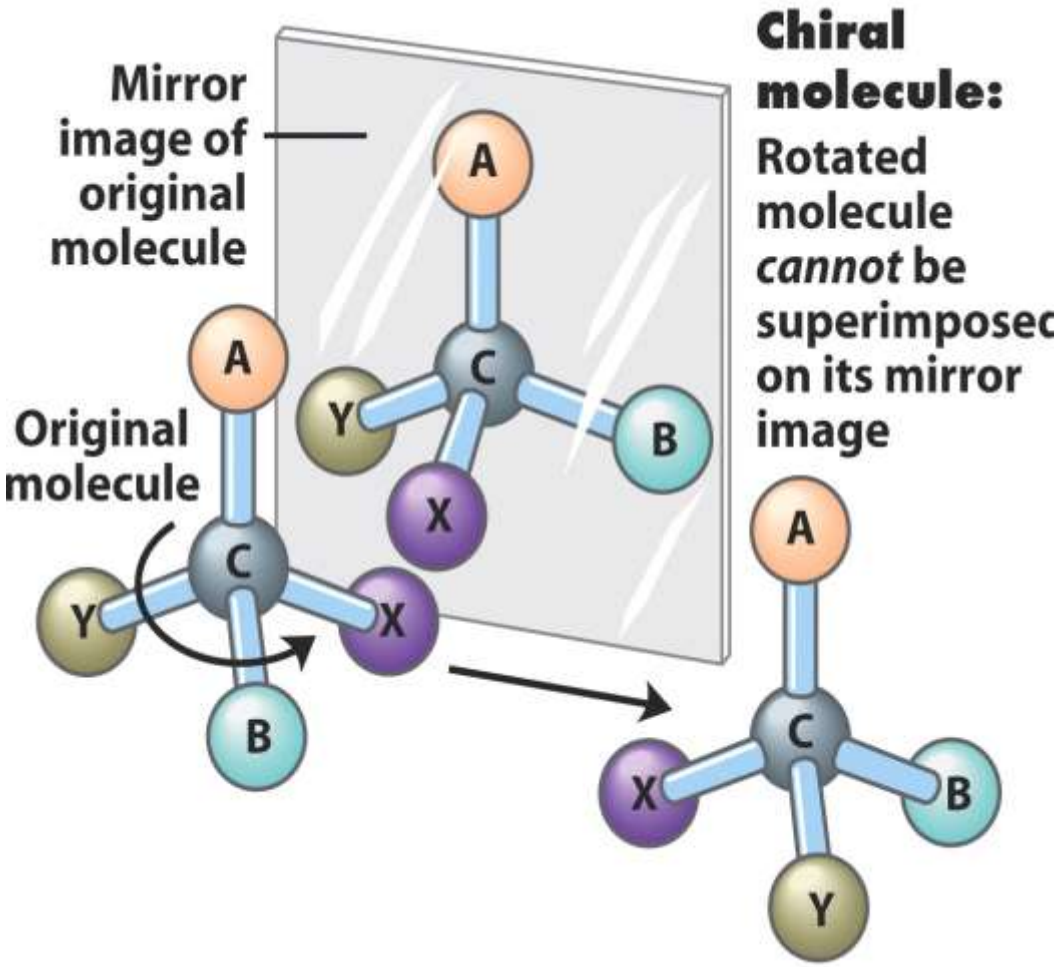
Isomerism

- The identifying characteristic of stereoisomers is that they cannot be interconverted without temporarily breaking one or more covalent bonds

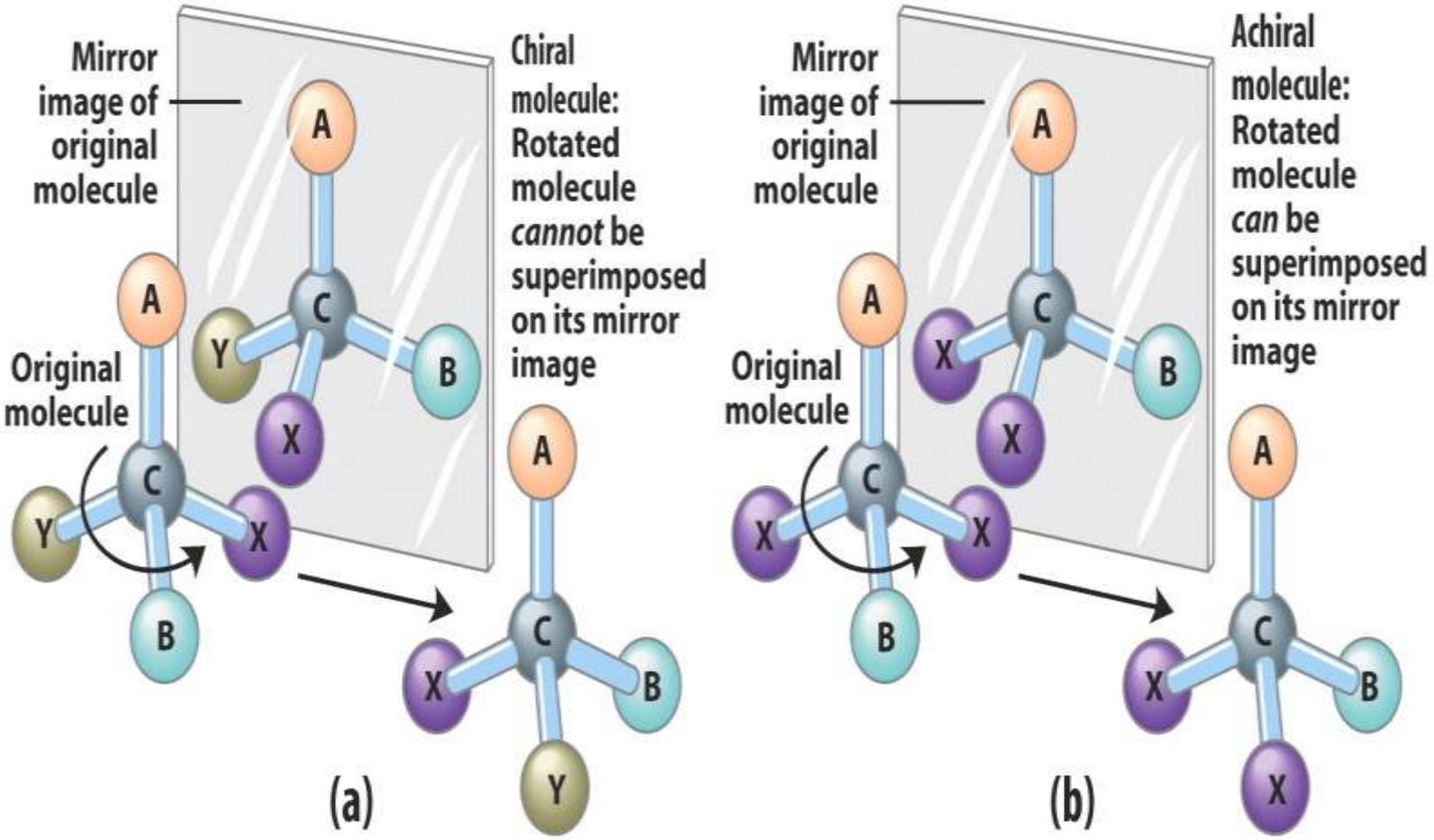
Carbohydrates

- A carbon atom to which four different atoms or groups of atoms are attached is said to be **asymmetric** or ***chiral***

Carbohydrates



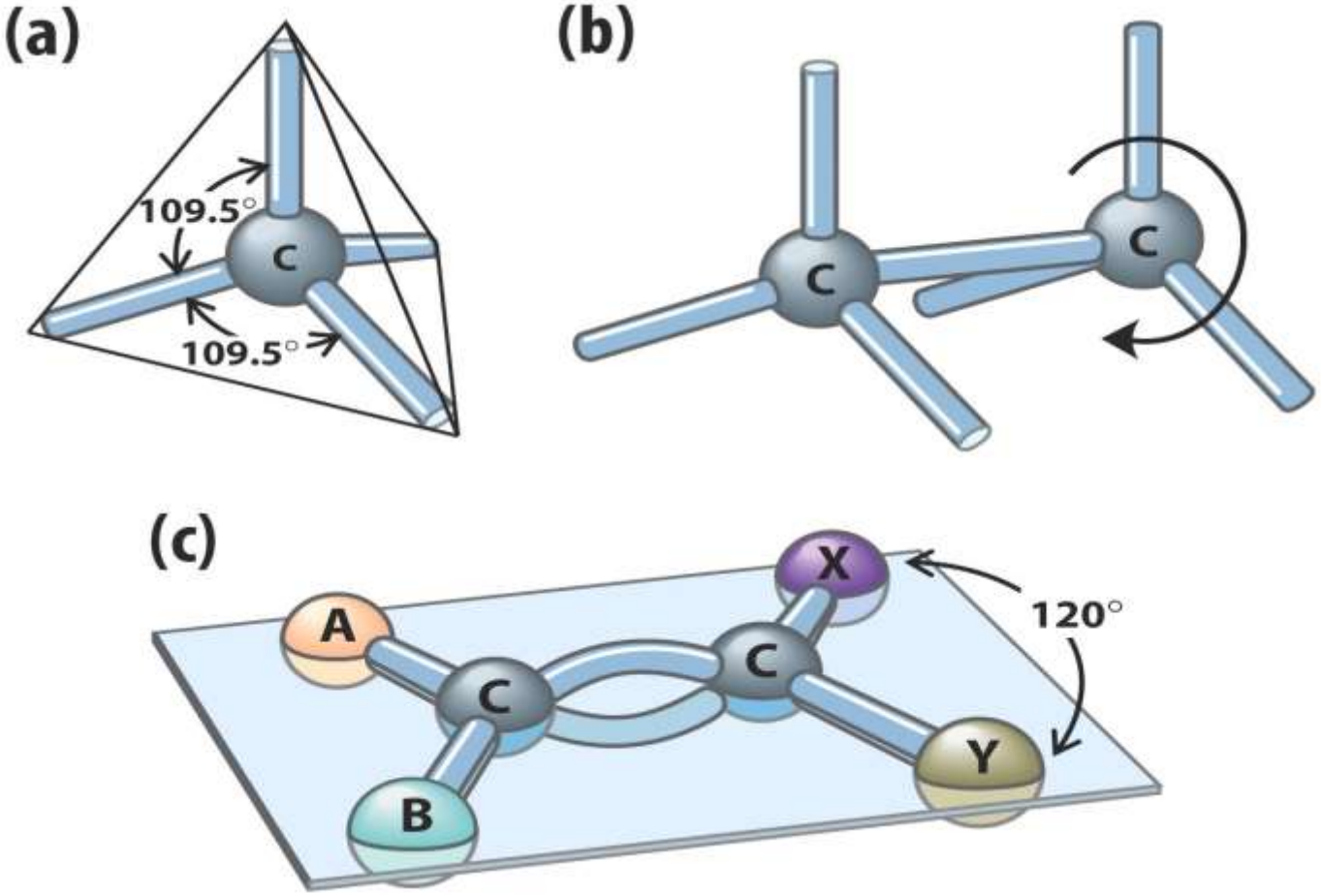
Carbohydrates



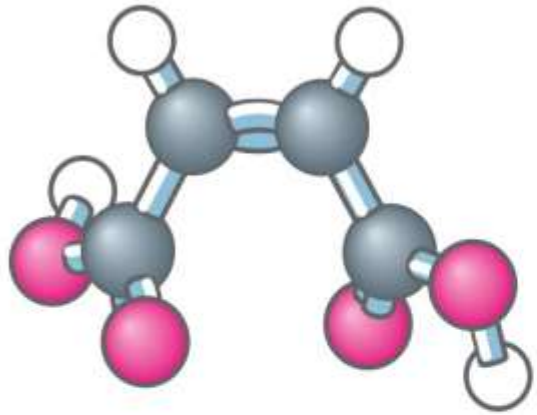
Carbohydrates

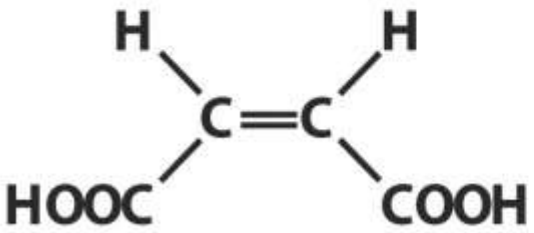
- Configuration is conferred by the presence of either
  - (a) ***chiral centers,*** around which substituent groups are arranged in a specific orientation
  - (b) ***double bonds,*** around which there is no freedom of rotation

Carbohydrates

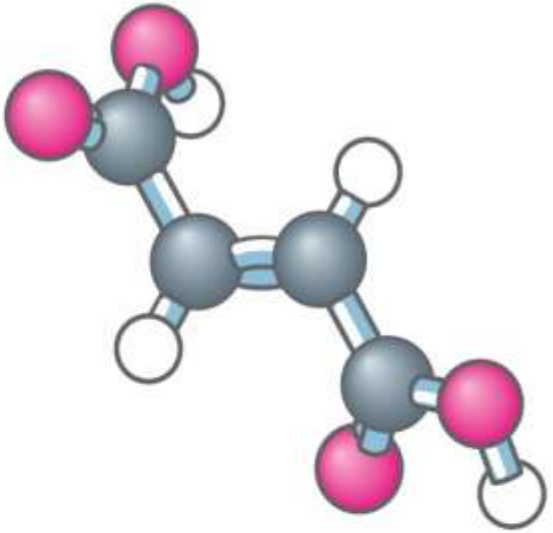


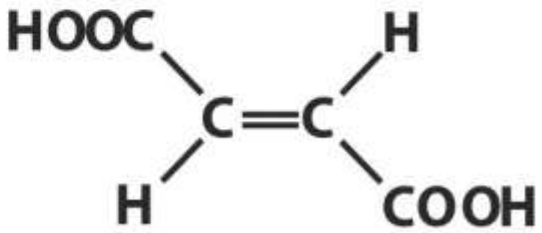
Carbohydrates





Maleic acid (cis)





Fumaric acid (trans)

Carbohydrates

- **Geometric isomers;** differ in the arrangement of their substituent groups with respect to the nonrotating double bond (each is a well-defined compound that can be separated from the other, and each has *its own unique chemical properties*)

Carbohydrates

- The identifying characteristic of configurational isomers is that they cannot be interconverted without temporarily breaking one or more covalent bonds.

Carbohydrates

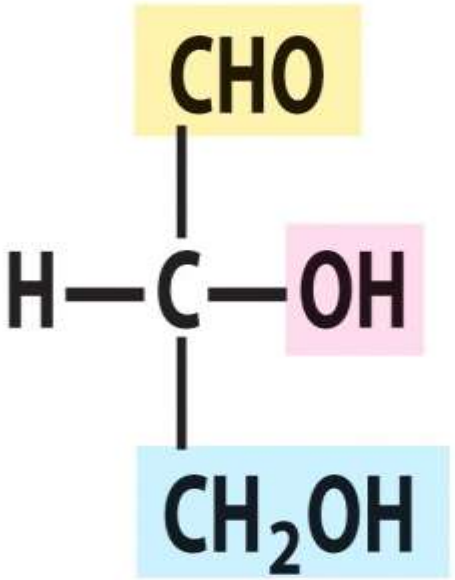
- The configurations of maleic acid and its isomer, fumaric acid.
- These compounds are **geometric,** or **cis-trans, isomers;**
- They differ in the arrangement of their substituent groups with respect to the nonrotating double bond

Carbohydrates

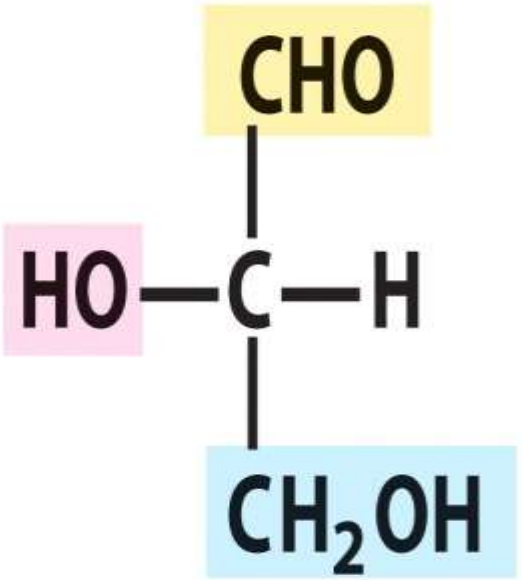
D and L isomerism:

- The designation of a sugar isomer as the D form or as the L form is determined by its spatial relationship to the parent compound of the carbohydrates, the glyceraldehyde

Carbohydrates



D-Glyceraldehyde



L-Glyceraldehyde

Fischer projection formulas

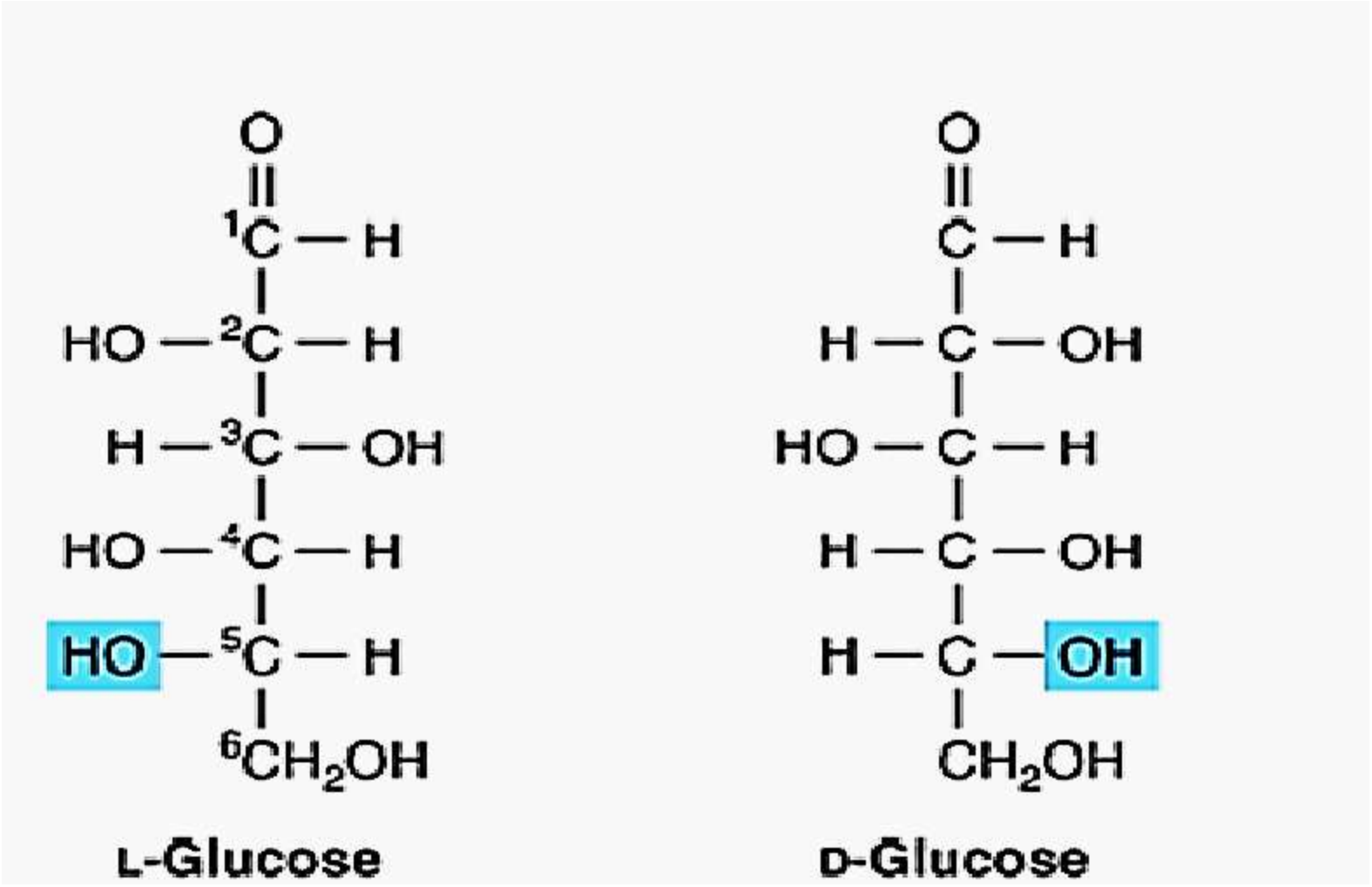
Carbohydrates

- The orientation of the —H and —OH groups around the ***carbon atom adjacent to the terminal primary alcohol or carbon most distant from the carbonyl carbon – the penultimate carbon***

Carbohydrates

- ***(carbon 5 in glucose)*** determines whether the sugar belongs to the D or L series.
- When the —OH group on this carbon is on the right the sugar is the D isomer;
- when it is on the left, it is the L isomer

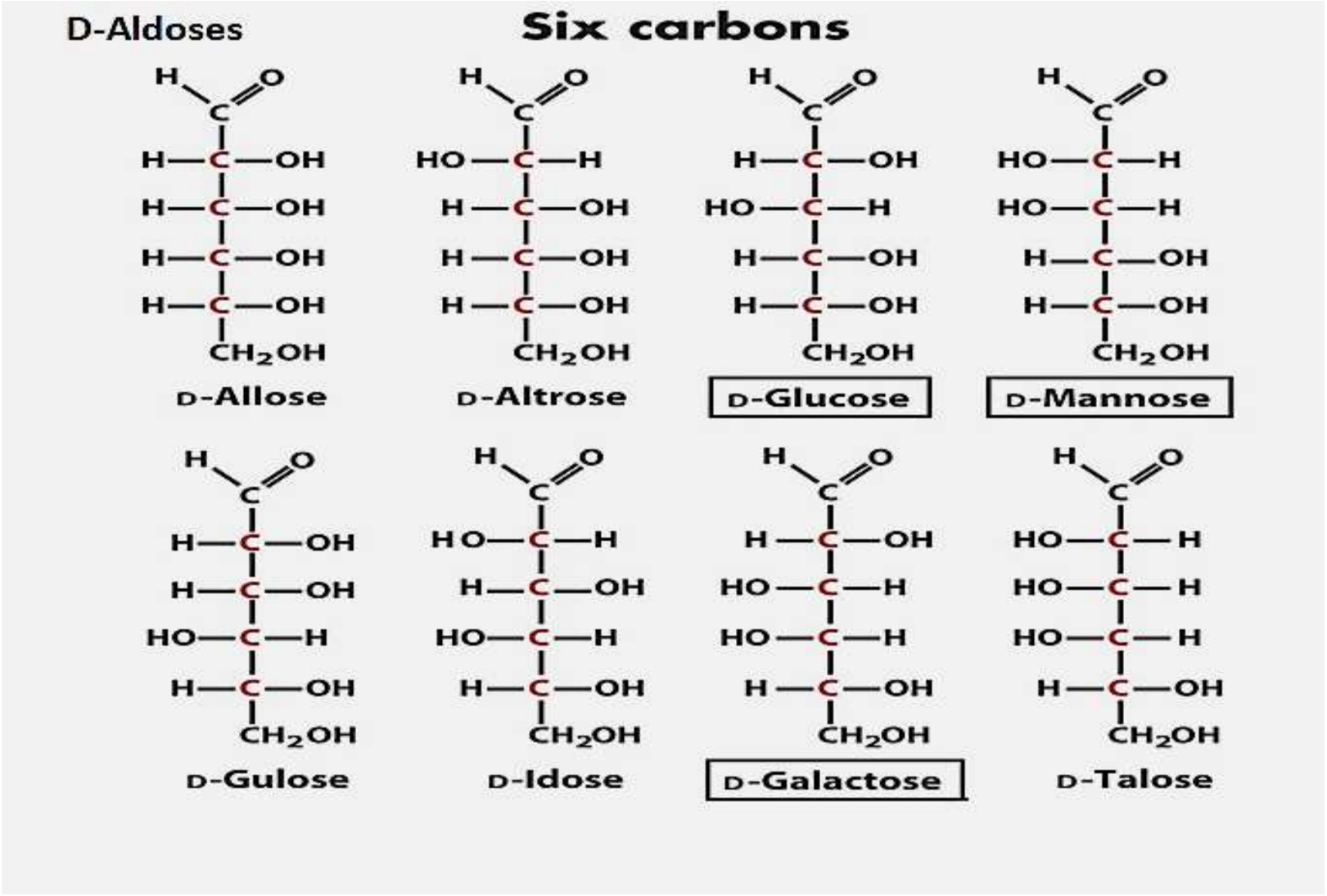
Carbohydrates



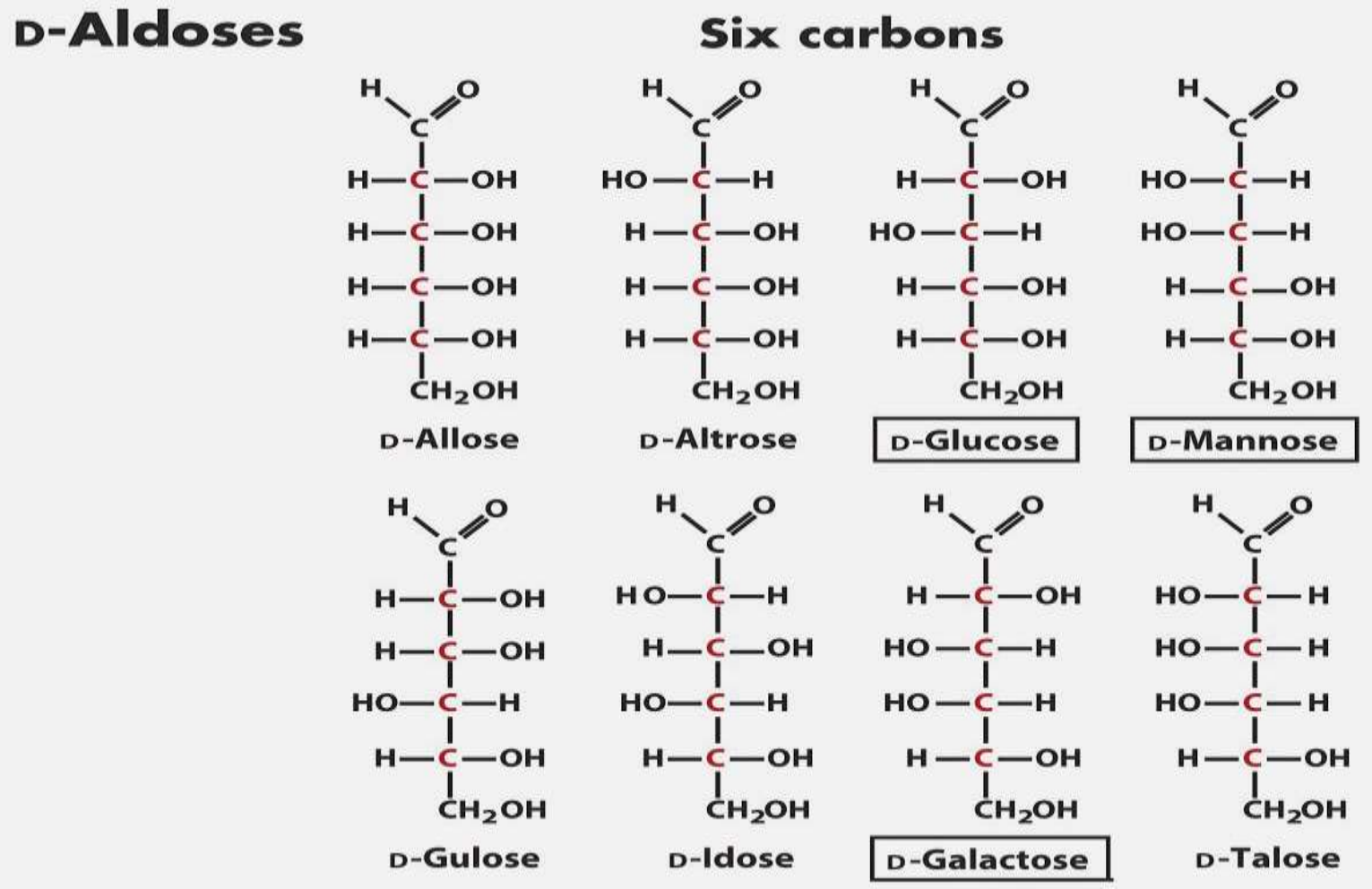
Carbohydrates

- Of the 16 possible aldohexoses eight are D forms and eight are L
- Most of the hexoses of living organisms are D isomers and the enzymes responsible for their metabolism are specific for this configuration

Carbohydrates



Carbohydrates



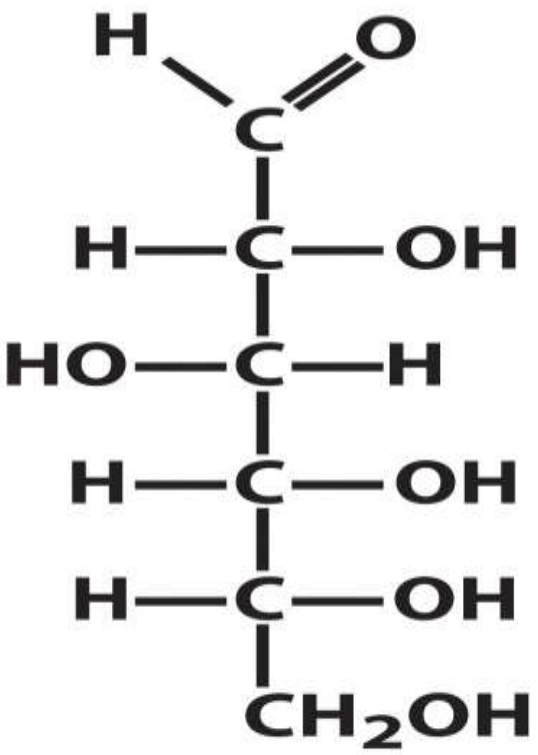
Carbohydrates

- Four different substituents bonded to a chiral carbon atom may be arranged in two different ways in space, that is, have two stereoisomers (with similar or identical chemical properties ***but differing in certain physical and biological properties)***

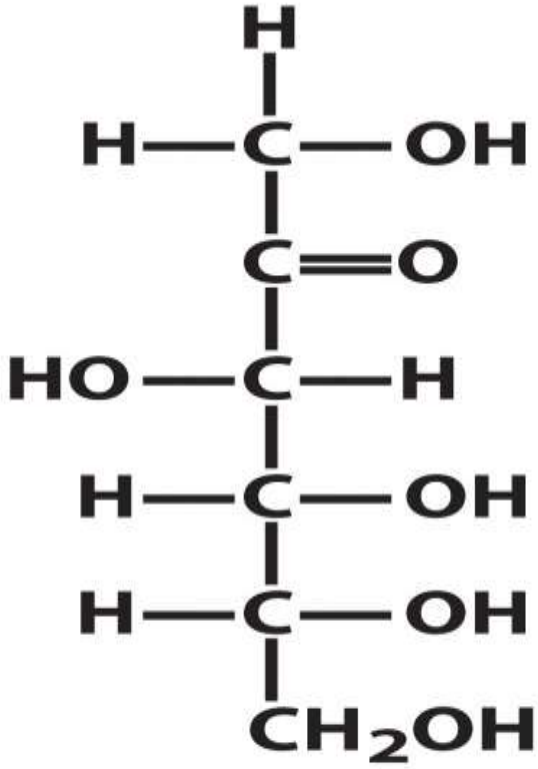
Carbohydrates

- A molecule with only one chiral carbon can have two stereoisomers; when two or more (n) chiral carbons are present, there can be 2^n stereoisomers

Carbohydrates



D-Glucose,
an aldohexose



D-Fructose,
a ketohexose

Carbohydrates

- **Enantiomers** have nearly identical chemical properties but differ in a characteristic physical property:
  - their interaction with plane-polarized light

Carbohydrates

- Thus the presence of asymmetric carbon atoms also confers **optical activity** on the compound

Carbohydrates

- When a beam of plane-polarized light is passed through a solution of an **optical isomer,** it rotates either to the right, dextrorotatory (+), or to the left, levorotatory (–)

Carbohydrates

- The direction of rotation of plane polarized light is independent of the stereochemistry of the sugar, so it may be designated D(–), D(+), L(–), or L(+)

Carbohydrates

- For example, the naturally occurring form of fructose is the D(–) isomer
- In solution, glucose is dextrorotatory, and glucose solutions are sometimes known as **dextrose**

Carbohydrates

- When equal amounts of dextrorotatory and levorotatory isomers are present, the resulting mixture has no optical activity

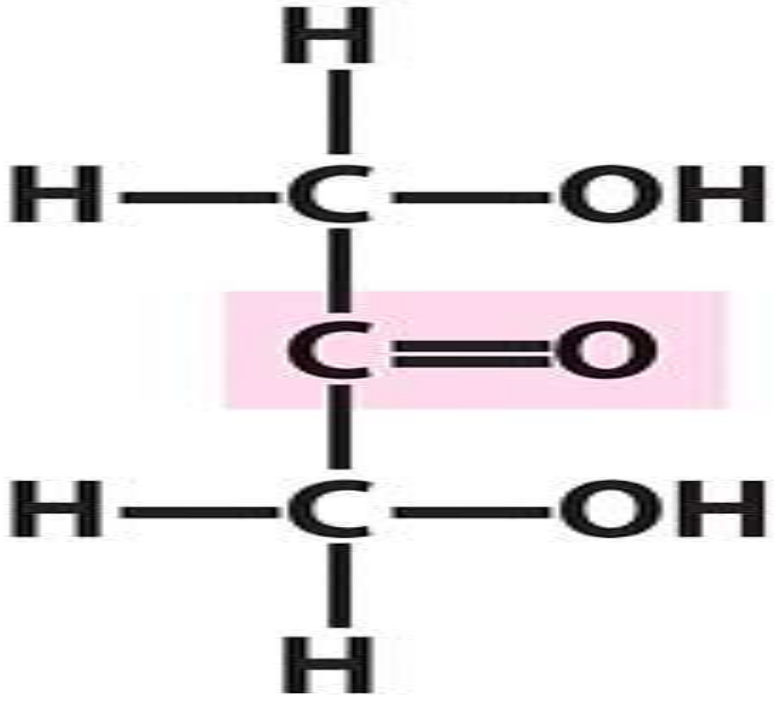
Carbohydrates

- Such a mixture is said to be ***racemic*** and separation of optically active isomers from a racemic mixture is called ***resolution***

Carbohydrates

- All the monosaccharides except dihydroxyacetoe contain one or more asymmetric (chiral) carbon atoms and thus occur in optically active isomeric forms

Carbohydrates

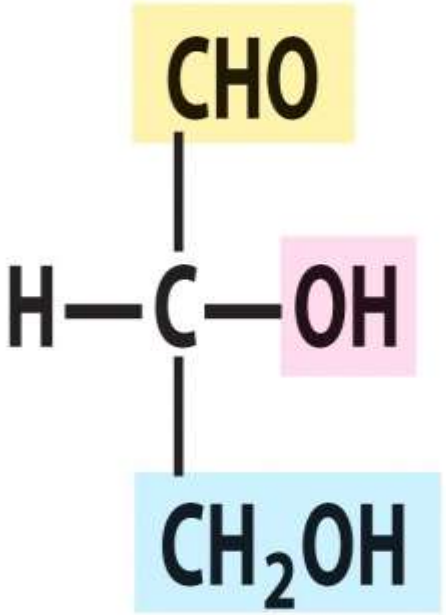


Dihydroxyacetone, a ketotriose

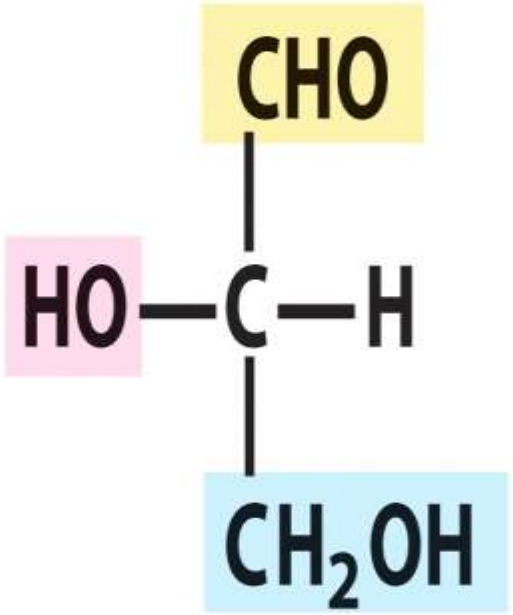
Carbohydrates

- The simplest aldose, glyceraldehyde contains one chiral center (the middle carbon atom) and therefore has two different optical isomers

Carbohydrates



D-Glyceraldehyde



L-Glyceraldehyde

Fischer projection formulas

Carbohydrates

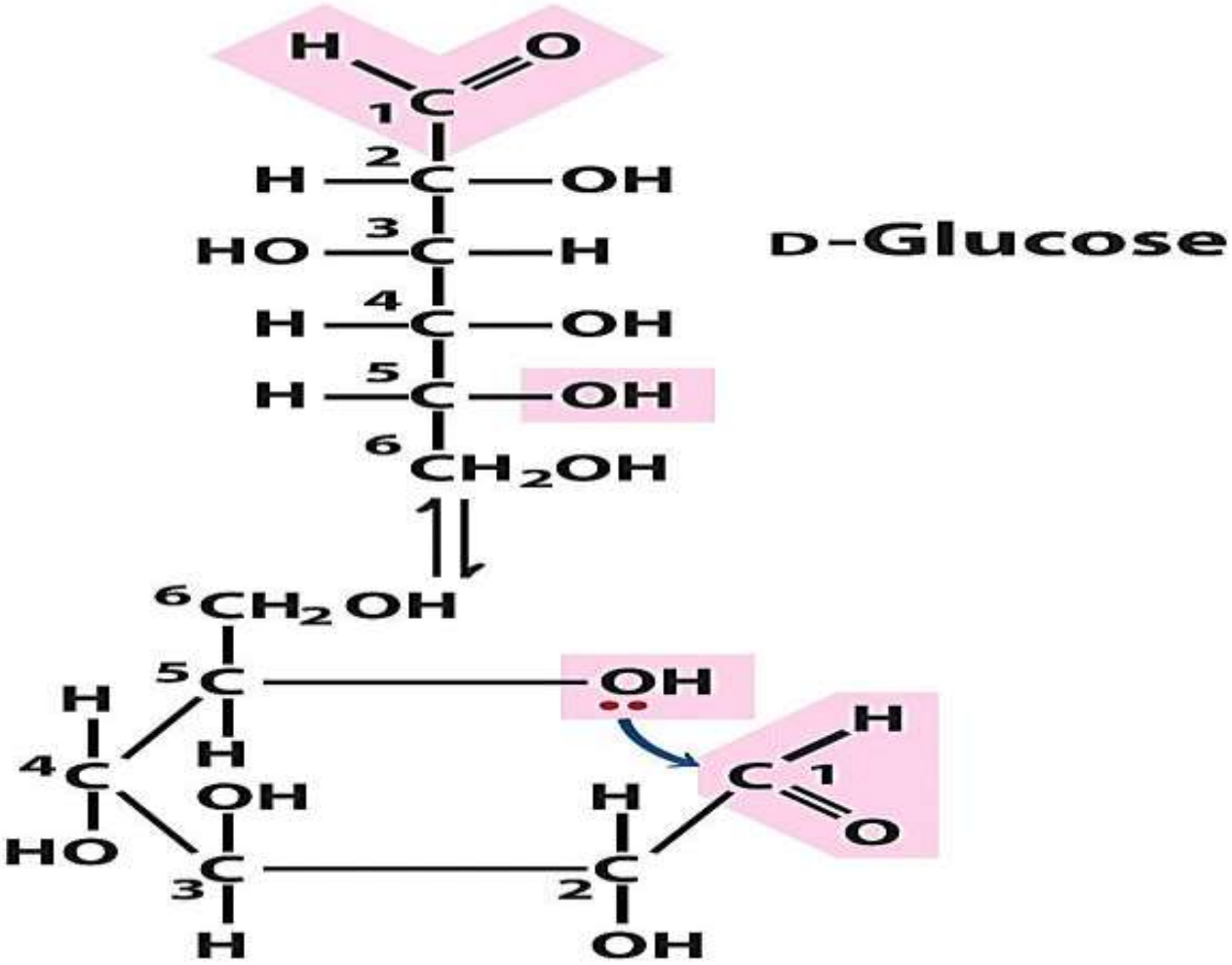
- In aqueous solution, aldotetroses and all monosaccharides with five or more carbon atoms in the backbone occur predominantly as ***cyclic (ring) structures***.

Carbohydrates

- In the ring structure the carbonyl group has formed a **covalent bond** with the oxygen of a hydroxyl group along the chain.

- Therefore forming an ether linkage

Carbohydrates



Carbohydrates

- As the two reacting groups, the aldehyde and the alcoholic group, belong to the same molecule, a cyclic structure forms

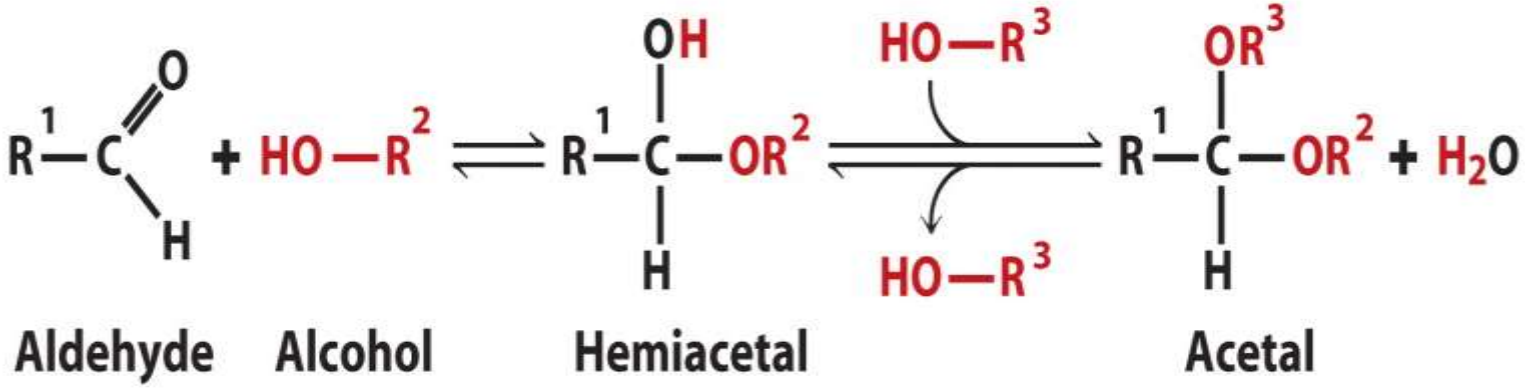
Carbohydrates

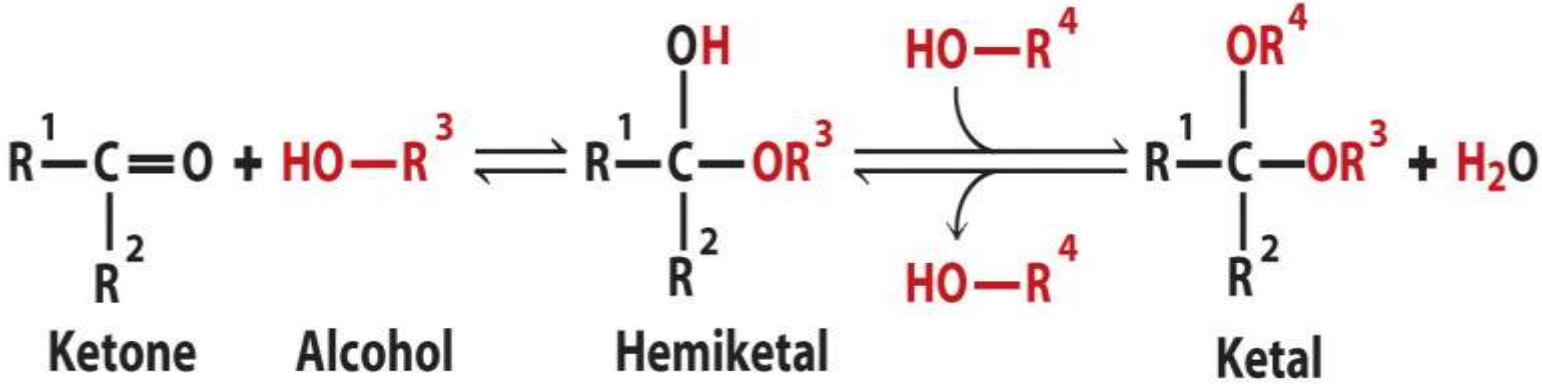
- The formation of these ring structures is the result of a general reaction between alcohols and aldehydes or ketones to form derivatives called ***hemiacetals or hemiketals.***

Carbohydrates

- Which contain an additional ***asymmetric carbon atom*** and thus can exist in two stereoisomeric forms

Carbohydrates





Carbohydrates

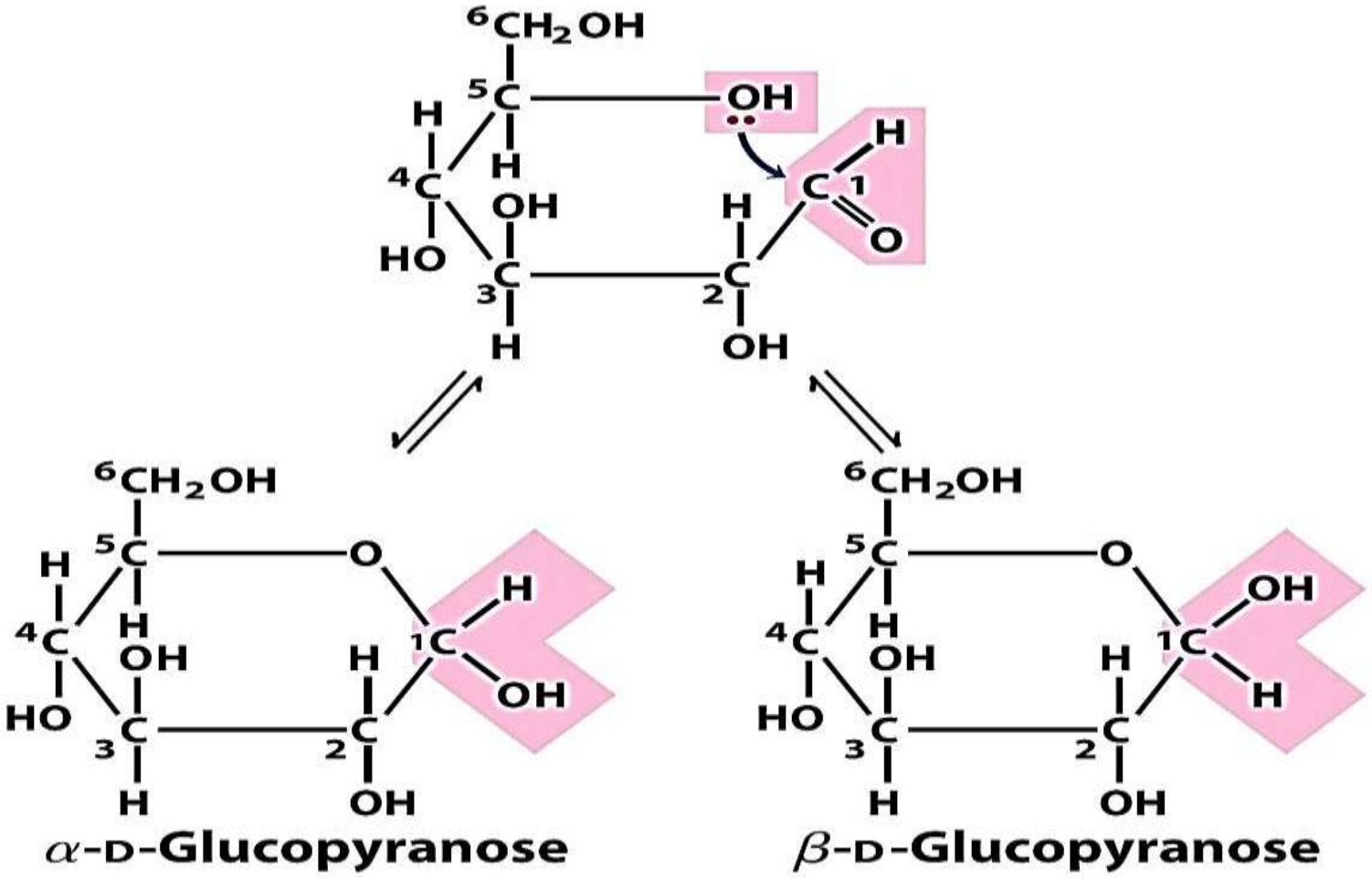
- For example,D-glucose exists in solution as an **intramolecular hemiacetal** in which the free hydroxyl group at C-5 has reacted with the aldehydic C-l,

Carbohydrates

- Therefore rendering the latter carbon (aldehydic C-l)

  asymmetric
- And producing two stereoisomers, designated ***α and β***

Carbohydrates

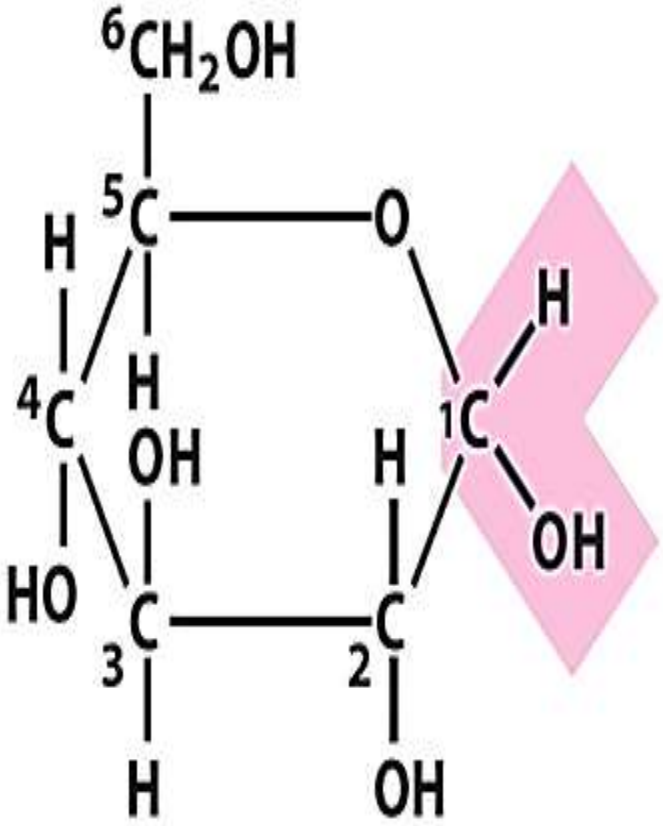


Carbohydrates

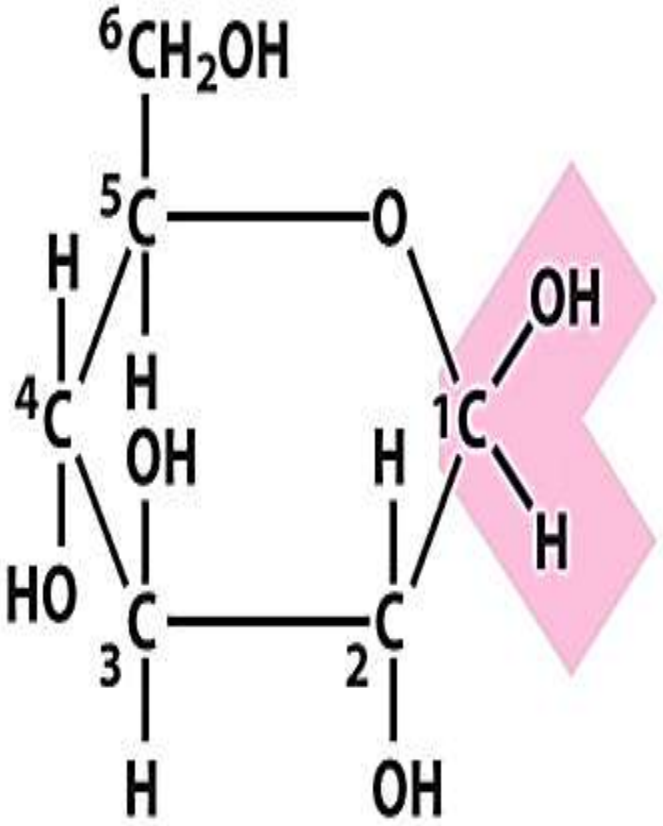
- The designation **α** indicates that the hydroxyl group at the C-1 is **trans** to the CH_2O.
- Whereas **β** indicates that the hydroxyl group at the C-1 is **cis** to the CH_2O

Carbohydrates

- The ordinary crystalline glucose happens to be in the α form



α-D-Glucopyranose



β-D-Glucopyranose

Carbohydrates

- Carbon 1, after cyclization has four different groups attached to it and thus it also becomes ‘asymmetric’

Carbohydrates

- *I*someric forms of monosaccharides that differ only in their configuration about the hemiacetal or hemiketal carbon atom are called **anomers**

Carbohydrates

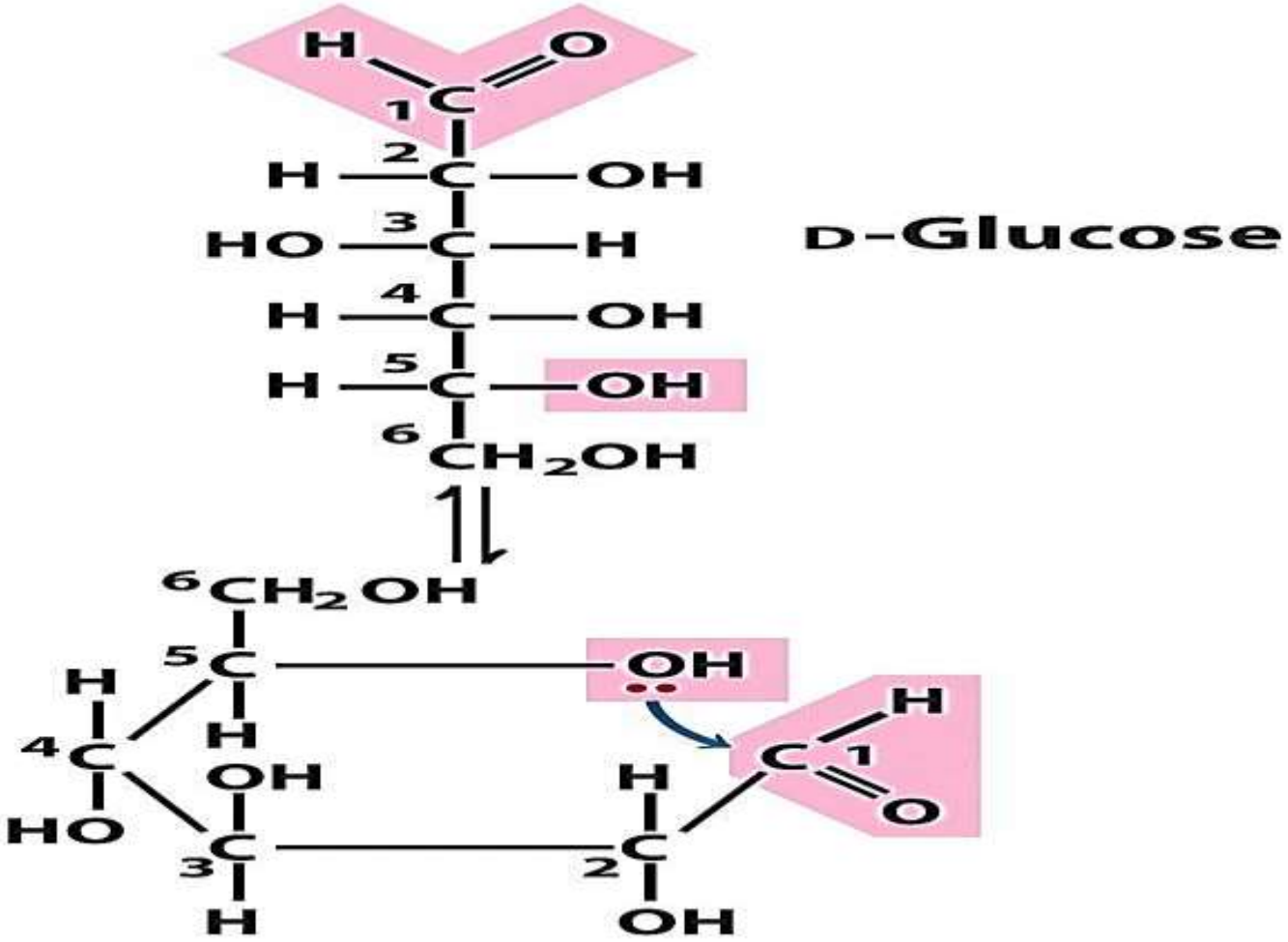
- The carbonyl carbon atom is called the ***anomeric carbon***

Carbohydrates

Ring Structure of Carbohydrates

- For example,D-glucose exists in solution as an **intramolecular hemiacetal** in which the free hydroxyl group at C-5 has reacted with the aldehydic C-l,

Carbohydrates



Carbohydrates

- Therefore rendering the latter carbon (aldehydic C-l)

  asymmetric
- And producing two stereoisomers, designated ***α and β***

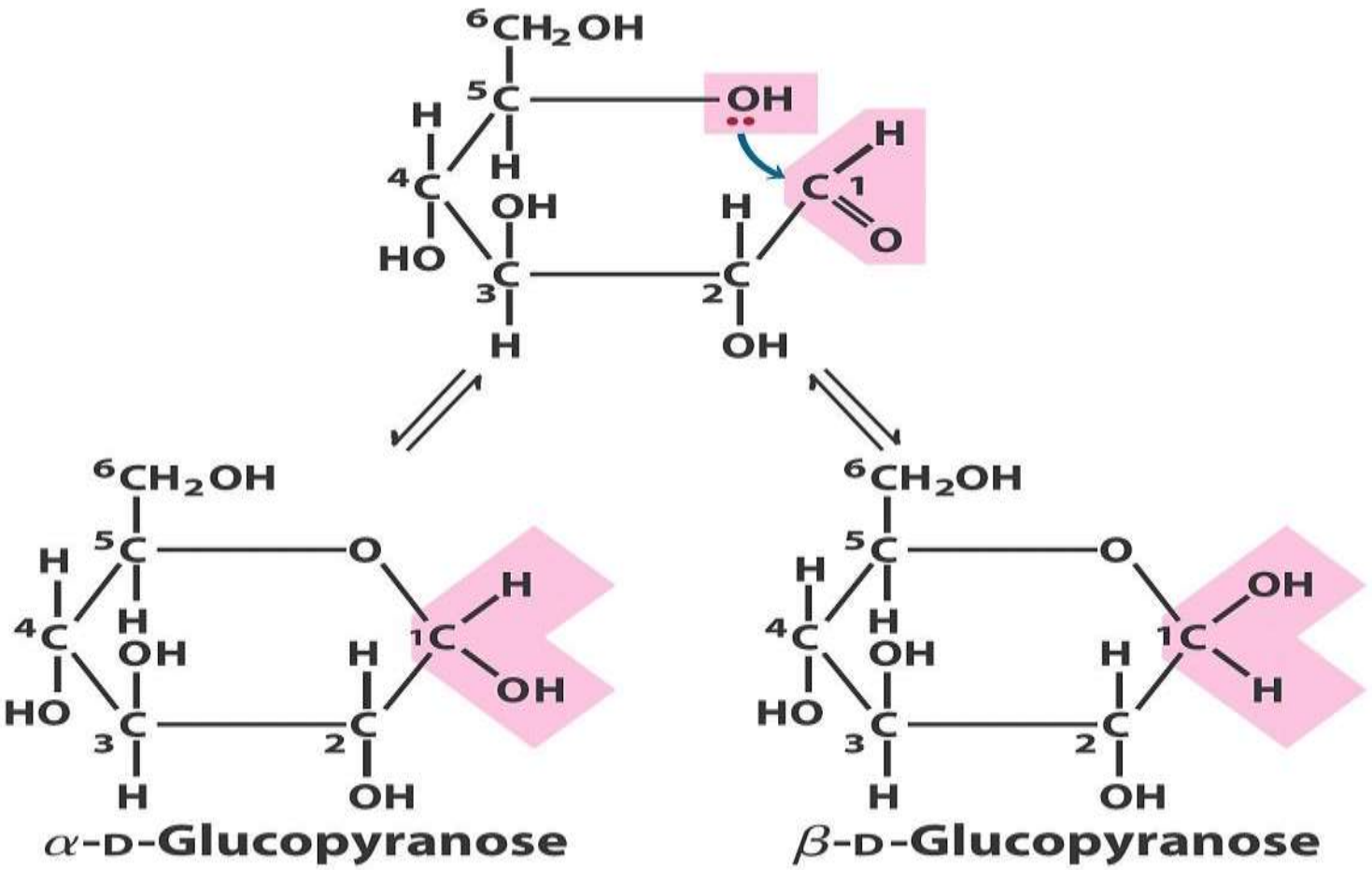
Carbohydrates

- Carbon 1, after cyclization has four different groups attached to it and thus it also becomes ‘asymmetric’
- The newly created assymetric or chiral carbon is called the ***anomeric carbon***

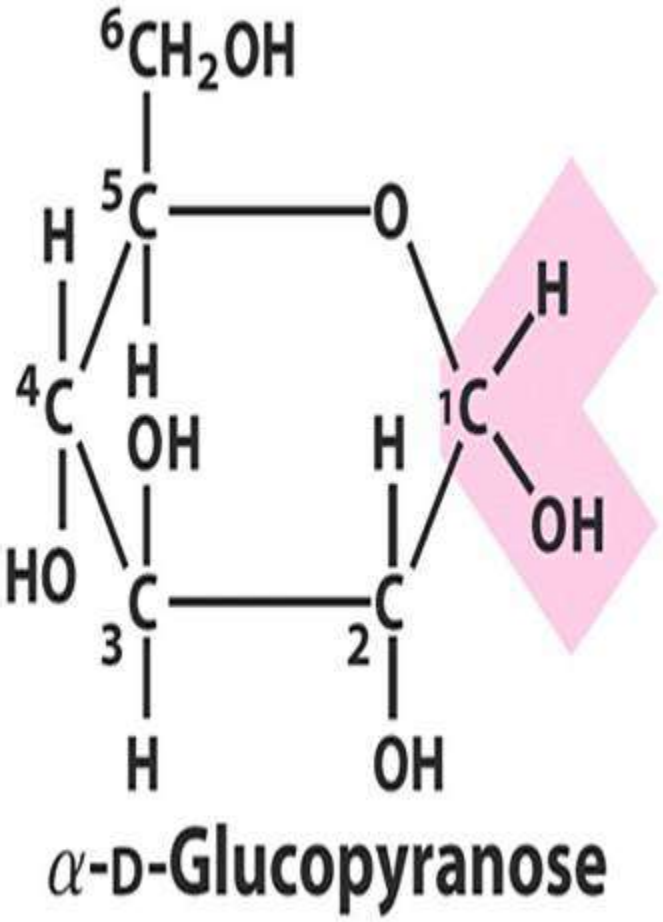
Carbohydrates

- Isomeric forms of monosaccharides that differ only in their configuration about the hemiacetal or hemiketal carbon atom are called **anomers**

Carbohydrates



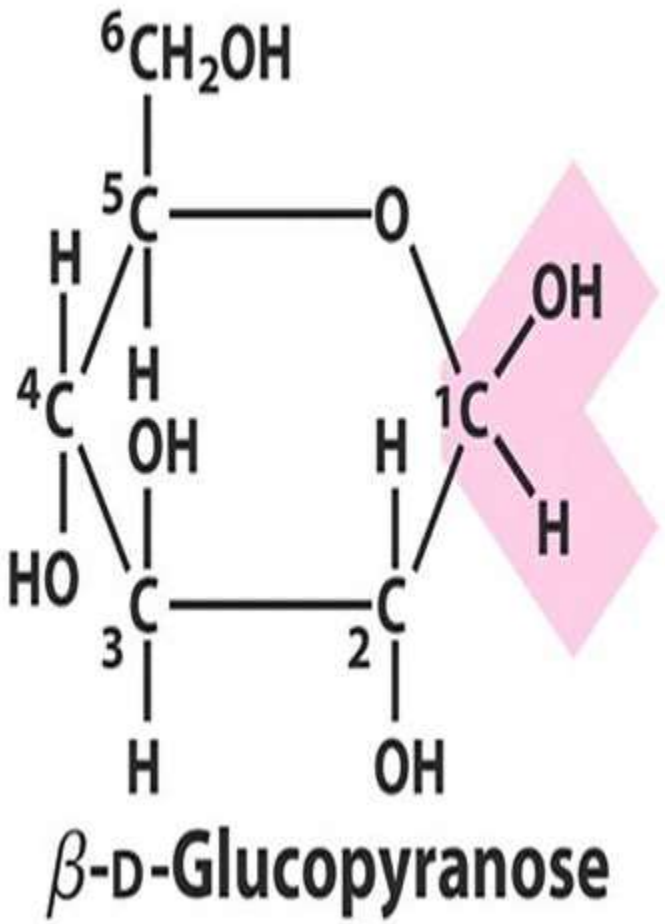
Carbohydrates



α-D-Glucopyranose

- The designation **α** indicates that the hydroxyl group at the C-1 is **trans** to the CH_2O.

Carbohydrates



β-D-Glucopyranose

- Whereas **β** indicates that the hydroxyl group at the C-1 is **cis** to the CH_2O

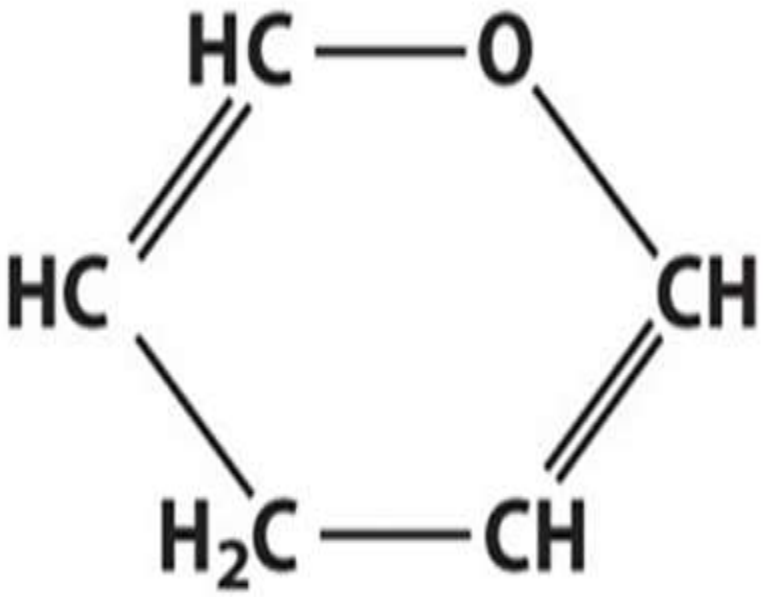
Carbohydrates

- The ordinary crystalline glucose happens to be in the α form

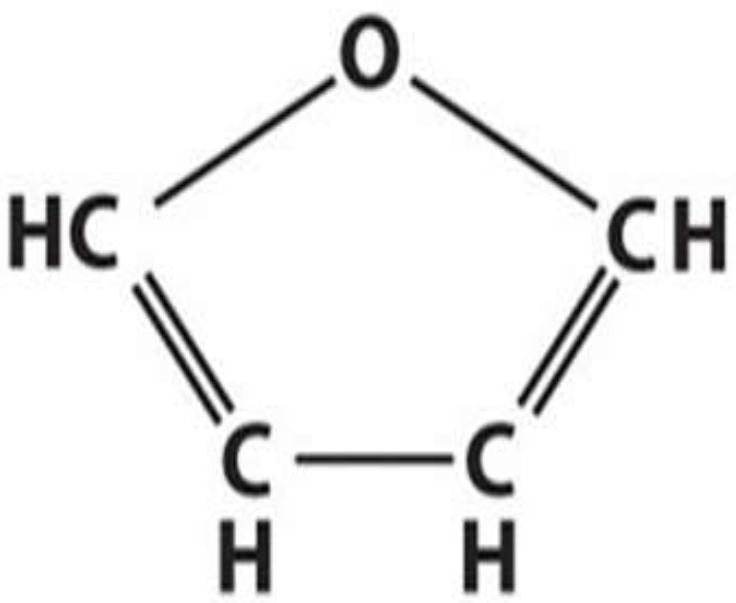
Carbohydrates

- The ring structures of monosaccharides are similar to the ring structures of either pyran (six-membered ring) and are called ***pyranoses*** or furan (a five-membered ring), which are known as ***furanoses***

Carbohydrates



Pyran

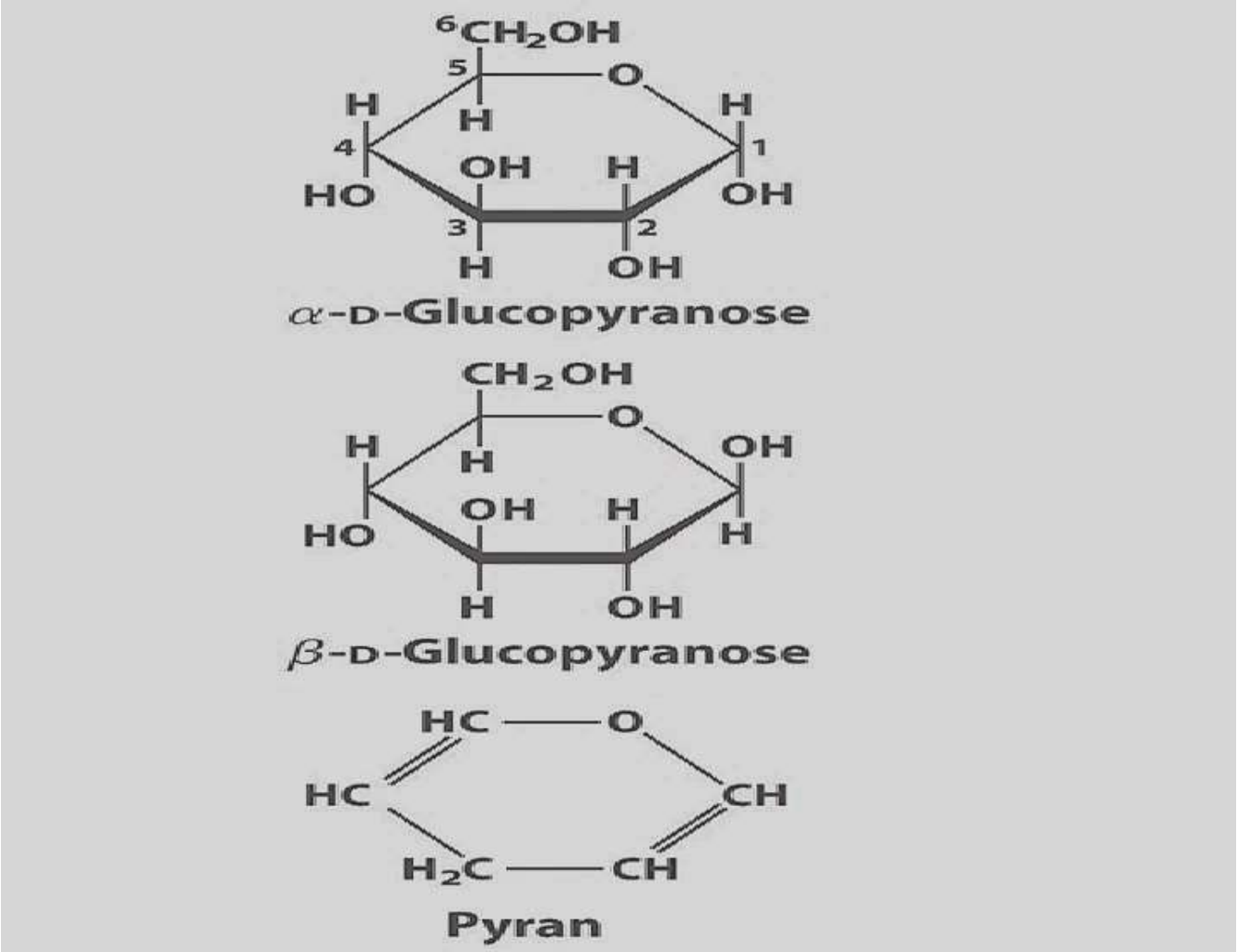


Furan

Carbohydrates

- The six-membered aldopyranose ring is much more stable than the aldofuranose ring
- Therefore predominates in aldohexose and aldopentose solutions
- Similarly Crystalline glucose is **α -D - glucopyranose**

Carbohydrates



Carbohydrates

- The α and *β* anomers of D-glucose interconvert in aqueous solution by a process called **mutarotation.**
- Thus, a solution of *α*-D-glucose and a solution of *β*-D-glucose eventually form identical equilibrium mixtures having identical optical properties.

Carbohydrates

- This mixture consists of about;
- one-third *α*-D-glucose,
- two-thirds *β*-D-glucose, and very small amounts of the linear and five-membered ring (glucofuranose) forms.

Carbohydrates

- α and β forms are NOT mirror images and are referred to as ***diastereomers***
- Enzymes are able to distinguish between these two structures and use one or the other preferentially

Carbohydrates

- For example, glycogen is synthesized from α-D-glucopyranose, whereas cellulose is synthesized from β-D-glucopyranose

Carbohydrates

- Ketohexoses also occur in α and β anomeric forms
- In these compounds the hydroxyl group at C-5 (or C-6) reacts with the keto group at C-2, forming a furanose (or pyranose) ring containing a hemiketal linkage

Carbohydrates

- D-Fructose readily forms the furanose ring ; the more common anomer of this sugar in combined forms or in derivatives is *β*-D-fructofuranose

Carbohydrates

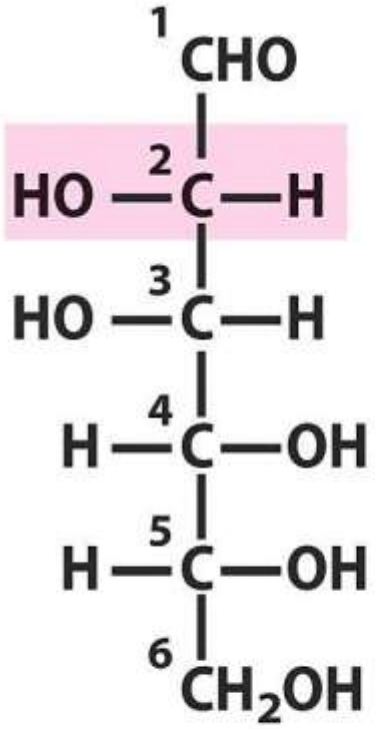
Epimers

- Two sugars which differ from one another only in configuration around a single carbon atom are termed ***epimers***
- *Glucose and Galactose are an example of an epimeric pair which differ only with respect to* C_4

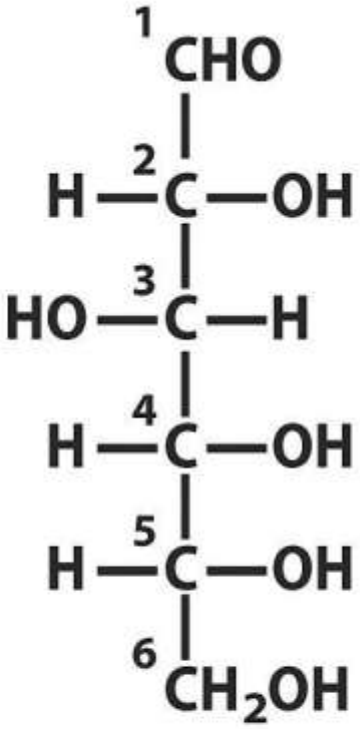
Carbohydrates

- *Similarly Glucose and mannose are epimers with respect to C_2*
- However, Galactose and mannose are NOT epimers as they differ in the position of –OH groups at two carbons (2 and 4), and are therefore defined only as isomers

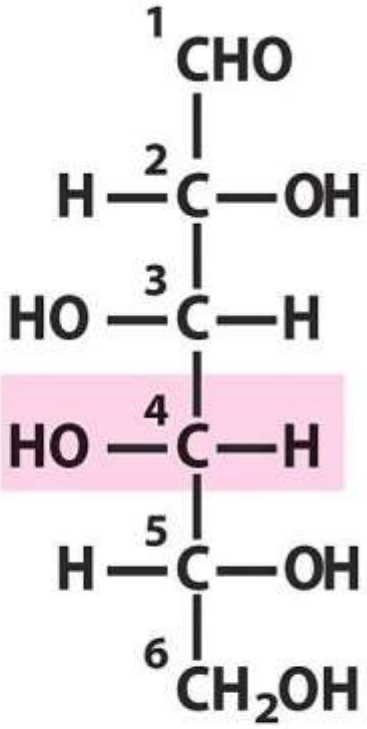
Carbohydrates



D-Mannose
(epimer at C-2)



D-Glucose



D-Galactose
(epimer at C-4)

Carbohydrates

***Trioses*:**

- Both **D-Glyceraldehyde & Dihydroxyacetone** occur as intermediates in glycolysis
- They are also the precursors of glycerol

Carbohydrates

Tetroses:

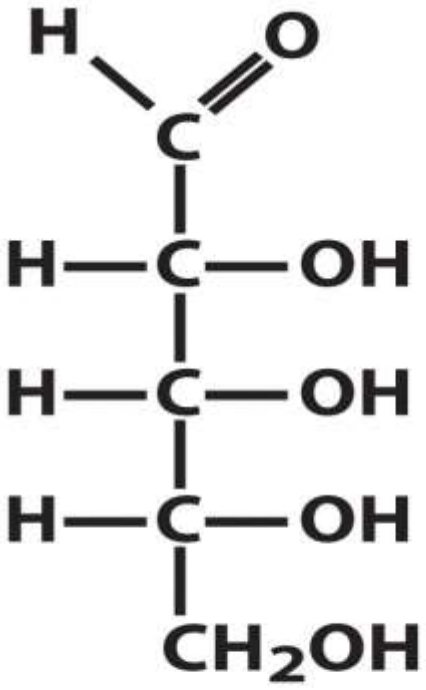
- **Erythrose** occurs as an intermediate in the Hexose Mono Phosohate shunt which is an alternative pathway for glucose oxidation

Carbohydrates

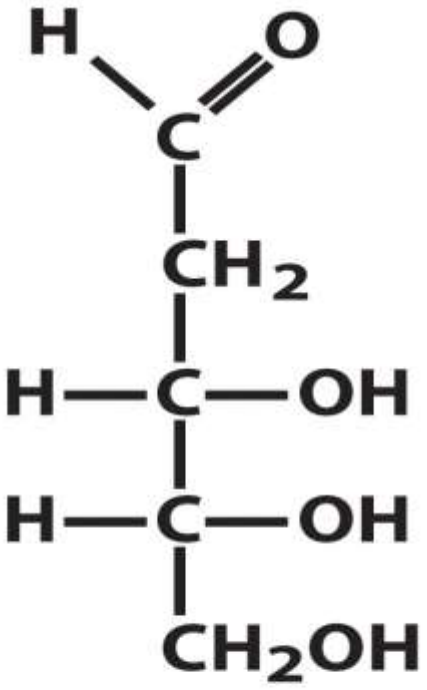
Pentoses*:*

- D-Ribose is a constituent of nucleic acid *RNA* and also certain co-enzymes like *FAD, NAD & co-enzyme A*
- D-2-Deoxyribose is a constituent of *DNA*

Carbohydrates



D-Ribose,
an aldopentose



2-Deoxy-D-ribose,
an aldopentose

Carbohydrates

Hexoses

- D-Glucose (Dextrose, Grape sugar):
- Most dietary carbohydrate is absorbed into the bloodstream as glucose formed by hydrolysis of dietary starch and disaccharides, and other sugars are converted to glucose in the liver

Carbohydrates

- Sources include fruit juices, hydrolysis of starch, cane or beet sugar, maltose and lactose
- Glucose is the major metabolic fuel of mammals

Carbohydrates

- It is the chief physiological sugar present in normal blood at a fairly constant level
- All tissues utilize glucose for energy
- Erythrocyte utilize glucose solely for energy purposes and Brain cells rely heavily on glucose

Carbohydrates

- It is the precursor for synthesis of all the other carbohydrates in the body, including
  - **glycogen** for storage;
  - **ribose** and **deoxyribose** in nucleic acids;
  - **galactose** in lactose of milk, in glycolipids,

Carbohydrates

- And in combination with protein in glycoproteins and proteoglycans
- Stored as Glycogen in liver and muscles only
- Excreted in the urine (glucosuria) in poorly controlled diabetes mellitus as a result of hyperglycemia

Carbohydrates

D-Galactose:

- Occurs as a constituent of milk sugar lactose
- It is an epimer of glucose and differs in orientation of H and OH around C-4
- Readily metabolized to glucose;

Carbohydrates

- Synthesized in the mammary gland for synthesis of lactose in milk
- A constituent of glycolipids and glycoproteins
- **Hereditary galactosemia** as a result of failure to metabolize galactose leads to cataracts

Carbohydrates

D-Fructose:

- Commonly known as *fruit sugar* as it occurs free in fruits, honey etc
- Also occurs as a constituent of sucrose
- It is levorotatory and hence is also called Levulose

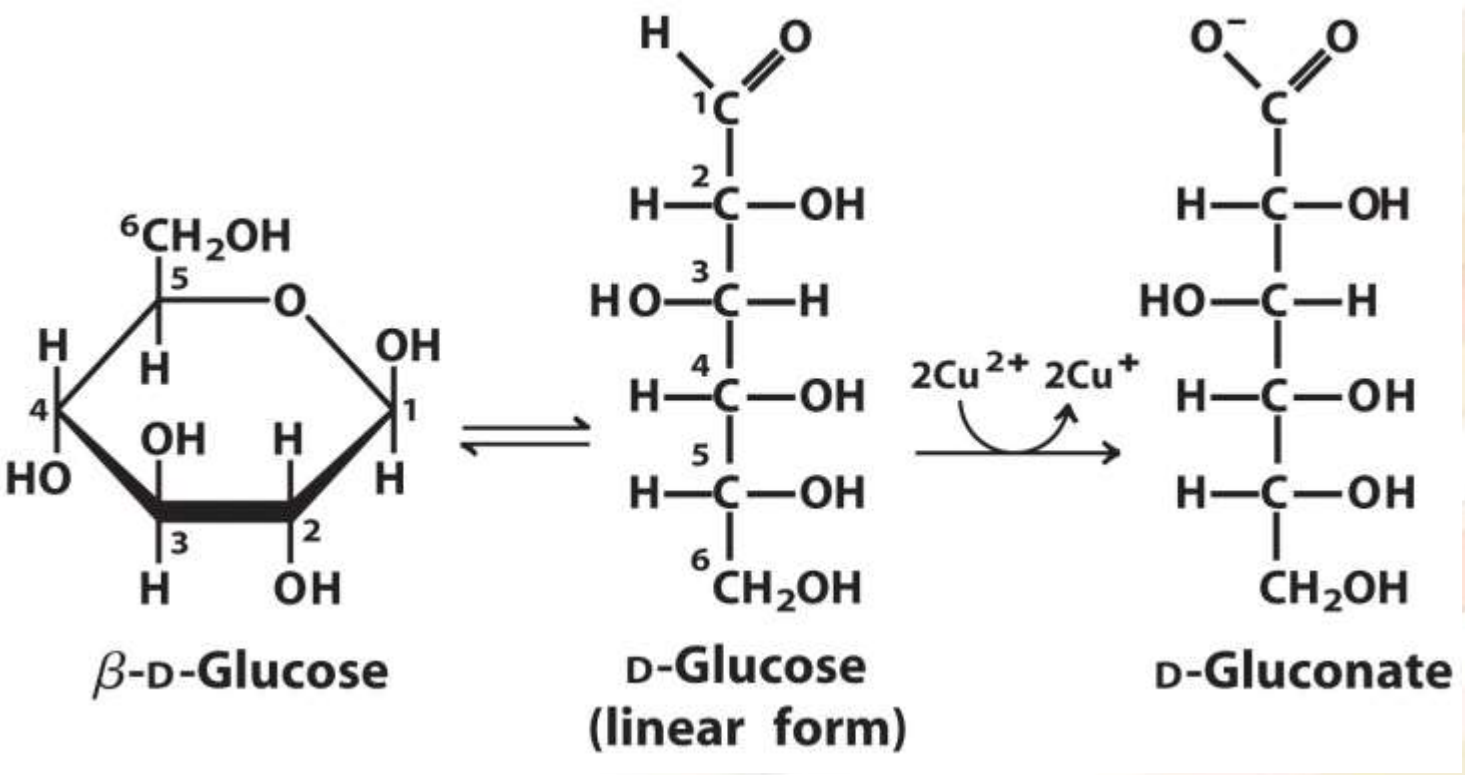
Carbohydrates

- *Seminal fluid is rich in fructose and sperm utilize fructose for energy*
- Readily metabolized either via glucose or directly
- **Hereditary fructose intolerance** leads to fructose accumulation

Carbohydrates

- **Properties of Monosaccharides**
- **1. Oxidation** of the carbonyl (aldehyde) carbon of glucose to the carboxyl level produces ***gluconic acid***
- Other aldoses yield other ***aldonic acids***

Carbohydrates



Carbohydrates

- Oxidation of the carbon at the other end of the carbon chain---C-6 of glucose, galactose, or mannose-forms the corresponding ***uronic acid,*** glucuronic, galacturonic, or mannuronic acid

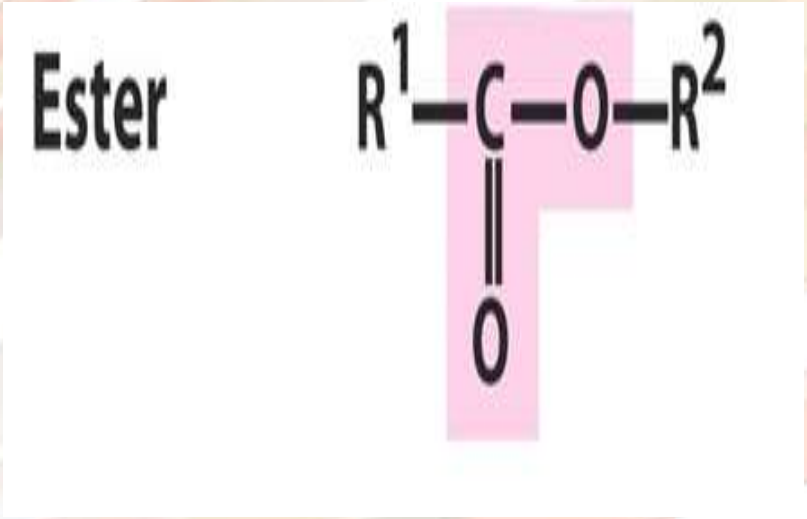
Carbohydrates

- The carboxylic acid groups of the acidic sugar derivatives are ionized at pH 7, and the compounds are therefore correctly named as the carboxylates-glucuronate, galacturonate, and so forth

Carbohydrates

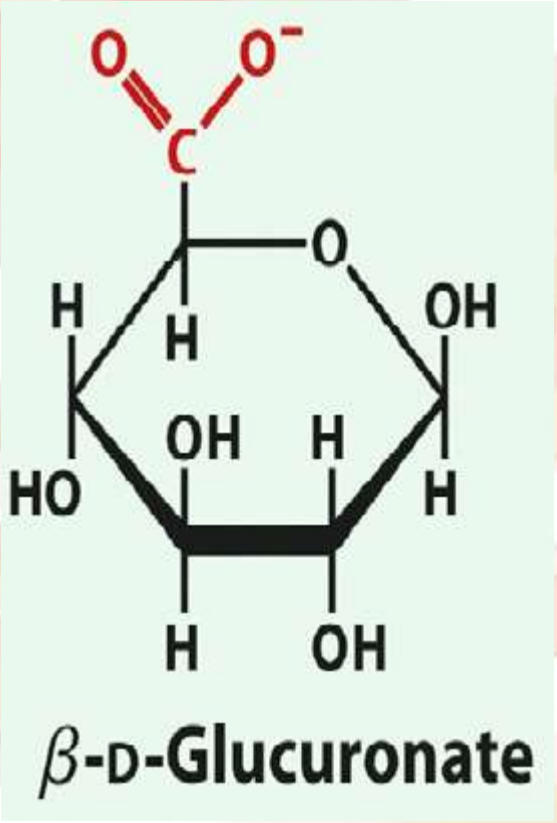
- Both aldonic and uronic acids form stable intramolecular esters called ***lactones***
- D-Glucono-δ-lactone results from formation of an ester linkage between the C-I carboxylate group and the C-5 (also known as the δ carbon) hydroxyl group of D-gluconate

Carbohydrates



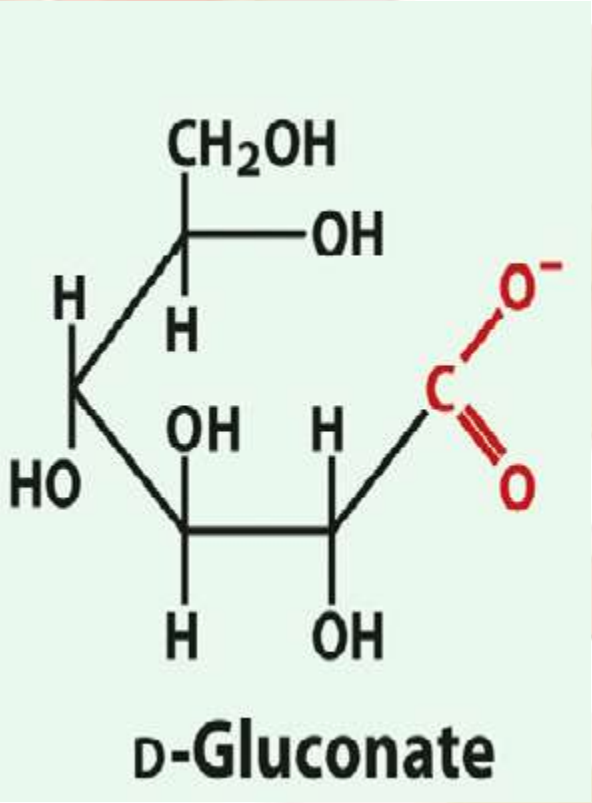
Carbohydrates

Glucuronic acid(Oxidation at C-1)



Carbohydrates

Gluconic acid(Oxidation at C-1)



Carbohydrates



Carbohydrates

- **Properties of Monosaccharides**
- Glucono delta-lactone (GDL) is commonly found in honey, fruit juices, personal lubricants, and wine . GDL is neutral.

Carbohydrates

- But hydrolyses in water to gluconic acid which is acidic, adding a tangy taste to foods
- It is metabolized to glucose; one gram of GDL yields roughly the same amount of metabolic energy as one gram of sugar.

Carbohydrates

- In the body, D-Glucuronic acid is formed from glucose in liver
- It occurs as a constituent of certain polysaccharides
- It is of importance in that it conjugates toxic substances, drugs, hormones and even bilirubin.

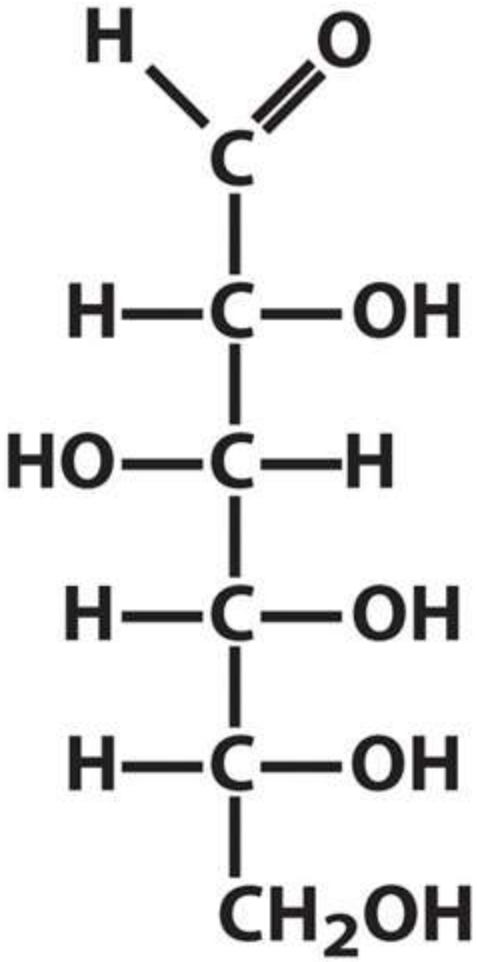
Carbohydrates

- And converts them to a soluble non toxic substance, a glucuronide, which is excreted in urine
- Oxidation of the carbon at the both ends of the carbon chain.

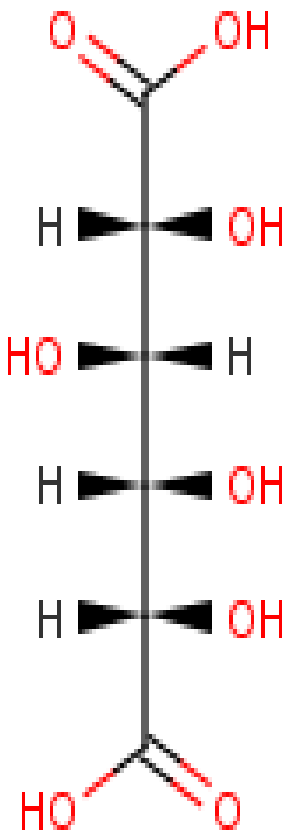
Carbohydrates

- **The terminal methyl & the carbonyl carbon** of glucose, galactose, or mannose- forms the corresponding **aldaric acid**
- Glucose forms **Glucaric acid**

Carbohydrates



D-Glucose,



Carbohydrates

- Monosacharides can be oxidized by relatively mild oxidizing agents such as cupric (Cu^{2+}) ion
- The carbonyl carbon is oxidized to a carboxyl group
- Glucose and other sugars capable of reducing cupric ion are called ***reducing sugars***

Carbohydrates

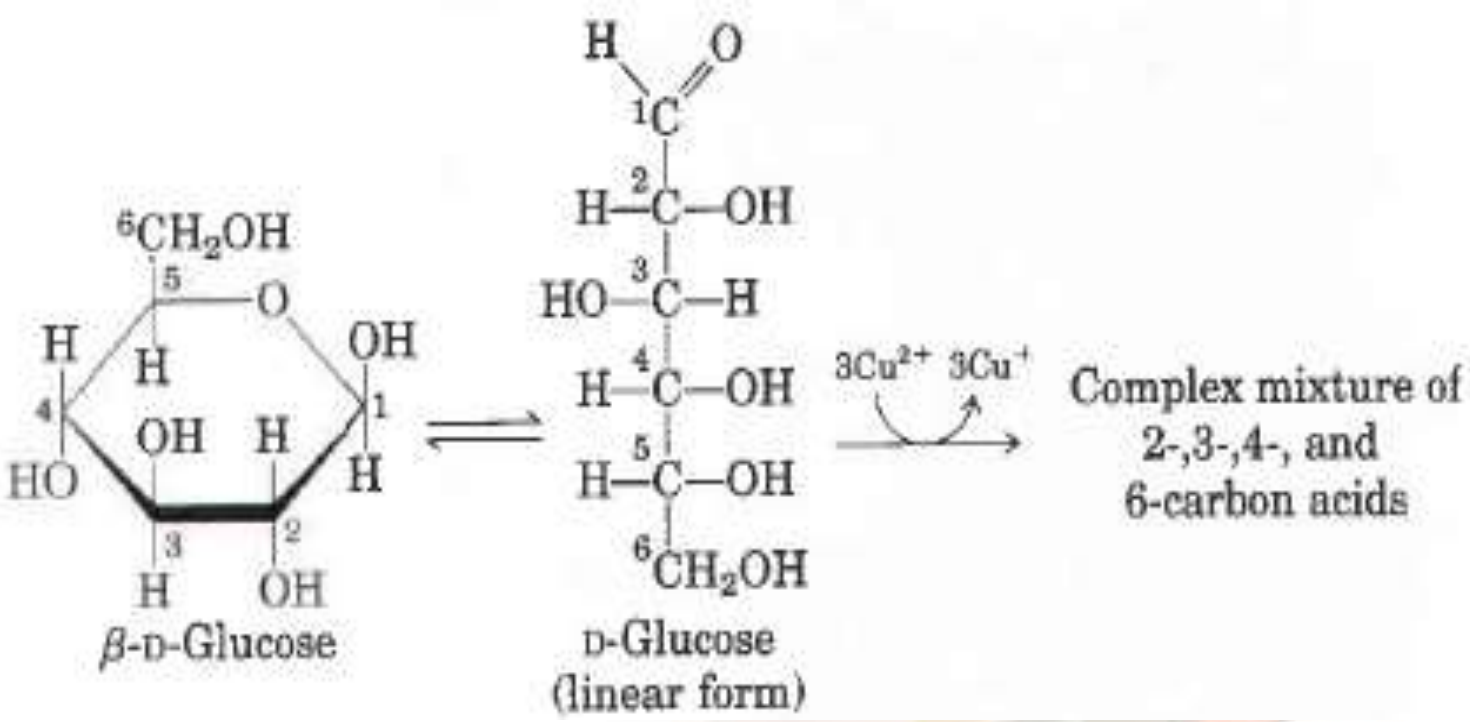
Sugars as reducing agents

- Oxidation of the anomeric carbon (and probably the neighboring carbon) of glucose and other sugars under alkaline conditions is the basis for
- Fehling's reaction
- Benedicts reaction

Carbohydrates

- The cuprous ion (Cu^{+}) produced forms a red cuprous oxide precipitate

Carbohydrates



Carbohydrates

- By measuring the amount of oxidizing agent reduced by a solution of a sugar, it is also possible to estimate the concentration of that sugar

Carbohydrates

- For many years this test was used to detect and measure elevated glucose levels in blood and urine in the diagnosis of diabetes mellitus

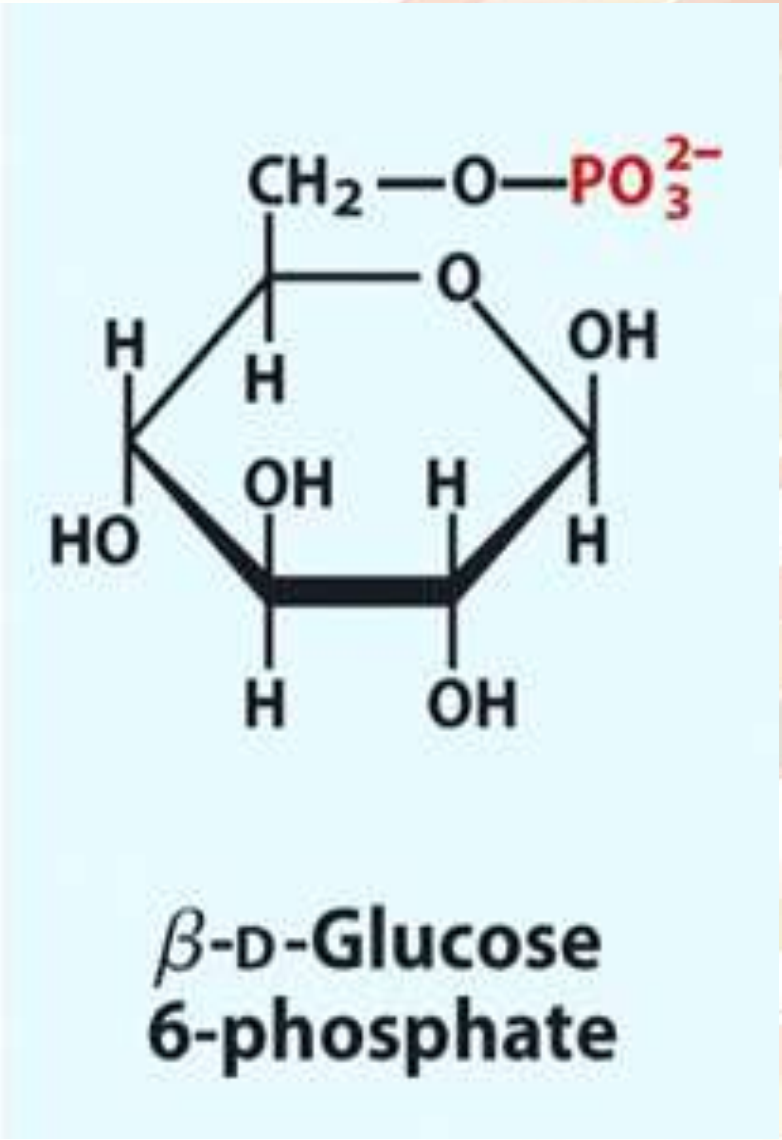
Carbohydrates

- **Properties of Monosaccharides**
- 2. Condensation of phosphoric acid with one of the hydroxyl groups of a sugar forms a ***phosphate ester***, as in glucose 6-phosphate

Carbohydrates

- Sugar phosphates are relatively stable at neutral pH and bear a negative charge

Carbohydrates



Carbohydrates

- In fact, one of the strategies of glycolysis is to form three carbon intermediates, that can transfer their phosphate groups to ADP, to achieve a net synthesis of ATP

Carbohydrates

- In the synthesis and metabolism of carbohydrates, the intermediates are very often not the sugars themselves but their phosphorylated derivatives

Carbohydrates

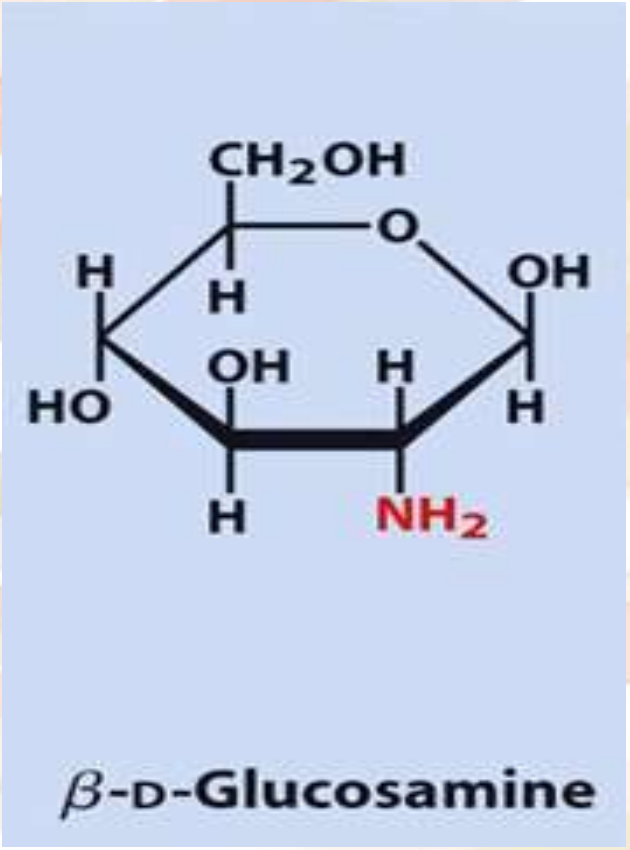
- Another function of phosphorylation

is the creation of reactive intermediates for the formation of O- and N-glycosidic linkages

Carbohydrates

- **Properties of Monosaccharides**
- 3.Synthesis of Amino Sugar
- If the hydroxyl at C-2 of the parent monosaccharide is replaced with an amino group, the corresponding ***amino sugar*** is formed

Carbohydrates

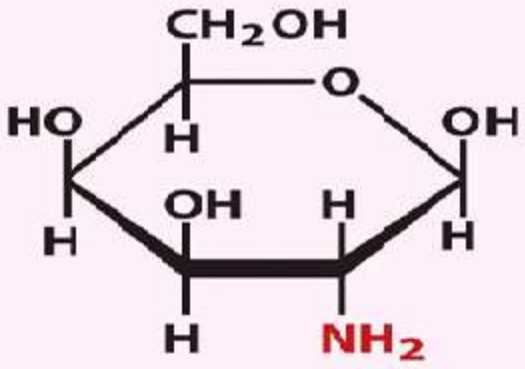


Carbohydrates

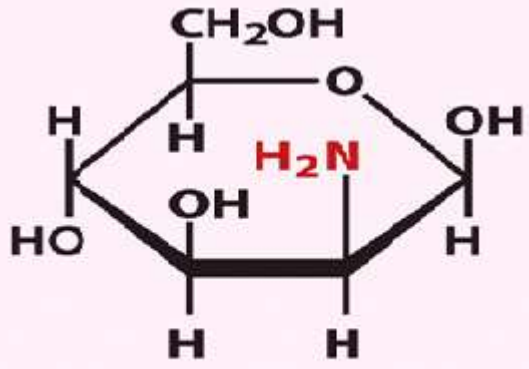
- The amino sugars include:
  - ***D -glucosamine***, a constituent of hyaluronic acid
  - ***D -galactosamine*** (also known as *chondrosamine*), a constituent of chondroitin
  - ***D -mannosamine***

Carbohydrates

Amino sugars



β-D-Galactosamine



β-D-Mannosamine

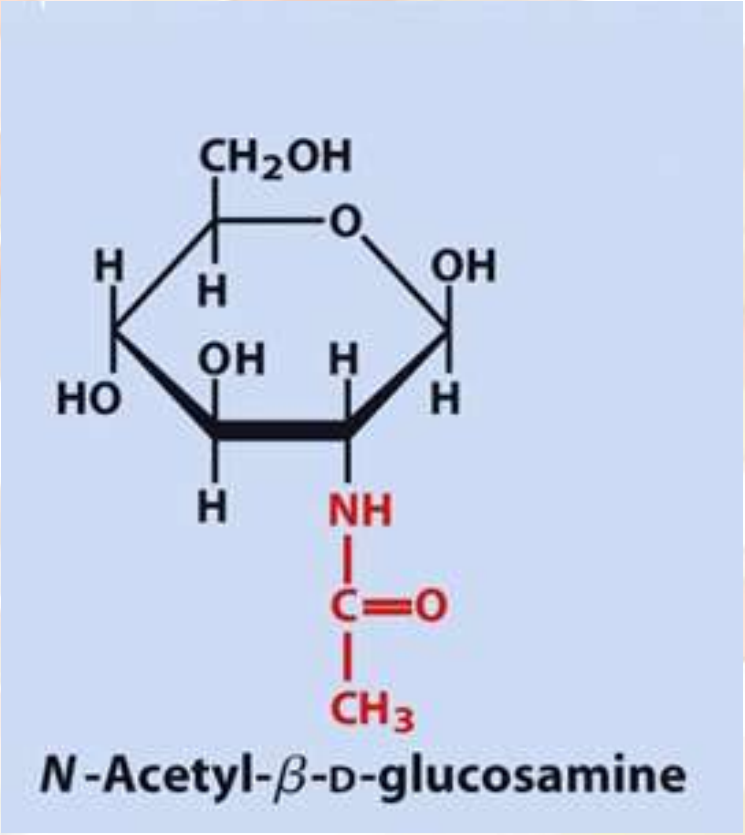
Carbohydrates

- The amino group is nearly always condensed with ***acetic acid***, as in N-acetylglucosamine
- Both glucosamine and galactosamine occur as *N*-acetyl derivatives in more complex carbohydrates.

Carbohydrates

- For example glycoproteins and is part of many structural polymers, including those of the bacterial cell wall

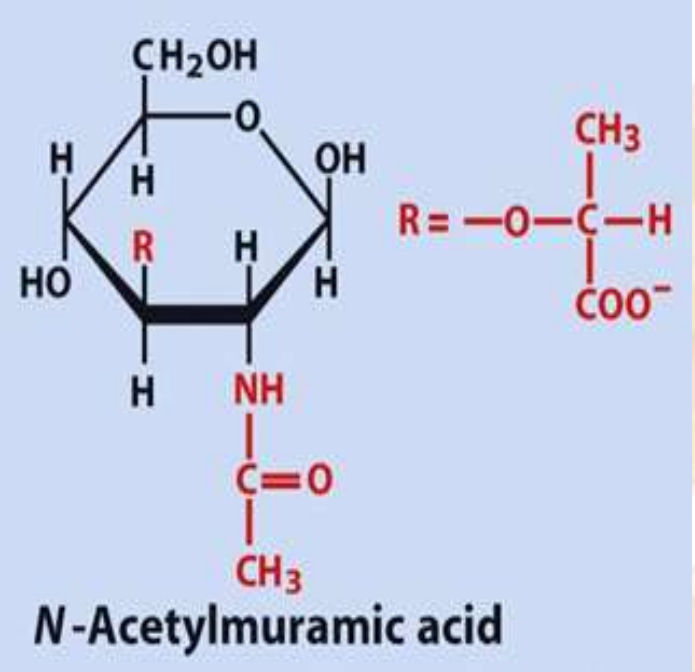
Carbohydrates



Carbohydrates

- Bacterial cell walls also contain a derivative of glucosamine, ***N-acetylmuramic acid***, in which lactic acid (a three carbon carboxylic acid) is ether-linked to the oxygen at C-3 of N-acetylglucosamine

Carbohydrates



Carbohydrates

- The Amino Sugars (Hexosamines) are Components of Glycoproteins, Gangliosides, & Glycosaminoglycans
- Several **antibiotics** (e.g, erythromycin) contain amino sugars, which are important for their antibiotic activity

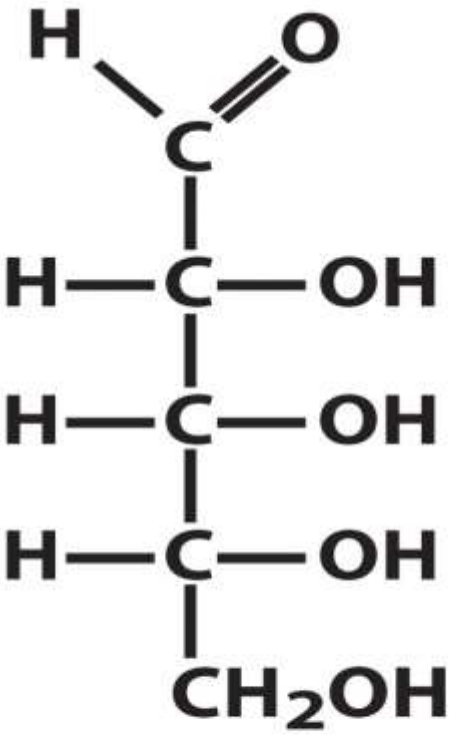
Properties of Monosaccharides

- 4. When oxygen of a –OH group is removed in a monosaccharide, leaving behind only hydrogen, ***deoxy sugars*** are formed

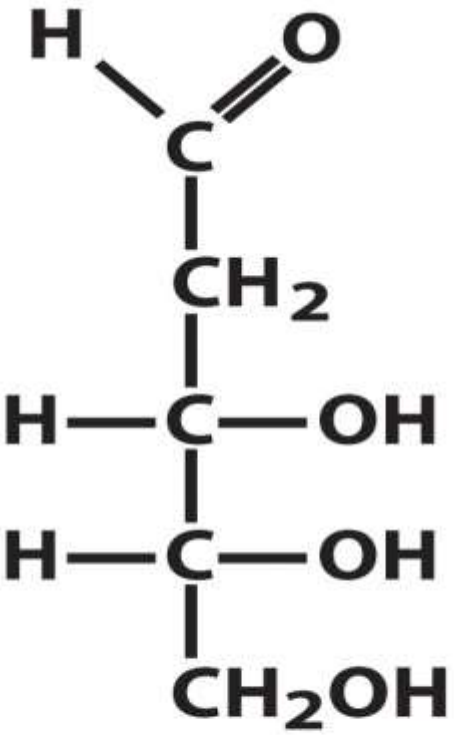
Properties of Monosaccharides

- 2-Deoxy-D-Ribose is derived from D-Ribose and is an important constituent of DNA

Properties of Monosaccharides



D-Ribose,
an aldopentose



2-Deoxy-D-ribose,
an aldopentose

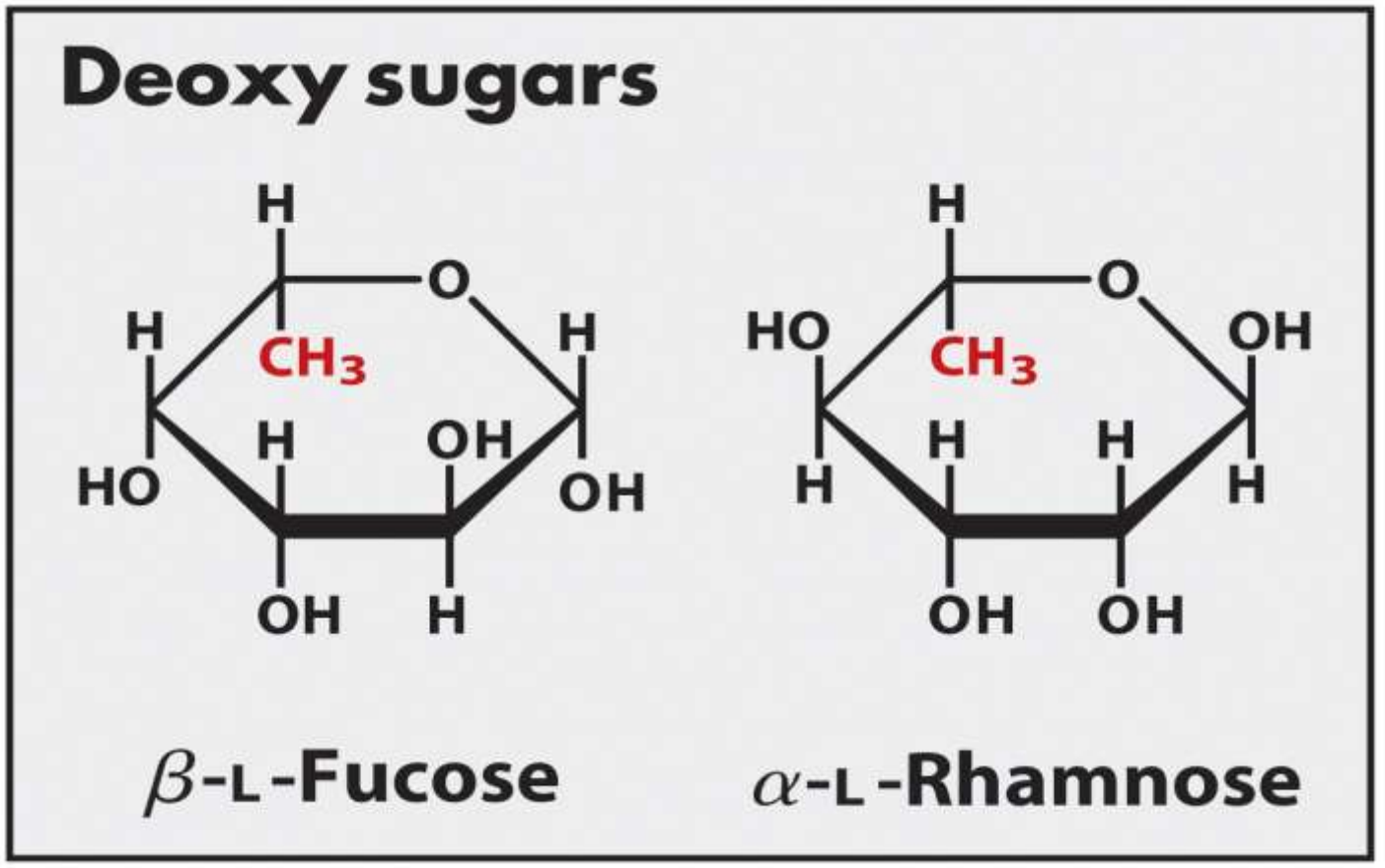
Properties of Monosaccharides

- The substitution of a hydrogen for the hydroxyl group at C-6 of ;
- L-galactose &
- L-mannose produces;
- L-fucose and
- L-rhamnose, respectively.

Properties of Monosaccharides

- ***L-Fucose*** is found in the complex oligosaccharide components of
- glycoproteins and glycolipids;
- **L-Rhamnose** is found in plant polysaccharide

Properties of Monosaccharides



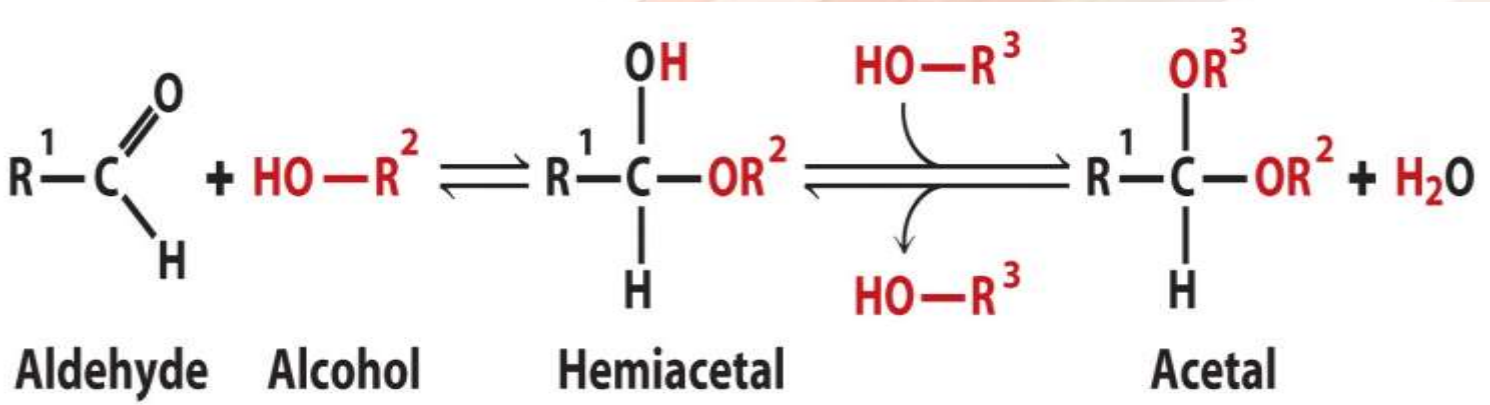
Properties of Monosaccharides

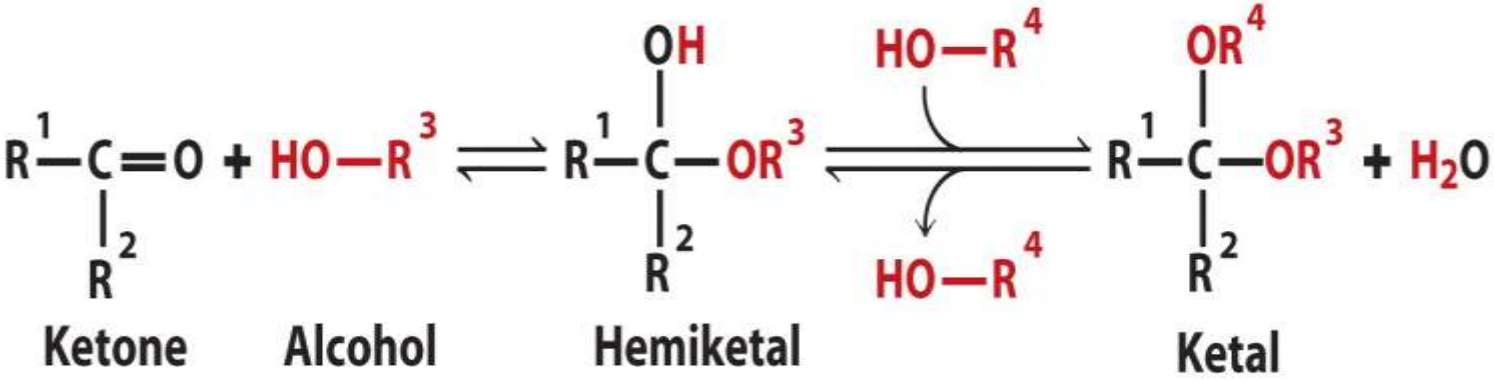
- **5. Glycosides** are formed by condensation between the hydroxyl group of the anomeric carbon of a monosaccharide, and a second compound that may or may not (in the case of an **aglycone**) be another monosaccharide

Properties of Monosaccharides

- If the second group is a ***hydroxyl,*** the ***O-glycosidic*** bond is an **acetal link** because it results from a reaction between a hemiacetal group (formed from an aldehyde and an —OH group) and another —OH group

Properties of Monosaccharides

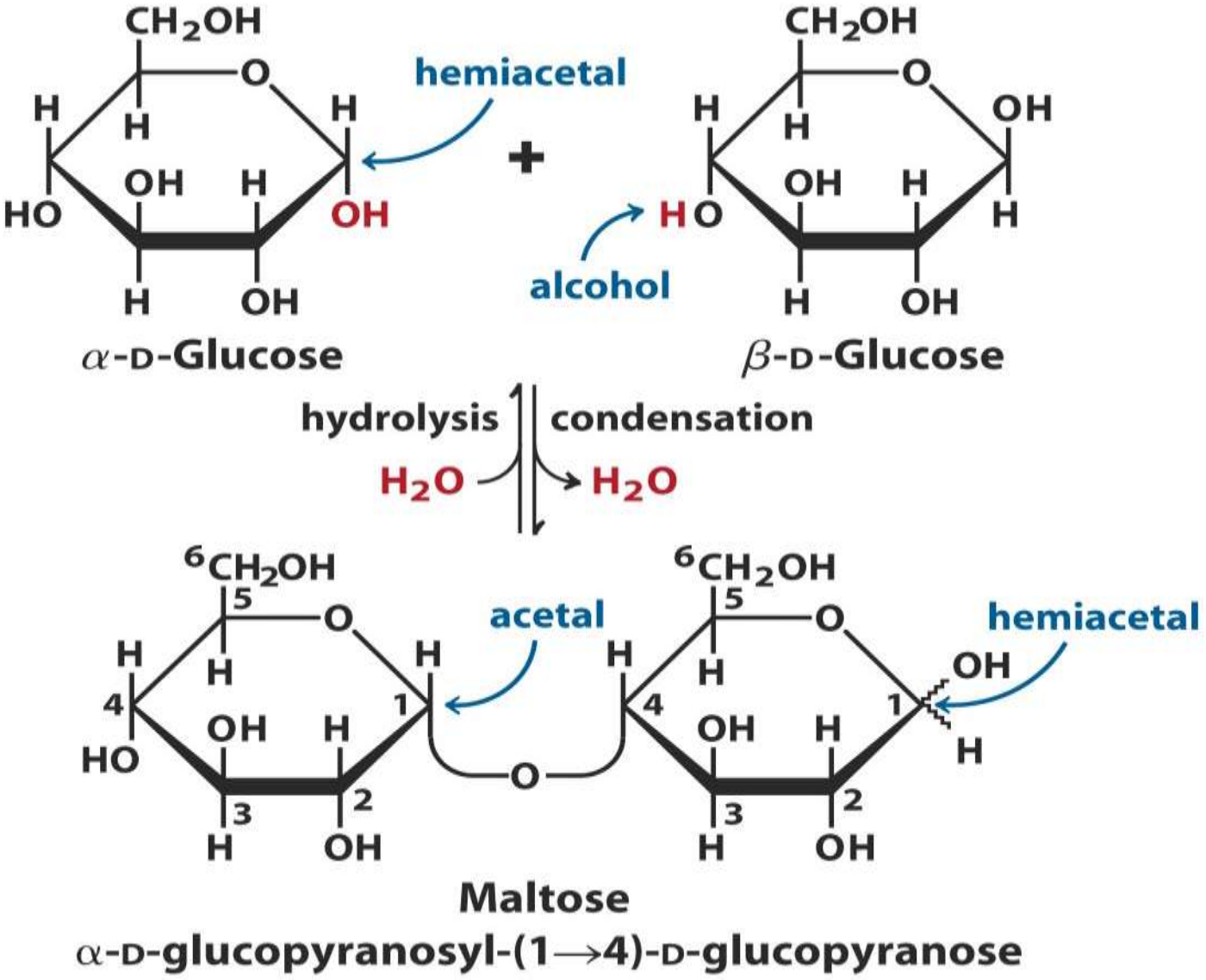




Properties of Monosaccharides

- If the hemiacetal portion is glucose, the resulting compound is a **glucoside**
- If galactose, a **galactoside,** and so on

Properties of Monosaccharides



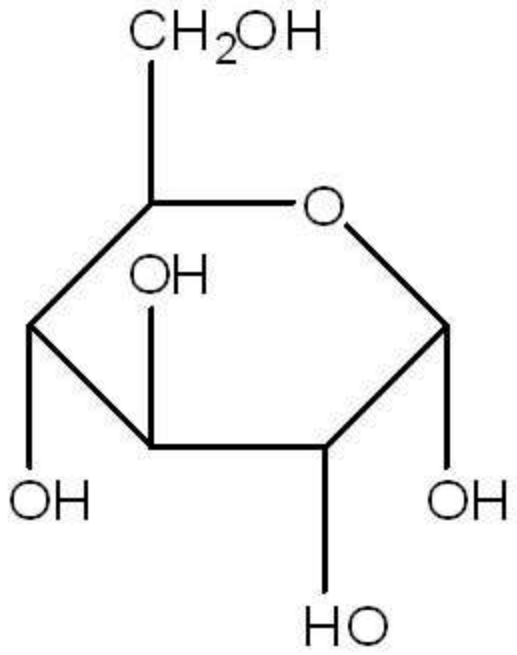
Properties of Monosaccharides

6. The monosaccharides may be reduced to form their ***corresponding alcohols*** by reducing agents;

Properties of Monosaccharides

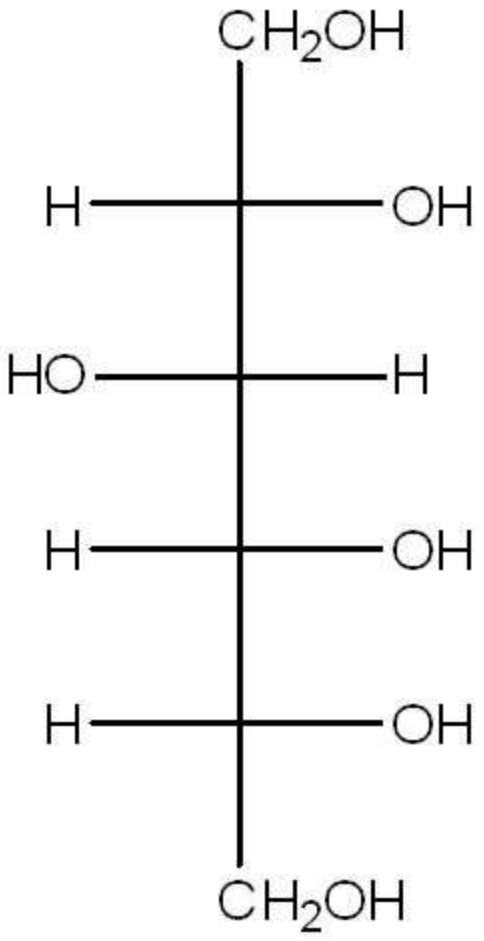
- Examples are
  - D-Glucose yields D-Sorbitol
  - D-Galactose yields D-Dulcitol
  - D-Mannose yields D-Mannitol
  - Ribose to ribitol

Properties of Monosaccharides



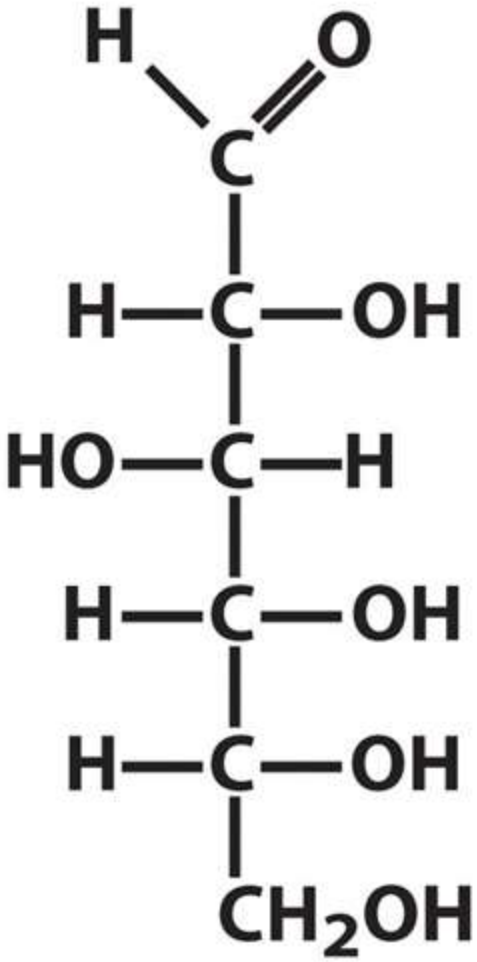
D-Glucose



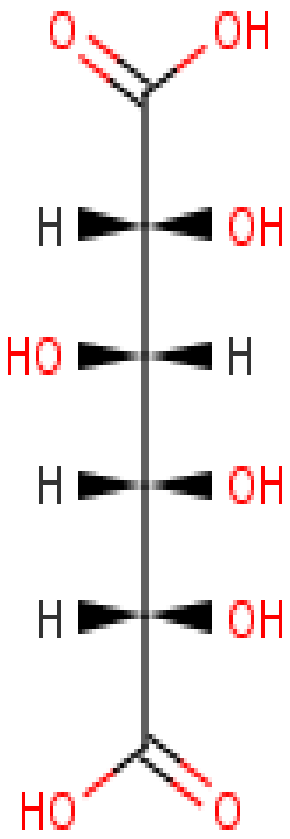


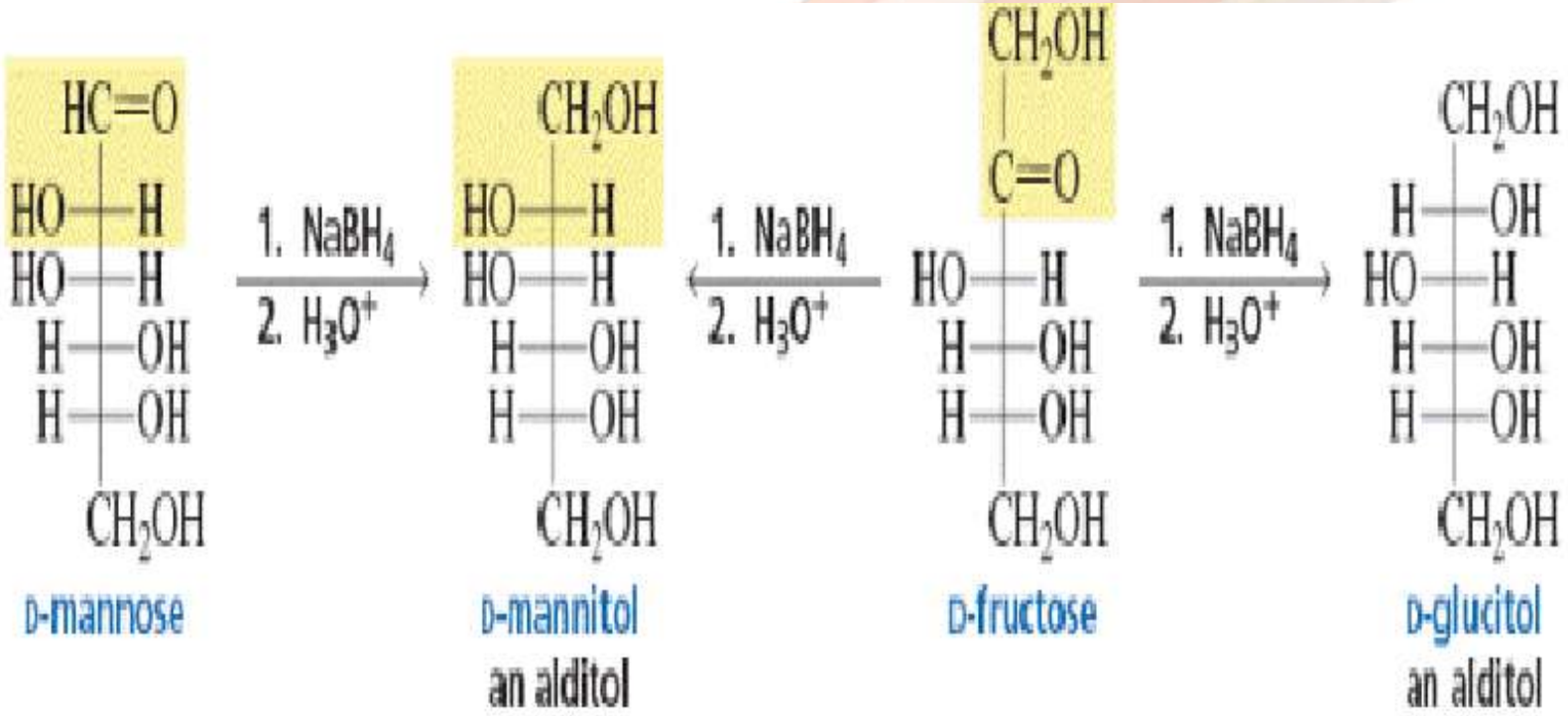
D-Sorbitol

Carbohydrates



D-Glucose,





D-mannose

D-mannitol
an alditol

D-fructose

D-glucitol
an alditol

Properties of Monosaccharides

Sorbitol:

- 35-60% sweetness of glucose
- Used as an artificial sweetner
- Accumulates in tissues such as the eye lens in diabetes mellitus

Properties of Monosaccharides

Mannitol:

- Administered intravenously as an osmotic diuretic in acute renal failure
- Also used to relieve raised intracranial pressure by forced diuresis

Properties of Monosaccharides

7. Tautomerization

- The process of shifting a hydrogen atom from one carbon atom to another to produce **enediols** is known as tautomerization

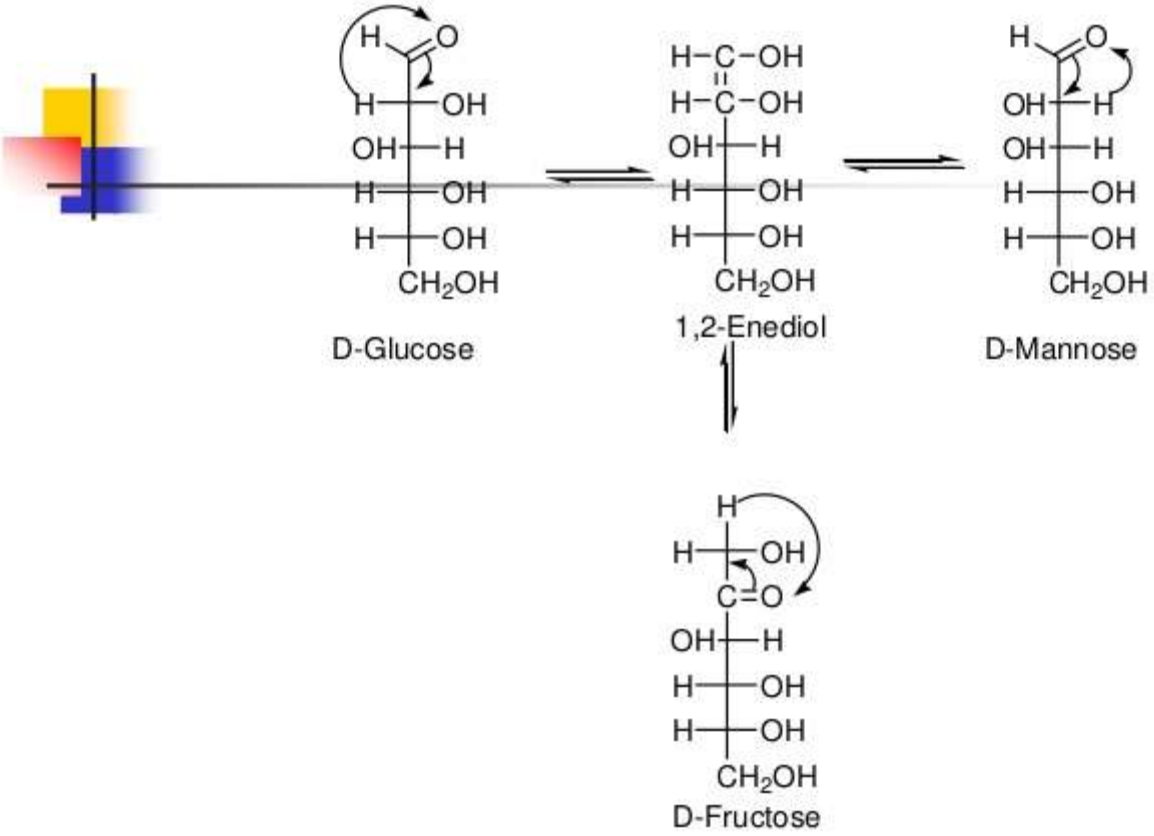
Properties of Monosaccharides

Sugars possessing anomeric carbon atom undergo

tautomerization in alkaline solutions

- When glucose is kept in alkaline solution for several hours, it undergoes isomerization to form D-fructose and D-mannose

Properties of Monosaccharides



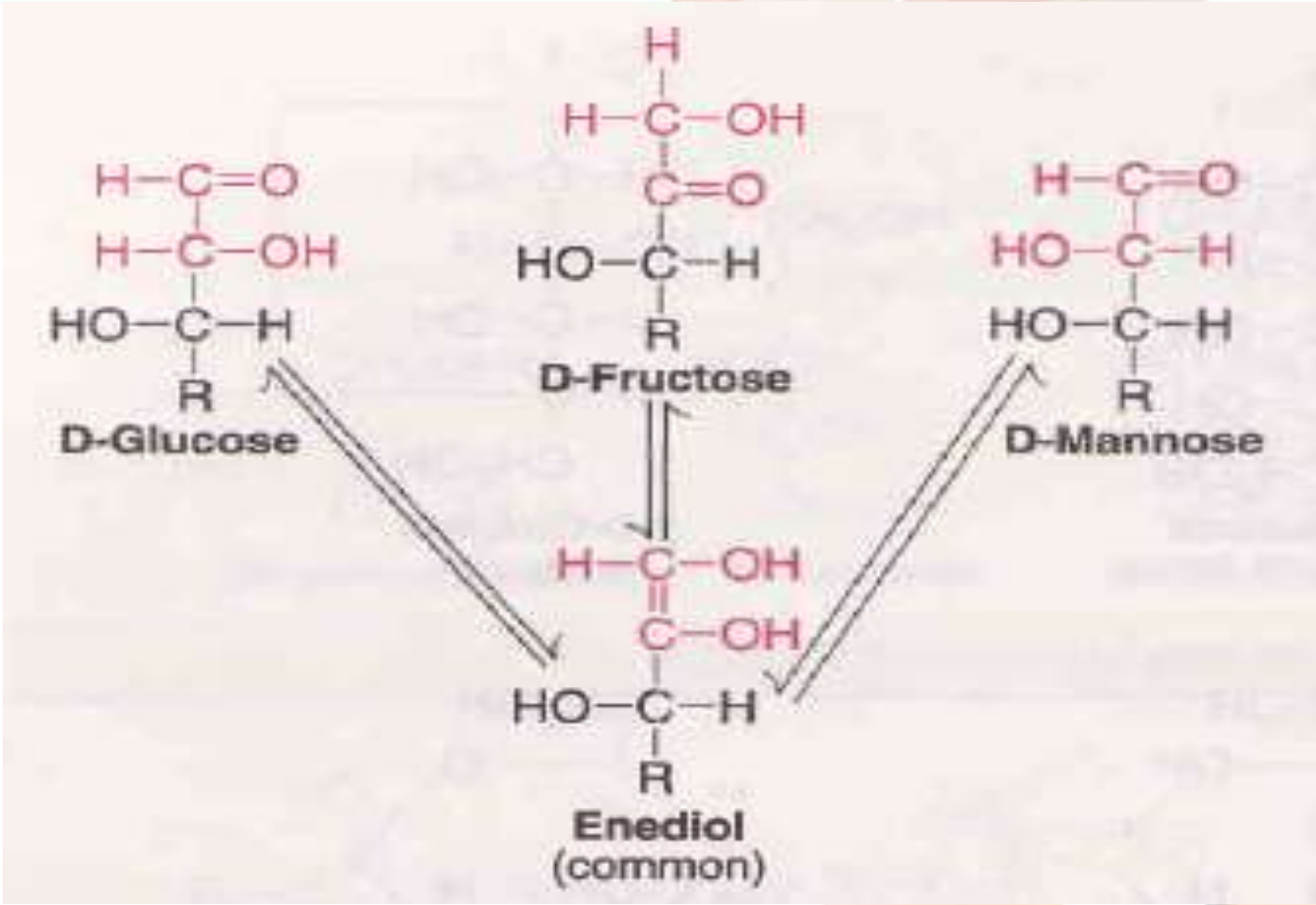
Properties of Monosaccharides

This reaction results in the formation of a

common intermediate-namely enediol--for all the three sugars

- The enediols are highly reactive, hence sugars in alkaline solution are powerful reducing agents

Properties of Monosaccharides



Properties of Monosaccharides

8. ***Osazone formation***:

- It is a useful means of preparing *crystalline derivatives* of sugars called ***osazones***
- They are obtained by adding a mixture of ***phenyl hydrazine hydrochloride*** and ***sodium acetate*** to a sugar solution and heating in water bath for 45 mins

Properties of Monosaccharides

- Crystals are formed after the solution is cooled slowly
- Reaction involves only the carbonyl carbon and the adjacent carbon

Properties of Monosaccharides

- First phenyl hydrazone is formed and then reacts with an additional molecule of phenylhydrazine to form the osazones.
- As is evident from the reaction, the first two carbons (C1 and C2) are involved in osazone formation

Properties of Monosaccharides

- The sugars that differ in their configuration on these two carbons give the same type of osazones, since the difference is masked by binding with phenylhydrazine

Properties of Monosaccharides

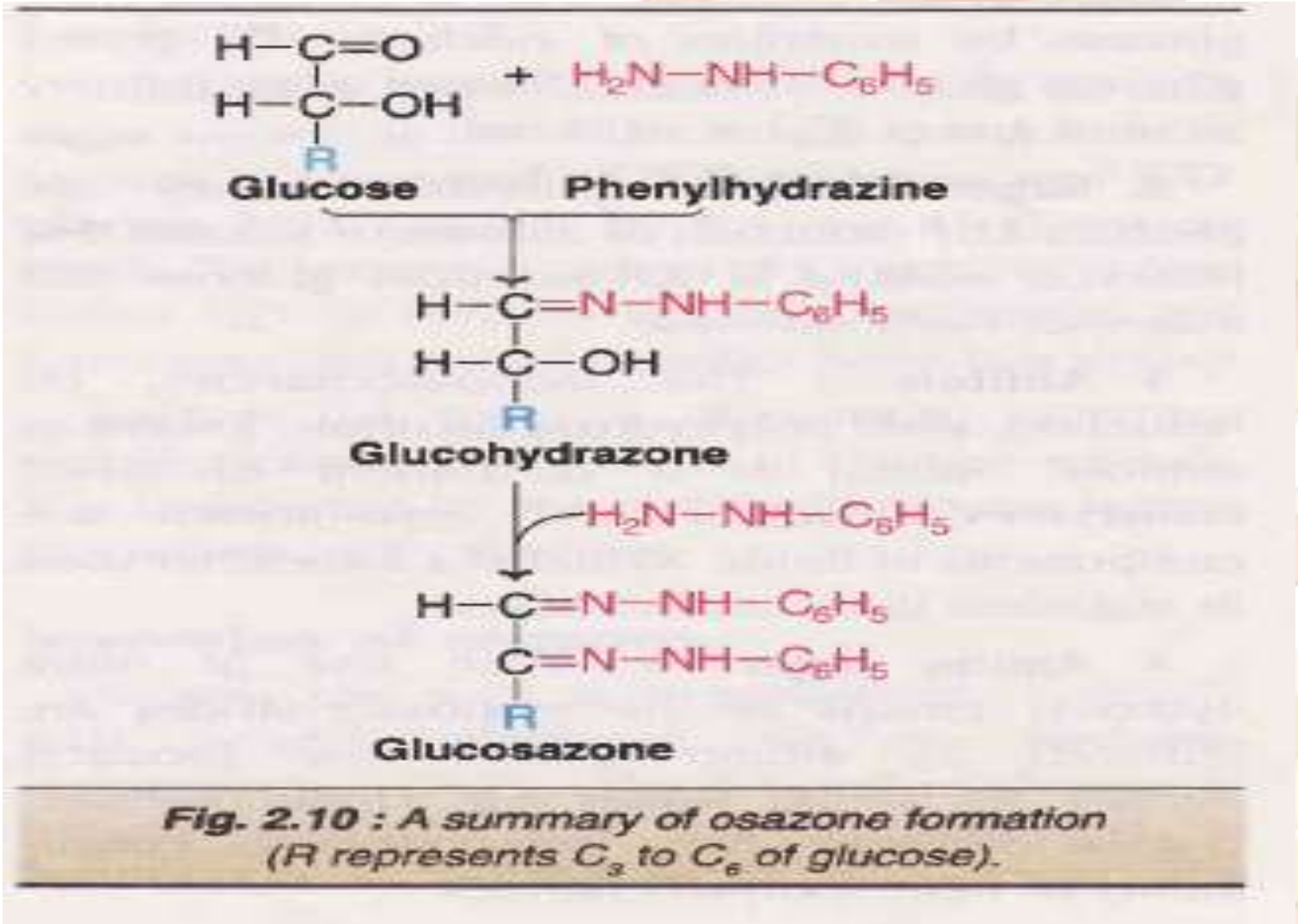
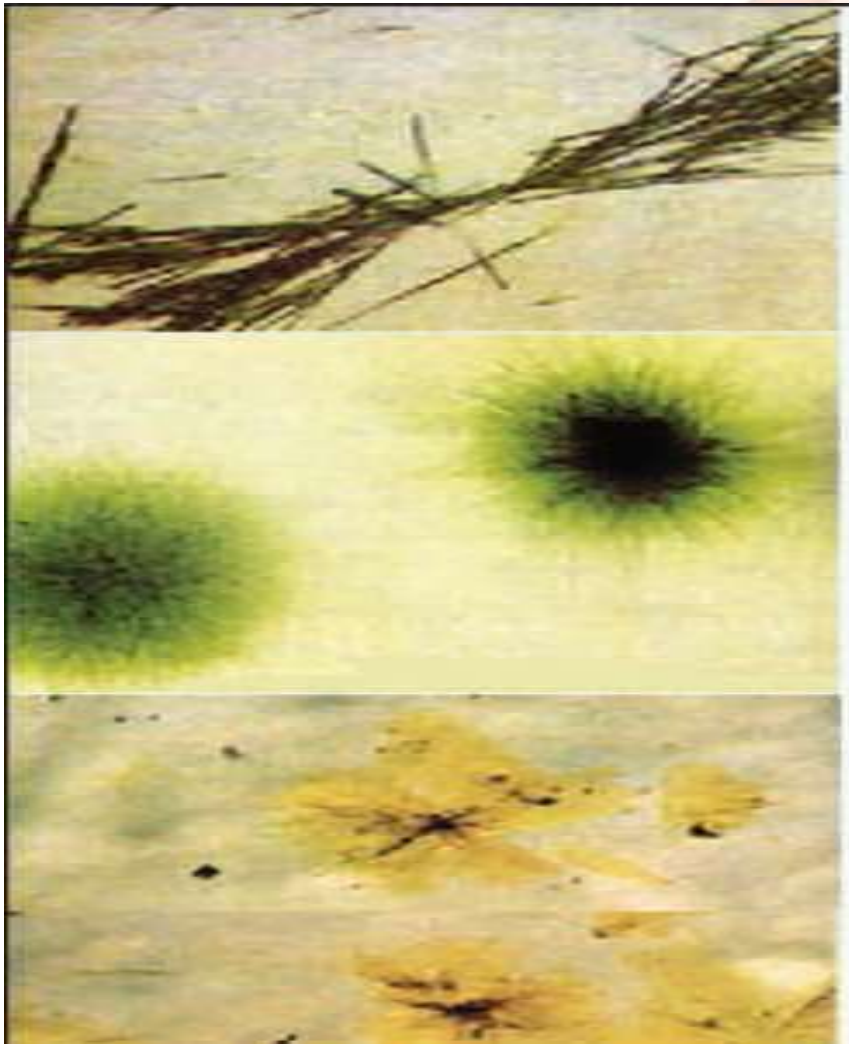


Fig. 2.10 : A summary of osazone formation (R represents C_3 to C_6 of glucose).

Properties of Monosaccharides

- Thus glucose, fructose and mannose give the same type (needle-shaped) osazones
- Reducing disaccharides also give osazones
- Maltose sunflower-shaped
- And lactose powder puff shaped

Properties of Monosaccharides



Needle shaped crystals arranged like a broom
Glucososazone

Hedgehog or "pincushion with pins" or flower of "touch-me-not-plant"
Lactososazone

Sunflower shaped or petal shaped crystals of
Maltosazone

Properties of Monosaccharides

- Osazones have characteristic
  - Melting points
  - Crystal structures
  - Precipitation time
  - and thus are used in identifying the sugars

Properties of Monosaccharides

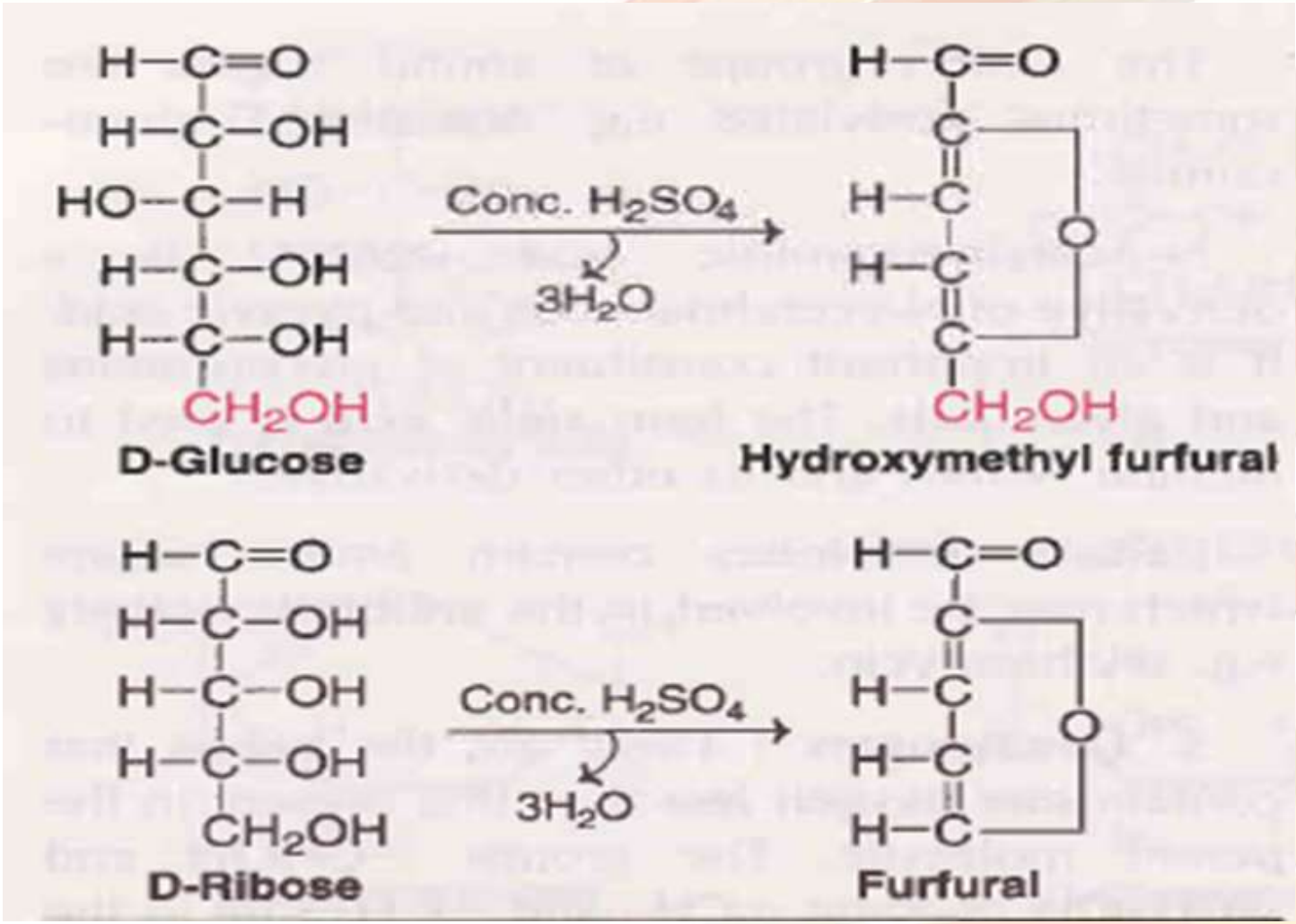
9. Action of acids on carbohydrates

- Polysaccharides are hydrolyzed into their constituent monosaccharides by boiling with **dilute** acids
- With **concentrated** acids the monosaccharides are decomposed

Properties of Monosaccharides

- Pentoses yield a cyclic aldehyde *furfural* with 12% HCL
- Hexoses are decomposed by hot strong acids to give *hydroxy methyl furfural*

Properties of Monosaccharides



Properties of Monosaccharides

- The furfural products thus formed can condense with certain organic phenols (α-naphthol) to form compounds having characteristic colours
- This forms basis of certain tests used for detection of sugars

Properties of Monosaccharides

- Molisch,s and
- Selivanoff,s are examples of such tests

Disacharrides

- **Disaccharides** (such as maltose, lactose, and sucrose) consist of two monosaccharides joined covalently by an **O-glycosidic bond.**
- Which is formed when a hydroxyl group of one sugar reacts with the anomeric carbon of the other

Disacharrides

- When the anomeric carbon is involved in **a glycosidic bond**, that sugar residue is not capable of reducing cupric ions and therefore becomes a non-reducing sugar

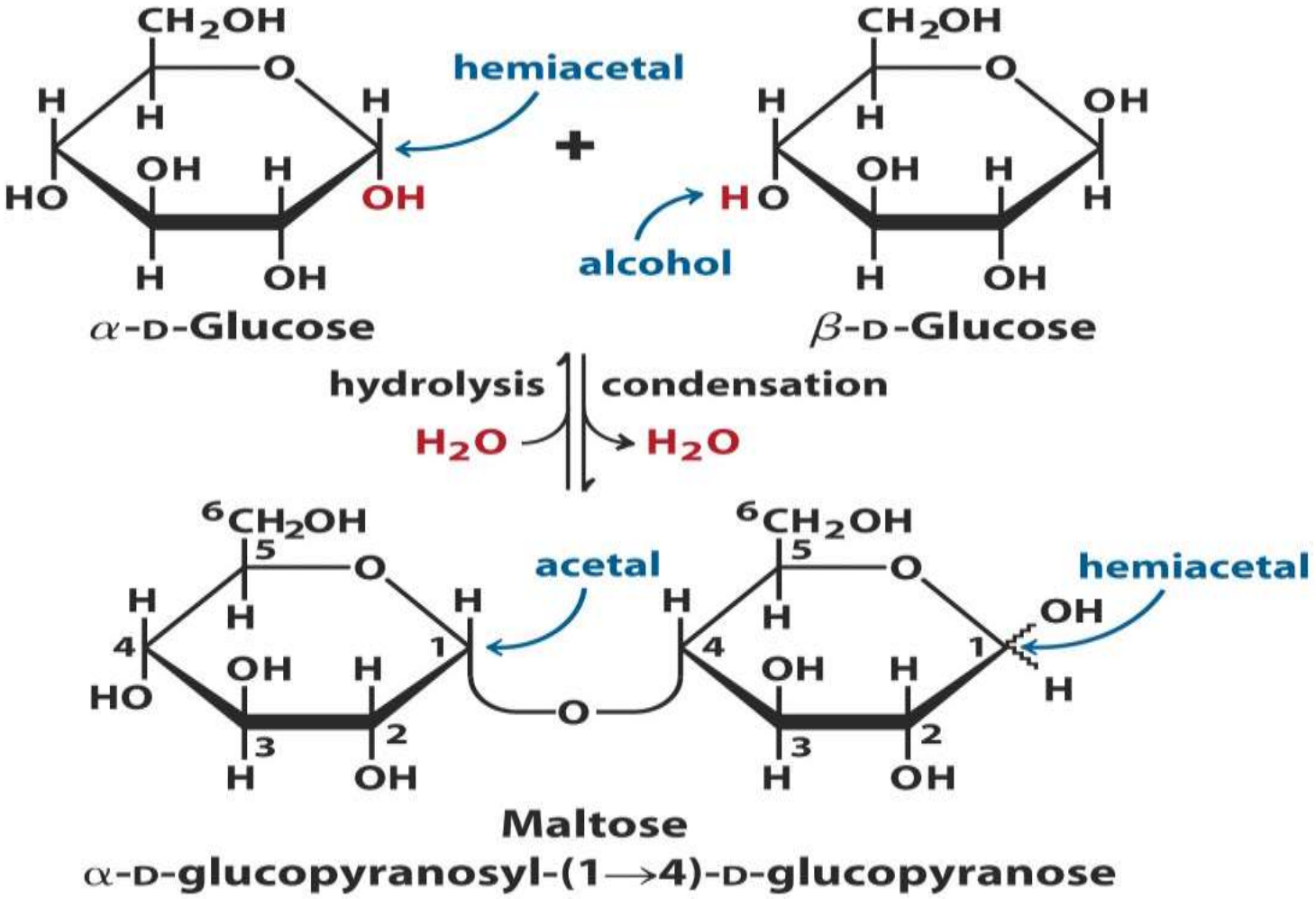
Disacharrides

- In describing disaccharides or polysaccharides, the end of a chain with a free anomeric carbon (one not involved in a glycosidic bond) is commonly called the reducing end

Disacharrides

- **Maltose** has two glucose units joined by an **α(1→4)** glycosidic linkage
- Since it has one aldehyde ‘free’, it has ***reducing properties*** and also exhibits mutarotation

Disacharrides



Disacharrides

- Maltose comes from the hydrolysis of starch (by **amylase**)
- It is in turn hydrolysed to glucose by **maltase**, which is located on the intestinal brush border

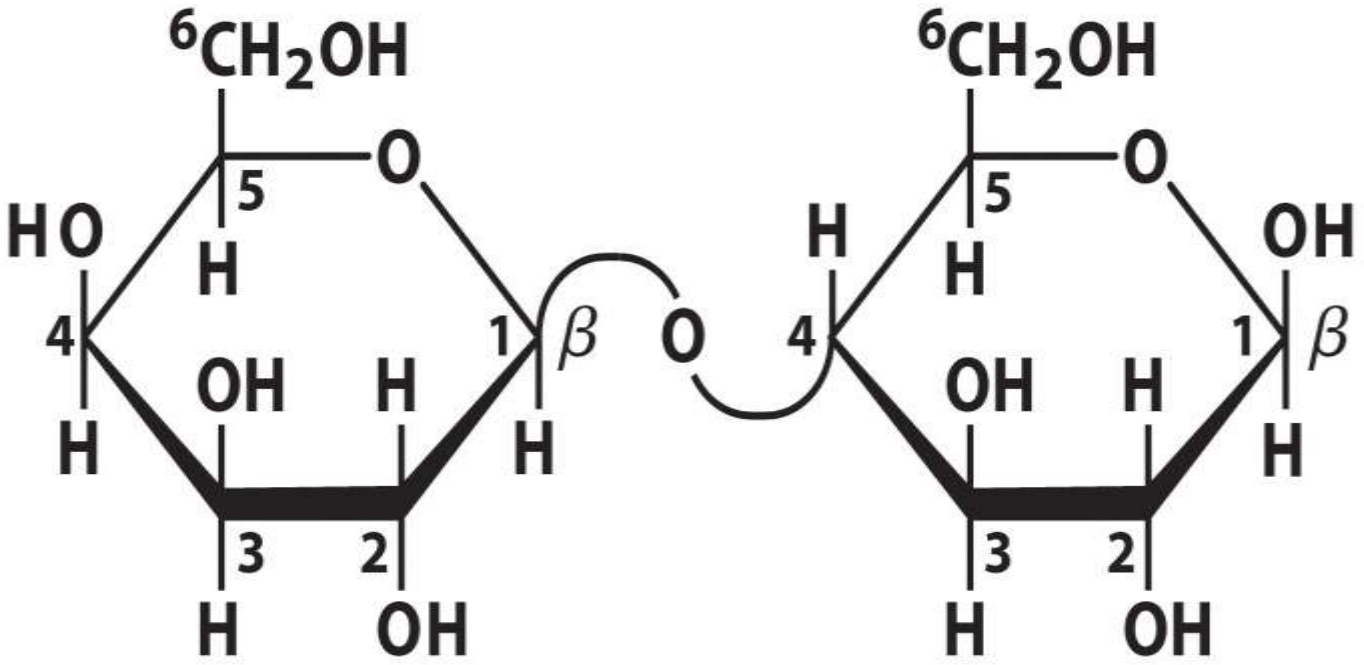
Disacharrides

- From nutritional point of view, maltose is easily digestible
- Various food preparations, such as baby and invalid foods, are produced by hydrolysis of grains and contain large amounts of maltose

Disacharrides

- **Lactose,** the disaccharide of milk consists of galactose joined to glucose by **β (1→4)** glycosidic linkage
- It is dextrorotatory
- As one of the aldehyde group is free, it has reducing properties and exhibits mutarotation

Disacharrides



Lactose (β form)

β-D-galactopyranosyl-(1$\rightarrow$4)-β-D-glucopyranose

Gal(β1$\rightarrow$4)Glc

Disacharrides

- In lactating mammary gland, the lactose is synthesized from glucose by the duct epithelium
- Many organisms that are found in milk, e.g., *E.Coli* convert lactose of milk to lactic acid (by ***β* galactosidase**) thus causing souring of milk

Disacharrides

- **Lactase**, the specific enzyme which hydrolyses lactose is present in the intestinal brush border

Disacharrides

- **Lactose intolerance** refers to a condition of pain, nausea, and flatulence after the ingestion of foods containing lactose, most notably dairy products
- Although it is often caused by
  - low levels of lactase
  - it also can because by intestinal injury

Disacharrides

- In a **lactase deficient** person, lactose accumulates in the lumen of the small intestine after ingestion of milk

Disacharrides

- The large osmotic effect of the unabsorbed lactose leads to an influx of fluid into the small intestine

Disacharrides

Lactose Intolerance



- Lactose transits through the small intestine undigested due to insufficient lactase activity on the intestinal surface
- Undigested lactose transits to the large intestine where unadapted colonic bacteria attempts to ferment (metabolize) the lactose
- Lactose fermentation results in large quantities of gas, causing flatulence, distention and acute diarrhea



Disacharrides

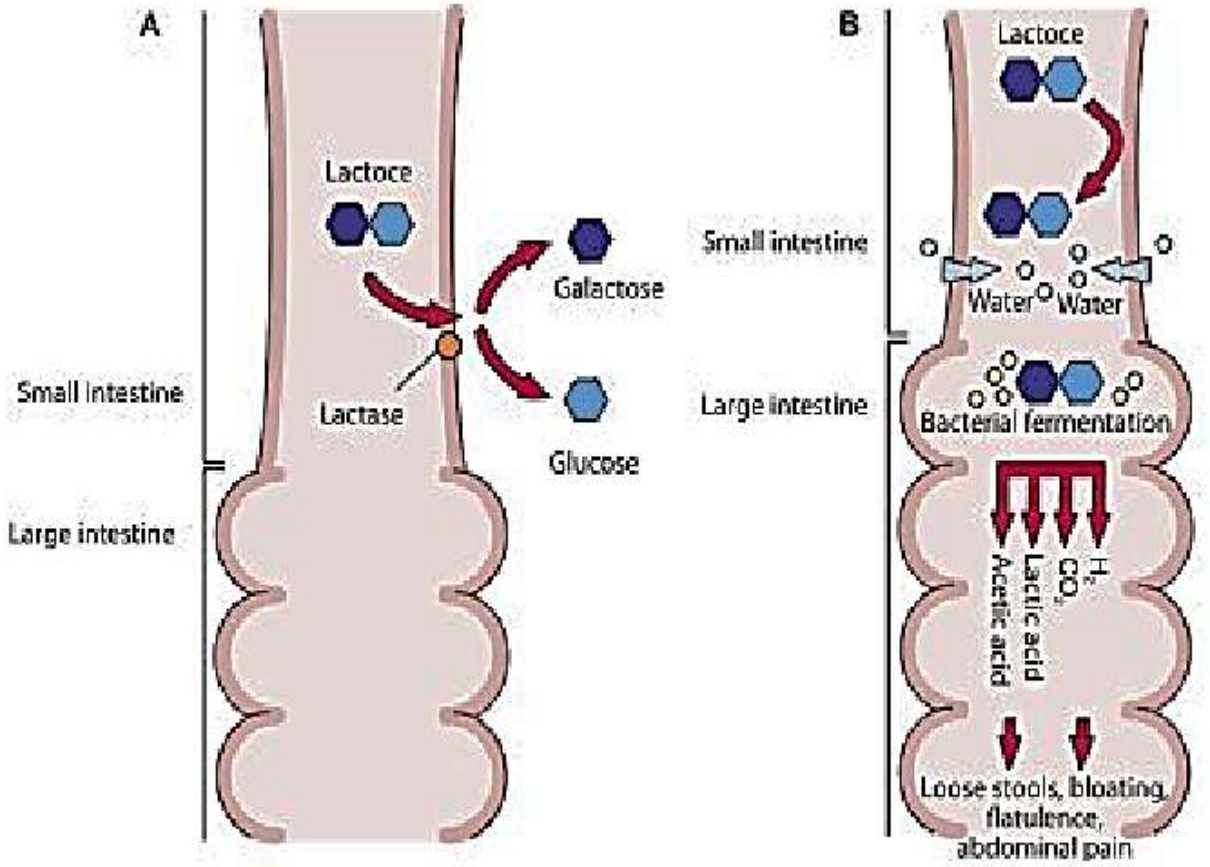


Figure 3. Picture A: Lactase degrades lactose. No symptoms of lactose intolerance. Picture B: Unabsorbed lactose in the large intestine causes lactose intolerance symptoms.

Disacharrides

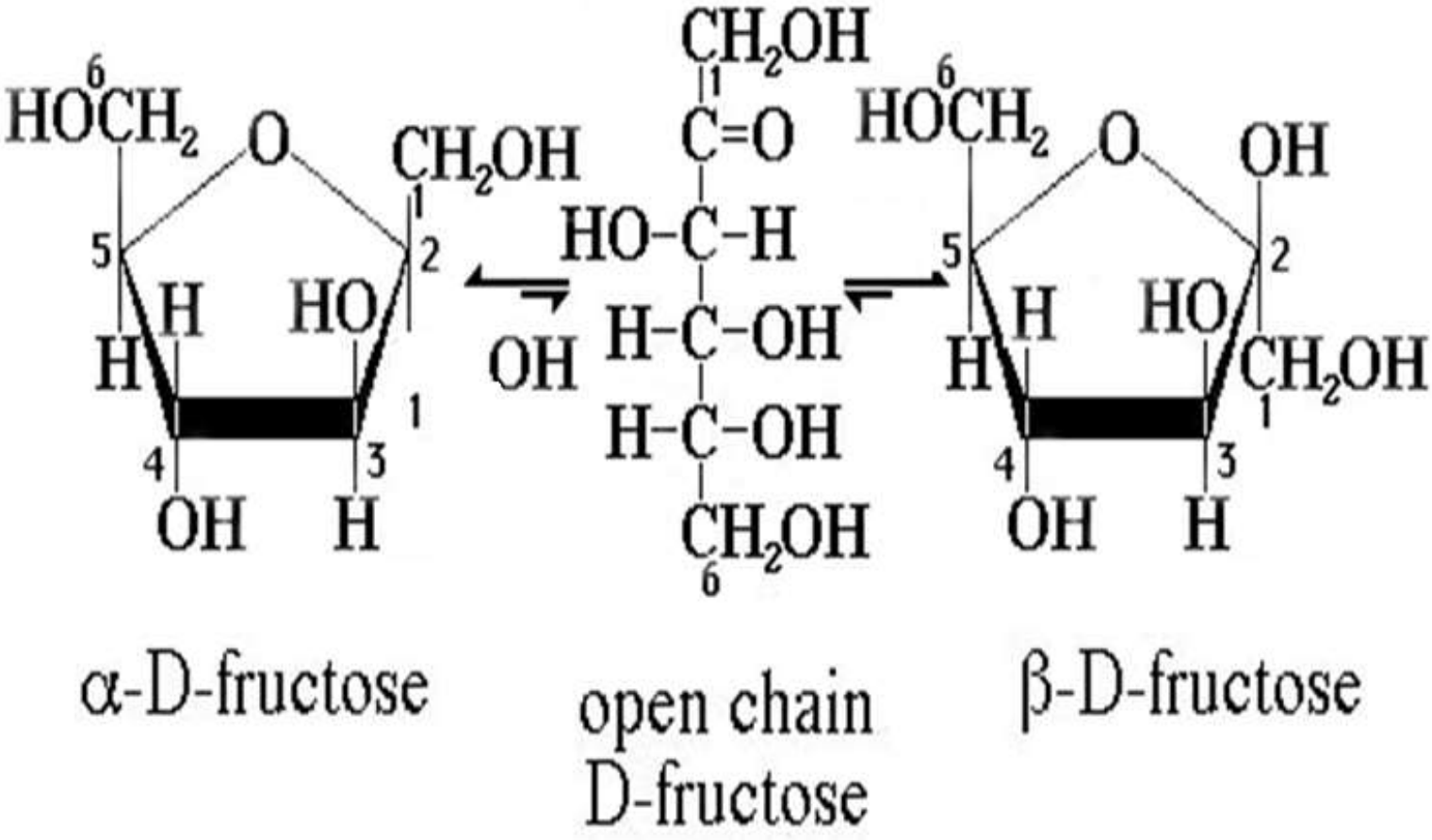
- Hence, the clinical symptoms of **lactose intolerance** are:
  - abdominal distention,
  - nausea,
  - cramping,
  - pain
  - a watery diarrhea

End

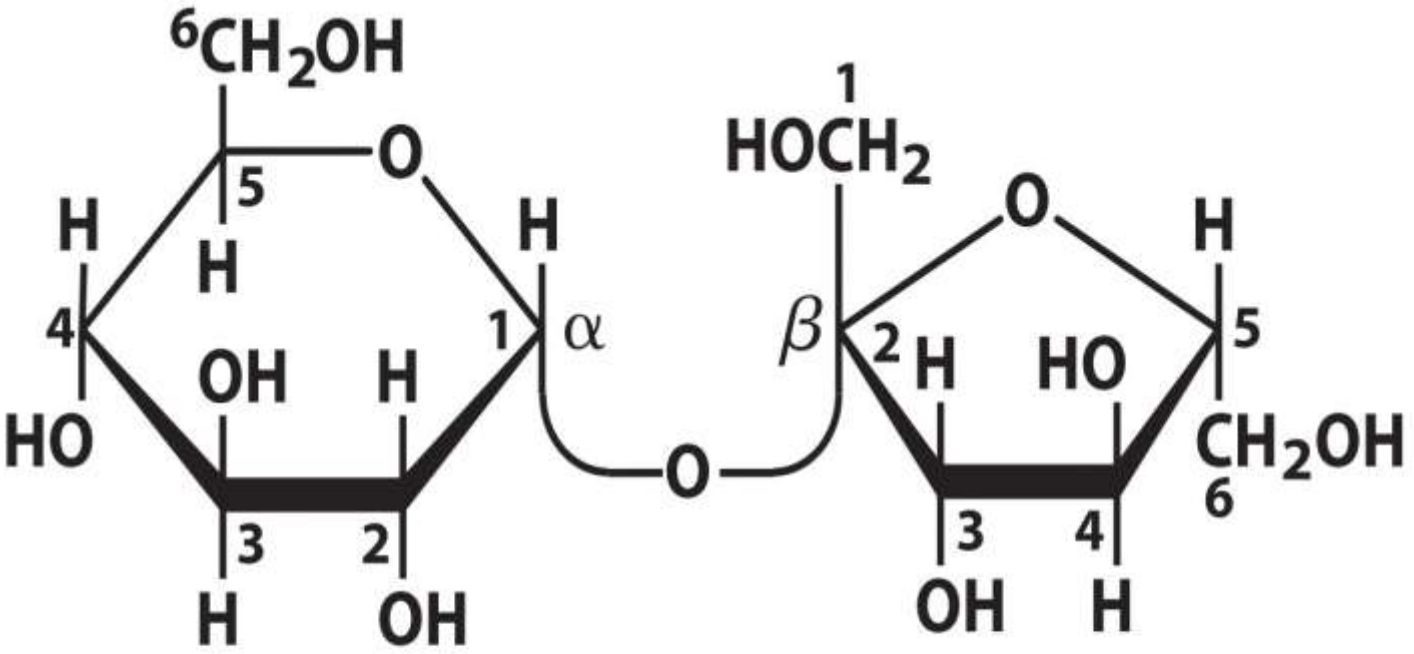
Disacharrides

- **Sucrose** (Ordinary table sugar) is obtained commercially from cane or beet
- The anomeric carbon atoms of a glucose unit and a fructose unit are joined in this disaccharide and the configuration of this glycosidic linkage is α for glucose and β for fructose

Disacharrides



Disacharrides



Sucrose

α-D-glucopyranosyl β-D-fructofuranoside

Glc(α1↔2β)Fru

Disacharrides

- Consequently, sucrose lacks a free reducing group (an aldehyde or ketone end), in contrast with most other sugars
- Sucrose is therefore a non-reducing sugar and does not exhibit mutarotation

Disacharrides

- Sucrose is dextrorotatory but its hydrolytic products are levorotatory as fructose has a levorotation greater than the dextrorotation of glucose

Disacharrides

- As the hydrolytic products invert the rotation, the resulting mixture is called as **invert sugar** and the process is called as **inversion**

Disacharrides

- The hydrolysis of sucrose to glucose and fructose is catalyzed by **sucrase** (also called **invertase**), which is also present in the intestinal brush border like lactase and maltase

Disacharrides

- Honey is largely invert sugar and the presence of fructose accounts for the greater sweetness of honey

END

Oligosaccharides

- **Oligosaccharides** are short polymers of three to ten monosaccharides joined by glycosidic bonds.
- Most are not digested by human enzymes

Oligosaccharides

- At one end of the chain, the reducing end, is a monosaccharide unit with its anomeric carbon not involved in a glycosidic bond

Oligosaccharides

Biomedical Importance

- Integral membrane proteins contain covalently attached carbohydrate units, oligosaccharides, on their extracellular face
- Many secreted proteins such as antibodies and coagulation factors also contain oligosaccharide units

Oligosaccharides

- The oligosaccharides participate in molecular targeting and cell to cell recognition
- They also mark the passage of time and determine when the proteins should be taken out of circulation

Oligosaccharides

- The human ABO blood groups illustrate the effects of glycosyl-transferases
- Carbohydrates are attached to glycoproteins and glycolipids on the surfaces of red blood cells

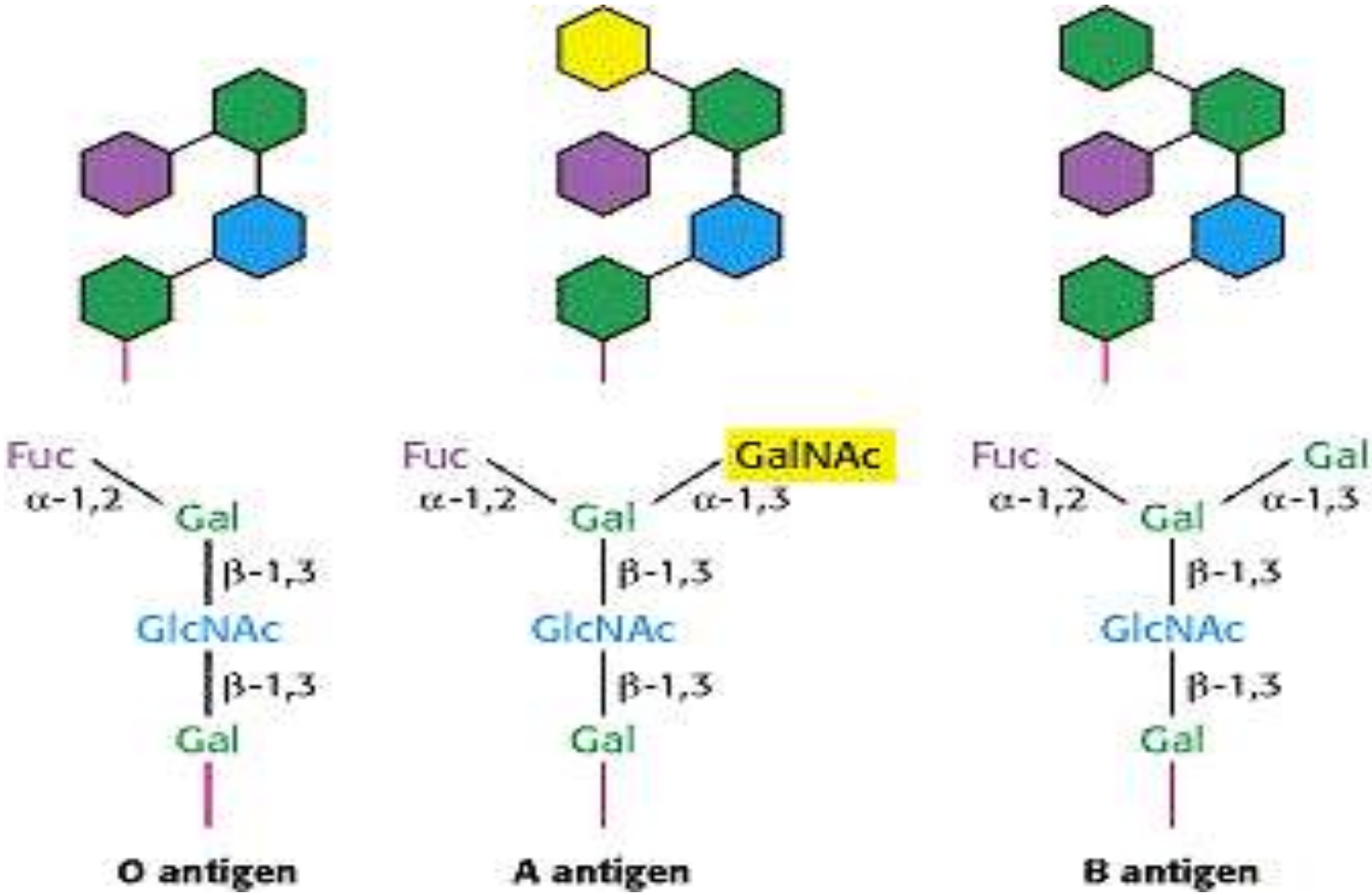
Oligosaccharides

- For one type of blood group, one of the three different structures, termed A, B, and O, may be present
- These structures have in common an oligosaccharide foundation called the O (or sometimes H) antigen

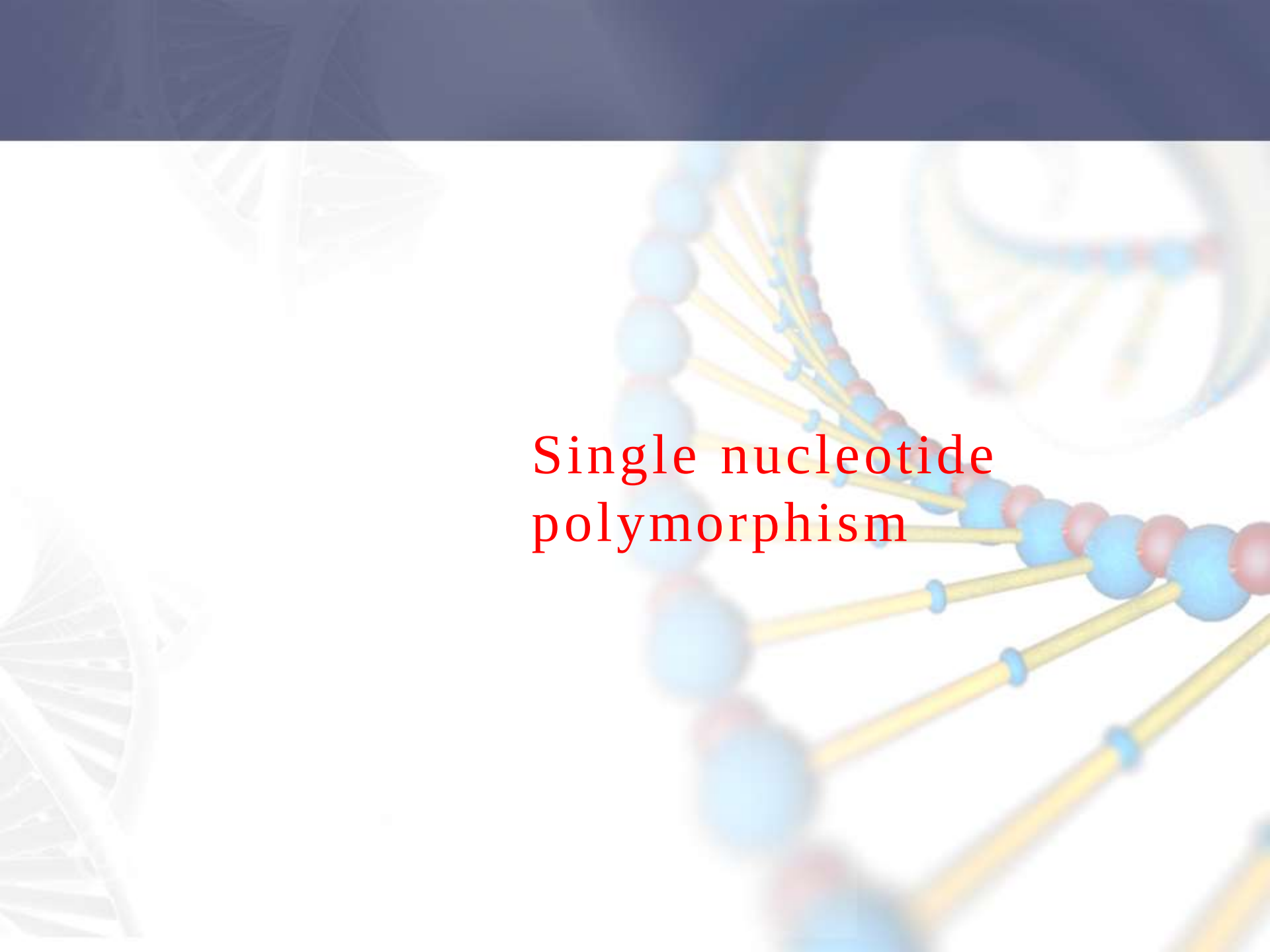
Oligosaccharides

- The A and B antigens differ from the O antigen by the addition of one extra monosaccharide, either *N*-acetylgalactosamine (for A) or galactose (for B) through an α-1,3 linkage to a galactose moiety of the O antigen.

Oligosaccharides

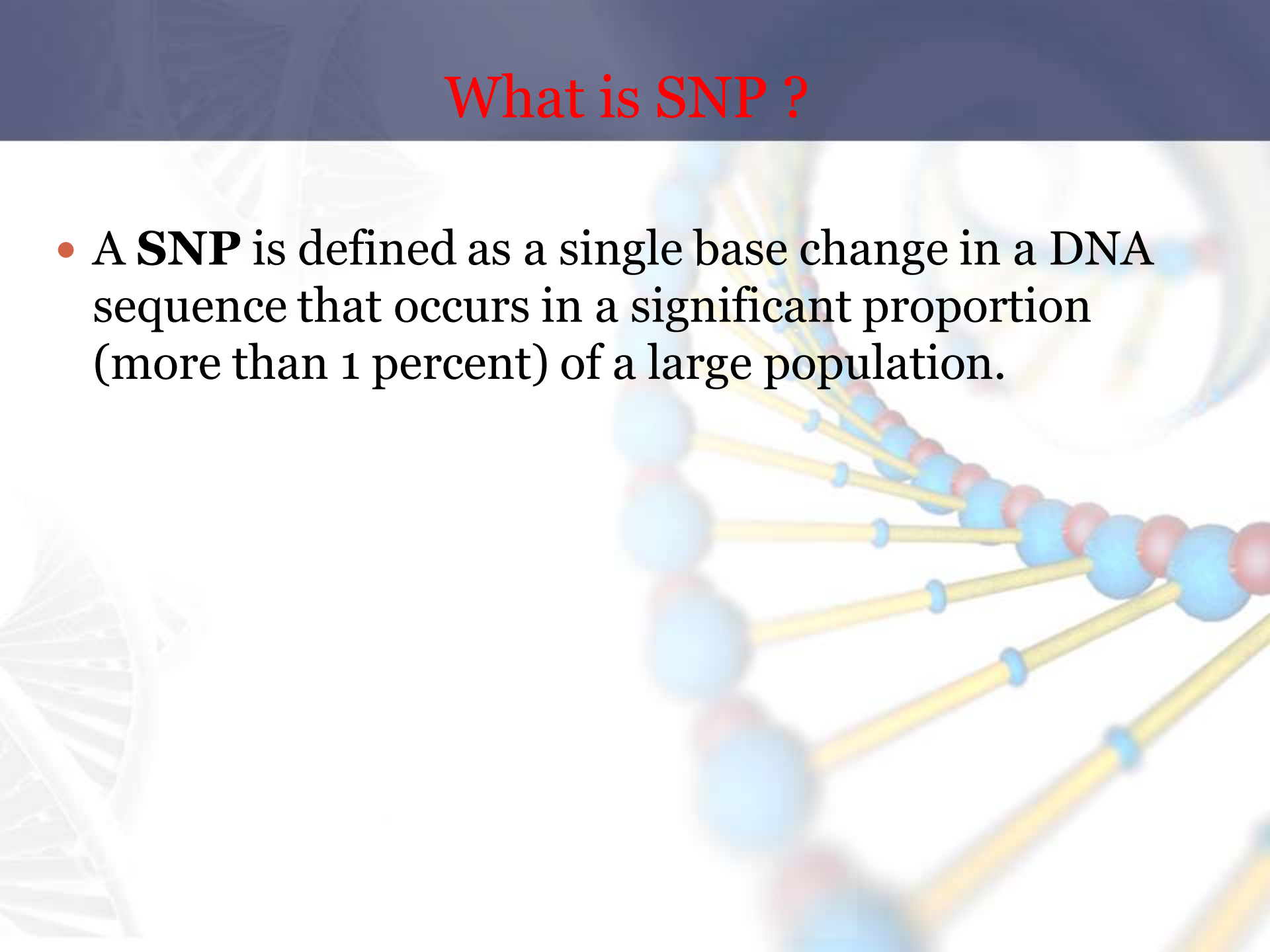


Single nucleotide polymorphism



What is SNP ?

- A **SNP** is defined as a single base change in a DNA sequence that occurs in a significant proportion (more than 1 percent) of a large population.



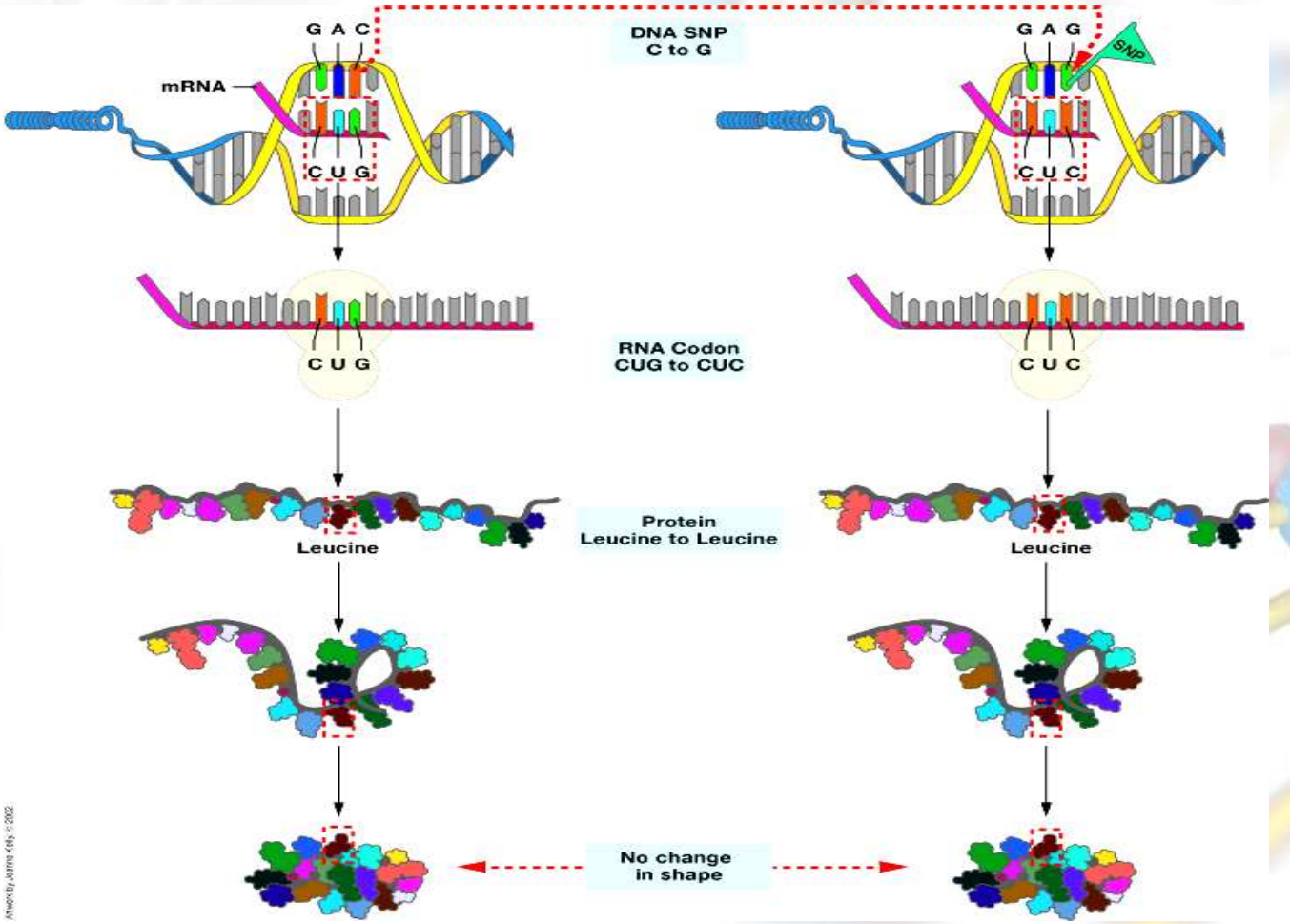
Some Facts

- In human beings, 99.9 percent bases are same.
- Remaining 0.1 percent makes a person unique.
  - Different attributes / characteristics / traits
    - how a person looks,
    - diseases he or she develops.

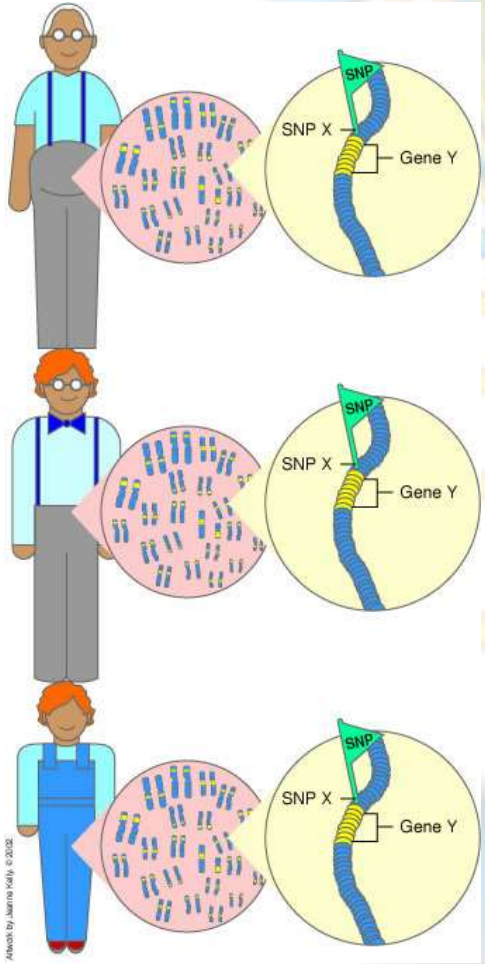
- These variations can be:
  - Harmless (change in phenotype)
  - Harmful (diabetes, cancer, heart disease, Huntington's disease, and hemophilia)
  - Latent (variations found in coding and regulatory regions, are not harmful on their own, and the change in each gene only becomes apparent under certain conditions e.g. susceptibility to lung cancer)

- SNPs are found in
  - coding and (mostly) noncoding regions.
- Occur with a very high frequency
  - about 1 in 1000 bases to 1 in 100 to 300 bases.
- The abundance of SNPs and the ease with which they can be measured make these genetic variations significant.
- SNPs close to particular gene acts as a marker for that gene.
- SNPs in coding regions may alter the protein structure made by that coding region.

SNPs may / may not alter protein structure



SNPs act as gene markers



Proteins

- Proteins mediate virtually every process that takes place in a cell, exhibiting an almost endless diversity of functions.

Proteins

- To explore the molecular mechanism of a biological process, a biochemist almost inevitably studies one or more protein

Proteins

- Proteins are the most abundant biological macromolecules occurring ;
- in all cells and
- all parts of cells

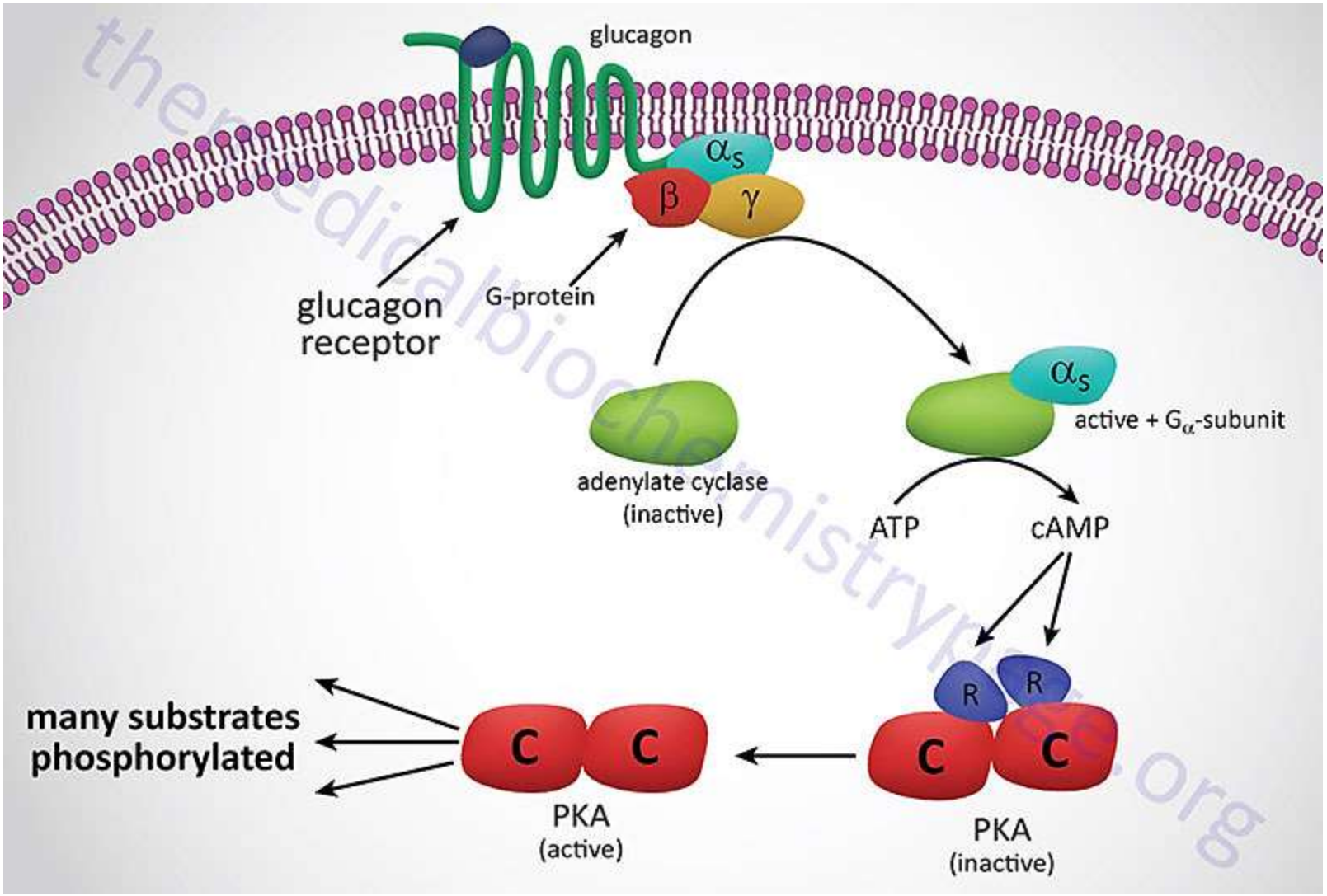
Proteins

- Proteins also occur in great variety; thousands of different kinds may be found in a single cell

Proteins

- Moreover, proteins exhibit enormous diversity of biological function and are the most important final products of the information pathways
- Example signal transduction

Proteins


glucagon
α_s
β
γ
glucagon receptor
G-protein
adenylate cyclase (inactive)
α_s
active + G_α-subunit
ATP
cAMP
R
R
C
C
C
C
many substrates phosphorylated
PKA (active)
PKA (inactive)

Biomedical importance

- Biochemical catalysts known as **enzymes** are proteins
- **Immunoglobulins** that serve as the first line of defense against bacteria and viruses and other foreign agents are proteins

Biomedical importance

- Several **hormones** are proteins
- **Structural proteins** provide **mechanical support** e.g collagen
- **Contractile proteins** help in the **movement** of muscle fiber and microvilli

Biomedical importance

- Some proteins present in the cell membrane, cytoplasm and nucleus act as **receptors**
- Certain other proteins in the cell membrane act as **channels and transporters**

Biomedical importance

- **The transport proteins** carry out the function of transporting specific substances across the membrane or in the body fluids
- **Storage proteins** bind with specific substances and store them

Biomedical importance

- Under certain conditions, proteins can be catabolized to **provide energy**
- Proteins help in the maintenance of water and electrolyte balance in the body by **exerting osmotic pressure**

Biomedical importance

- Hence proteins are the most versatile macromolecules in living systems and serve crucial functions in essentially all biological processes

Biomedical importance

- They function as catalysts, they transport and store other molecules such as oxygen,
- They provide mechanical support and immune protection

Biomedical importance

- They generate movement, and they control growth and differentiation

END

Properties of proteins

- Several key properties enable proteins to participate in such a wide range of functions

1. Proteins are linear polymers built of monomer units called amino acids

Properties of proteins

- The function of a protein is directly dependent on its three dimensional structure

Properties of proteins

- Remarkably, proteins spontaneously fold up into three-dimensional structures that are determined by the sequence of amino acids in the protein polymer

Properties of proteins

2. Proteins contain a wide range of functional groups

These functional groups include

- alcohols,
- thiols,
- thioethers,
- carboxylic acids,
- carboxamides,
- and a variety of basic groups

Properties of proteins

- When combined in various sequences, this array of functional groups accounts for the broad spectrum of protein function

Properties of proteins

3. Proteins can interact with one another and with other biological macromolecules to form complex assemblies

Properties of proteins

- The proteins within these assemblies can act synergistically to generate capabilities not afforded by the individual component proteins

Properties of proteins

- These assemblies include macromolecular machines that carry out;
- the accurate replication of DNA,
- the transmission of signals within cells, and many other essential processes

Properties of proteins

4. Some proteins are quite rigid, whereas others display limited flexibility

- Rigid units can function as structural elements in the cytoskeleton (the internal scaffolding within cells) or in connective tissue

Properties of proteins

- Parts of proteins with limited flexibility may act as
- hinges,
- springs, and
- levers

Properties of proteins

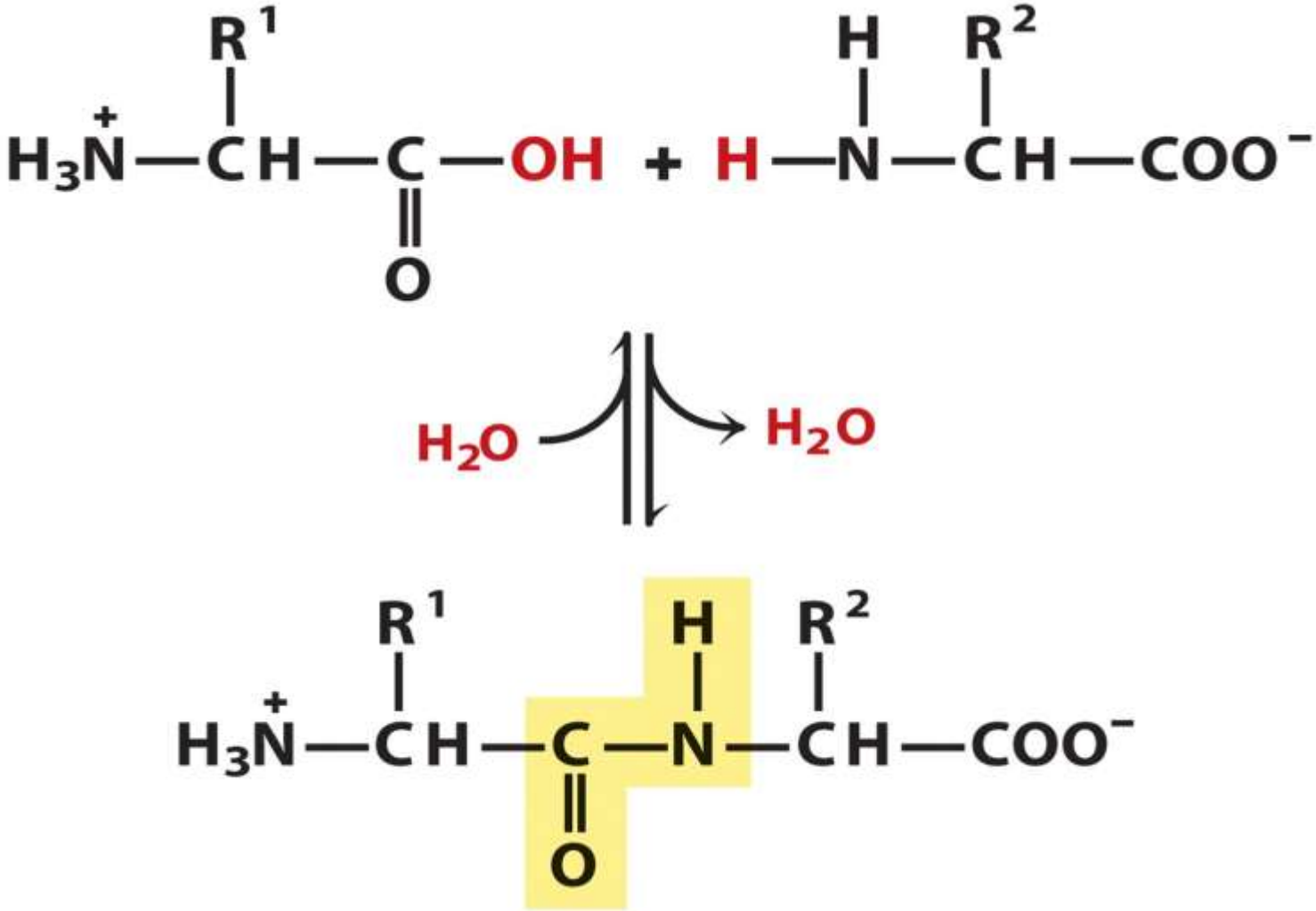
- That are crucial to protein function, to the assembly of proteins with one another and with other molecules into complex units
- And to the transmission of information within and between cells

END

Composition of Proteins

- Proteins are polymers of amino acids, with each amino acid residue joined to its neighbor by a specific type of covalent bond termed the **peptide bond**

Composition of Proteins



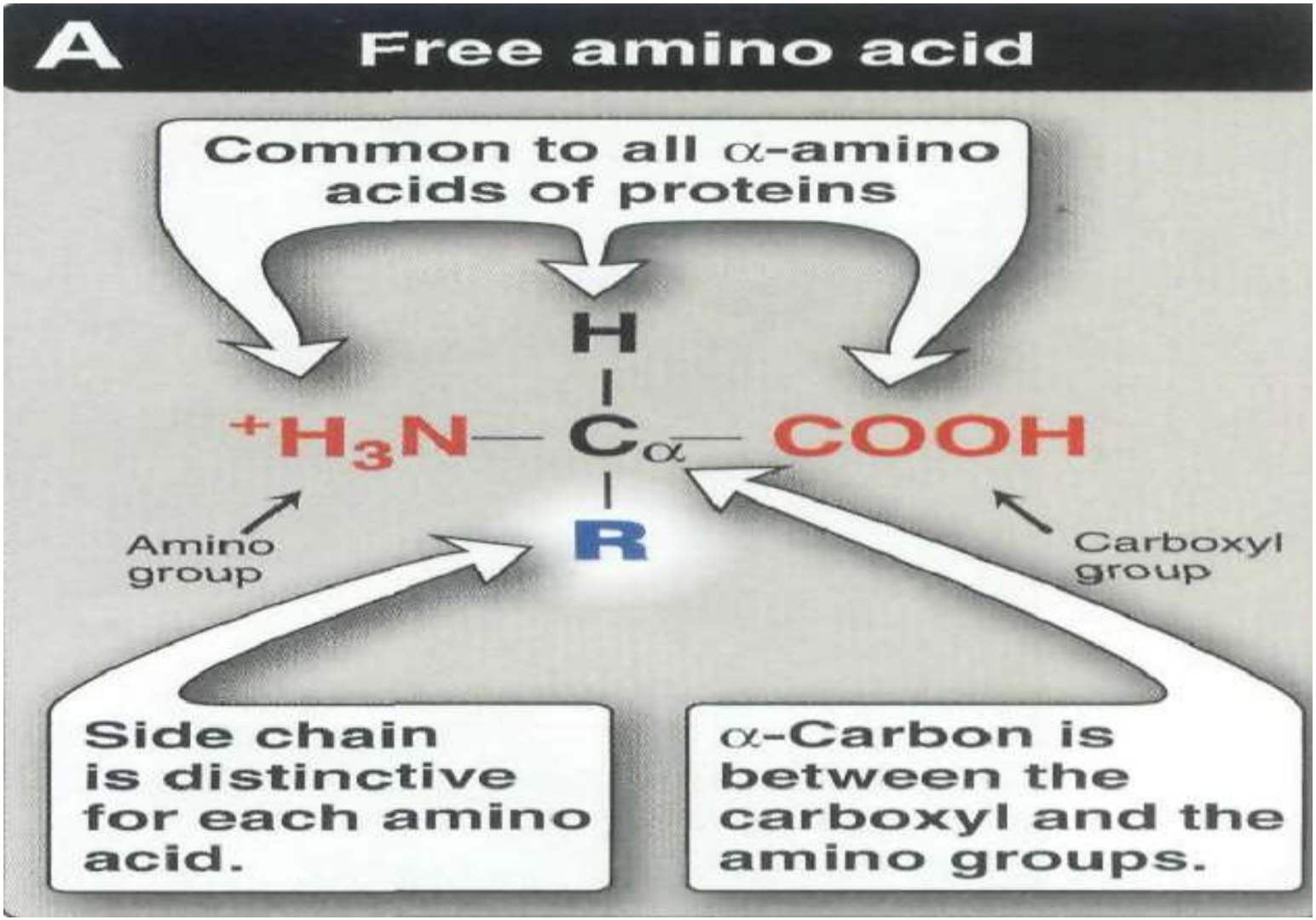
Composition of Proteins

- The term "residue" reflects the loss of the elements of water when one amino acid is joined to another
- Twenty different amino acids are commonly found in proteins

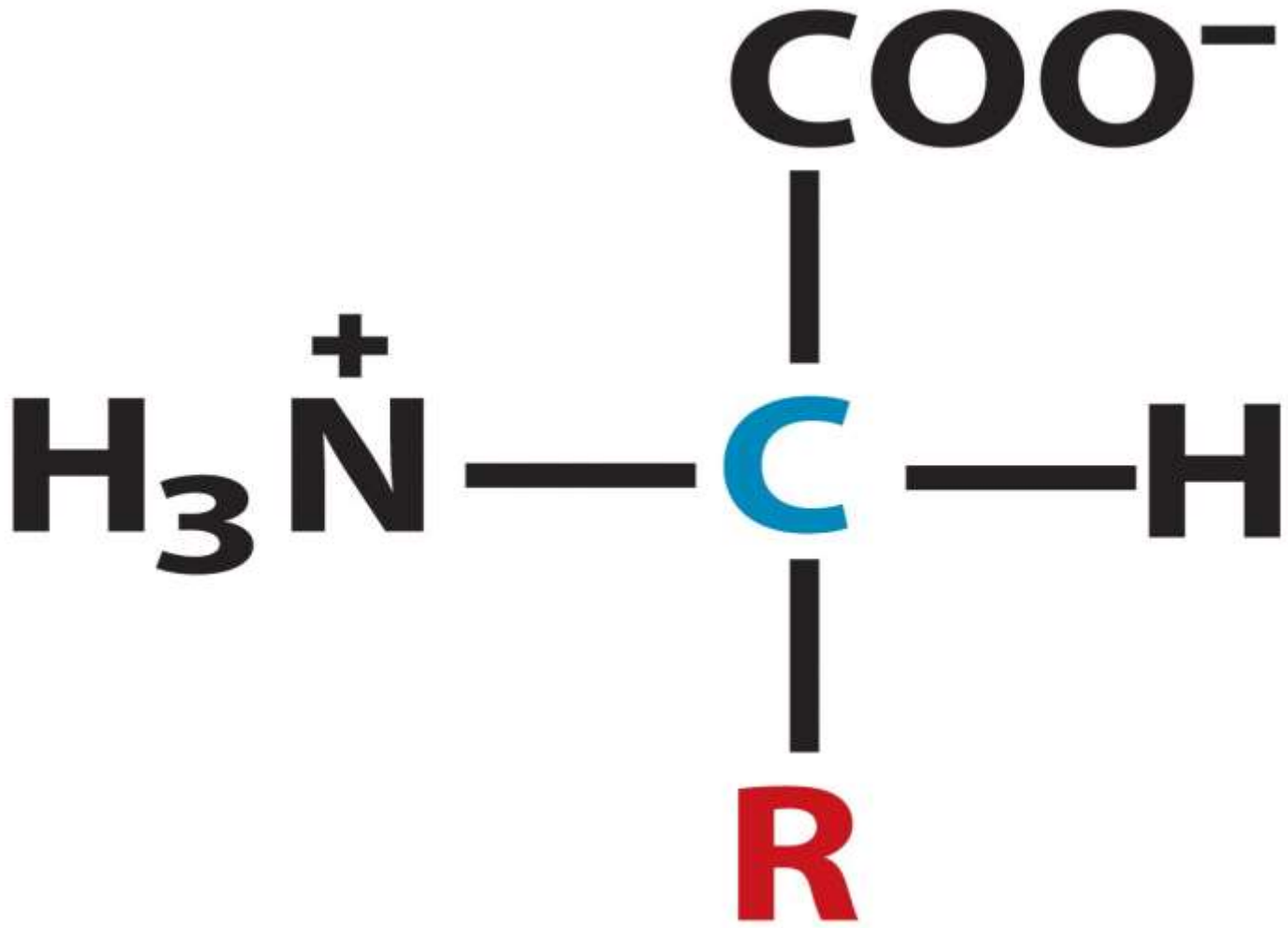
Composition of Proteins

- All of the 20 common amino acids are **α amino acids**
- They have a carboxyl group and an amino group bonded to the same carbon atom (the α carbon)

Composition of Proteins



Composition of Proteins



Composition of Proteins

- They differ from each other in their side chains, or R groups, which vary in
  - structure,
  - size,
  - electric charge,
  - and which influence the solubility of the amino acids in water

Composition of Proteins

Standard amino acids

- The 20 amino acids that constitute the monomer units of proteins are **the standard amino acids** as the genetic code specifies only these 20 L -α-amino acids

Composition of Proteins

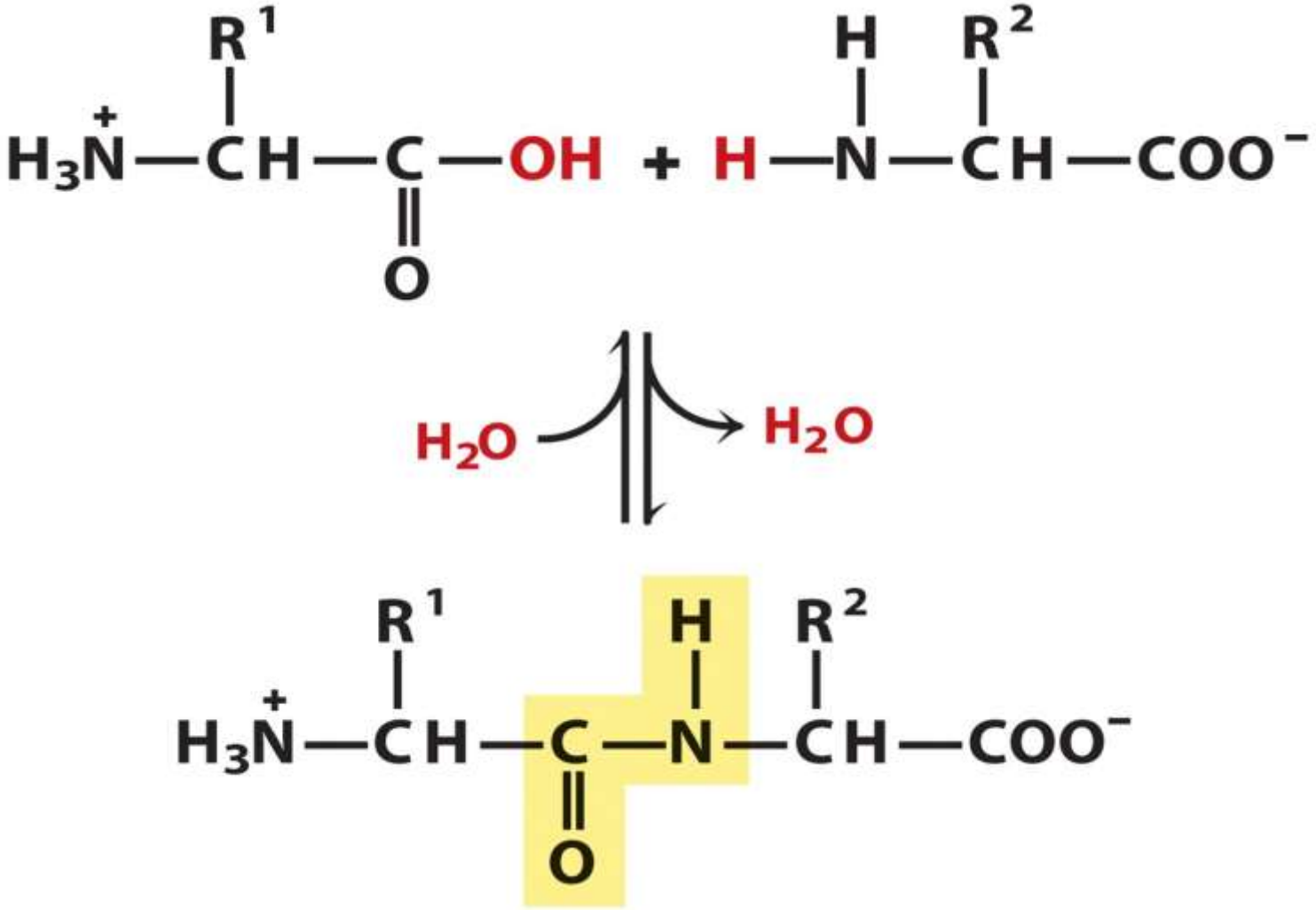
- In addition to these 20 amino acids there are many less common ones i.e., **the non standard amino acids**

END

Composition of Proteins

- Proteins are polymers of amino acids, with each amino acid residue joined to its neighbor by a specific type of covalent bond termed the **peptide bond**

Composition of Proteins



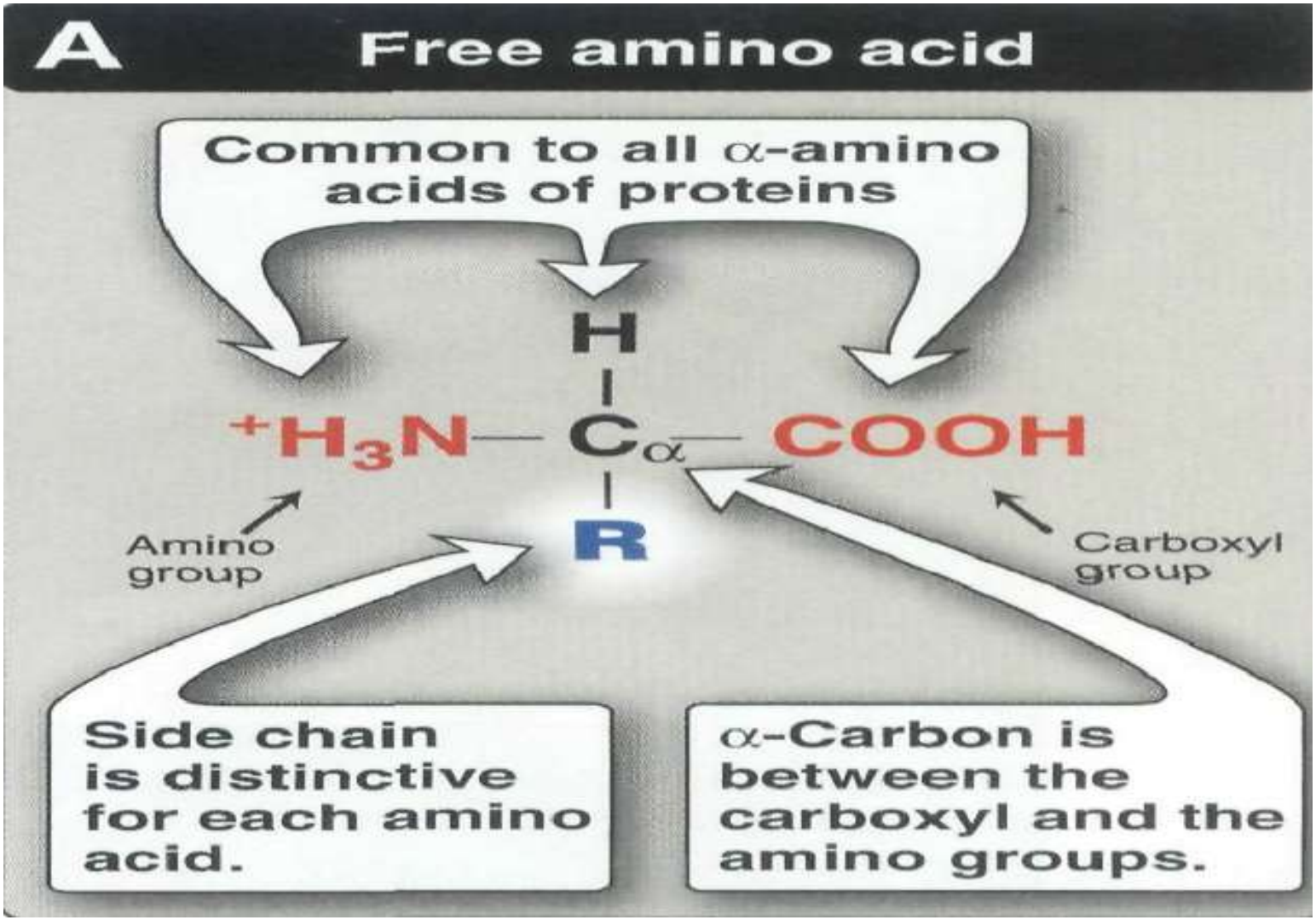
Composition of Proteins

- The term "residue" reflects the loss of the elements of water when one amino acid is joined to another
- Twenty different amino acids are commonly found in proteins

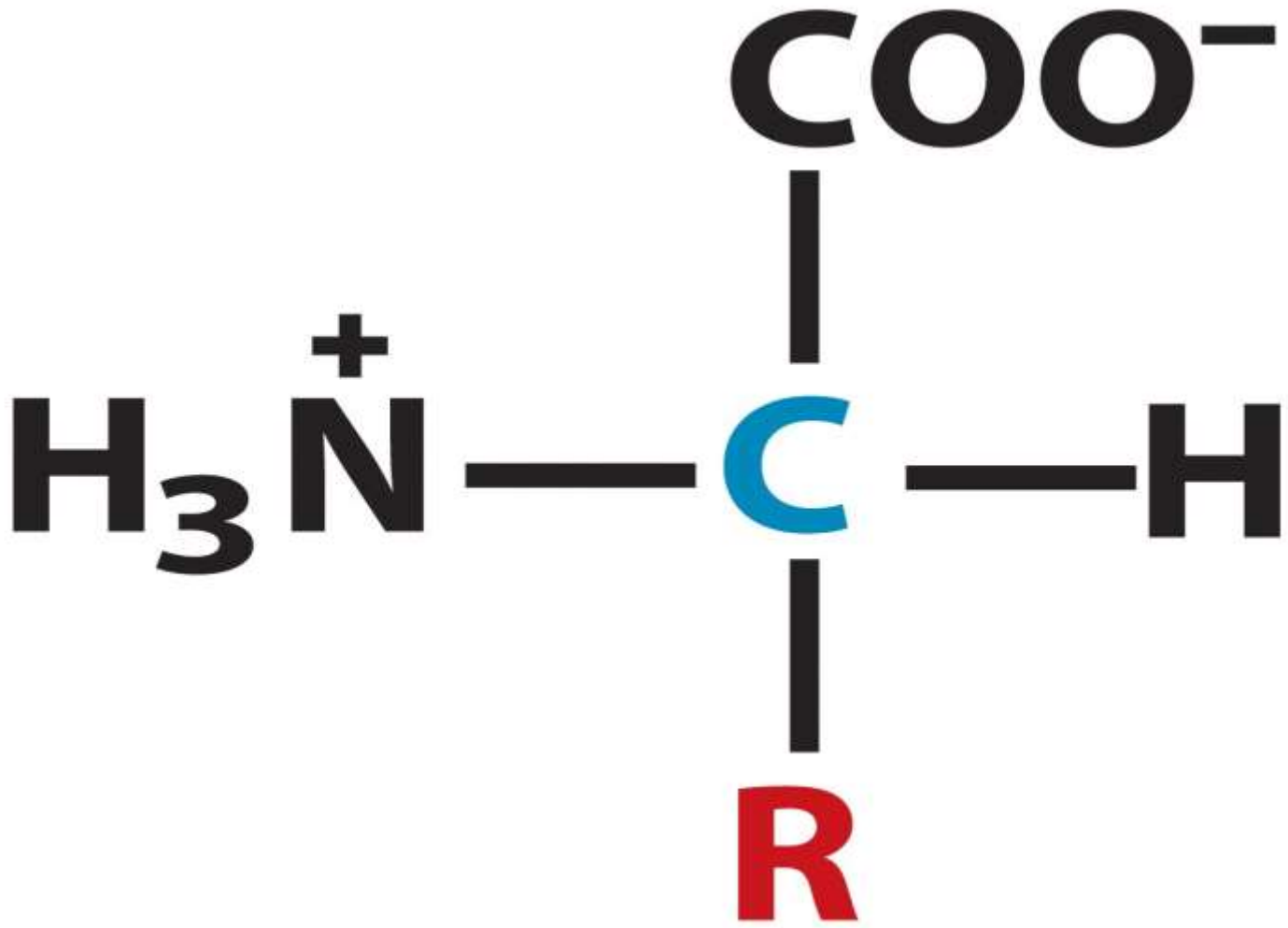
Composition of Proteins

- All of the 20 common amino acids are **α amino acids**
- They have a carboxyl group and an amino group bonded to the same carbon atom (the α carbon)

Composition of Proteins



Composition of Proteins



Composition of Proteins

- They differ from each other in their side chains, or R groups, which vary in
  - structure,
  - size,
  - electric charge,
  - and which influence the solubility of the amino acids in water

Composition of Proteins

Standard amino acids

- The 20 amino acids that constitute the monomer units of proteins are **the standard amino acids** as the genetic code specifies only these 20 L -α-amino acids

Composition of Proteins

- In addition to these 20 amino acids there are many less common ones i.e., **the non standard amino acids**

END

Composition of Proteins

- In addition to the 20 common amino acids, proteins may contain residues created by modification of common residues already incorporated in to a polypeptide

Composition of Proteins

- Some proteins contain additional amino acids that arise by modification of an amino acid already present in a peptide i.e., after the protein has been synthesized

Composition of Proteins

- Examples include
  - conversion of peptidyl proline and lysine to
  - 4-hydroxyproline and
  - 5-hydroxylysine
  - the conversion of peptidyl glutamate to carboxyglutamate

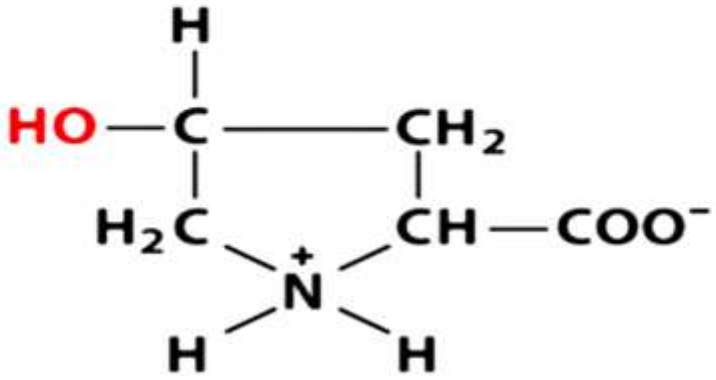
Composition of Proteins

- the methylation,
- formylation,
- acetylation,
- prenylation, and
- phosphorylation of certain aminoacyl residues

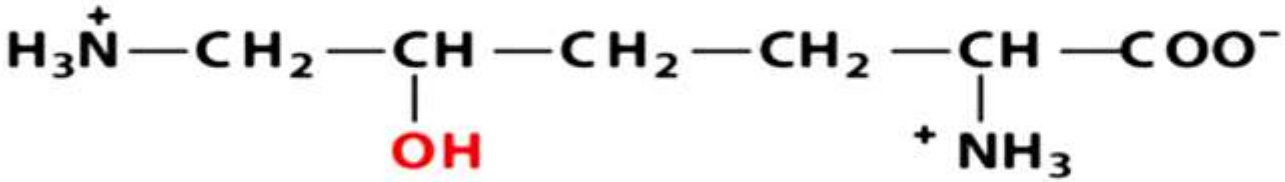
Composition of Proteins

- Among these uncommon amino acids
  - **4-hydroxyproline**, a derivative of proline
  - **5-hydroxylysine**, derived from lysine
- The former is found in plant cell wall proteins, and both are found in collagen

Composition of Proteins



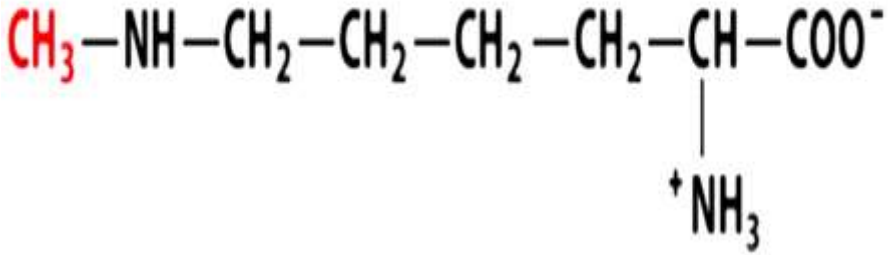
4-Hydroxyproline



5-Hydroxylysine

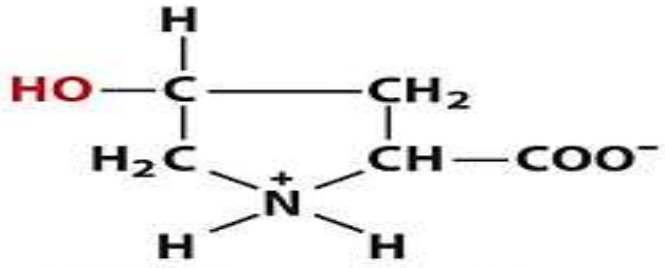
Composition of Proteins

- **6-N-Methyllysine** is a constituent of myosin, a contractile protein

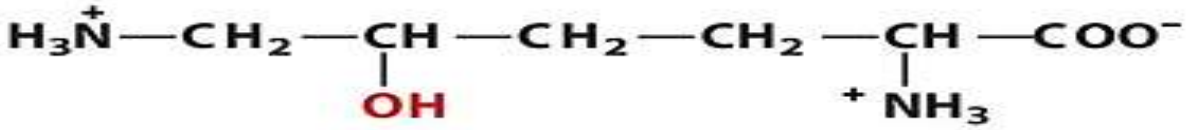


6-*N*-Methyllysine

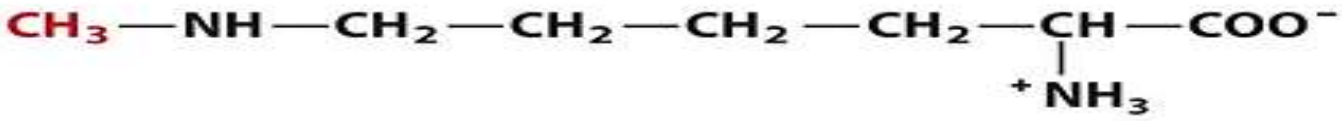
Composition of Proteins



4-Hydroxyproline



5-Hydroxylysine

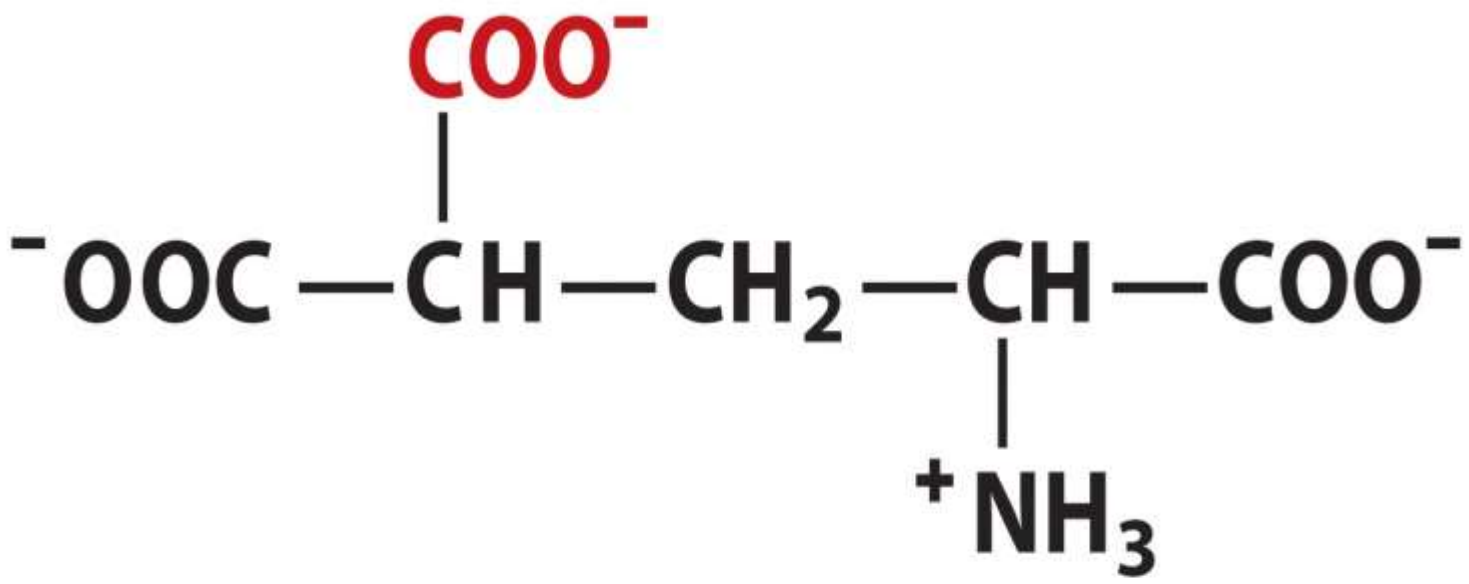


6-*N*-Methyllysine

Composition of Proteins

- Another important uncommon amino acid is;
- **γ -carboxyglutamate,** found in the blood clotting protein prothrombin and in certain other proteins that bind Ca^{2+} as part of their biological function

Composition of Proteins



γ-Carboxyglutamate

Composition of Proteins

- Some 300 additional amino acids have been found in cells but not as constituents of proteins
- Examples include
  - Ornithine and citrulline (intermediates in urea cycle)

Composition of Proteins

- Taurine (found in bile acids)
- δ-Aminolevulinic acid (intermediate in haem synthesis)

Composition of Proteins

D- AMINO ACIDS

- D-amino acids are also non standard amino acids that occur **naturally** and include
  - free D- serine, and D-aspartate in brain tissue

Composition of Proteins

  - D-alanine & D-glutamate in the cell walls of gram positive bacteria
- D-amino acids are also found in some **antibiotics**

Composition of Proteins

Selenocysteine, the 21st L-α-Amino Acid?

- Selenocysteine is an L-α-amino acid found in a handful of proteins, including certain **peroxidases and reductases** where it participates in the catalysis of electron transfer reactions

Composition of Proteins

- Its synthesis is not a post translational modification, but a modification to a **serine** that occurs while the serine is bound to a unique RNA
- The hydroxyl group of serine is replaced by a selenium atom

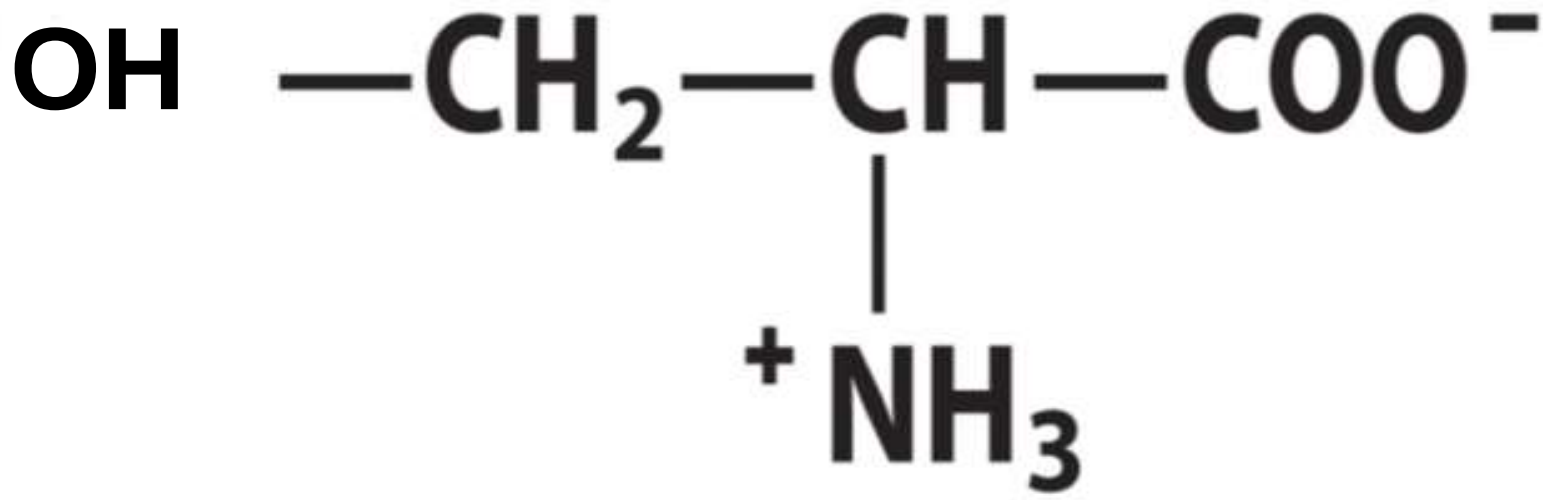
Composition of Proteins

- The selenocysteine is then inserted into a protein as it is being synthesized
- Since selenocysteine is inserted into polypeptides during translation, it is commonly referred to as the "21st amino acid"

Composition of Proteins

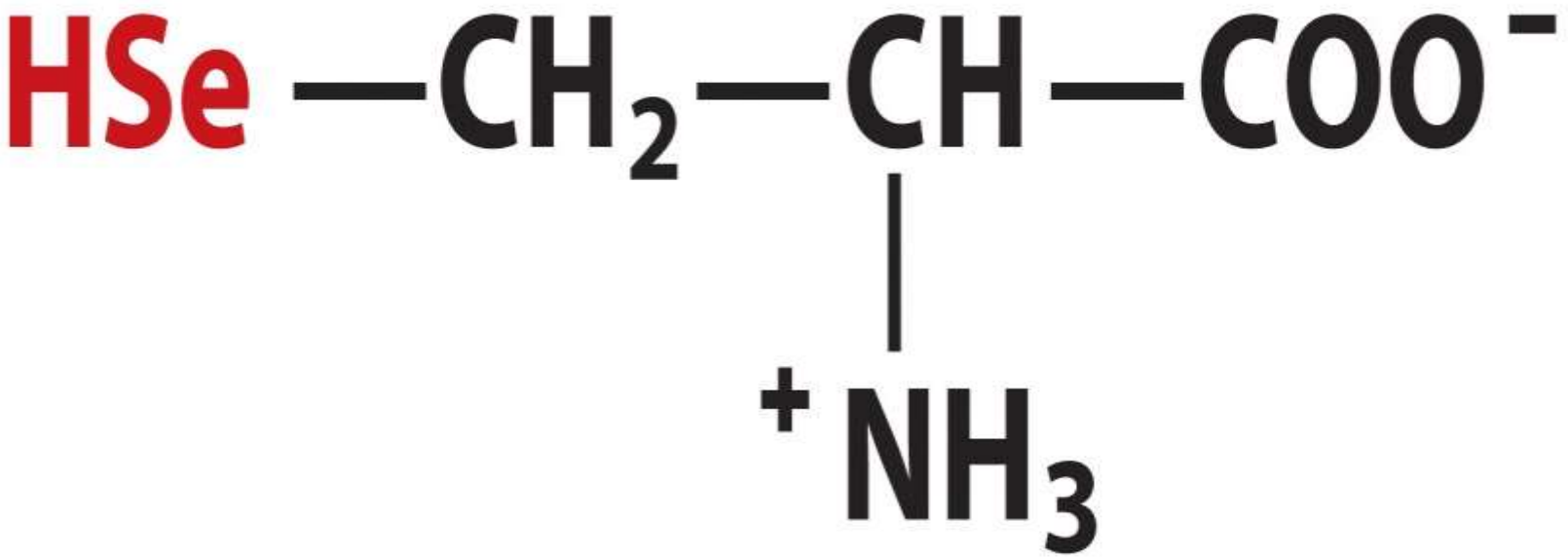
- However, unlike the other 20 genetically encoded amino acids, selenocysteine is not specified by a simple three-letter codon

Composition of Proteins



Serine

Composition of Proteins



Selenocysteine

Composition of Proteins

- **22nd Amino Acid**
- Pyrrolysine (abbreviated as Pyl or O) is a naturally occurring, genetically coded amino acid used by some methanogenic archaea.

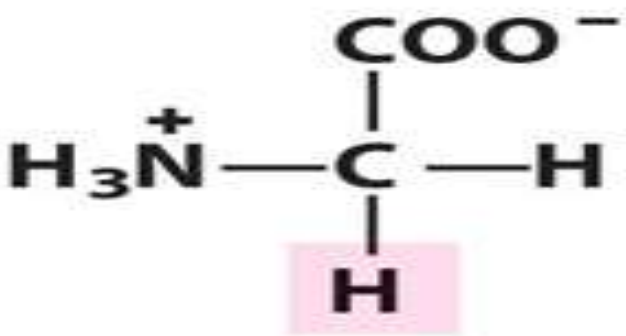
END

Composition of Proteins

Chiral centers and isomerism

- For all the common amino acids except glycine (the R group is another hydrogen atom), the α carbon is bonded to four different groups:

Composition of Proteins

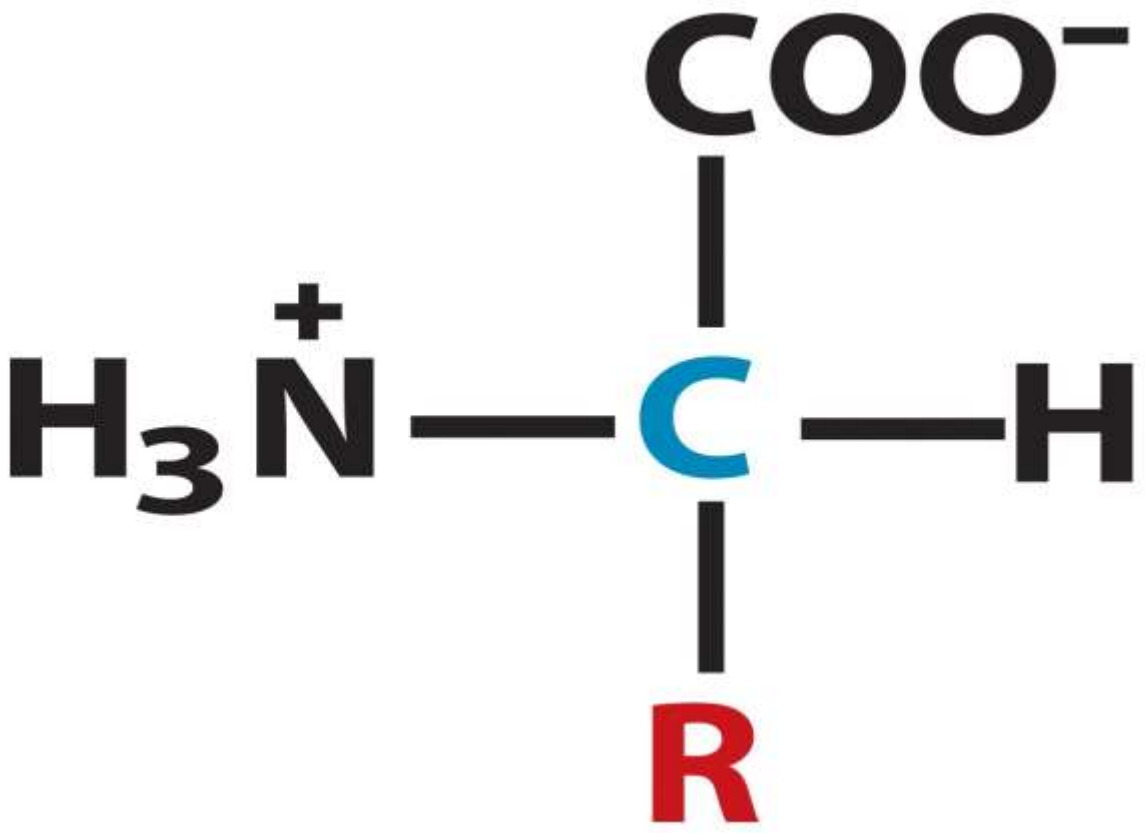


Glycine

Composition of Proteins

- a carboxyl group,
- an amino group,
- an R group,
- a hydrogen atom

Composition of Proteins



Composition of Proteins

- The α-carbon atom is thus a chiral center and thus amino acids have two possible stereoisomers
- All molecules with a chiral center are also optically active-that is, they rotate plane polarized light

Composition of Proteins

- Special nomenclature has been developed to specify the absolute configuration of the four substituents of asymmetric carbon atom

Composition of Proteins

- The absolute configurations of simple sugars and amino acids are specified by the D, L system, based on the absolute configuration of the three-carbon sugar glyceraldehyde

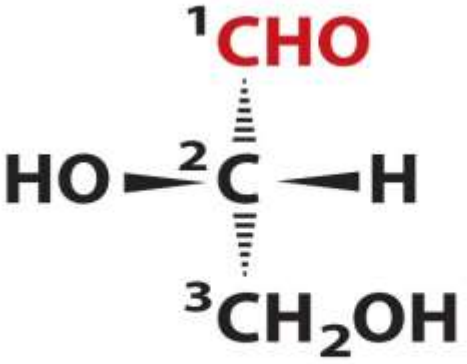
Composition of Proteins

- Thus the amino group of L-alanine occupies the same position about the chiral carbon as does the hydroxyl group of L-glyceraldehyde

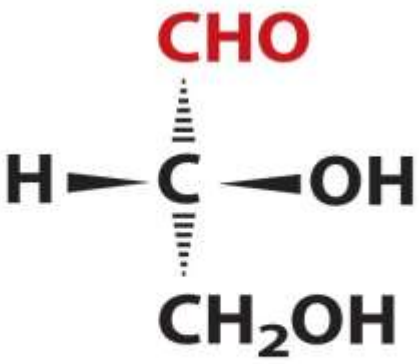
Composition of Proteins

- Since they are nonsuperposable mirror images of each other, the two forms represent a class of stereoisomers called **enantiomers**

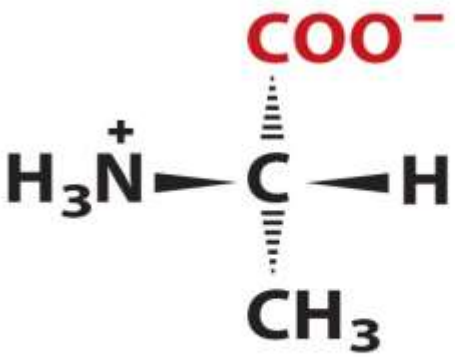
Composition of Proteins



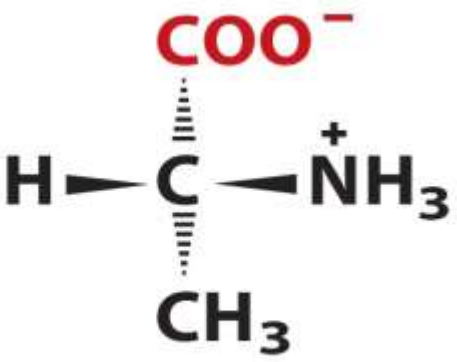
L-Glyceraldehyde



D-Glyceraldehyde



L-Alanine



D-Alanine

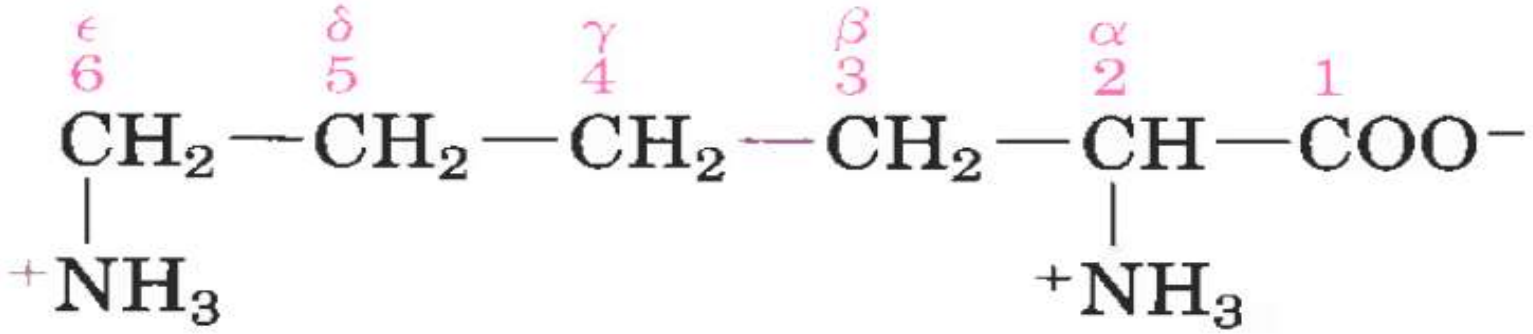
END

Conventions used to identify the carbons

Two conventions are used to identify the carbons in an amino acid

The additional carbons in an R group are commonly designated **β, γ, δ** and so forth, proceeding out from the **α carbon**

Conventions used to identify the carbons



Lysine

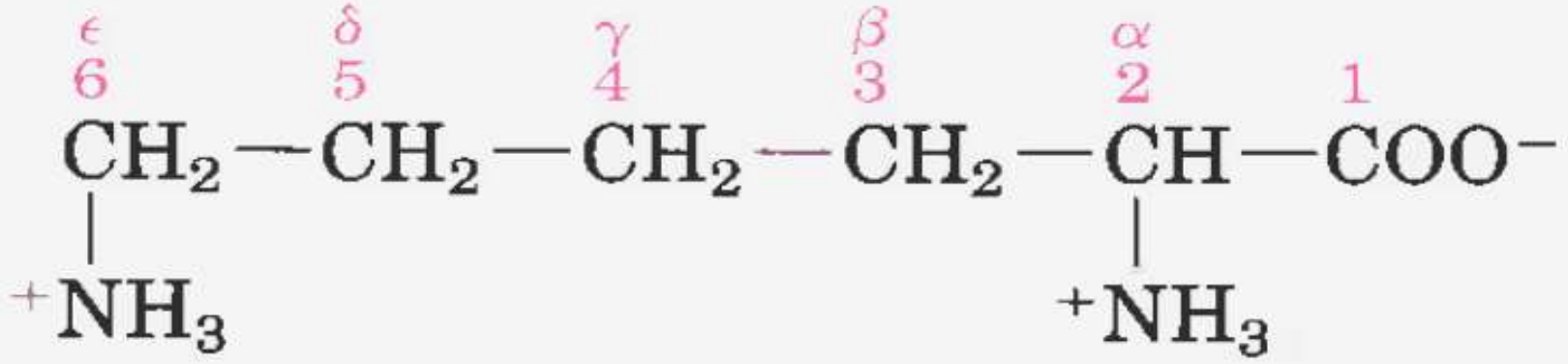
Conventions used to identify the carbons

- For most other organic molecules, carbon atoms are simply numbered from one end, giving highest priority (C-1) to the carbon with the substituent containing the atom of **highest atomic number**

Conventions used to identify the carbons

- Within this latter convention, the carboxyl carbon of an amino acid would be **C-1** and the **α** carbon would be **C-2**

Conventions used to identify the carbons

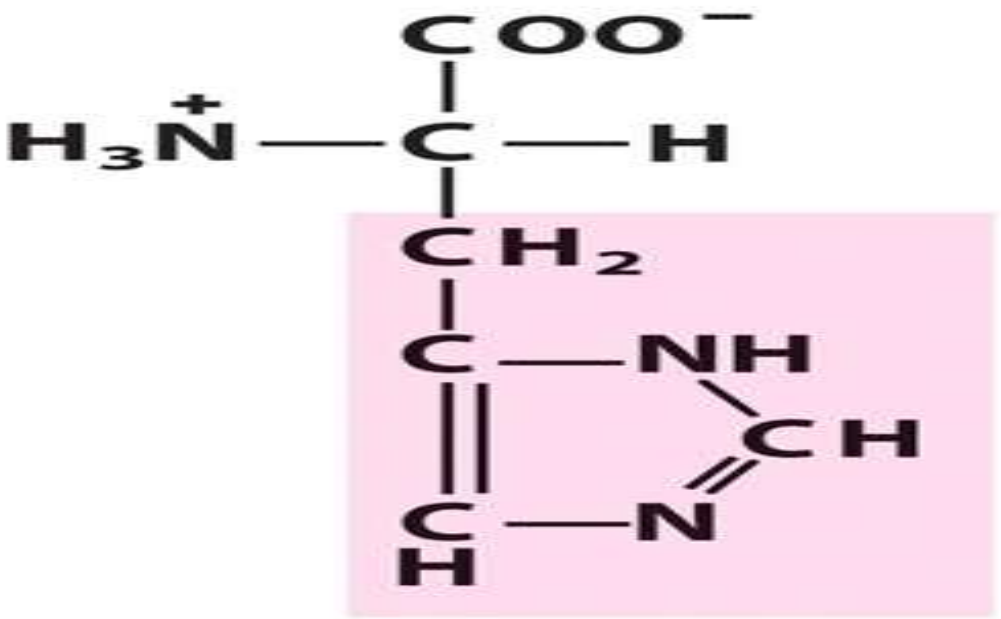


Lysine

Conventions used to identify the carbons

- In some cases, such as amino acids with heterocyclic R groups (such as histidine), the Greek lettering system is ambiguous and the numbering convention is therefore used

Conventions used to identify the carbons



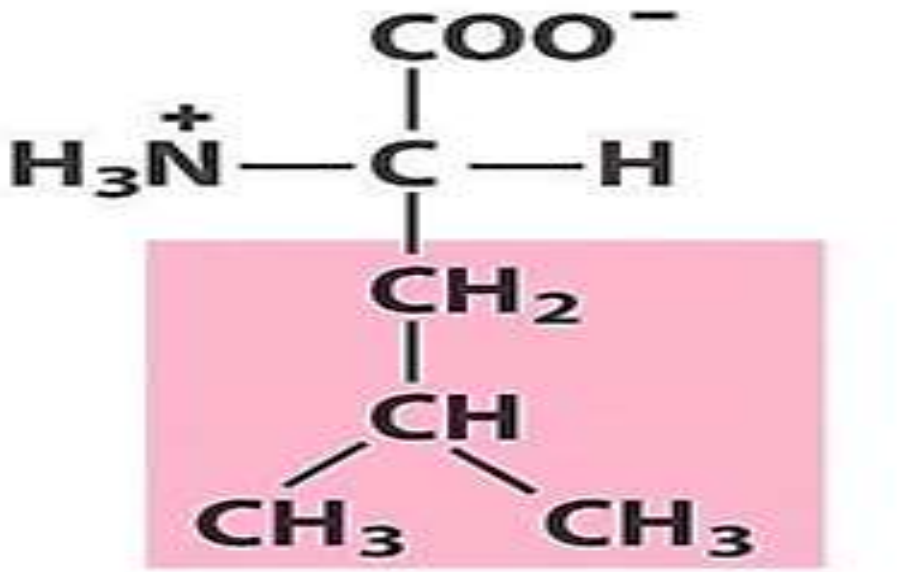
Histidine

Conventions used to identify the carbons

- For branched amino acid side chains, equivalent carbons are given numbers after the Greek letters

- Leucine thus has δ1 and δ2 carbons

Conventions used to identify the carbons



Leucine

END

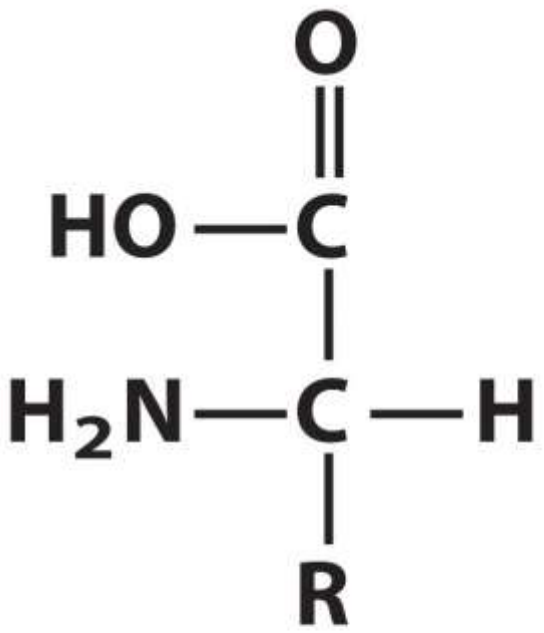
Acidic and basic properties of amino acids

- The form of an amino acid that has both a positive and a negative charge is called a **zwitterion**

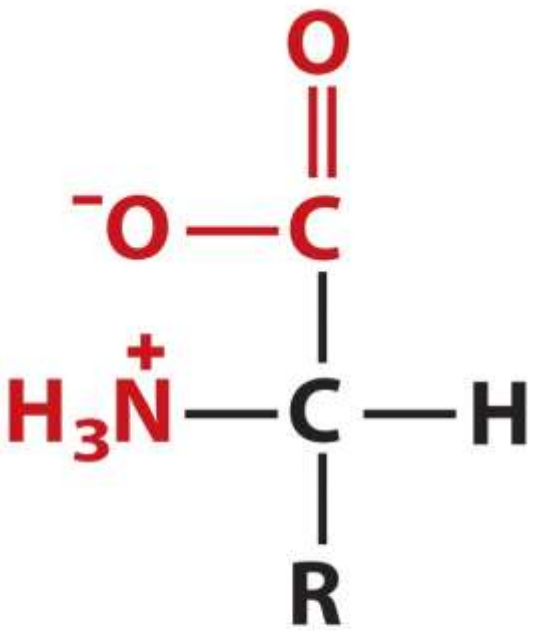
Acidic and basic properties of amino acids

- Amino acids in solution at neutral pH exist predominantly as dipolar ions **(zwitterions)**

Acidic and basic properties of amino acids



Nonionic form



Zwitterionic form

Acidic and basic properties of amino acids

- In this form, the amino group is protonated ($–NH_3^+$)
- and the carboxyl group is deprotonated ($–COO^-$)
- A zwitterion can act either as an acid (proton donor) or as a base (proton acceptor)

Acidic and basic properties of amino acids

- Substances having this dual nature are **amphoteric** and are often called **ampholytes**

Acidic and basic properties of amino acids

Acidic and basic properties of amino acids

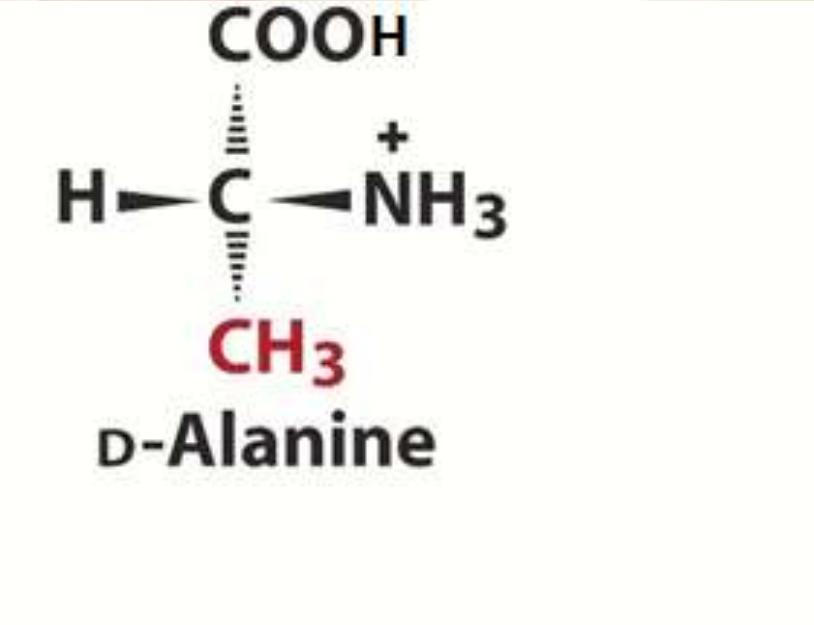
Acidic and basic properties of amino acids

- The ionization state of an amino acid varies with pH
- At physiological pH, the carboxyl group is dissociated , forming the negatively charged carboxylate ion ($-COO^-$) and the amino group is protonated (NH_3^+)

Acidic and basic properties of amino acids

- A simple monoamino monocarboxylic -amino acid, such as alanine, is a diprotic acid when fully protonated—it has two groups, the-COOH group and the - NH_3^+ group, that can yield protons

Acidic and basic properties of amino acids



END

Acidic and basic properties of amino acids

- Each acid has a characteristic tendency to lose its proton in an aqueous solution
- The stronger the acid, the greater its tendency to lose its proton

Acidic and basic properties of amino acids

- The tendency of any acid (HA) to lose a proton and form its conjugate base (A^-) is defined by the equilibrium constant (K_{eq}) for the reversible reaction
- $HA \rightleftharpoons A^- + H^+$

Acidic and basic properties of amino acids

- Equilibrium constants for ionization reactions are usually called ionization constants or acid dissociation constants, often designated pK_a

Acidic and basic properties of amino acids

- The stronger the tendency to dissociate a proton, **the stronger is the acid and the lower its pK_a**
- The p*K*a expresses, on a logarithmic scale, the relative strength of a weak acid or base:

Acidic and basic properties of amino acids

- $$pH = pK_a + \log \frac{[A^-]}{[HA]}$$

- p*K*a = $\log \frac{1}{Ka} = -LogKa$

- The stronger the acid, the lower its p*K*a;

Acidic and basic properties of amino acids

- The p*K*a can be determined experimentally;
- it is the pH at the midpoint of the titration curve for the acid or base

Acidic and basic properties of amino acids

- At the midpoint of the titration, the concentrations of the proton donor and proton acceptor are equal, and the pH is numerically equal to the p*K*a.

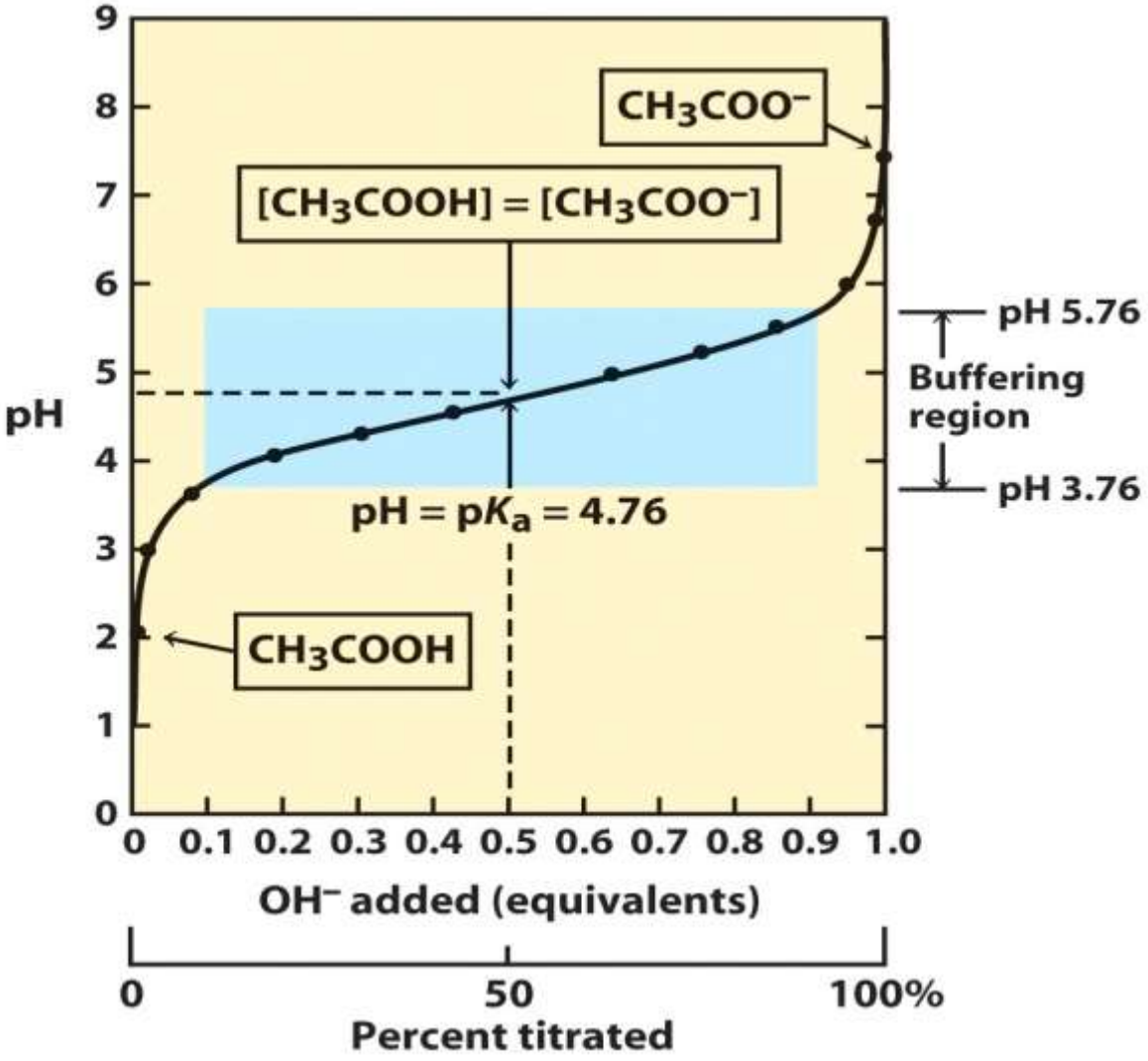
Acidic and basic properties of amino acids

- $$pH = pK_a + \log \frac{[A^-]}{[HA]}$$

- At that point, [HA] equals [A], and

- pH = p*K*a + log 1
- = p*K*a + 0 = p*K*a

Acidic and basic properties of amino acids



Acidic and basic properties of amino acids

- pKa is the measure of the tendency of a group to give up (donate) a proton with the tendency decreasing ten-fold as the pKa increases by one unit

Acidic and basic properties of amino acids

- Each titratable group has a pK_a that is numerically equal to the pH at which exactly one half of the protons have been removed from that group

Acidic and basic properties of amino acids

- The pK_a for the most acidic group (COOH) is pK_1, whereas the pK_a for the next most acidic group (NH_3^+) is pK_2
- **In acid solution** (e.g., pH 1), the amino group is protonated ($–NH_3^+$) and the carboxyl group is not dissociated (–COOH)

Acidic and basic properties of amino acids

- As the pH is raised, the carboxylic acid is the first group to give up a proton, as its p*Ka is near 2*
- The dipolar form persists until the pH approaches 9, when the protonated amino group loses a proton

END

Acidic and basic properties of amino acids

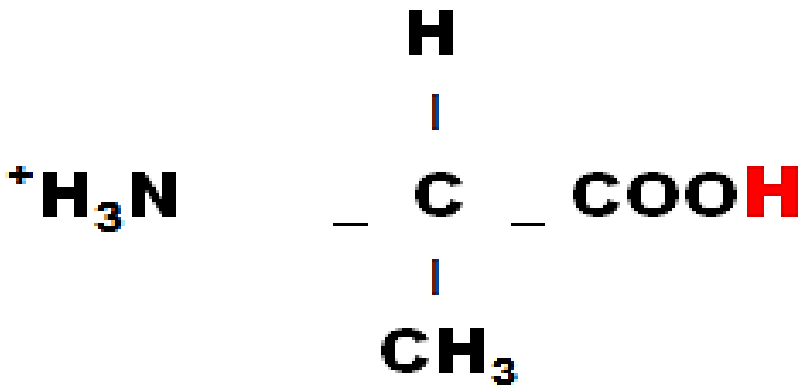
Titration of an amino acid Dissociation of the carboxyl group:

- The titration curve of an amino acid can be analyzed
- Consider alanine, for example, which contains both an α-carboxyl and an α-amino group

Acidic and basic properties of amino acids

- At a low (acidic) pH, both of these groups are protonated

Acidic and basic properties of amino acids



FORM I

Alanine in acid solution
(pH less than 2)

Net charge = + 1

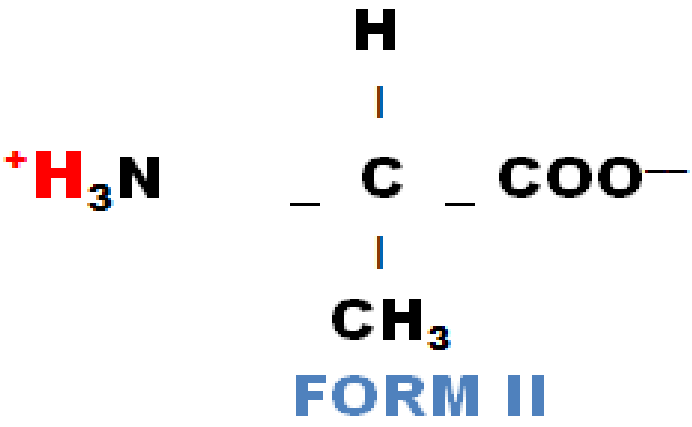
Acidic and basic properties of amino acids

- As the pH of the solution is raised, the -COOH group of form I can dissociate by donating a proton to the medium
- The release of a proton results in the formation of the carboxylate group $–COO^-$

Acidic and basic properties of amino acids

- This structure is shown as form II, which is the **dipolar form of the molecule**
- This form, also called a **zwitterion,** is the **isoelectric form** of alanine
- That is, it has an overall charge of zero

Acidic and basic properties of amino acids



FORM II

Alanine in neutral solution (pH approximately 6)

Net charge = 0 (isoelectric form)

Acidic and basic properties of amino acids

- The dissociation constant of the carboxyl group of an amino acid is called K_1 rather than K_2 because the molecule contains a second titratable group

Acidic and basic properties of amino acids

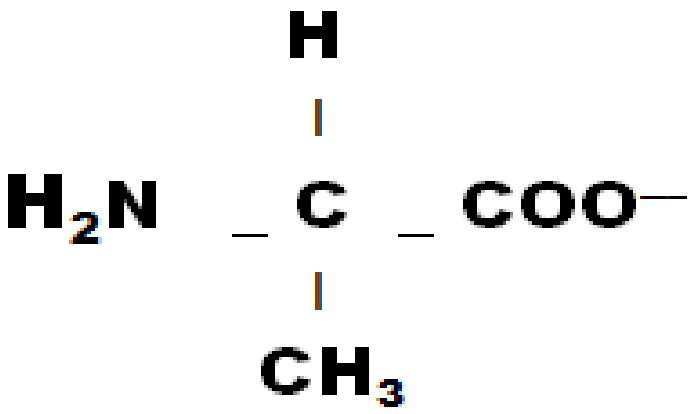
Dissociation of the amino group:

- The second titratable group of alanine is the amino group
- This is a much weaker acid than the -COOH group and, therefore, has a much smaller dissociation constant, K_2

Acidic and basic properties of amino acids

- Release of a proton from the protonated amino group of form II results in the fully deprotonated form of alanine, form III

Acidic and basic properties of amino acids

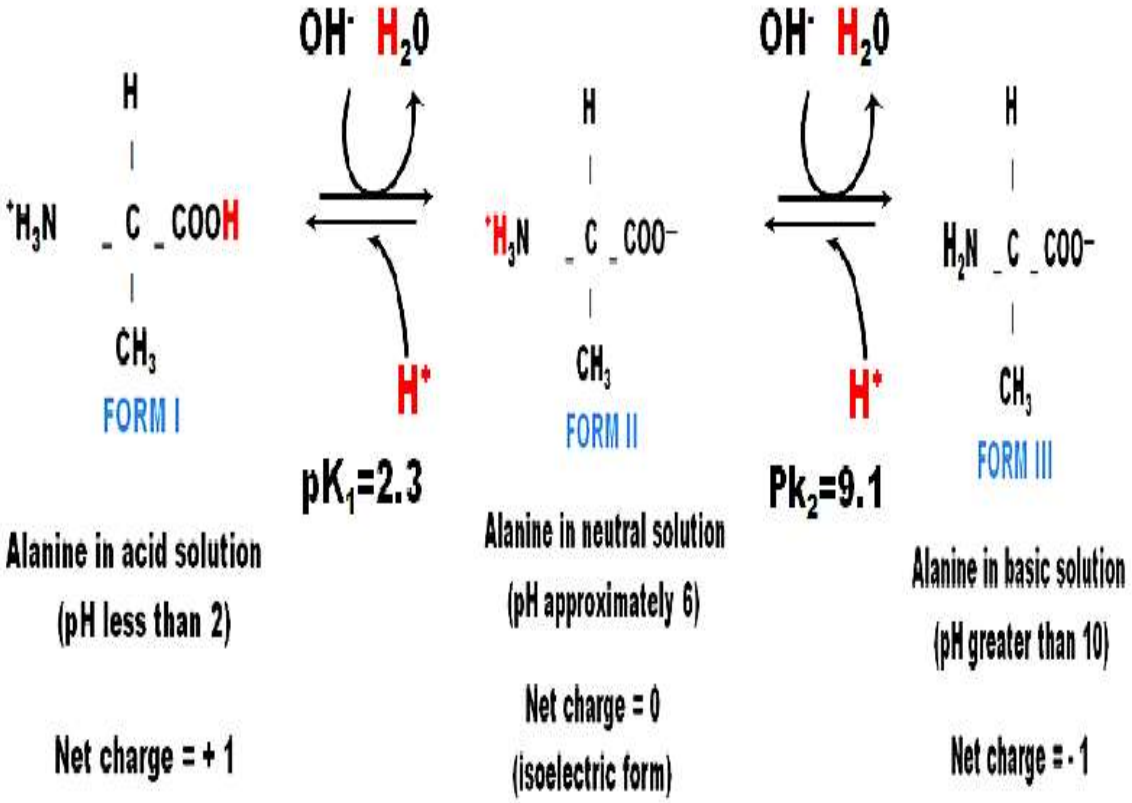


FORM III

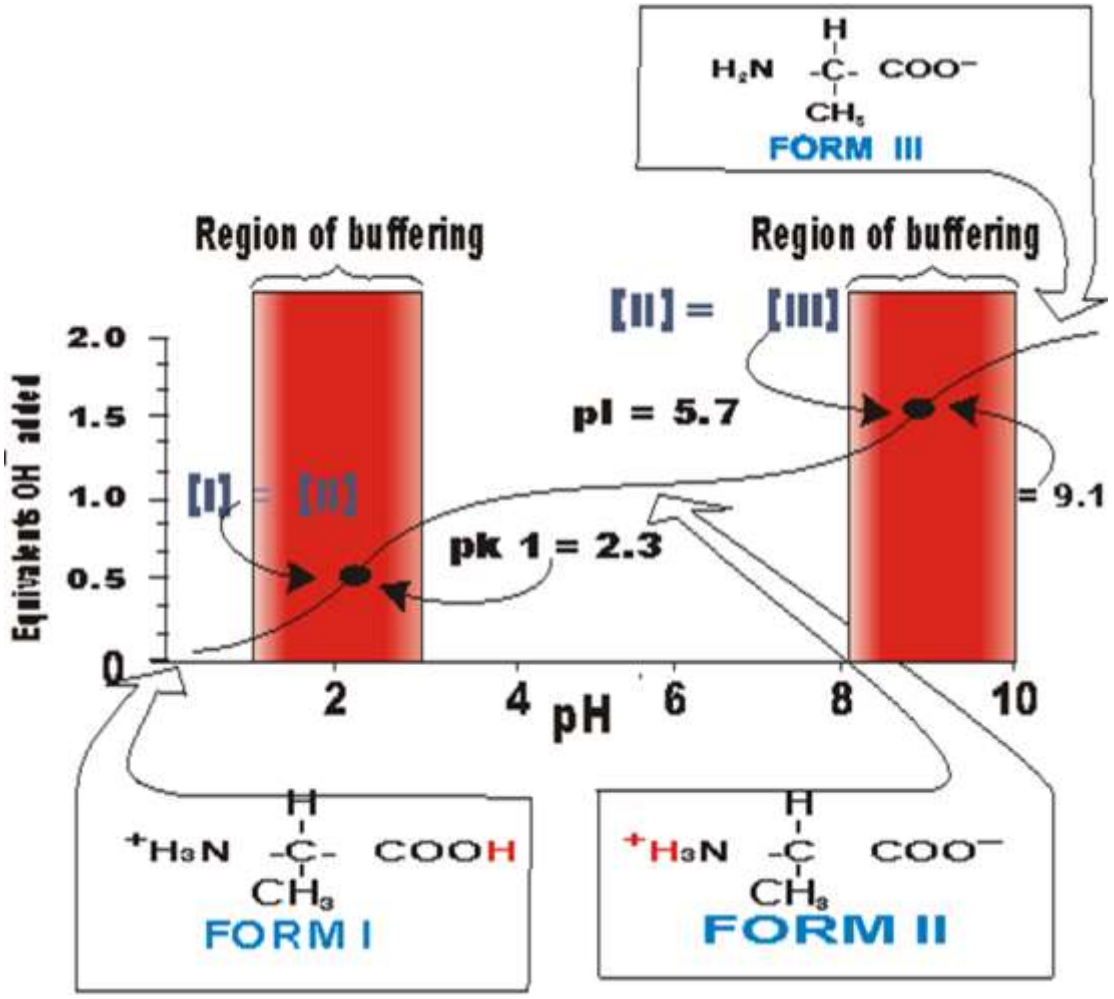
Alanine in basic solution
(pH greater than 10)

Net charge = - 1

Acidic and basic properties of amino acids



Acidic and basic properties of amino acids



END

Acidic and basic properties of amino acids

- A simple amino acid e.g. **glycine** is a diprotic acid when fully protonated
- This means that it has two groups, the COOH and the NH_3^+ group that can yield protons

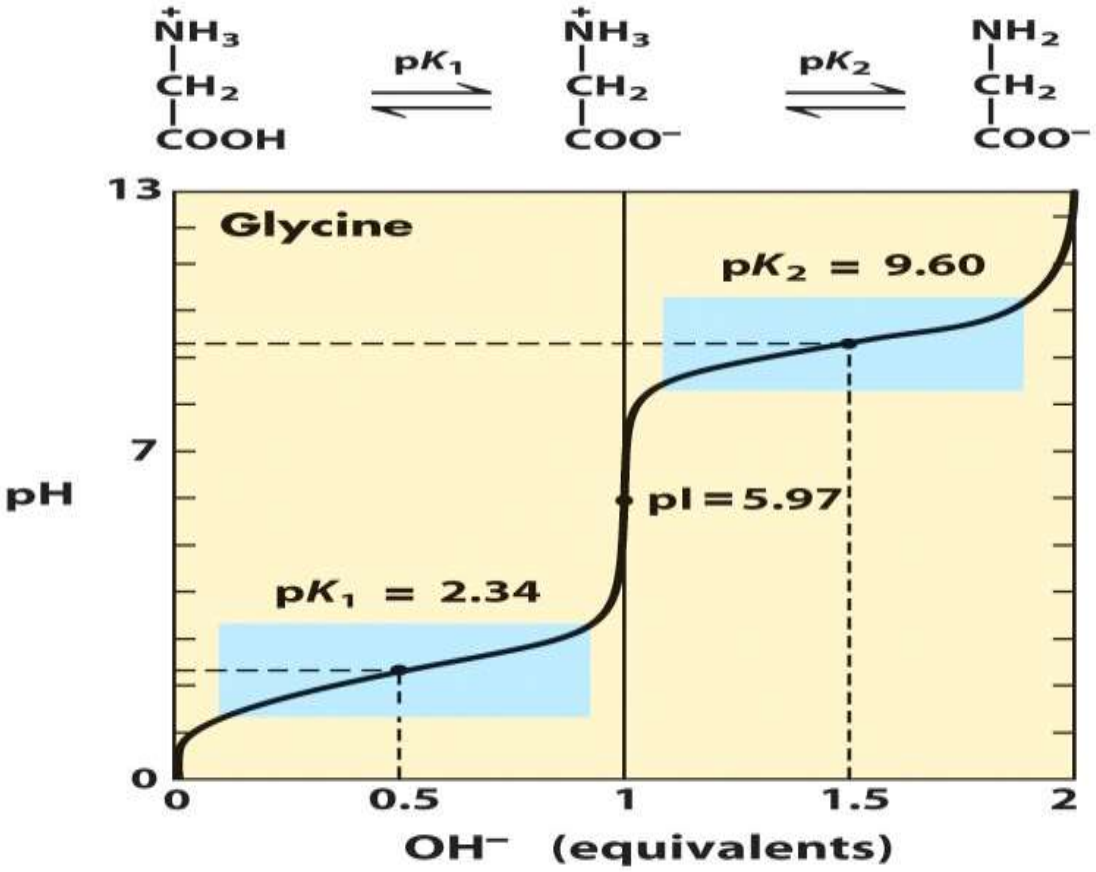
Acidic and basic properties of amino acids

- Figure shows the titration curve of the diprotic form of glycine
- The two ionizable groups of glycine, the carboxyl group and the amino group, are titrated with a strong base

Acidic and basic properties of amino acids

- The plot has two distinct stages, corresponding to deprotonation of two different groups on glycine

Acidic and basic properties of amino acids



Acidic and basic properties of amino acids

- At very low pH, the predominant ionic species of glycine is **the fully protonated form**,

$^{+}H_3N\text{-}CH_2\text{-}COOH$

Acidic and basic properties of amino acids

- **At the midpoint** of any titration, a point is reached where the **pH is equal to the pK_a** of the protonated group being titrated

Acidic and basic properties of amino acids

- At the midpoint in the first stage of the titration, in which the --COOH group of glycine loses its proton, equimolar concentrations of the protonated ($^{+}H_3N$-CH_2-COOH) and the deprotonated ($^{+}H_3N$-CH_2-COO^{-}) species are present

Acidic and basic properties of amino acids

- As the titration proceeds, another important point is reached at pH 5.97
- This is a point, at which removal of the first proton is essentially complete and removal of the second has just begun

Acidic and basic properties of amino acids

- At this pH glycine is present largely as the dipolar ion (zwitterion)

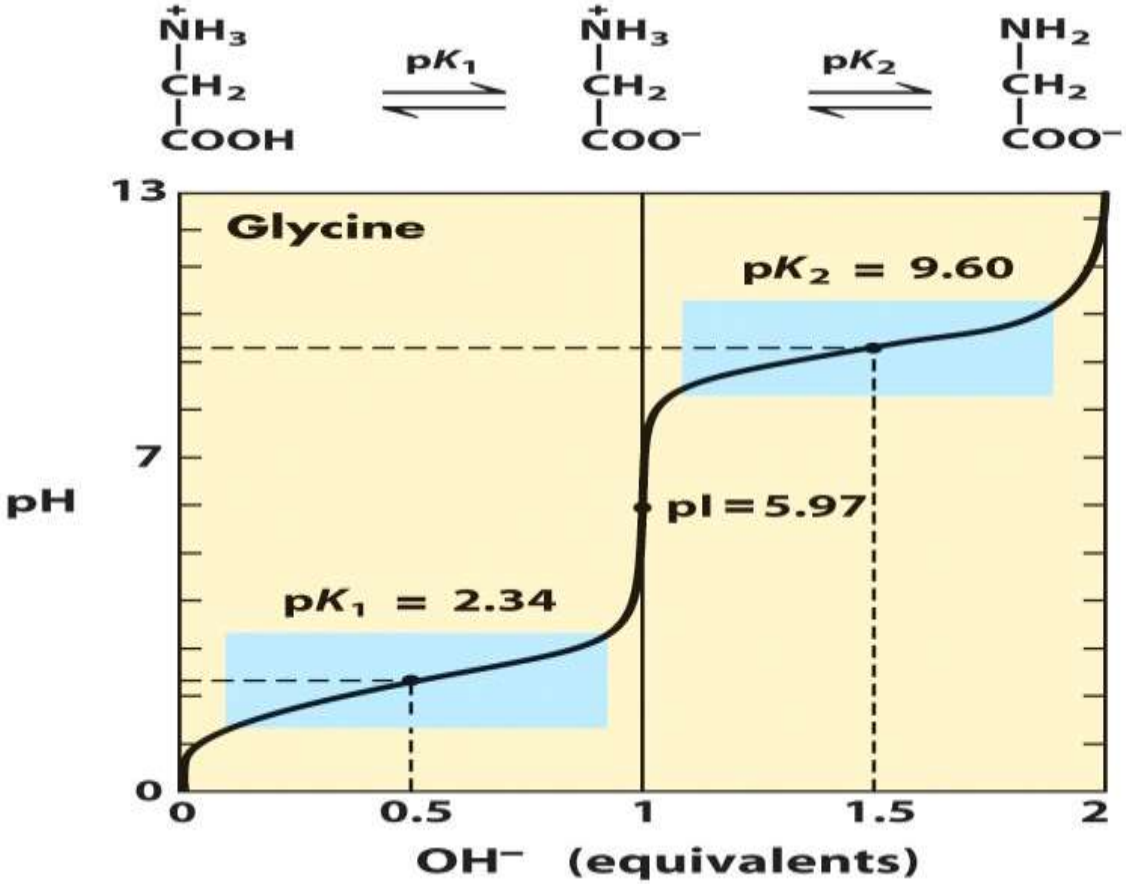
  $^{+}H_3N\text{-}CH_2\text{-}COO^{-}$

- The second stage of the titration corresponds to the removal of a proton from the $–NH_3^{+}$ group of glycine

Acidic and basic properties of amino acids

- The pH at the midpoint of this stage is 9.60, equal to the pK_a for the $–NH_3$ group
- The titration is essentially complete at a pH of about 12, at which point the predominant form of glycine is
- $H_2N\text{-}CH_2\text{-}COO^-$

Acidic and basic properties of amino acids

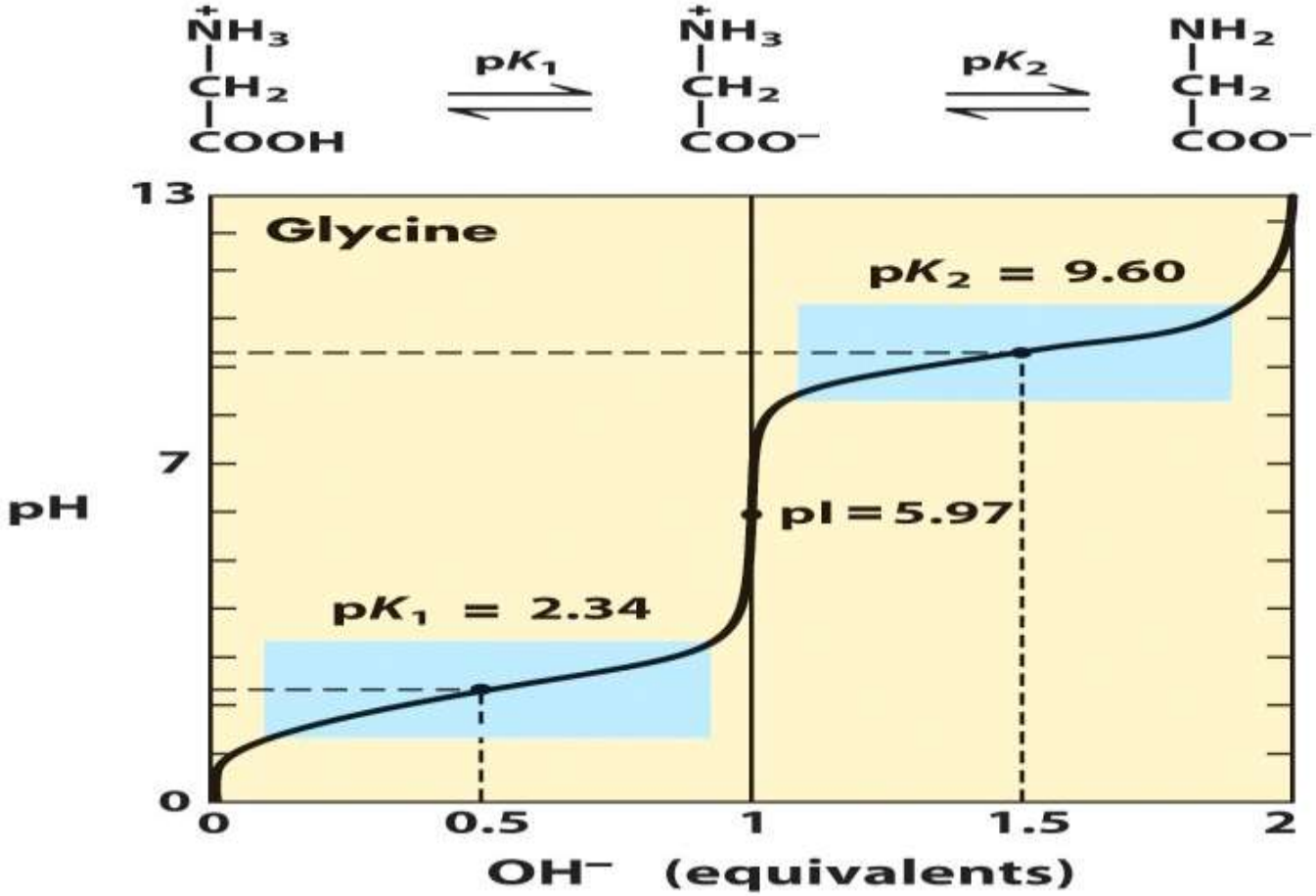


END

Acidic and basic properties of amino acids

- From the titration curve of glycine we can derive several important pieces of information

Acidic and basic properties of amino acids



Acidic and basic properties of amino acids

- First, it gives a quantitative measure of the pK_a, of each of the two ionizing groups:
  - 2.34 for the -COOH group
  - 9.60 for the $-NH_3^+$ group

Acidic and basic properties of amino acids

- The second piece of information is that this amino acid has **two regions of buffering power**

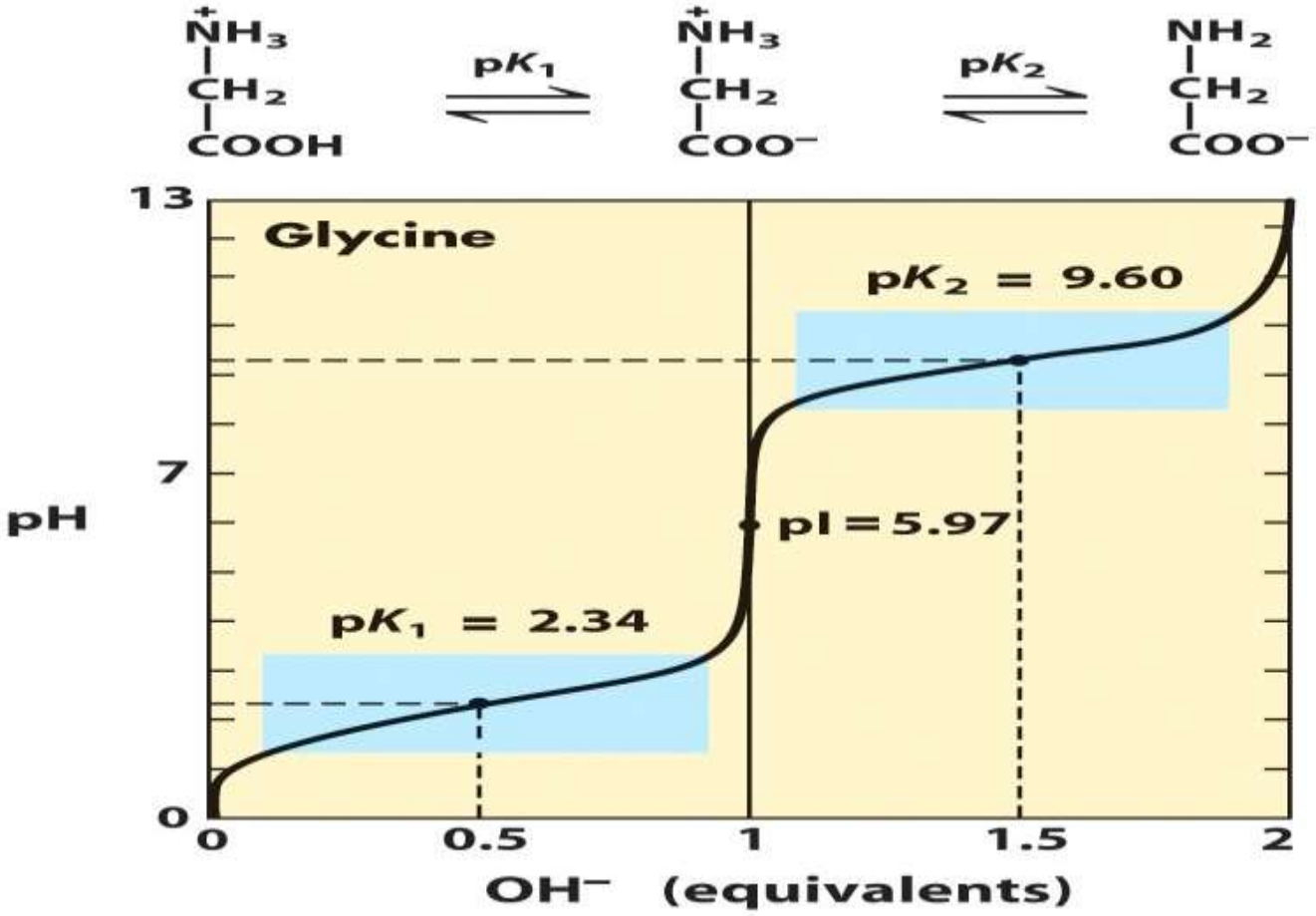
Acidic and basic properties of amino acids

- One of these is the relatively flat portion of the curve, extending for approximately 1 pH unit on either side of the first pKa, of 2.6, indicating that glycine is a good buffer near this pH

Acidic and basic properties of amino acids

- The other buffering zone is centered around pH 9.60
- Note that glycine is not a good buffer at the pH of intracellular fluid or blood, about 7.4

Acidic and basic properties of amino acids



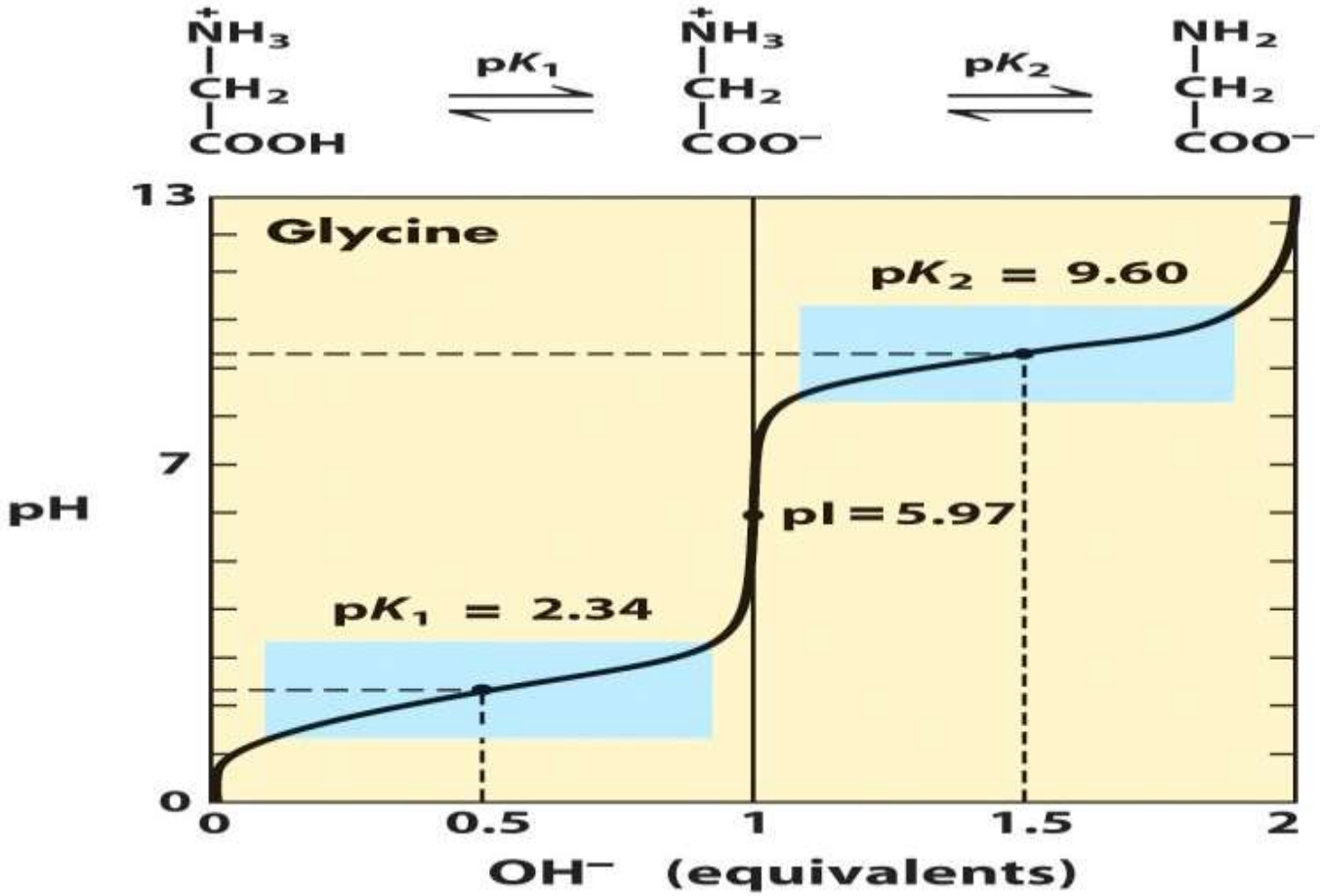
Acidic and basic properties of amino acids

- Another important piece of information derived from the titration curve of an amino acid is the relationship between its net charge and the pH of the solution

Acidic and basic properties of amino acids

- At pH 5.97, glycine is present predominantly as its dipolar form, fully ionized but with no net electric charge

Acidic and basic properties of amino acids



Acidic and basic properties of amino acids

- The characteristic pH at which the net electric charge is zero is called the **isoeleetric point or isoelectric pH, designated pI**

Acidic and basic properties of amino acids

- For glycine, which has no ionizable group in its side chain;
- the pI is simply the arithmetic mean of the two pKa values
- pI = ½ (pK1+pK2)
- = ½ (2.34+9.60) = 5.97

Acidic and basic properties of amino acids

- The farther the pH of a glycine solution is from its pI, the greater the net electric charge of the population of glycine molecules

Acidic and basic properties of amino acids

- At any pH below its pI, glycine has a **net positive charge** and will move toward the negative electrode (cathode)

Acidic and basic properties of amino acids

- At any pH above its pI, glycine will have a net negative charge and thus will move towards the positive electrode (anode)

END

Acidic and basic properties of amino acids

- All amino acids with a
- single α-amino group,
- a single α-carboxyl group, and
- an R group that does not ionize
- have titration curves resembling that of glycine

Acidic and basic properties of amino acids

- These amino acids have very similar, although not identical, pKa values:
- pKa of the –COOH group in the range of 1.8 to 2.4

Acidic and basic properties of amino acids

- pKa of the -NH3+ group in the range of 8.8 to 1 1
- The differences in these pKa values reflect the effects of the R groups
- Alanine has only two dissociable hydrogens (one from α carboxyl & one from α amino group)

Acidic and basic properties of amino acids

- Its pI can be calculated as follows:
- pI = pK1 + pK2 ÷ 2
- pK1 = 2.3 (for carboxyl group)
- pK2 = 9.1 (for amino group)
- pI = 2.3 + 9.1 ÷ 2 = 5.7

Acidic and basic properties of amino acids

- Amino acids with an ionizable R group have more complex titration curves, with three stages corresponding to the three possible ionization steps and thus they have **three pKa values**

Acidic and basic properties of amino acids

- The additional stage for the titration of the ionizable R group merges to some extent with the other two
- In addition, each of the acidic and basic amino acids contains an ionizable group in its side chain

Acidic and basic properties of amino acids

- Thus, both free amino acids and some amino acids combined in peptide linkages can act as **buffers**
- A buffer is a solution that resists change in pH following the addition of an acid or base

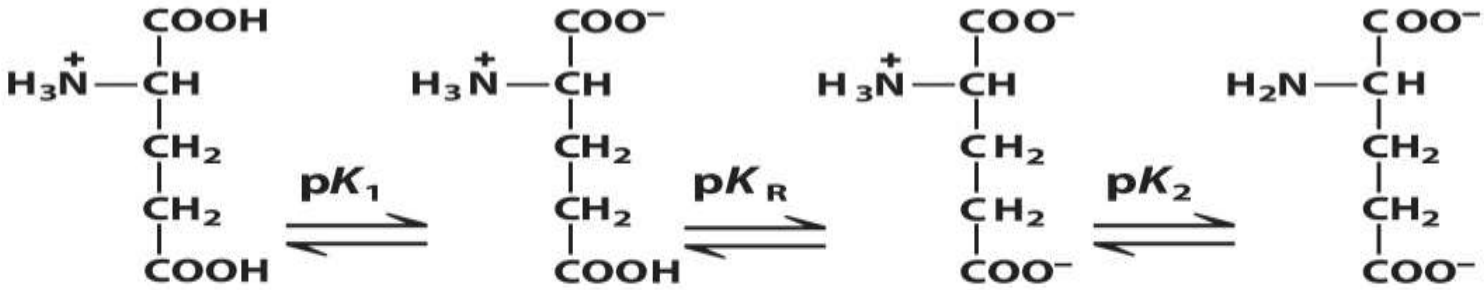
Acidic and basic properties of amino acids

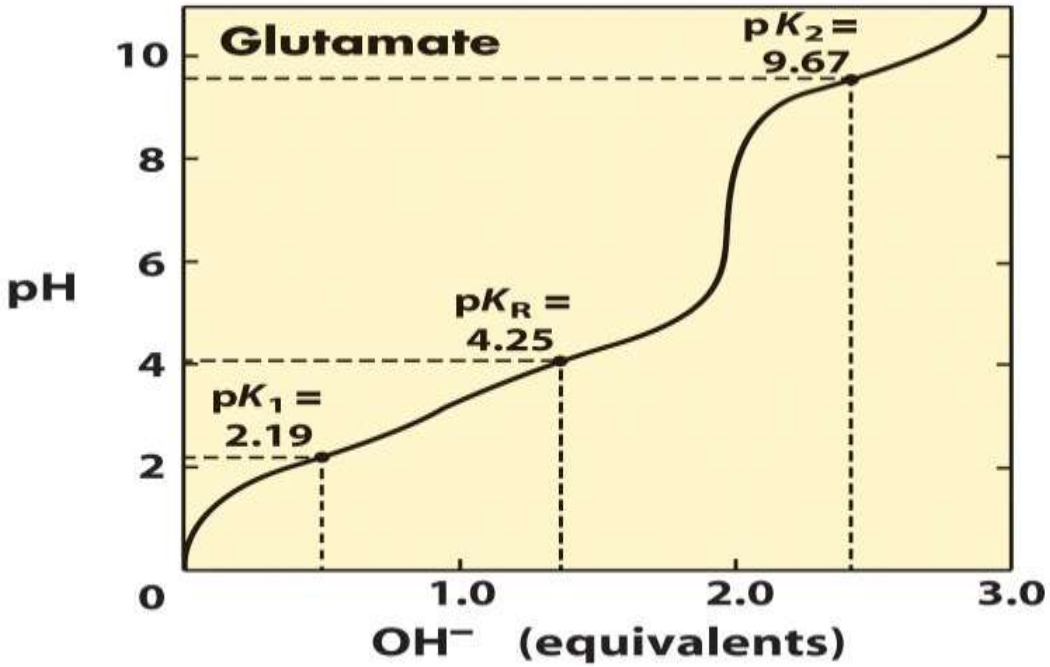
- The isoelectric points reflect the nature of the ionizing R groups present
- For example, glutamate has a pI of 3.22, considerably lower than that of glycine

Acidic and basic properties of amino acids

- This is due to the presence of two carboxyl groups, with an average of their pKa values of 3.22

Acidic and basic properties of amino acids





Acidic and basic properties of amino acids

- Histidine has got 3 dissociable hydrogens
  - one from carboxyl, pK1=1.8,
  - one from imidazole group, pK2=6.0
  - one from amino group, pK3=9.2

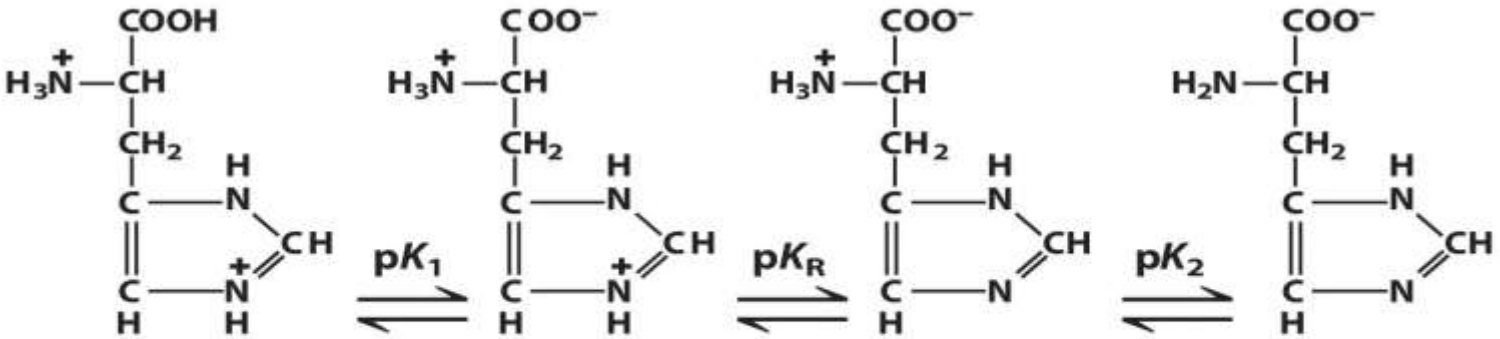
Acidic and basic properties of amino acids

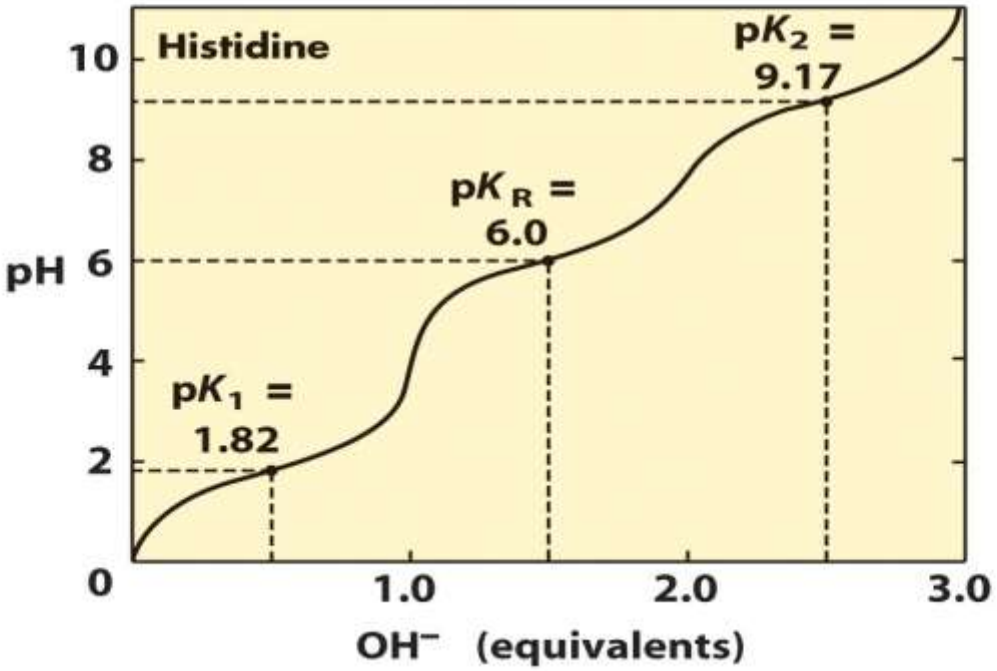
- The pI of Histidine, with two groups that are positively charged when protonated, is 7.59 (the average of the pKa values of the amino and imidazole groups)

Acidic and basic properties of amino acids

- Only histidine has an R group (pKa = 6.0) providing significant buffering power near the neutral pH usually found in the intracellular and extracellular fluids of most animals

Acidic and basic properties of amino acids





END

Acidic and basic properties of amino acids

- All amino acids with a
- single α-amino group,
- a single α-carboxyl group, and
- an R group that does not ionize
- have titration curves resembling that of glycine

Acidic and basic properties of amino acids

- These amino acids have very similar, although not identical, pKa values:
- pKa of the –COOH group in the range of 1.8 to 2.4

Acidic and basic properties of amino acids

- pKa of the -NH3+ group in the range of 8.8 to 1 1
- The differences in these pKa values reflect the effects of the R groups
- Alanine has only two dissociable hydrogens (one from α carboxyl & one from α amino group)

Acidic and basic properties of amino acids

- Its pI can be calculated as follows:
- pI = pK1 + pK2 ÷ 2
- pK1 = 2.3 (for carboxyl group)
- pK2 = 9.1 (for amino group)
- pI = 2.3 + 9.1 ÷ 2 = 5.7

Acidic and basic properties of amino acids

- Amino acids with an ionizable R group have more complex titration curves, with three stages corresponding to the three possible ionization steps and thus they have **three pKa values**

Acidic and basic properties of amino acids

- The additional stage for the titration of the ionizable R group merges to some extent with the other two
- In addition, each of the acidic and basic amino acids contains an ionizable group in its side chain

Acidic and basic properties of amino acids

- Thus, both free amino acids and some amino acids combined in peptide linkages can act as **buffers**
- A buffer is a solution that resists change in pH following the addition of an acid or base

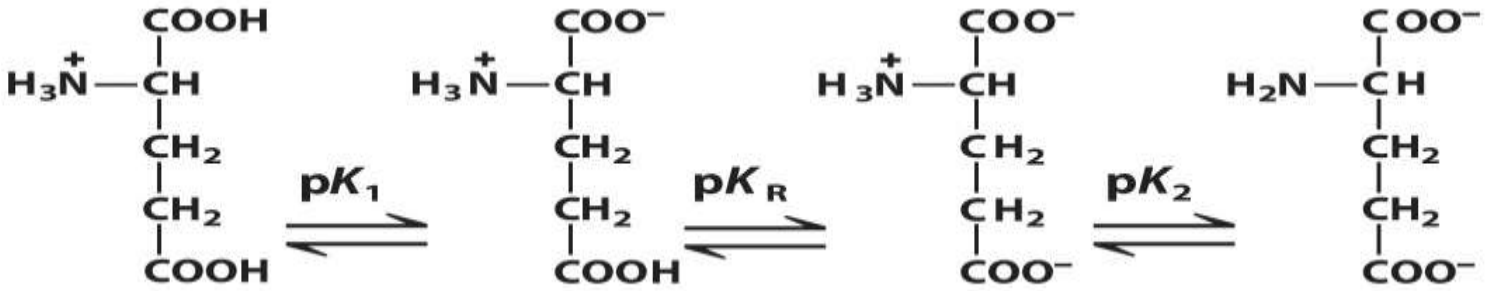
Acidic and basic properties of amino acids

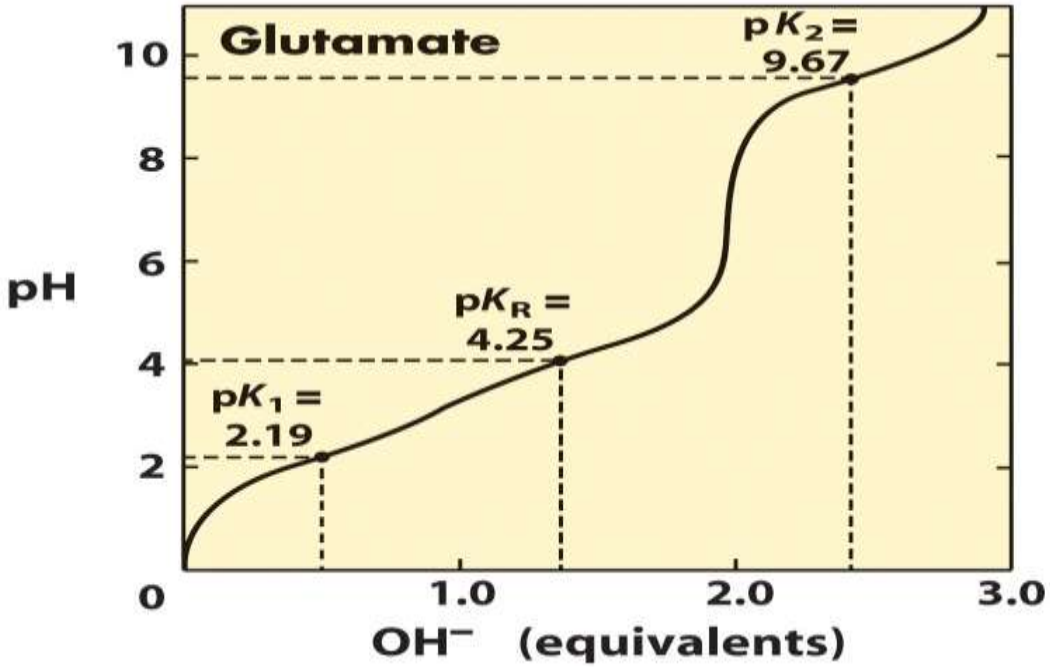
- The isoelectric points reflect the nature of the ionizing R groups present
- For example, glutamate has a pI of 3.22, considerably lower than that of glycine

Acidic and basic properties of amino acids

- This is due to the presence of two carboxyl groups, with an average of their pKa values of 3.22

Acidic and basic properties of amino acids





Acidic and basic properties of amino acids

- Histidine has got 3 dissociable hydrogens
  - one from carboxyl, pK1=1.8,
  - one from imidazole group, pK2=6.0
  - one from amino group, pK3=9.2

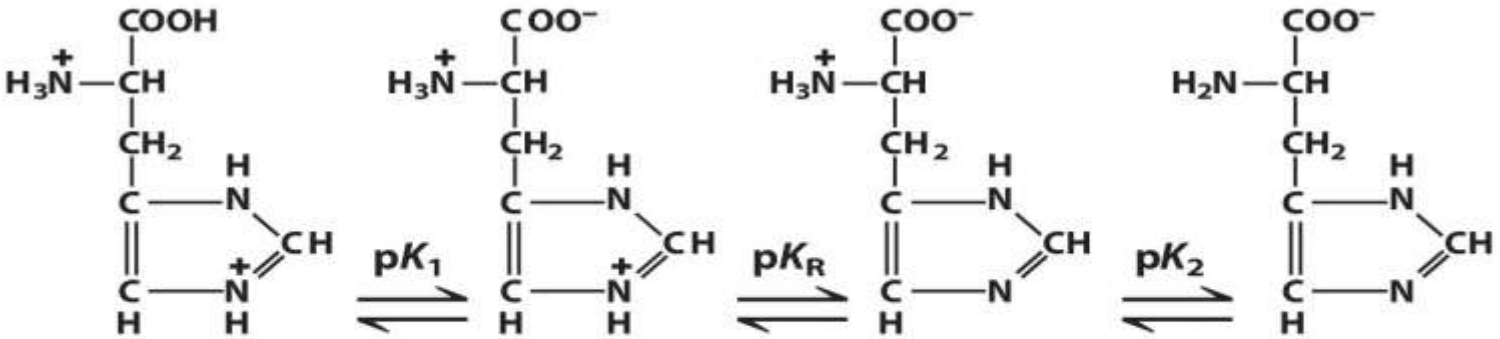
Acidic and basic properties of amino acids

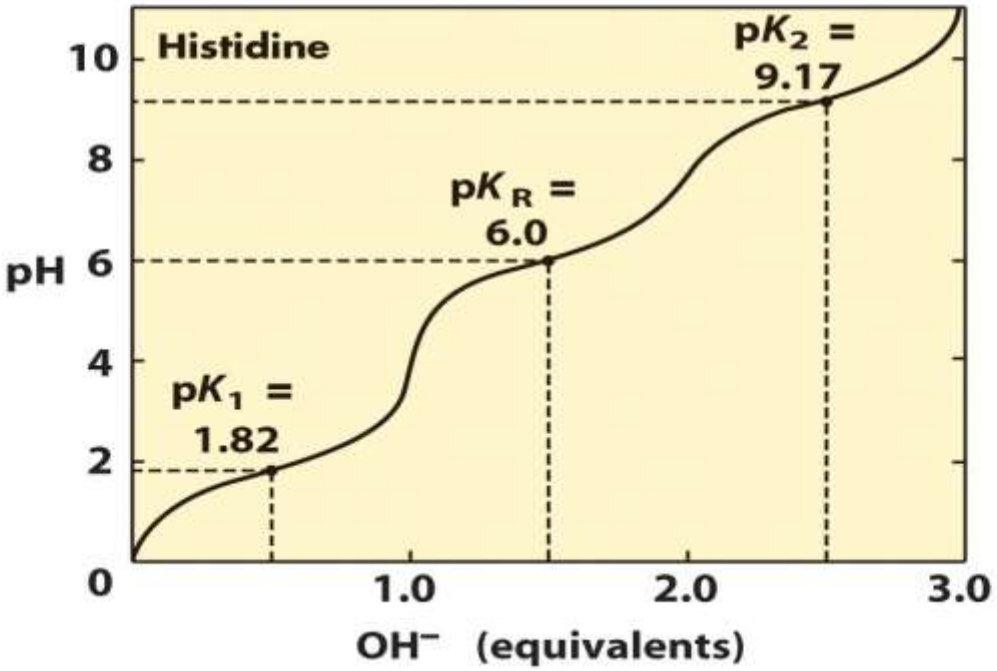
- The pI of Histidine, with two groups that are positively charged when protonated, is 7.59 (the average of the pKa values of the amino and imidazole groups)

Acidic and basic properties of amino acids

- Only histidine has an R group (pKa = 6.0) providing significant buffering power near the neutral pH usually found in the intracellular and extracellular fluids of most animals

Acidic and basic properties of amino acids





END

Abbreviations and symbols of amino acids

- **Abbreviations and symbols for the commonly occurring amino acids**
- Each amino acid name has an associated three-letter abbreviation and a one-letter symbol

Abbreviations and symbols of amino acids

- The one-letter codes are determined by the following rules:

1. Unique first letter:

- If only one amino acid begins with a particular letter, then that letter is used as its symbol, for example,
- I = isoleucine

1 Unique first letter:

Cysteine	=	Cys	=	C
Histidine	=	His	=	H
Isoleucine	=	Ile	=	I
Methionine	=	Met	=	M
Serine	=	Ser	=	S
Valine	=	Val	=	V

Abbreviations and symbols of amino acids

- **2. Most commonly occurring amino acids have priority:**
- If more than one amino acid begins with a particular letter, the most common of these amino acids receives this letter as its symbol

Abbreviations and symbols of amino acids

- For example, glycine is more common than glutamate,
- So G = glycine

2 Most commonly occurring amino acids have priority:

Alanine	=	Ala	=	A
Glycine	=	Gly	=	G
Leucine	=	leu	=	L
Proline	=	Pro	=	P
Threonine	=	Thr	=	T

Abbreviations and symbols of amino acids

- **Similar sounding names:**
- Some one-letter symbols sound like the amino acid they represent
- For example,
- F = phenylalanine, or W= tryptophan (“twyptophan”)

3 Similar sounding names:

Arginine	=	Arg	=	R	("aRginine")
Asparagine	=	Asn	=	N	(contains N)
Aspartate	=	Asp	=	D	("asparDic")
Glutamate	=	Glu	=	E	("glutEmate")
Glutamine	=	Gin	=	Q	("Q-tamine")
Phenylalanine	=	Phe	=	F	("Fenylalanine")
Tyrosine	=	Tyr	=	Y	("tYrosine")
Tryptophan	=	Trp	=	W	(dpuble ring in the molecule)

Abbreviations and symbols of amino acids

Letter close to initial letter:

- For the remaining amino acids, a one-letter symbol is assigned that is as close in the alphabet as possible to the initial of the amino acid

Abbreviations and symbols of amino acids

Amino acid	*Abbreviation/ symbol*	
Nonpolar, aliphatic R groups		
Glycine	Gly	G
Alanine	Ala	A
Proline	Pro	P
Valine	Val	V
Leucine	Leu	L
Isoleucine	Ile	I
Methionine	Met	M
Aromatic R groups		
Phenylalanine	Phe	F
Tyrosine	Tyr	Y
Tryptophan	Trp	W

Abbreviations and symbols of amino acids

Amino acid	*Abbreviation/ symbol*	
Polar, uncharged R groups		
Serine	Ser	S
Threonine	Thr	T
Cysteine	Cys	C
Asparagine	Asn	N
Glutamine	Gln	Q
Positively charged R groups		
Lysine	Lys	K
Histidine	His	H
Arginine	Arg	R
Negatively charged R groups		
Aspartate	Asp	D
Glutamate	Glu	E

Classification of amino acids

- The topic can be simplified by grouping the amino acids into five main classes based on the properties of their R groups, in particular, their **polarity,** or tendency to interact with water at biological pH (near pH 7.0).

Classification of amino acids

- The polarity of the R groups varies widely, from nonpolar and hydrophobic (water-insoluble) to highly polar and hydrophilic (water-soluble)

Classification of amino acids

A. Nonpolar, Aliphatic R Groups

- These are hydrophobic
- The amino acids in this group have non polar side chains that do not bind or give off protons

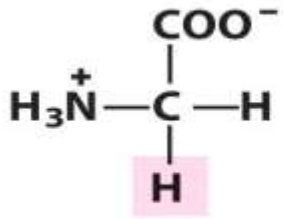
Classification of amino acids

- They also do not participate in hydrogen bonds
- They are ‘oily’ or lipid like and promote hydrophobic interactions
- These side chains cluster together in the interior of the protein

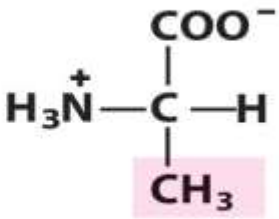
Classification of amino acids

- This class includes
  - Glycine
  - Alanine
  - Proline
  - Valine
  - Leucine
  - Isoleucine
  - Methionine

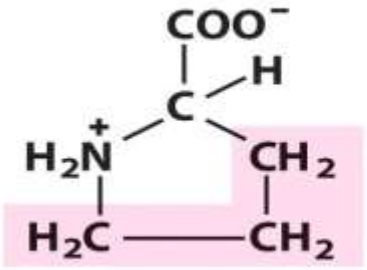
Nonpolar, aliphatic R groups



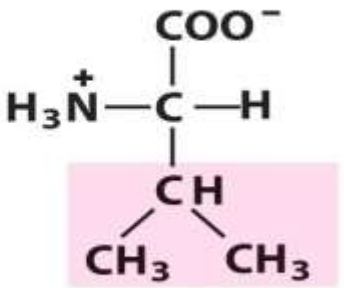
Glycine



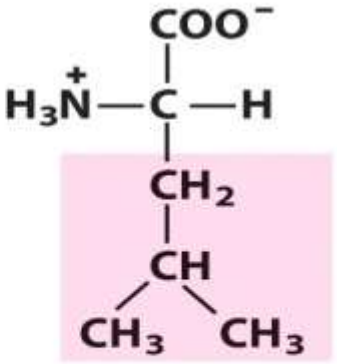
Alanine



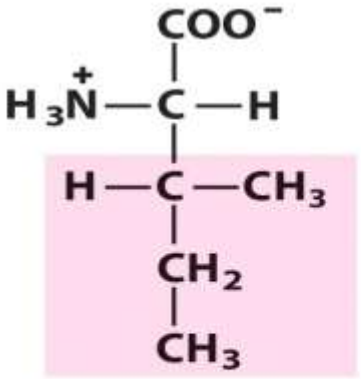
Proline



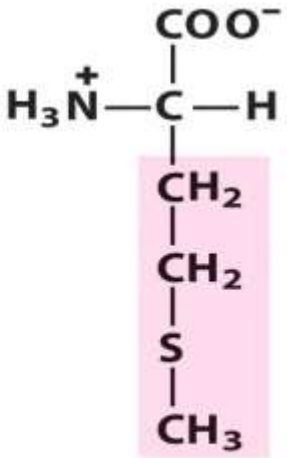
Valine



Leucine



Isoleucine



Methionine

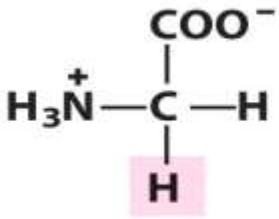
Classification of amino acids

- Electrons are shared equally between the carbon and hydrogen atoms in these side chains, so that they can not hydrogen bond with water

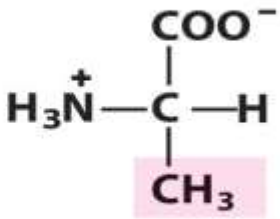
Classification of amino acids

- Within proteins, these amino acid side chains will cluster together to form hydrophobic cores

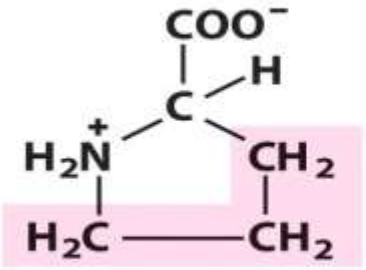
Nonpolar, aliphatic R groups



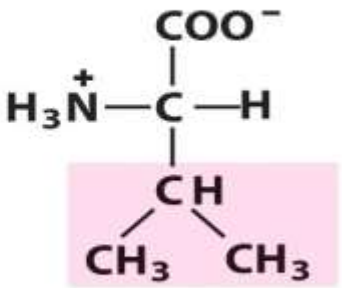
Glycine



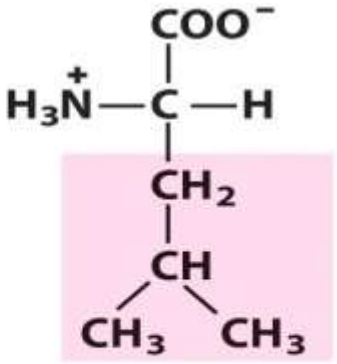
Alanine



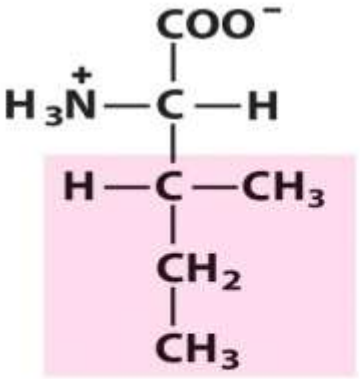
Proline



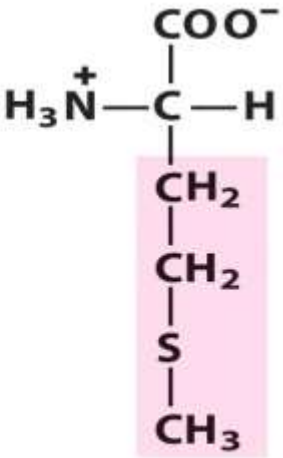
Valine



Leucine



Isoleucine



Methionine

Classification of amino acids

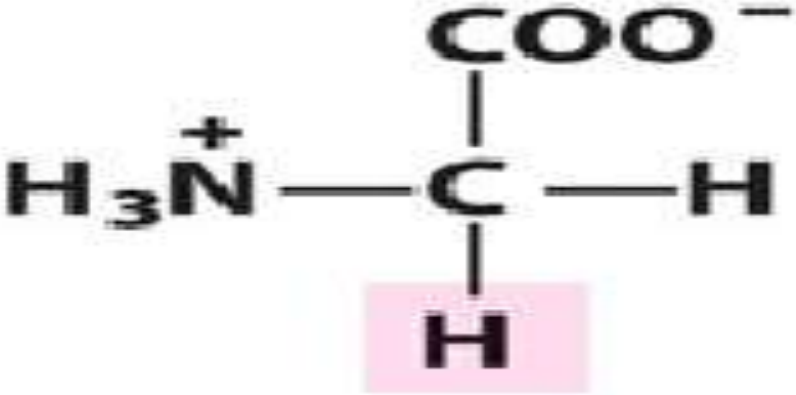
- **Glycine** has the simplest structure. Although it is formally nonpolar, its very small side chain makes no real contribution to hydrophobic interactions

Classification of amino acids

- It is the simplest amino acid, and it really does not fit well into any classification because its side chain is only a hydrogen atom
- Because of this, it causes the least amount of steric hindrance in a protein

Classification of amino acids

- Therefore glycine is often found in bends or in the tightly packed chains of fibrous proteins.



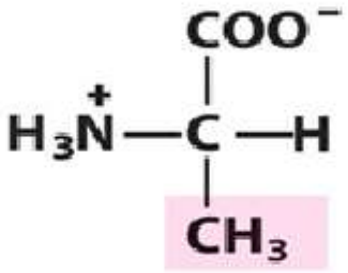
Glycine

Classification of amino acids

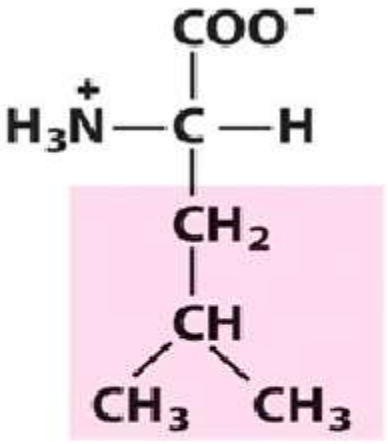
- **Alanine** and the branched chain amino acids **(valine, leucine, and isoleucine)** have bulky, nonpolar, *aliphatic(open chain* hydrocarbon) side chains

Classification of amino acids

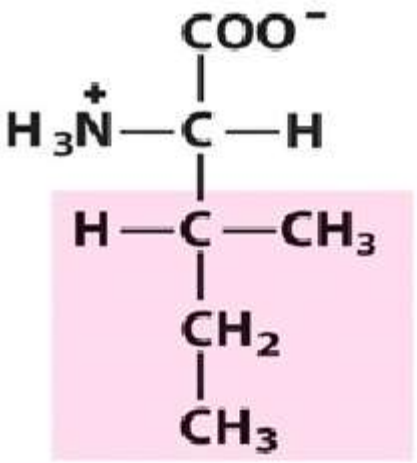
- The side chains of **alanine, valine, leucine,** and **isoleucine** tend to cluster together within proteins, stabilizing protein structure by means of hydrophobic interactions.



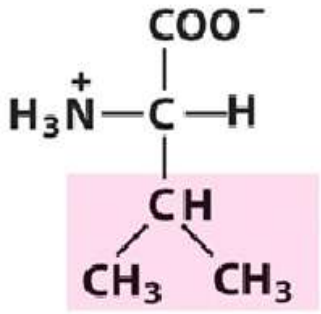
Alanine



Leucine



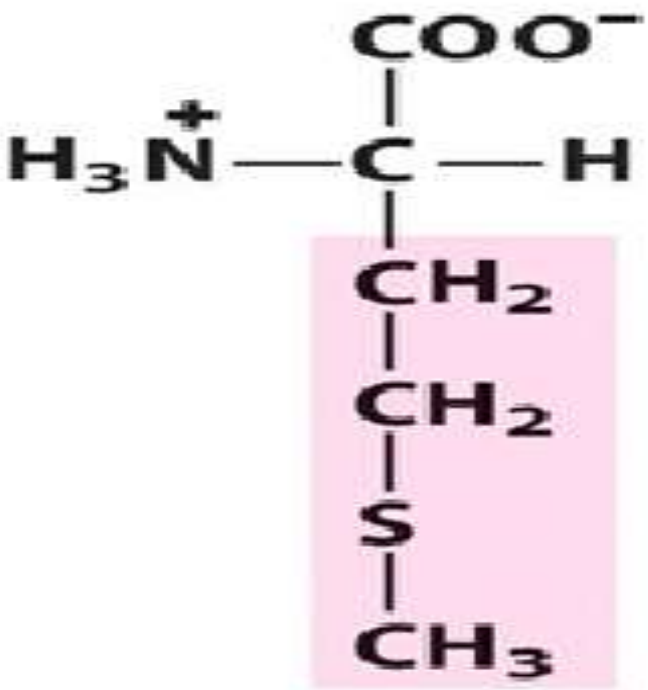
Isoleucine



Valine

Classification of amino acids

- **Methionine**, one of the two sulfur-containing amino acids, has a nonpolar thioether group in its side chain



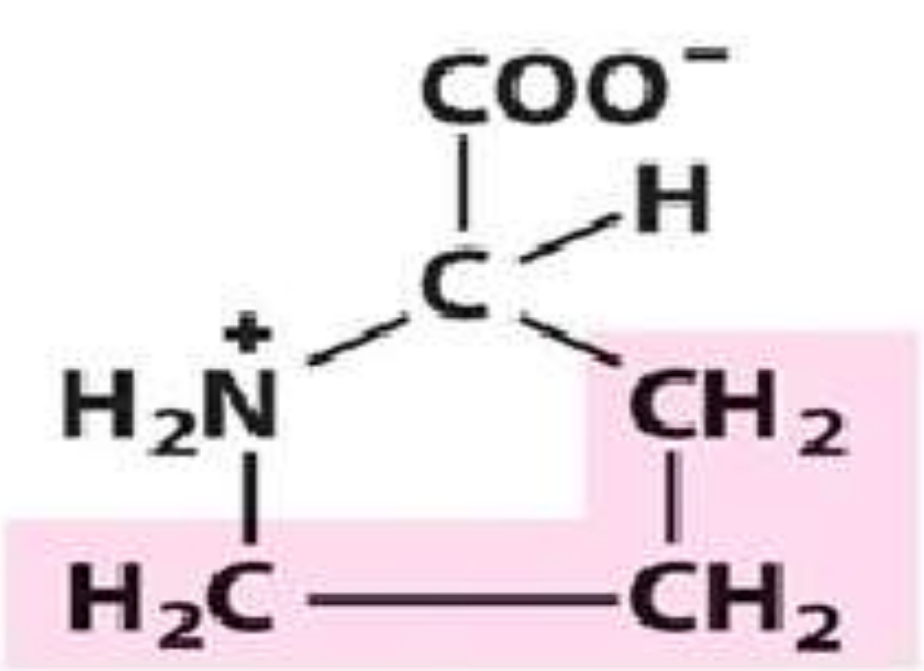
Methionine

Classification of amino acids

- The role of **proline** in amino acid structure differ from those of the nonpolar amino acids
- The amino acid proline contains a ring involving its α–carbon and its α-amino group, which are part of the peptide backbone

Classification of amino acids

- It is an imino acid
- This rigid ring causes a kink in the peptide backbone that prevents it from forming its usual configuration



Proline

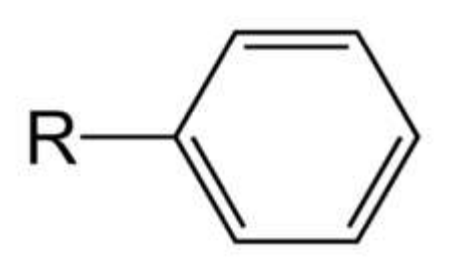
Classification of amino acids

B. Aromatic Amino Acids

- The aromatic amino acids have been grouped together because they all contain ring structures with similar properties, **but their polarity differs**

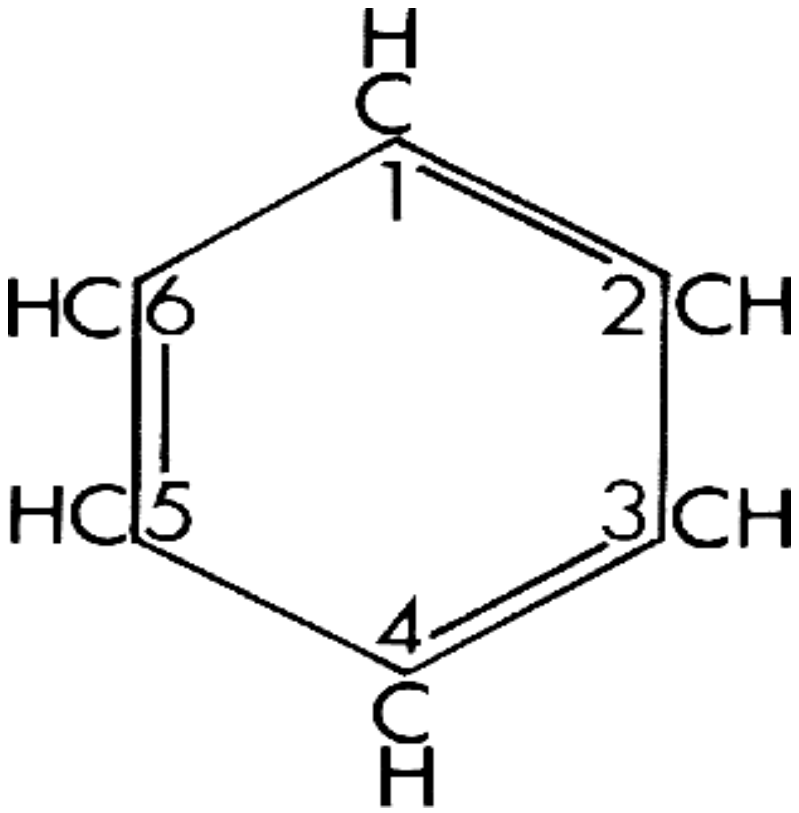
Classification of amino acids

- The aromatic ring is a six-membered carbon–hydrogen ring with three conjugated double bonds (the benzene ring or phenyl group)



Classification of amino acids

- Benzene is an organic chemical compound with the chemical formula C6H6. Its molecule is composed of 6 carbon atoms joined in a ring, with 1 hydrogen atom attached to each carbon atom



Classification of amino acids

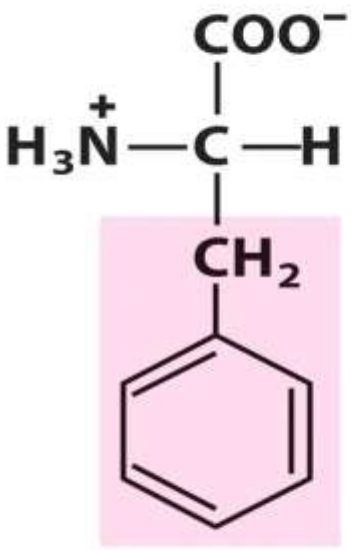
- The substituents on this ring determine whether the amino acid side chain engages in polar or hydrophobic interactions

Classification of amino acids

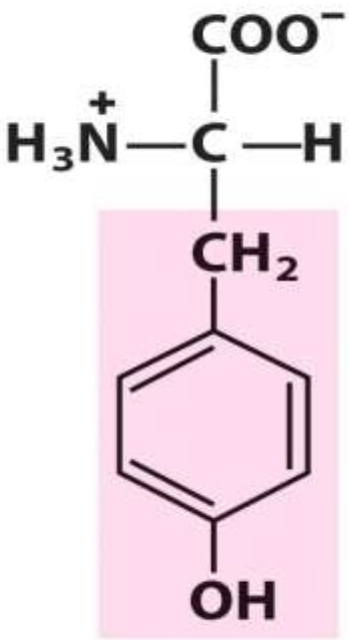
- This group includes
  - Phenylalanine
  - Tyrosine
  - Tryptophan
  - with their aromatic side chains, are relatively nonpolar (hydrophobic).
  - All can participate in hydrophobic interactions.

Classification of amino acids

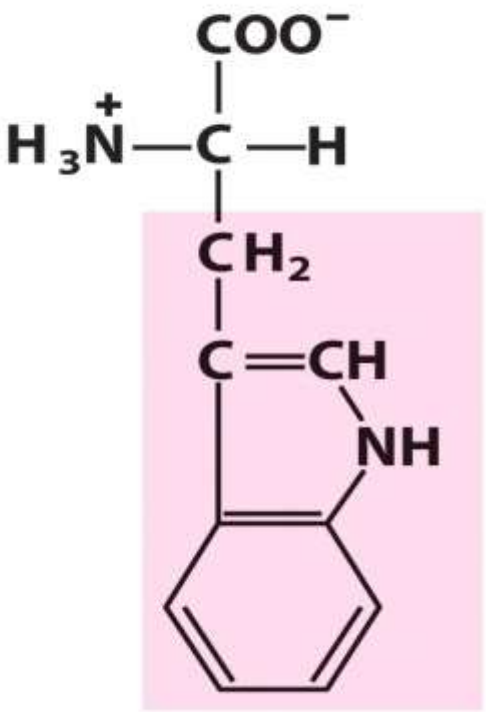
Aromatic R groups



Phenylalanine



Tyrosine



Tryptophan

Classification of amino acids

- In the amino acid **phenylalanine**, the ring contains no substituents, and the electrons are shared equally between the carbons in the ring, resulting in *a very* **nonpolar hydrophobic** structure in which the rings can stack on each other

Classification of amino acids

- In **tyrosine**, a hydroxyl group on the phenyl ring engages in hydrogen bonds, and the side chain is therefore more polar and **more hydrophilic**

Classification of amino acids

- The more complex ring structure in **tryptophan** is an indole ring with a nitrogen that can engage in hydrogen bonds

Classification of amino acids

- Tryptophan is therefore also more polar than phenylalanine
- Tyrosine and tryptophan are significantly more polar than phenylalanine, because of the tyrosine hydroxyl group and the nitrogen of the **tryptophan** indole ring

END

Classification of amino acids

C. Aliphatic, Polar, Uncharged R Groups

- The R groups of these amino acids are more soluble in water, or more hydrophilic, than those of the nonpolar amino acids.

Classification of amino acids

- Because they contain functional groups that form hydrogen bonds with water.

Classification of amino acids

- This class of amino acids includes
- **serine, threonine,**
- **cysteine,**
- **asparagine,** and
- **glutamine.**

Classification of amino acids

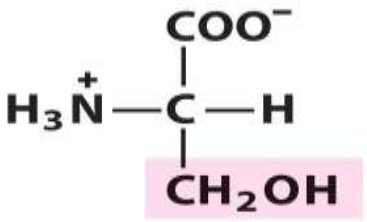
- These Amino acids have zero net charge at neutral pH
  - **Serine & threonine** of this class contain a polar hydroxyl group

Classification of amino acids

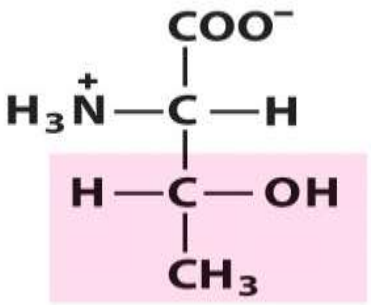
- Similarly side chains of **asparagine & glutamine** contain a carbonyl group & amide group, both of which can participate in hydrogen bonding

Classification of amino acids

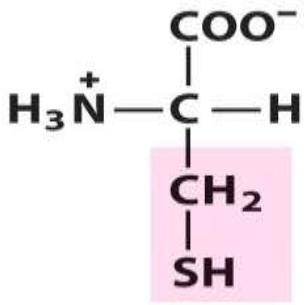
Polar, uncharged R groups



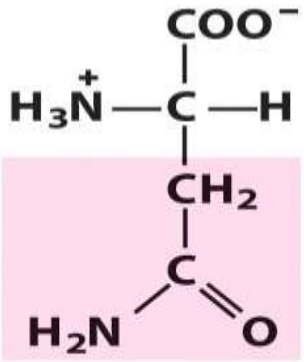
Serine



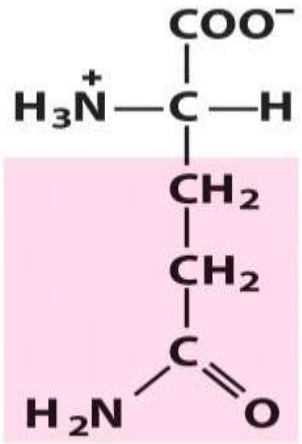
Threonine



Cysteine



Asparagine



Glutamine

Classification of amino acids

- These side chains are sites of attachment for other compounds
- The polar hydroxyl groups of **serine & threonine** serve as a site of attachment for phosphate groups

Classification of amino acids

- The amide group of **asparagine** & the hydroxyl group of **serine & threonine** serve as a site of attachment for oligosaccharide chains in glycoproteins

Classification of amino acids

- Amino acids with side chains that contain an amide group (asparagine and glutamine) or a hydroxyl group (serine and threonine) can be classified as **aliphatic, polar, uncharged amino acids**

Classification of amino acids

- Asparagine and glutamine are amides of the amino acids aspartate and glutamate
- As a consequence of their hydrophilicity, these amino acids are frequently found on the surface of water-soluble globular proteins

Classification of amino acids

- Cysteine, which is sometimes included in this class of amino acids, has been separated into the class of sulfur-containing amino acids

END

Classification of amino acids

D. Positively Charged (Basic) R Groups

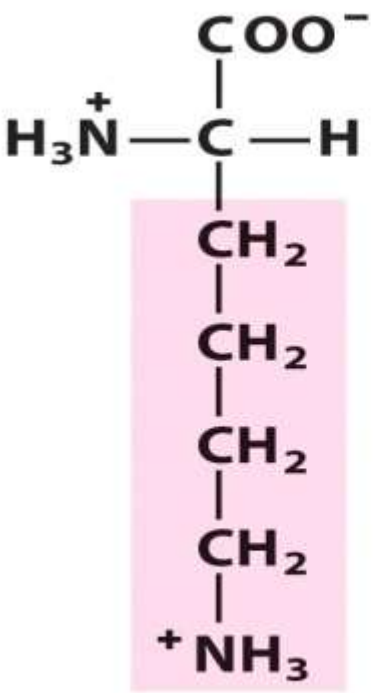
- The side chains of these amino acids are proton acceptors
- The amino acids in which the R groups have significant **positive charge** at pH 7.0 are:

Classification of amino acids

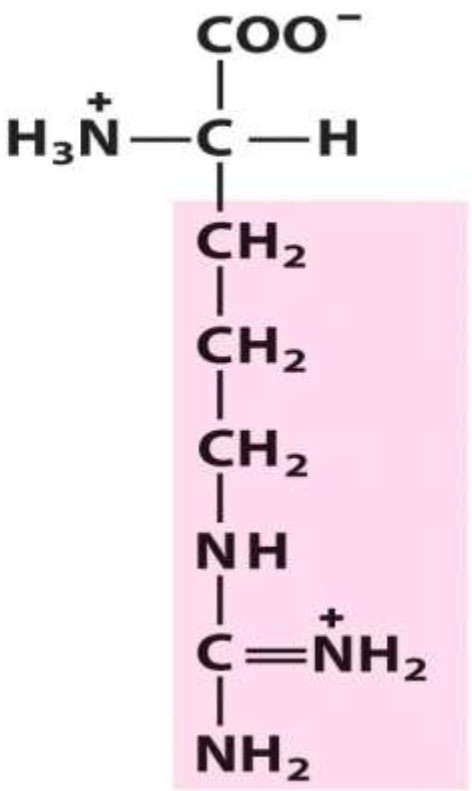
- **Lysine**, which has a second primary amino group at the ε position on its aliphatic chain
- **Arginine,** which has a positively charged guanidinium group
- **Histidine,** which has an aromatic imidazole group

Classification of amino acids

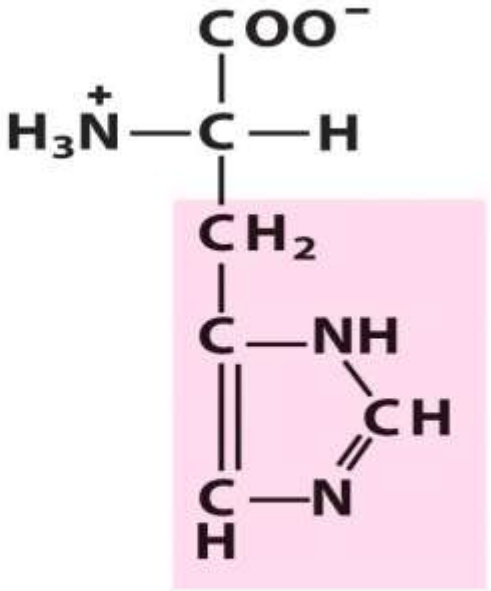
Positively charged R groups



Lysine



Arginine



Histidine

Classification of amino acids

- The side chains of the two basic amino acids, arginine and lysine, have pK_a values above 10, so that the positively charged form always predominates at physiologic pH

Classification of amino acids

- The side chain of histidine (pK_a ~ 6.0) dissociates near physiologic pH, so only a portion of the histidine side chains carry a positive charge

Classification of amino acids

- **Histidine** is therefore weakly basic & the *free* amino acid is mainly uncharged at physiologic pH

Classification of amino acids

- The amino acid side chains might have very different pK_as than those of the free amino acids if they are involved in hydrogen or ionic bonds with other amino acid side chains

Classification of amino acids

- Therefore, when histidine is incorporated into a protein, its side chain can be either positively charged or neutral depending on the ionic environment provided by the polypeptide chains of the protein

END

Classification of amino acids

E. Negatively Charged (Acidic) R Groups

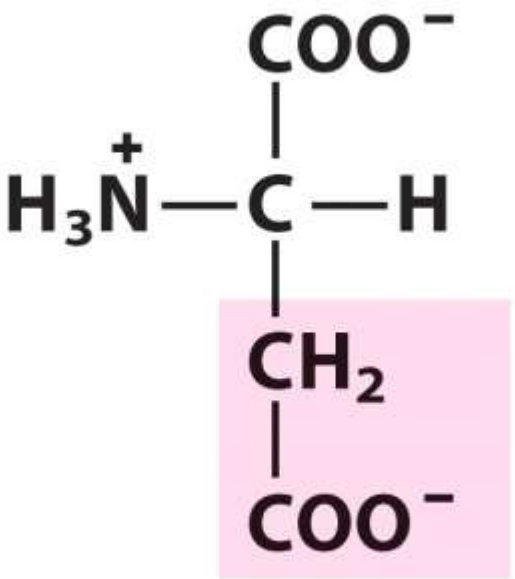
- These amino acids are proton donors
- At neutral pH, the side chains of these amino acids are fully ionized, containing a negatively charged carboxylate group
- (--COO^-)

Classification of amino acids

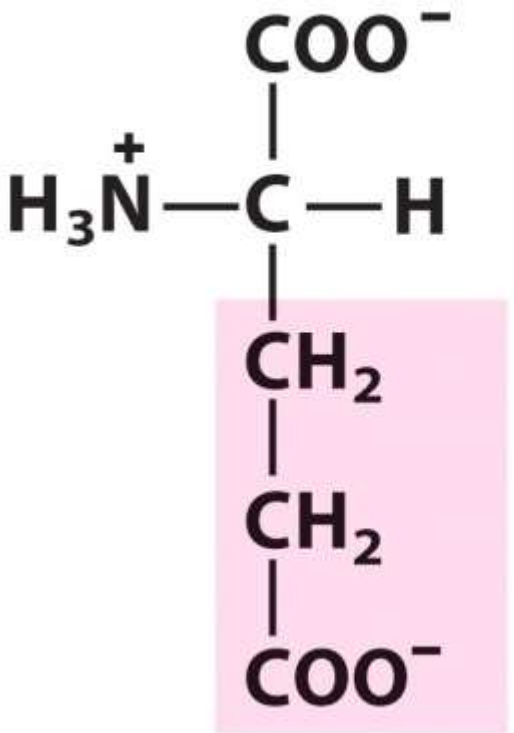
- Therefore these amino acids are negatively charged at phyisologic pH
- The two amino acids having R groups with a net negative charge at pH 7.0 are **aspartate and glutamate**, each of which has a second carboxyl group

Classification of amino acids

Negatively charged R groups



Aspartate



Glutamate

Classification of amino acids

- Humans have no dietary requirement for protein, per se, but, the protein in food does provide essential amino acids.

Classification of amino acids

- Nine of the twenty amino acids needed for the synthesis of body proteins are essential-that is, they cannot be synthesized in humans at an adequate rate

Classification of amino acids

- **Essential amino acids** can not be made by the body. As a result, they must come from food. The nine **essential amino acids** are: histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine.

Classification of amino acids

- Of these nine, some amino acids are essential at all times, whereas some are required only during periods of rapid tissue growth characteristic of childhood or recovery from illness

Classification of amino acids

	Glucogenic	Glucogenic And Ketogenic	Ketogenic
Nonessential	Alanine Arginine Asparagine Cysteine Glutamate Glutamine Glycine Proline Serine	Tyrosine	
Essentia	Histidine Methionine Valine	Threonine Isoleucine Phenyl-alanine Tryptophan	Leucine Lysine

Classification of amino acids

Essential	Conditionally Non-Essential	Non-Essential
Histidine	Arginine	Alanine
Isoleucine	Asparagine	Asparatate
Leucine	Glutamine	Cysteine
Methionine	Glycine	Glutamate
Phenylalanine	Proline	
Threonine	Serine	
Tryptophan	Tyrosine	
Valine		
Lysine		

Classification of amino acids

Essential	Conditionally Non-Essential	Non-Essential
Histidine	Arginine	Alanine
Isoleucine	Asparagine	Asparatate
Leucine	Glutamine	Cysteine
Methionine	Glycine	Glutamate
Phenylalanine	Proline	
Threonine	Serine	
Tryptophan	Tyrosine	
Valine		
Lysine		

Classification of amino acids

- Amino acids can be classified as **glucogenic or ketogenic** based on what intermediates are produced during their catabolism

Classification of amino acids

A. Glucogenic amino acids

- Amino acids whose catabolism yields **pyruvate are termed glucogenic or glycogenic**

Classification of amino acids

- These intermediates are substrates for **gluconeogenesis** and, therefore, can give rise to the net formation of glucose or glycogen in the liver and glycogen in the muscle

Classification of amino acids

B. Ketogenic amino acids

- Amino acids whose catabolism yields either **acetoacetate or its precursor, (acetyl CoA or acetoacetyl CoA)** are termed **ketogenic**

Classification of amino acids

- Acetoacetate is one of the ketone bodies which also include 3-hydroxybutyrate and acetone
- **Leucine and lysine** are the only exclusively ketogenic amino acids found in proteins

Classification of amino acids

- Their carbon skeletons are not substrates for gluconeogenesis and, therefore, cannot give rise to the net formation of glucose or glycogen in the liver, or glycogen in the muscles

Classification of amino acids

	Glucogenic	Glucogenic And Ketogenic	Ketogenic
Nonessential	Alanine Arginine Asparagine Cysteine Glutamate Glutamine Glycine Proline Serine	Tyrosine	
Essential	Histidine Methionine Valine	Threonine Isoleucine Phenyl-alanine Tryptophan	Leucine Lysine

Classification of amino acids

TABLE 14–4 Glucogenic Amino Acids, Grouped by Site of Entry

Pyruvate
Alanine
Cysteine
Glycine
Serine
Threonine
Tryptophan*

α-Ketoglutarate
Arginine
Glutamate
Glutamine
Histidine
Proline

Succinyl-CoA
Isoleucine*
Methionine
Threonine
Valine

Fumarate
Phenylalanine*
Tyrosine*

Oxaloacetate
Asparagine
Aspartate

Note: All these amino acids are precursors of blood glucose or liver glycogen, because they can be converted to pyruvate or citric acid cycle intermediates. Of the 20 common amino acids, only leucine and lysine are unable to furnish carbon for net glucose synthesis. *These amino acids are also ketogenic (see Fig. 18–21).

Classification of amino acids

TABLE 14–4 Glucogenic Amino Acids, Grouped by Site of Entry

Pyruvate
Alanine
Cysteine
Glycine
Serine
Threonine
Tryptophan*

α-Ketoglutarate
Arginine
Glutamate
Glutamine
Histidine
Proline

Succinyl-CoA
Isoleucine*
Methionine
Threonine
Valine

Fumarate
Phenylalanine*
Tyrosine*

Oxaloacetate
Asparagine
Aspartate

Note: All these amino acids are precursors of blood glucose or liver glycogen, because they can be converted to pyruvate or citric acid cycle intermediates. Of the 20 common amino acids, only leucine and lysine are unable to furnish carbon for net glucose synthesis. *These amino acids are also ketogenic (see Fig. 18–21).

END

Classification of Proteins

- Different classifications of proteins are based on their:

1. Shape and Size
2. Biological actions/ Functions
3. Solubility and physical properties
4. Quality

Functional Classification of Proteins

Catalytic Proteins:

These specialized proteins are called **enzymes** which catalyze the biochemical reactions

Functional Classification of Proteins

2. Protective Proteins

a) Immunoglobins (Igs)

- These freely circulating proteins protect the body from invading microbes such as bacteria or viruses by inactivating or killing them through various mechanisms.

Functional Classification of Proteins

b) Fibrinogen

- This forms fibrin clot and stops bleeding from wounds

Functional Classification of Proteins

3. Regulatory Proteins:

- Hormones control genetic expression, cellular signalling *and* biochemical reactions catalyzed by enzymes. Enzymes are either activated or inactivated through modification of their structure.

Functional Classification of Proteins

- Examples of protein hormones are
  - growth hormone
  - Insulin
  - Glucagon
  - Somatostatin

Functional Classification of Proteins

4. Structural Proteins:

- These proteins form various body structures e.g.
  - Collagen
  - Elastin
  - Keratin

Functional Classification of Proteins

5. Transport Proteins:

- These proteins transport various substances from one part of the body to the other e.g.
  - Hemoglobin transport O_2 from lungs to tissues and CO_2 from tissues to lungs
  - Transferrin transports iron

Functional Classification of Proteins

6. Contractile Proteins:

- These proteins are involved in muscle contraction and relaxation
  - Myosin of thick filaments
  - Actin of thin filaments of skeletal muscles

END

Functional Classification of Proteins

7. Respiratory Proteins:

- Heme containing proteins are involved in the function of respiration e.g.
  - Hemoglobin
  - Myoglobin
  - cytochromes

Functional Classification of Proteins

8. Digestive Proteins:

- These proteins are digestive enzymes which digest our food materials such as carbohydrates, proteins, lipids and include
  - Amylase
  - Pepsin
  - Lipases etc.

Functional Classification of Proteins

9. Toxin Proteins:

- These proteins are hydrolytic enzymes found in the venom of poisonous snakes, sting of bees and insects and hydrolyze the compounds forming the structure of the cell membrane

Functional Classification of Proteins

- The poisonous mushrooms also have such toxins

10. Storage Proteins:

- These proteins store some specific elements or compounds with them

Functional Classification of Proteins

- This is because of the presence of the many binding sites in them for the particular element e.g.
  - ferritin stores iron
  - ceruloplasmin stores copper

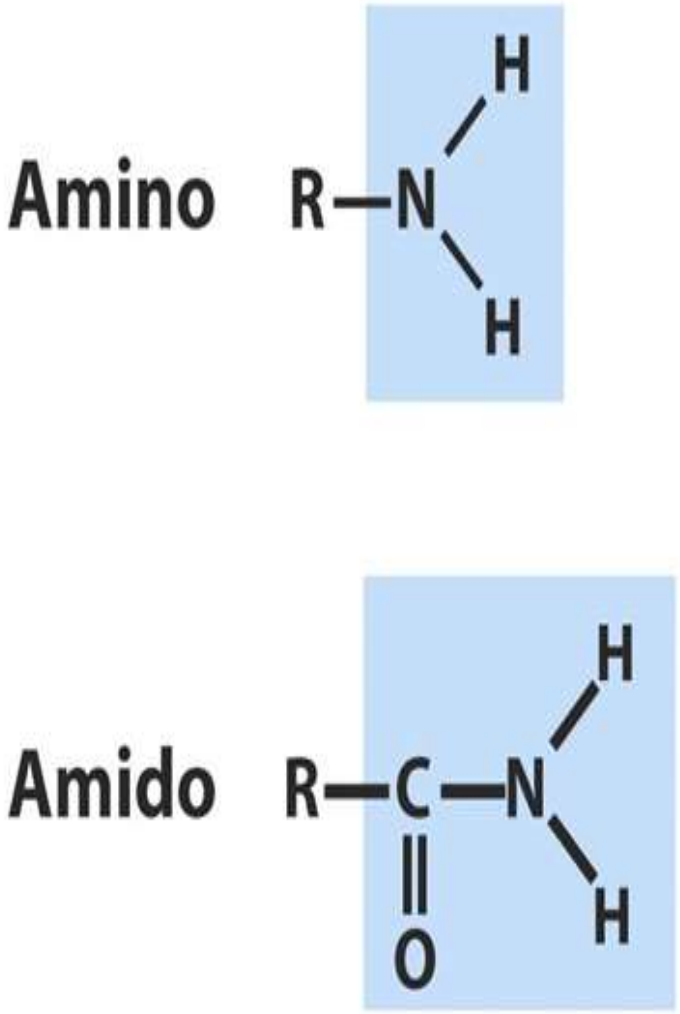
PEPTIDE BOND

- We now turn to polymers of amino acids, the peptides and proteins
- Biologically occurring polypeptides range in size from small to very large, consisting of two or three to thousands of linked amino acid residues

PEPTIDE BOND

- Two amino acid molecules can be covalently joined through **an amide linkage,** termed **a peptide bond**

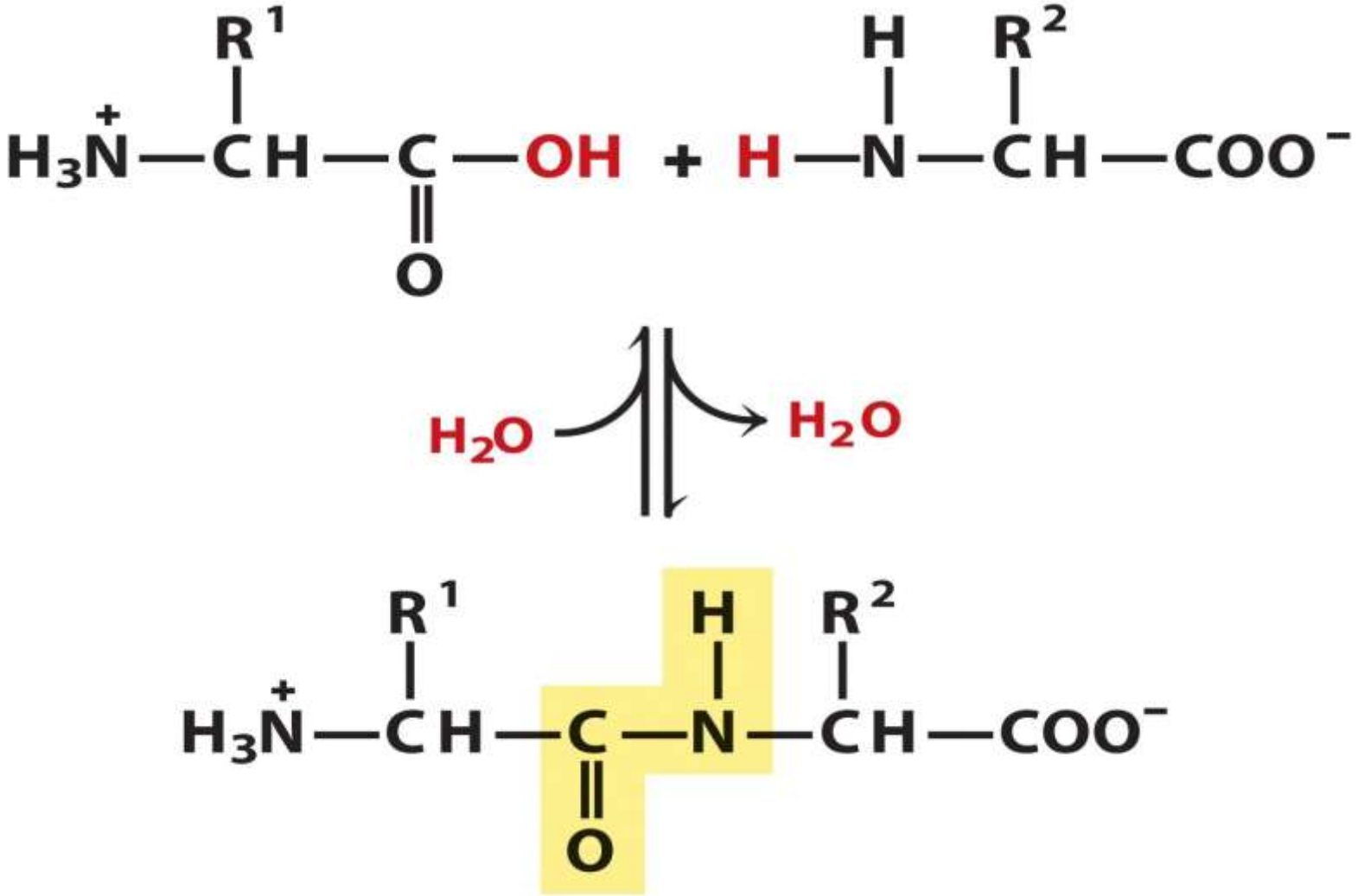
PEPTIDE BOND



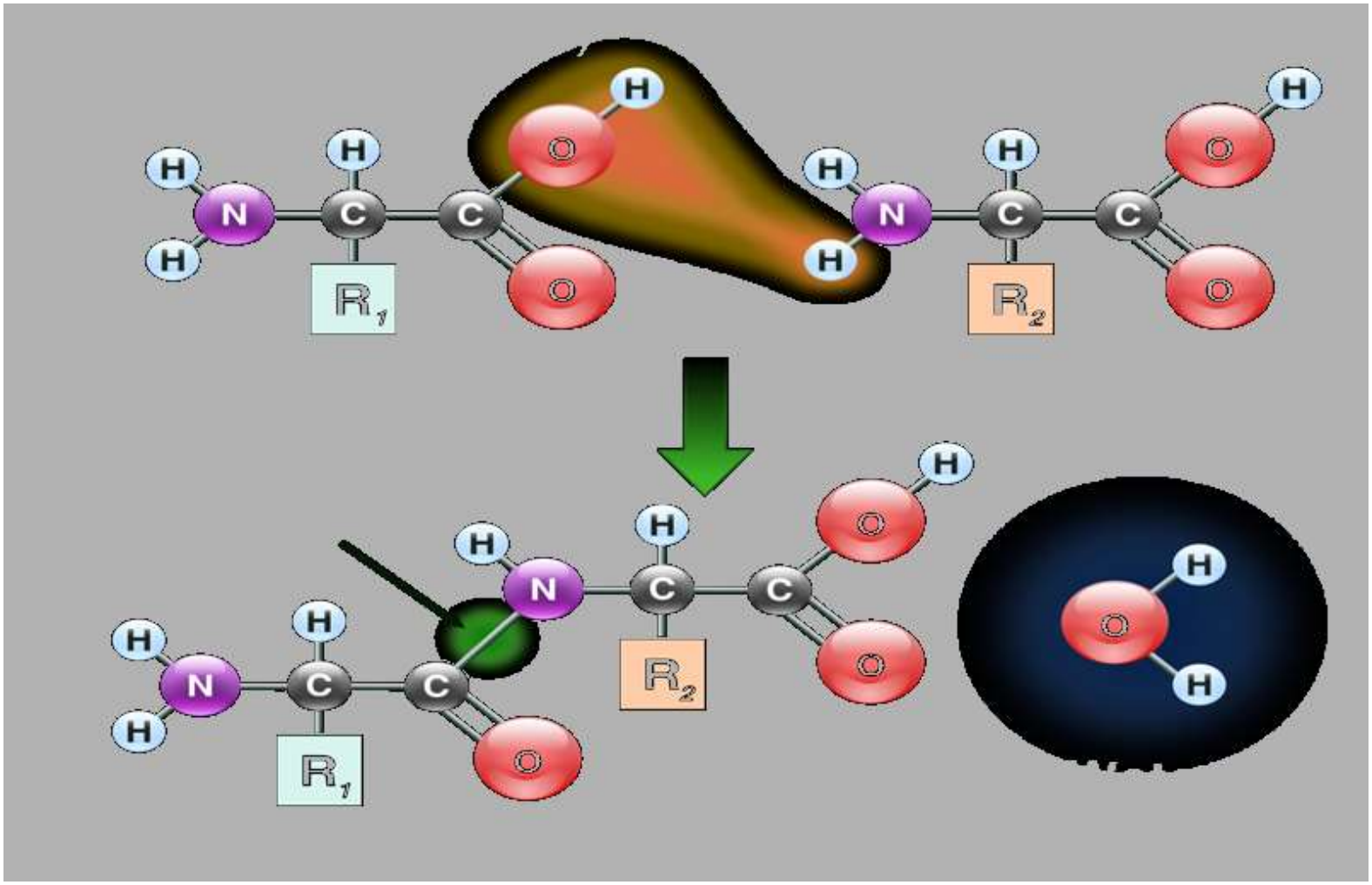
PEPTIDE BOND

- This linkage is formed by removal of the elements of water from the α-carboxyl group of one amino acid and the α-amino group of another

PEPTIDE BOND



PEPTIDE BOND



PEPTIDE BOND

- Two configurations are possible for a planar peptide bond
- In the trans configuration, the two -carbon atoms are on opposite sides of the peptide bond

PEPTIDE BOND

- In the cis con-, these groups are on the same side of the peptide bond.
- *Almost all peptide bonds in proteins are trans*

PEPTIDE BOND

- This preference for trans over cis can be explained by the fact that steric clashes between groups attached to the -carbon atoms hinder formation of the cis form but do not occur in the trans configuration

PEPTIDE BOND

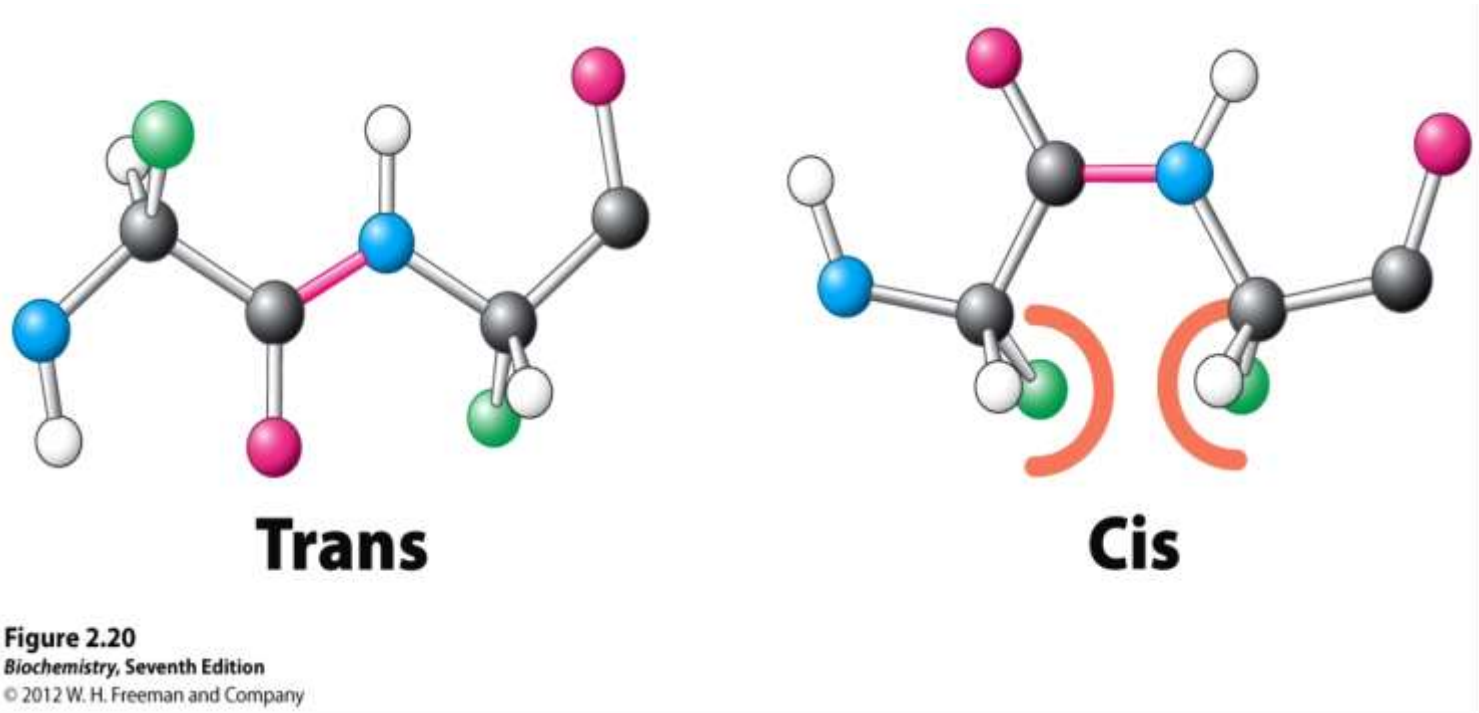


Figure 2.20
***Biochemistry*, Seventh Edition**
© 2012 W. H. Freeman and Company

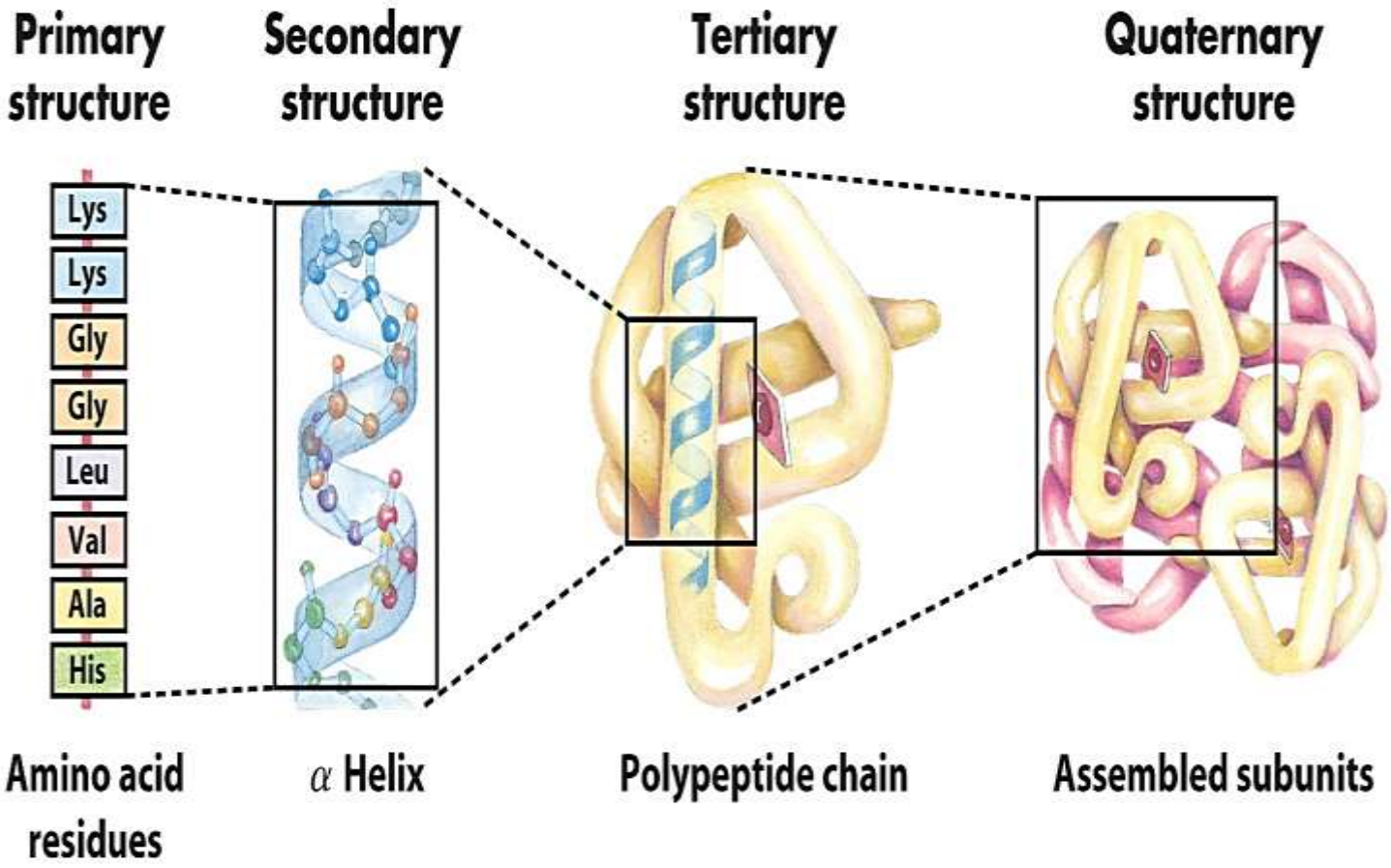
END

Structural Hierarchy of Proteins

The Structure of Proteins

- Four levels of protein structure are commonly defined

Structural Hierarchy of Proteins



Structural Hierarchy of Proteins

The Primary Structure

- A description of all covalent bonds (mainly **peptide bonds** and **disulfide bonds**) linking amino acid residues in a polypeptide chain is its primary structure

Structural Hierarchy of Proteins

- The most important element of primary structure is the **sequence** of amino acid residues

Structural Hierarchy of Proteins

Secondary structure

- Secondary structure refers to **particularly stable arrangements** of amino acid residues giving rise to recurring structural patterns

Structural Hierarchy of Proteins

Tertiary structure

- Tertiary structure describes all aspects of the **three-dimensional** folding of a polypeptide

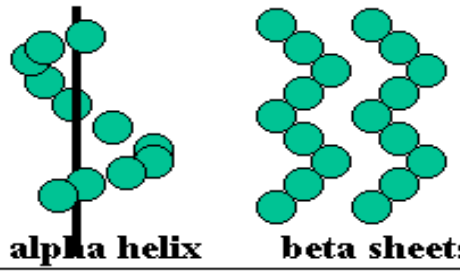
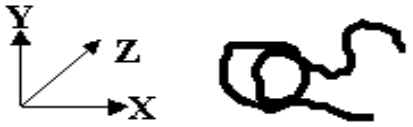
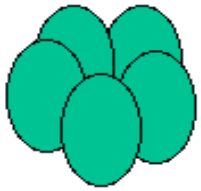
Structural Hierarchy of Proteins

Quaternary structure

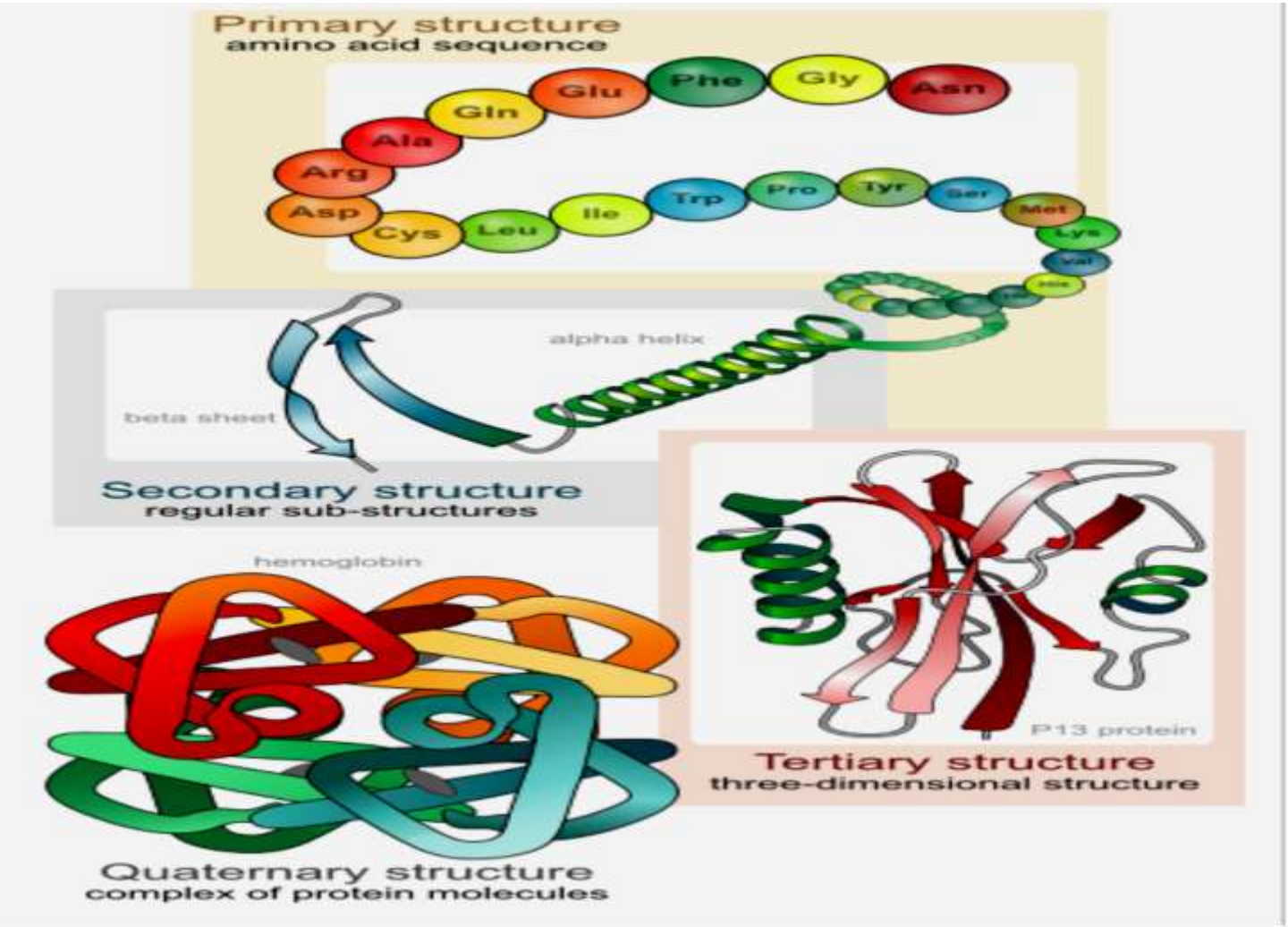
- When a protein has **two or more polypeptide** subunits, their arrangement in space is referred to as quaternary structure

Structural Hierarchy of Proteins

Protein Structure(Summary)

•Primary	The amino acid sequence	Glu-Arg-Phe-Gly
•Secondary	Characteristic structures that occur in many proteins (E.g. alpha helix , beta sheets)	
•Tertiary	Three dimensional structure of proteins	
•Quaternary	Three dimensional structure of proteins composed of multiple subunits	

Structural Hierarchy of Proteins



END

The Primary Structure

- The primary structure of a protein refers to the **linear sequence** of amino acids in the polypeptide chain.

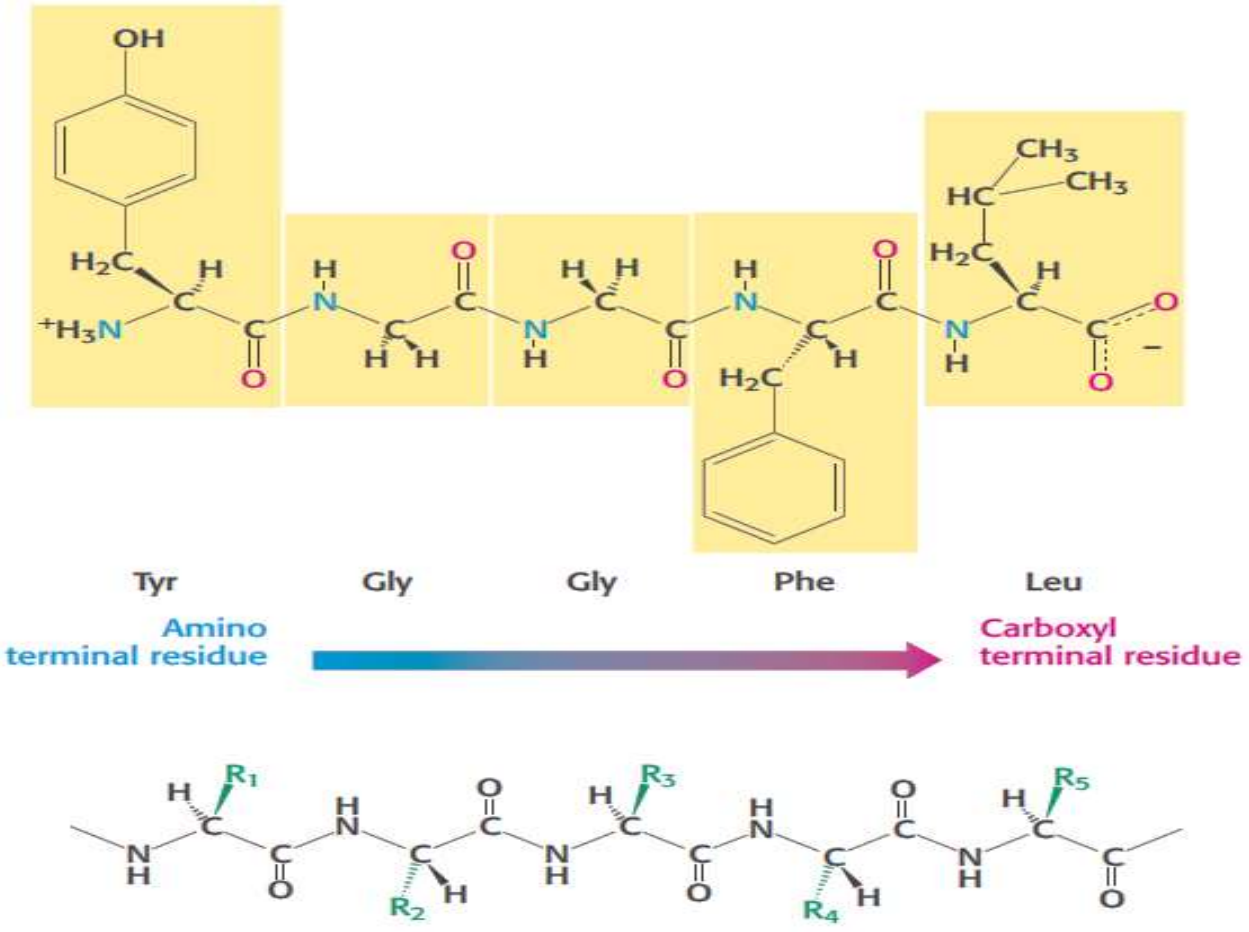
The Primary Structure

- The primary structure is held together by covalent bonds such as peptide bonds, which are made during the process of protein biosynthesis or translation.

The Primary Structure

- The two ends of the polypeptide chain are referred to as the **carboxyl terminus** (C-terminus) and the **amino terminus** (N-terminus) based on the nature of the free group on each extremity.

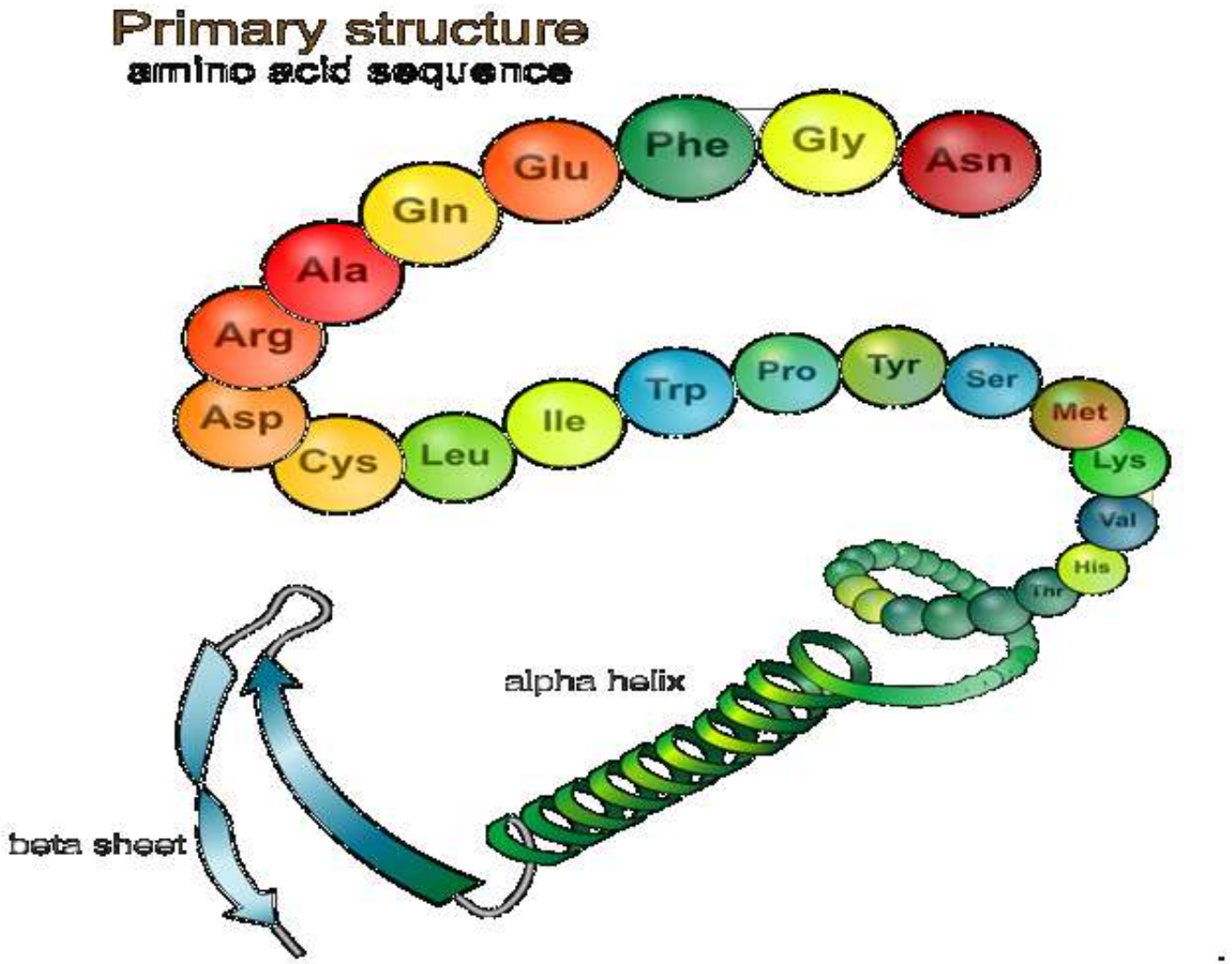
The Primary Structure



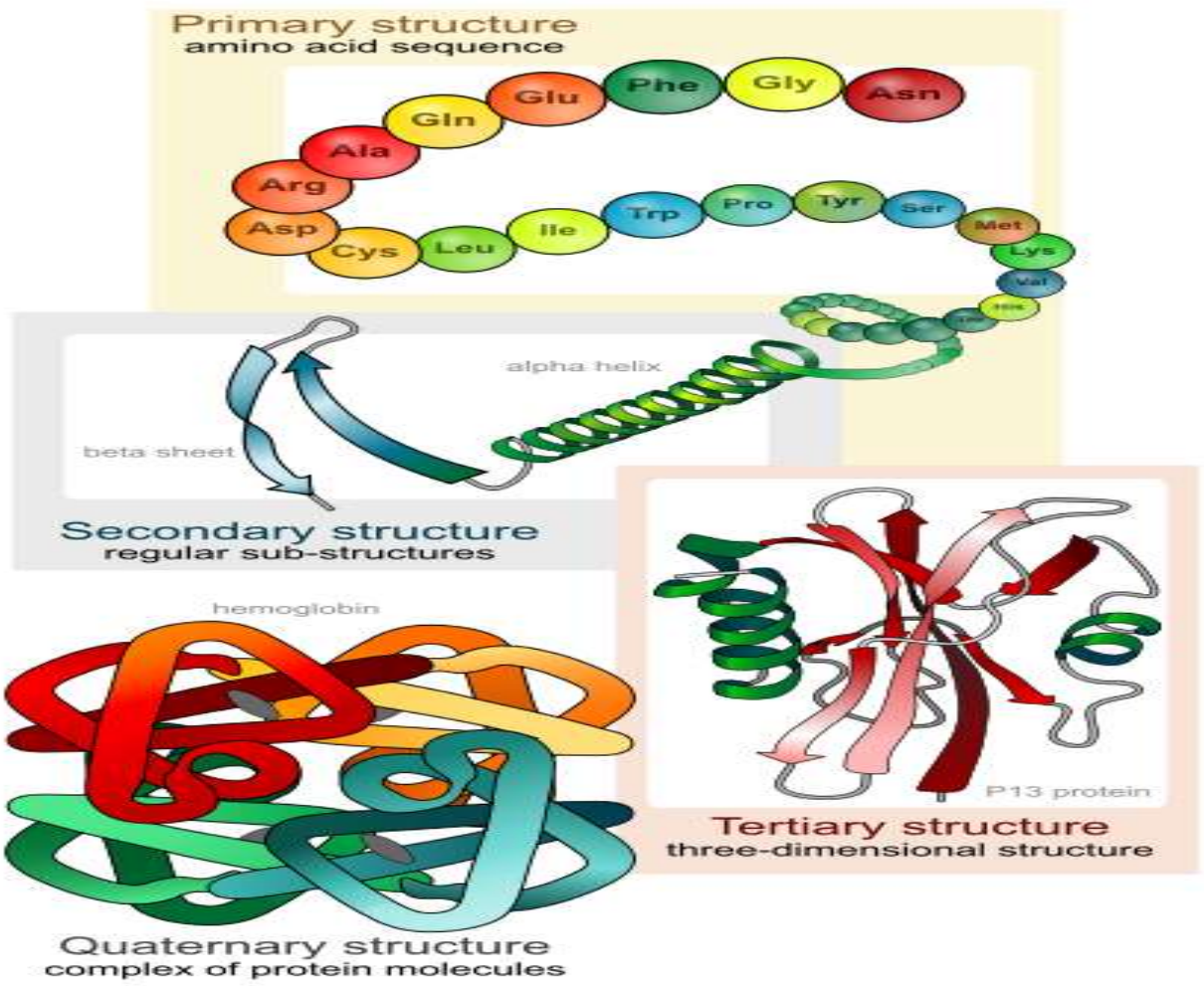
The Primary Structure

- The primary structure of a protein determines **how it folds up** into its unique three-dimensional structure, and this in turn determines the function of the protein

The Primary Structure



The Primary Structure



The Primary Structure

- The function of a protein therefore depends on its amino acid sequence

The Primary Structure

- Proteins have unique amino acid sequences, that are specified by genes.
- Each of the 20 amino acids incorporated in the proteins is encoded by one or more *specific* sequences of three nucleotides in DNA or RNA.

The Primary Structure

- Understanding of the primary structure of proteins is important because many genetic diseases result in proteins with abnormal amino acid sequences which cause loss or impairment of normal function.

The Primary Structure

- The defect can range from a single change in the amino acid sequence (as in **sickle cell anemia**) to deletion of a larger portion of the polypeptide chain (as in most cases of **Duchenne** muscular dystrophy)

The Primary Structure

- In Sickle cell anemia a **Glutamate** residue is replaced by **Valine** at position 6 of beta chain of Hemoglobin, resulting in decreased functioning and increased fragility of Hb.

The Primary Structure

- Duchenne Muscular dystrophy results from a large segment of gene deletion in X chromosome, resulting in small truncated **dystrophin** protein.

The Primary Structure

- Thus we know that if the primary structure is altered, the function of the protein may also be changed

END

The Secondary Structure

- Secondary structure, the folding of short (3- to 30-residue), contiguous segments of polypeptide into **geometrically ordered units**

The Secondary Structure

- The term secondary structure refers to any chosen segment of a polypeptide chain and describes the local spatial arrangement of its main-chain atoms, without regard to its relationship to other segments

The Secondary Structure

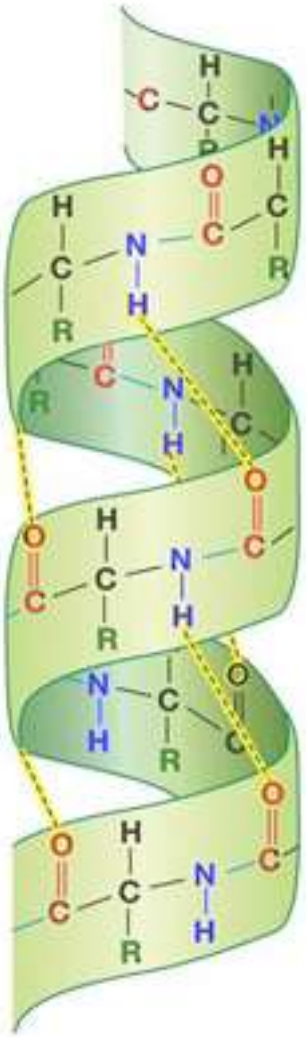
- There are a few types of secondary structure that are particularly stable and occur widely in proteins
- The most prominent are the **α helix and β conformations**

The Secondary Structure

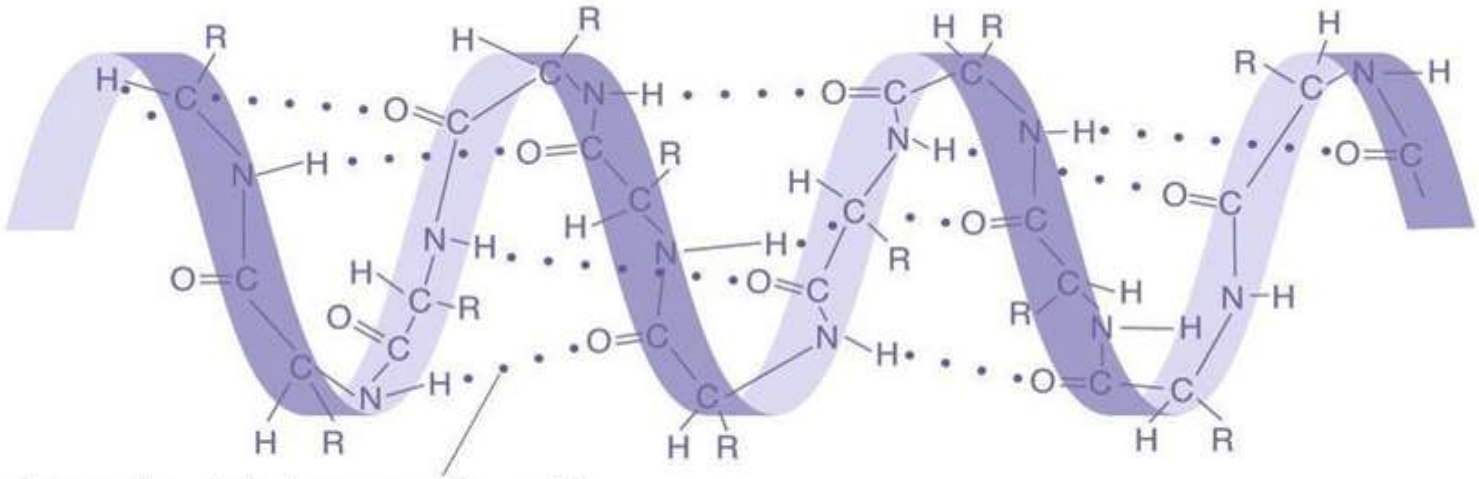
The α Helix is a common protein secondary structure

- The simplest arrangement the polypeptide chain can assume, given its rigid peptide bonds (but free rotation around its other, single bonds), is a helical structure, called the α helix

The Secondary Structure



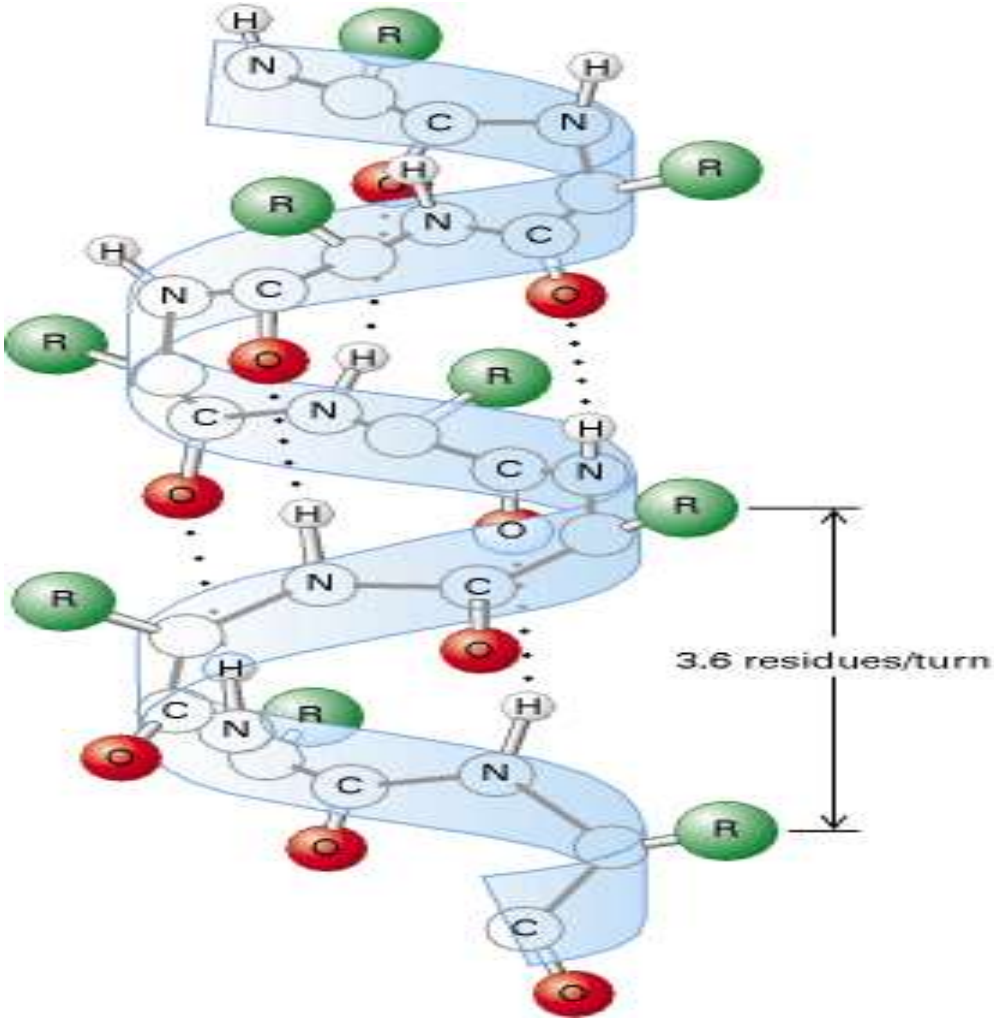
The Secondary Structure



Hydrogen bonds between amino acids at different locations in polypeptide chain

α helix

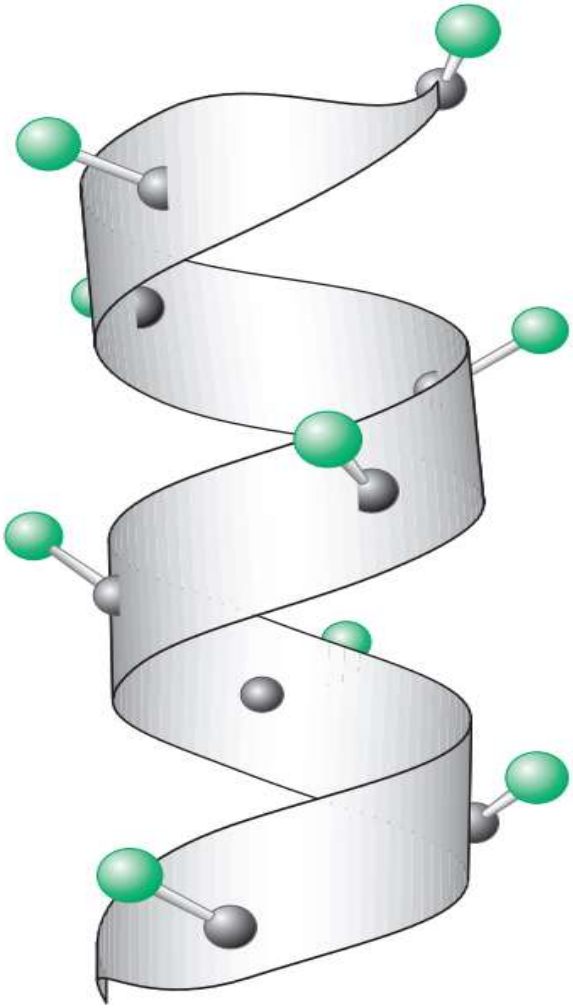
The Secondary Structure

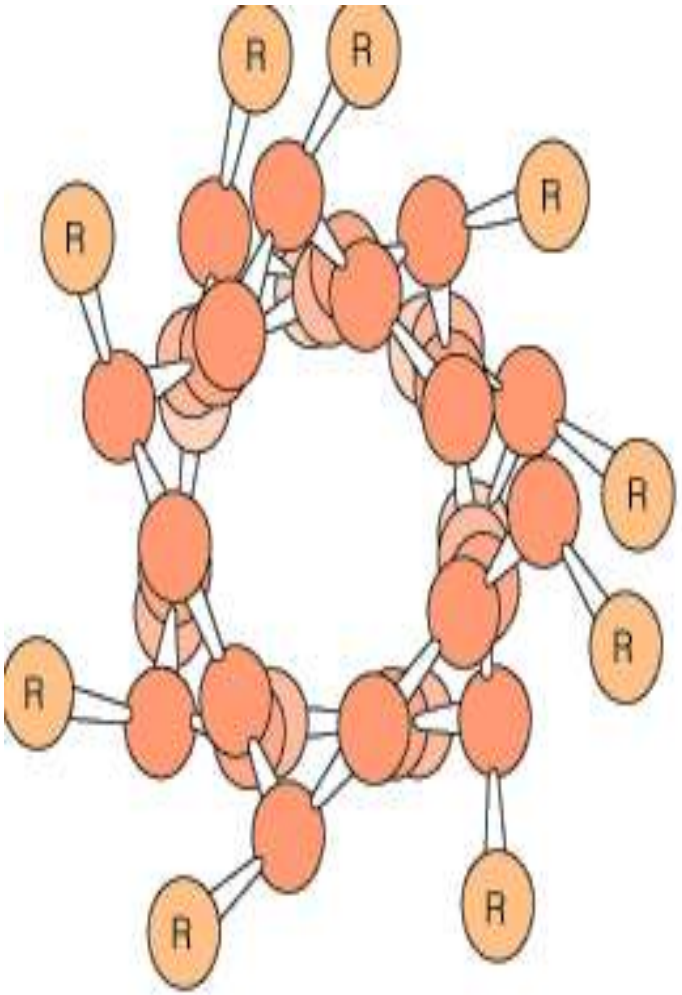


The Secondary Structure

- Within the α helix, every peptide bond (except those close to each end of the helix) participates in such hydrogen bonding

The Secondary Structure





END

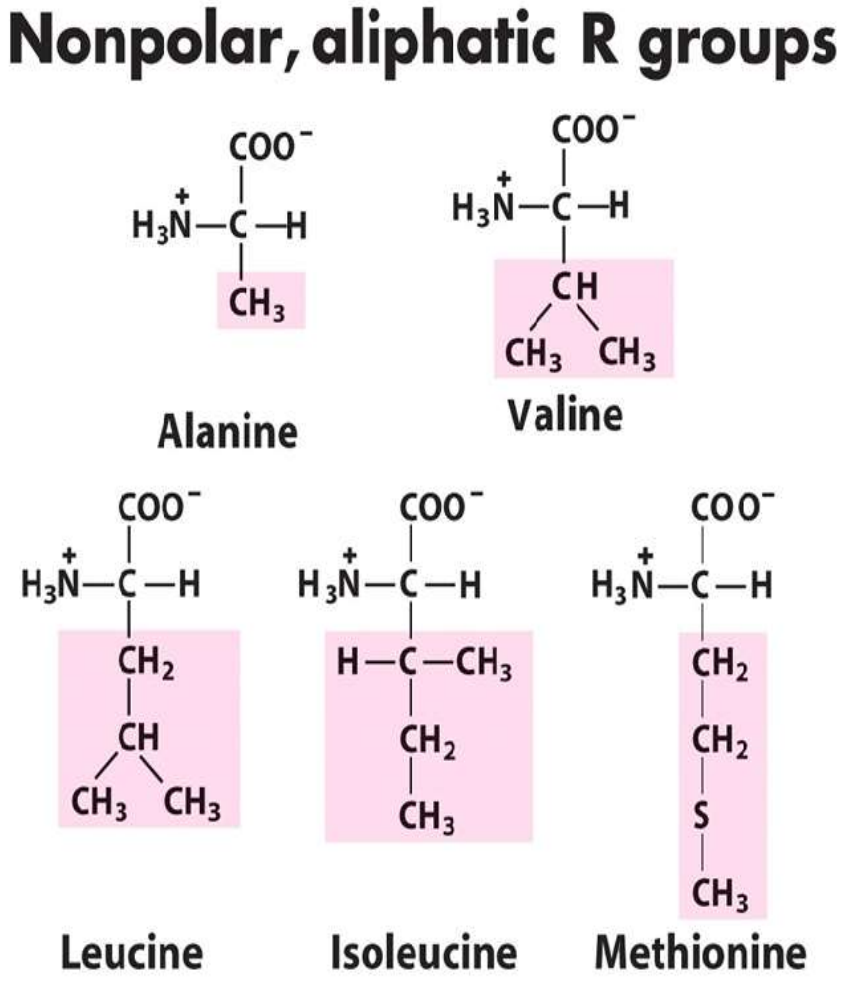
The Secondary Structure

- Amino acid residues included in alpha helix have an **intrinsic propensity** to form an alpha helix, reflecting the properties of the R group.

The Secondary Structure

- Amino acids most commonly found in alpha helices are **non-polar** with aliphatic side chains
- **Alanine** shows the greatest tendency to form α helices in most experimental model systems.

The Secondary Structure

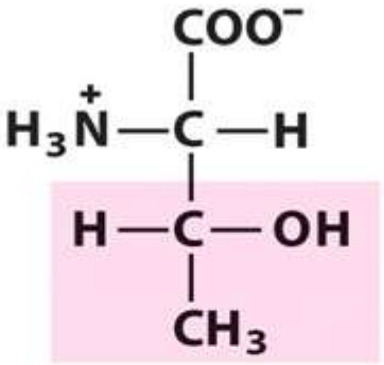


Amino acids most commonly found in alpha helices are **non-polar** with aliphatic side chains **Alanine** shows the greatest tendency to form α helices in most experimental model systems.

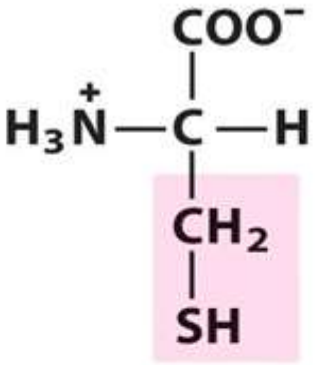
The Secondary Structure

- The bulk and shape of **Ser, Thr**, and **Cys** residues can destabilize an α helix if they are close together in the chain.

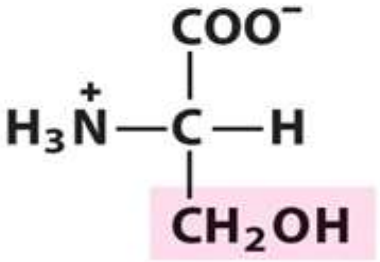
The Secondary Structure



Threonine



Cysteine



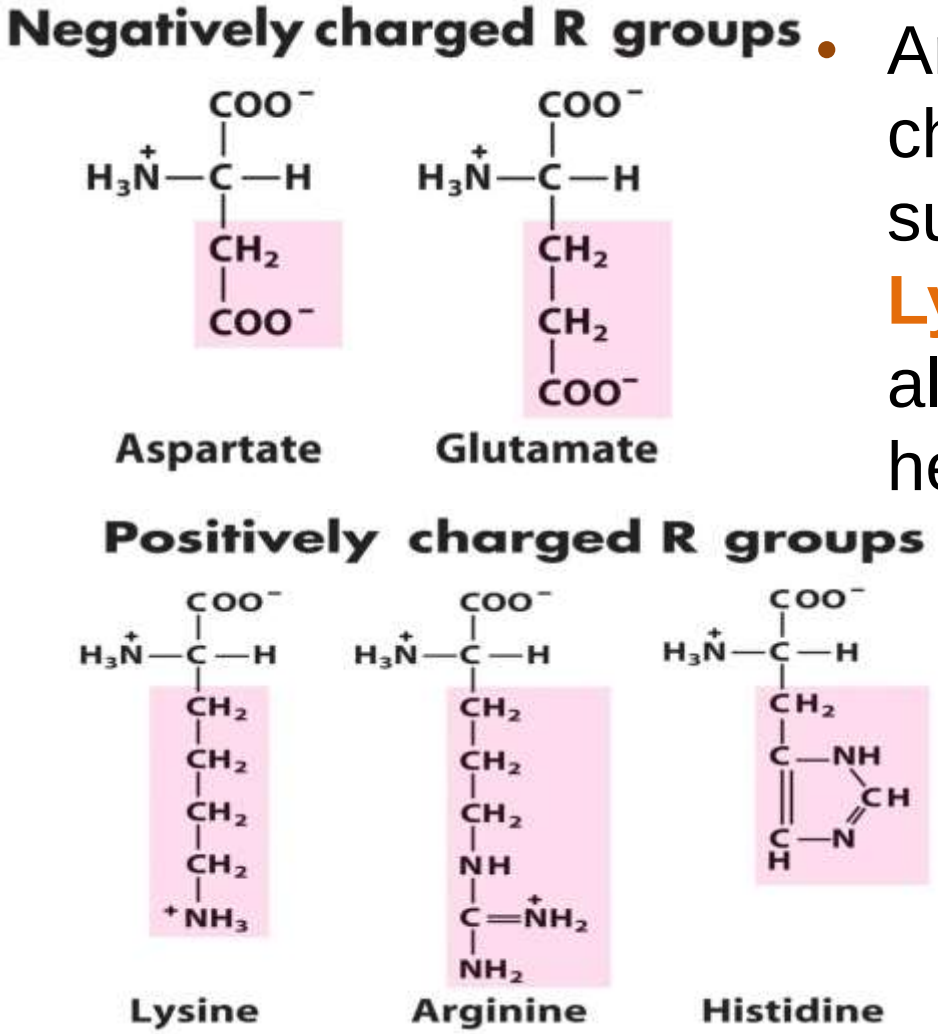
Serine

The bulk and shape of **Ser, Thr**, and **Cys** residues can destabilize an α helix if they are close together in the chain.

The Secondary Structure

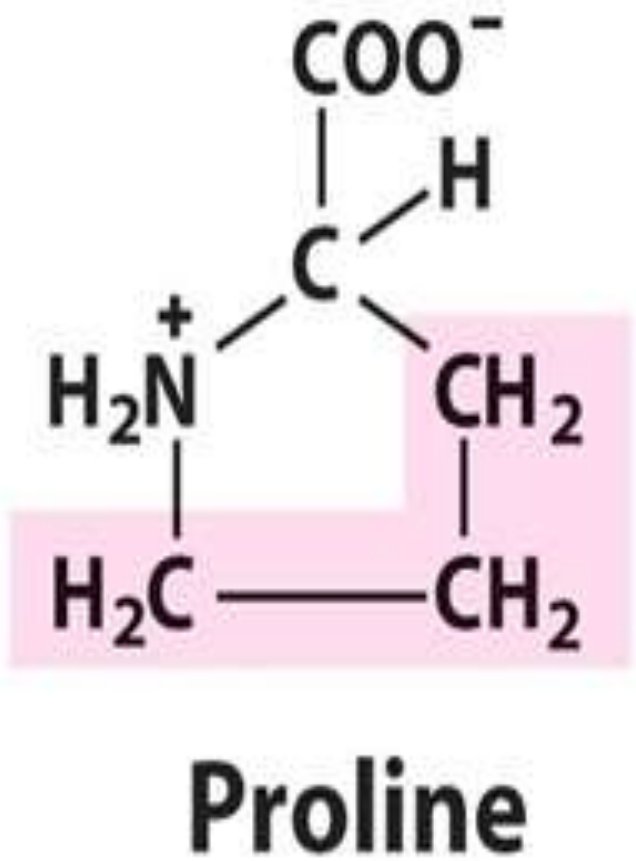
- Amino acids with charged R groups such as **Aspartate, Lysine** residues can also disrupt the helix.

The Secondary Structure



- Amino acids with charged R groups such as **Aspartate, Lysine** residues can also disrupt the helix.

The Secondary Structure



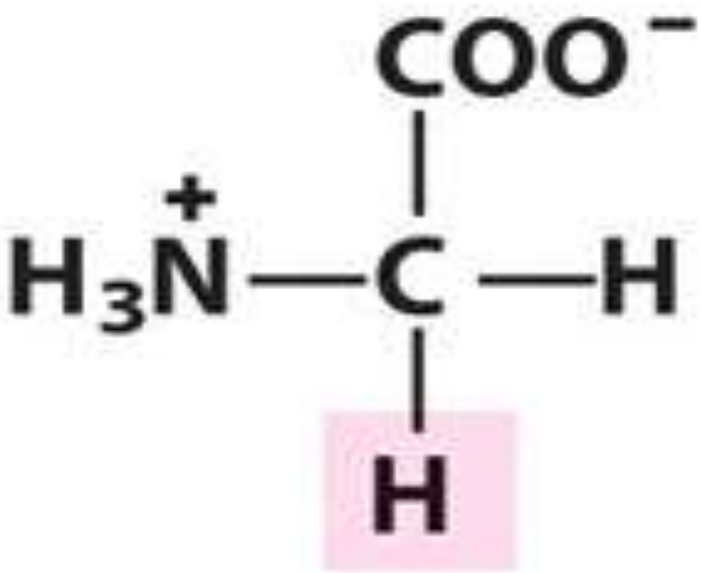
Proline

- In **proline**, the nitrogen atom is part of a rigid ring, and rotation about the N-C bond is not possible
- Thus, a proline residue introduces a destabilizing kink in an α helix

The Secondary Structure

- **Glycine** occurs infrequently in α helices for a different reason.
- It has more conformational flexibility than the other amino acid residues

The Secondary Structure



Glycine

Polymers of glycine tend to take up coiled structures quite different from an α helix

END

The Secondary Structure

- **A very diverse group of proteins contains α-helices**
- The alpha-helical content of proteins ranges widely, from nearly none to almost 100%

The Secondary Structure

- For example, the **keratins** are a family of closely related, fibrous proteins whose structure is nearly entirely α-helical.
- They are a major component of tissues such as hair, nails, hoofs and skin.

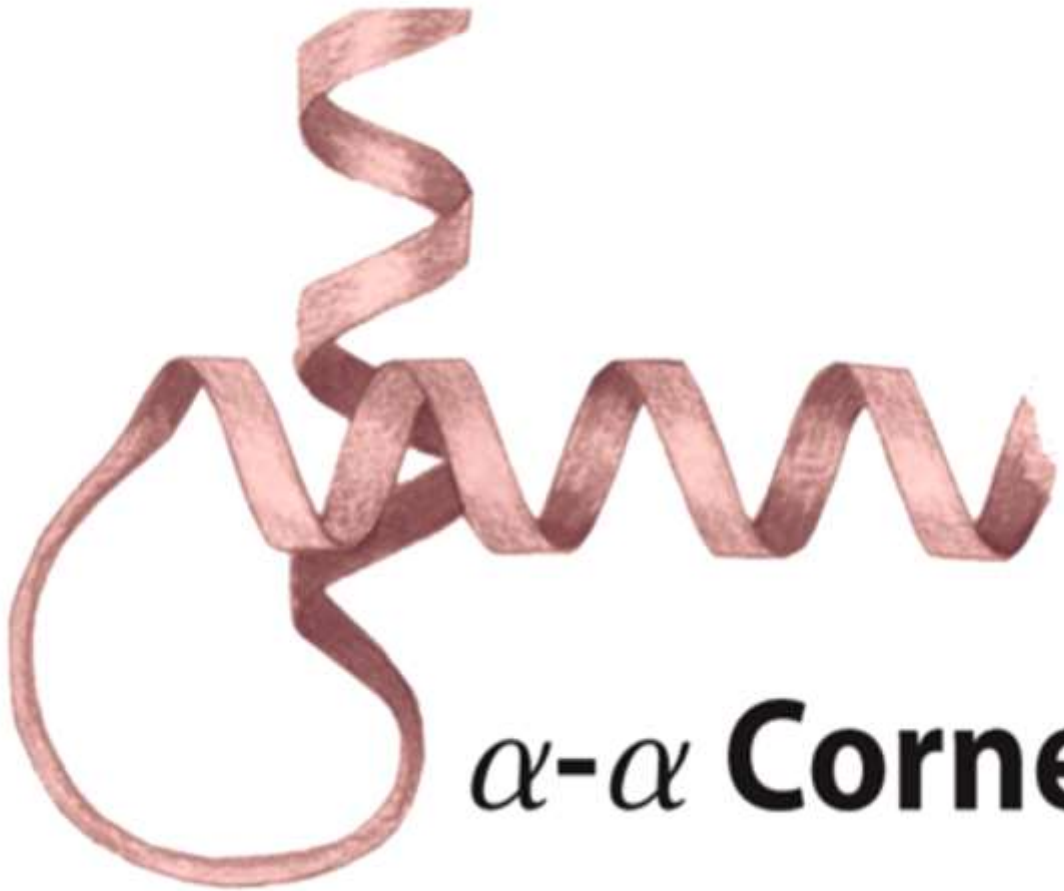
The Secondary Structure

- In contrast to keratin, myoglobin, whose structure is also highly α-helical, is a globular, flexible molecule.

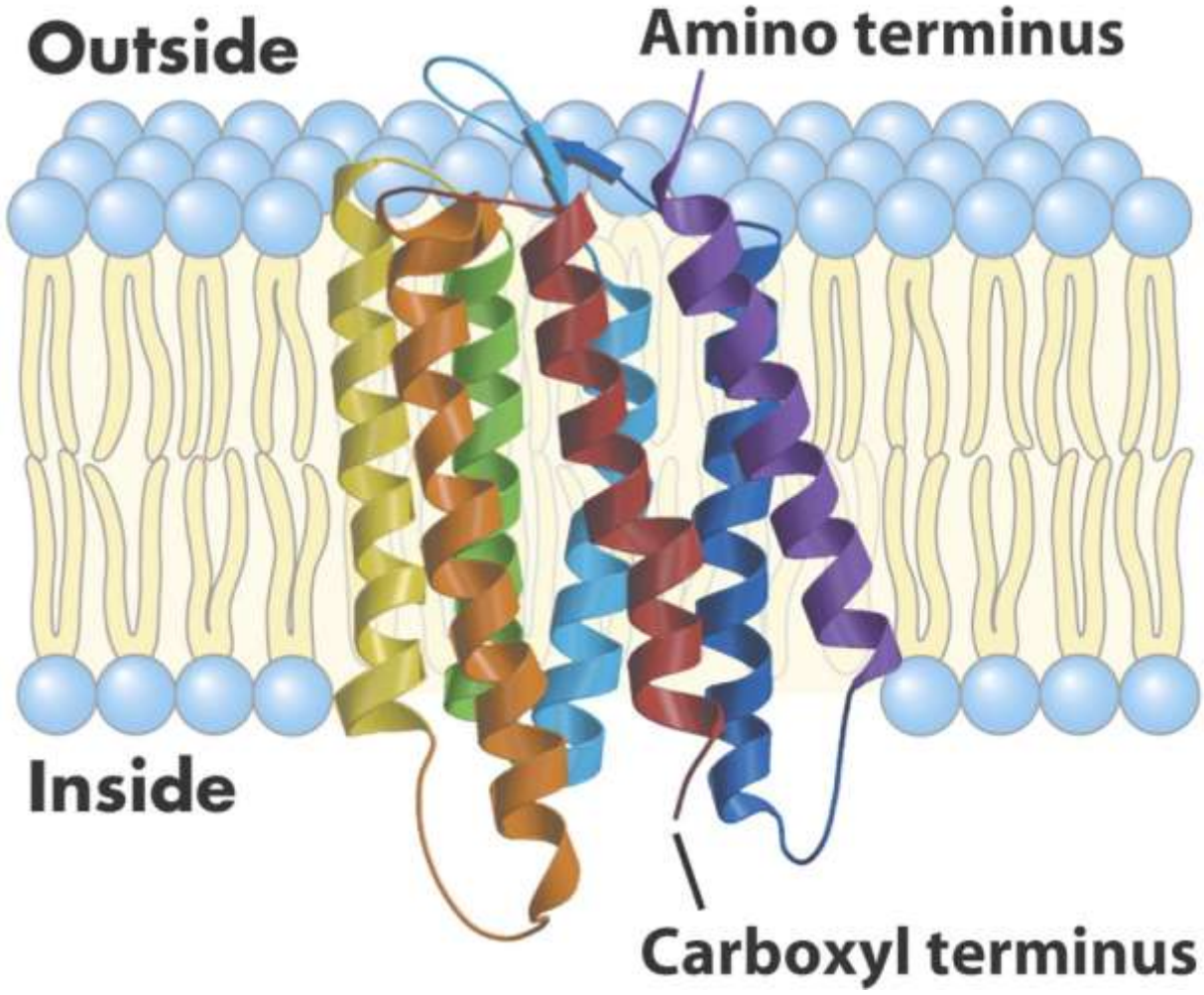
The Secondary Structure

- Nearly all trans membrane proteins contain alpha helices in their membrane spanning domains.

The Secondary Structure

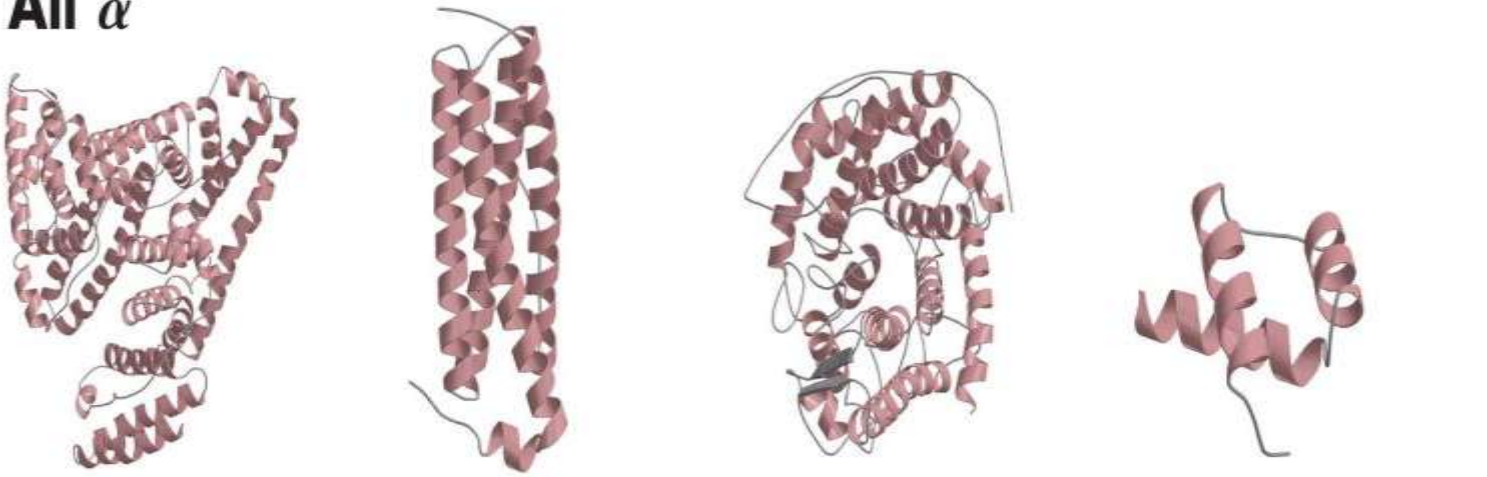


The Secondary Structure



The Secondary Structure

All α



1AO6	1BCF	1GAI	1ENH
Serum albumin	Ferritin-like	α/α toroid	DNA-binding 3-helical bundle
Serum albumin	Ferritin-like	Glycosyltransferases of the superhelical fold	Homeodomain-like
Serum albumin	Ferritin	Glucoamylase	Homeodomain
Serum albumin	Bacterioferritin (cytochrome b_1)	Glucoamylase	*engrailed* Homeodomain
Human (*Homo sapiens*)	*Escherichia coli*	*Aspergillus awamori*, variant x100	*Drosophila melanogaster*

The Secondary Structure

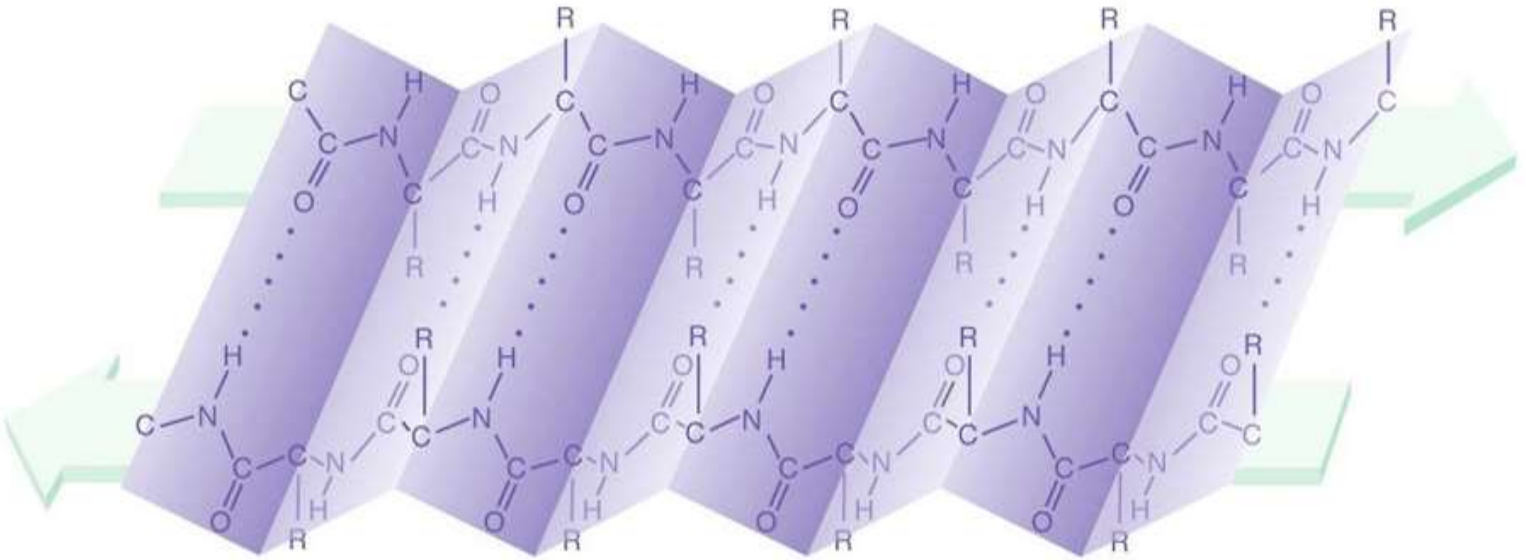
The β Conformation organizes polypeptide Chains in β sheets

β Sheets are a second type of regular secondary structure that maximizes hydrogen bonding between the peptide backbones.

The Secondary Structure

- The backbone of the polypeptide chain is extended into a zigzag rather than helical structure
- The zigzag polypeptide chains can be arranged side by side to form a structure resembling a series of pleats

The Secondary Structure



Pleated sheet

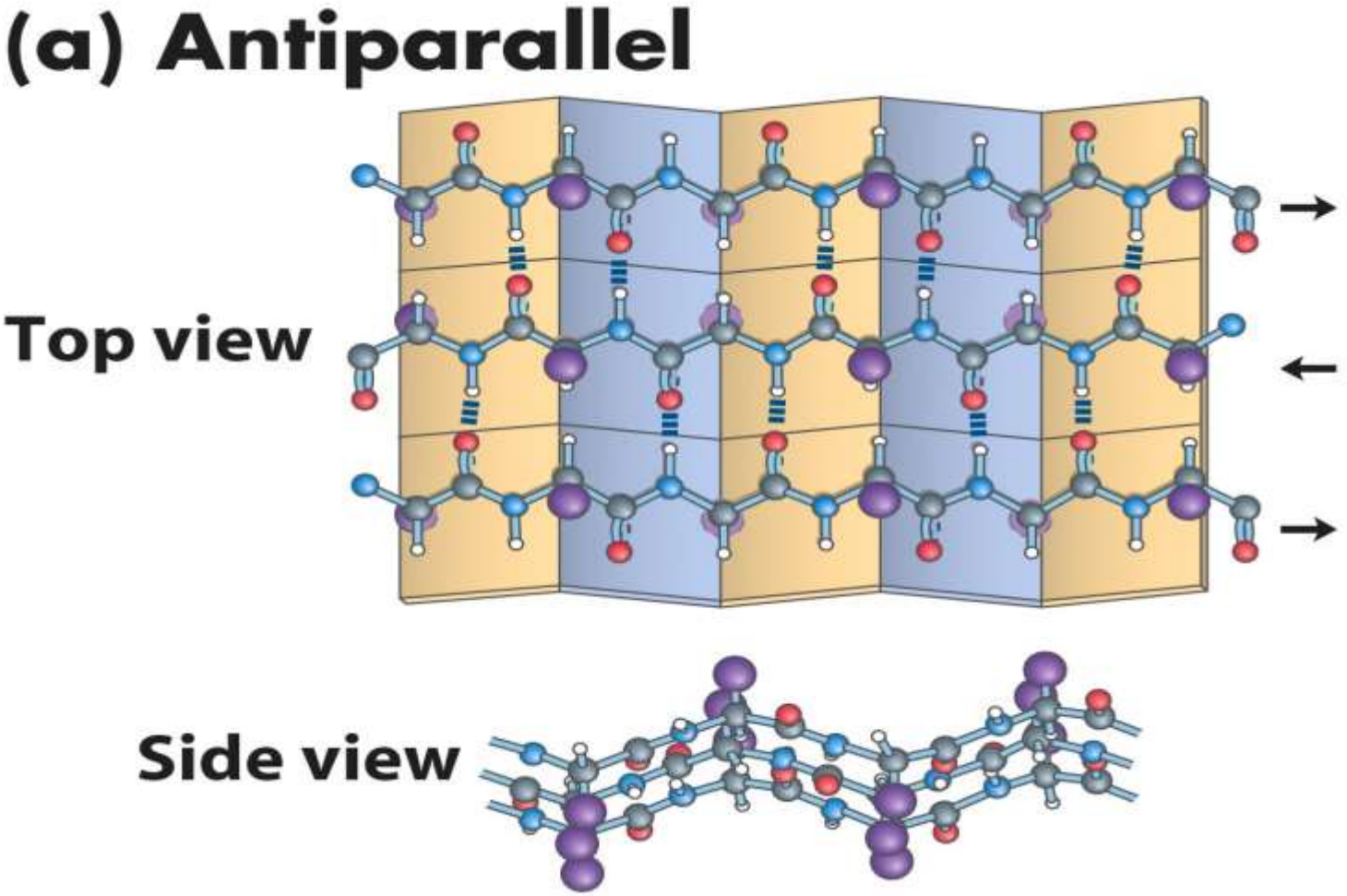
The Secondary Structure

- In this arrangement called a **β sheet**, hydrogen bonds form between adjacent segments of polypeptide chain

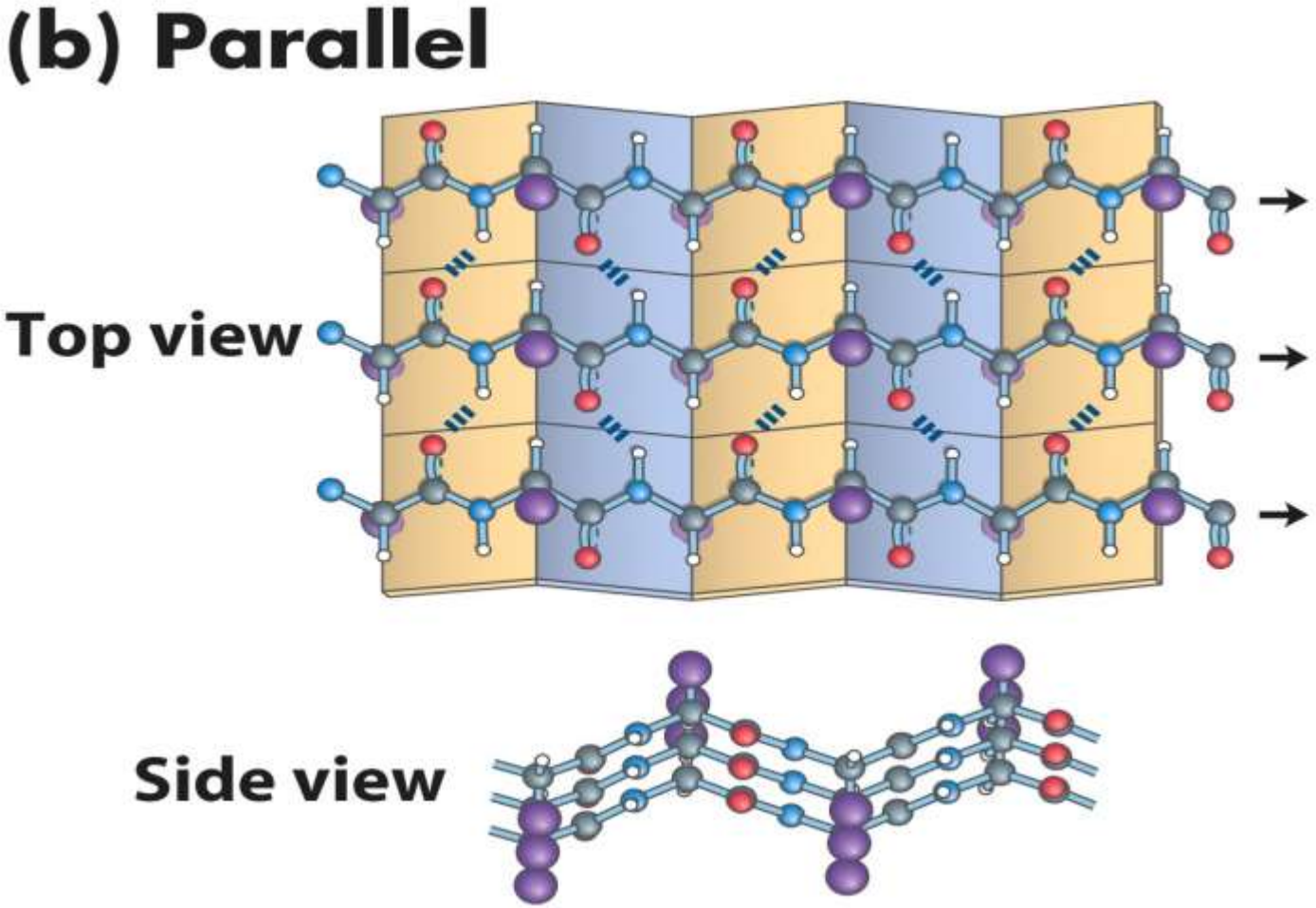
The Secondary Structure

- The -pleated sheet is described as **parallel** if the polypeptide strands run in the same direction (as defined by their amino and carboxy terminals) and **anti-parallel** if they run in opposite direction

The Secondary Structure



The Secondary Structure



The Secondary Structure

- **Antiparallel** strands are often the same polypeptide chain **folded back** on itself, with simple hairpin turns or long runs of polypeptide chain connecting the strands

The Secondary Structure

- The amino acid side chains(R groups) of each polypeptide strand alternate between extending above and below the plane of the β-sheet.
- Most sheets are not perfectly flat but tend to have a twist.

The Secondary Structure

- The individual segments that form a **β sheet** are usually nearby on the polypeptide chain, but can also be quite distant from each other in the linear sequence of the peptide and may even be in different polypeptide chains.

The Secondary Structure

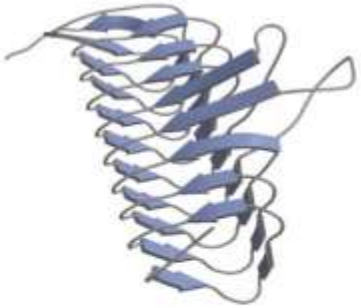
- Clusters of twisted strands of sheet form the core of many **globular proteins**.

The Secondary Structure

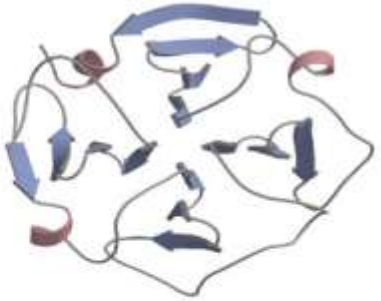
All β



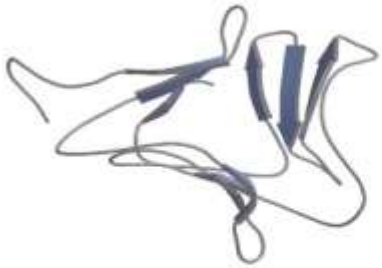
1HOE
α -Amylase inhibitor
α -Amylase inhibitor
α -Amylase inhibitor
HOE-467A
Streptomyces tendae 4158



1LXA
Single-stranded left-handed β helix
Trimeric LpxA-like enzymes
UDP *N*-acetylglucosamine acyltransferase
UDP *N*-acetylglucosamine acyltransferase
Escherichia coli



1PEX
Four-bladed β propeller
Hemopexin-like domain
Hemopexin-like domain
Collagenase-3 (MMP-13), carboxyl-terminal domain
Human (*Homo sapiens*)



1JPC
β-Prism II
α-D-Mannose-specific plant lectins
α-D-Mannose-specific plant lectins
Lectin (agglutinin)
Snowdrop (*Galanthus nivalis*)



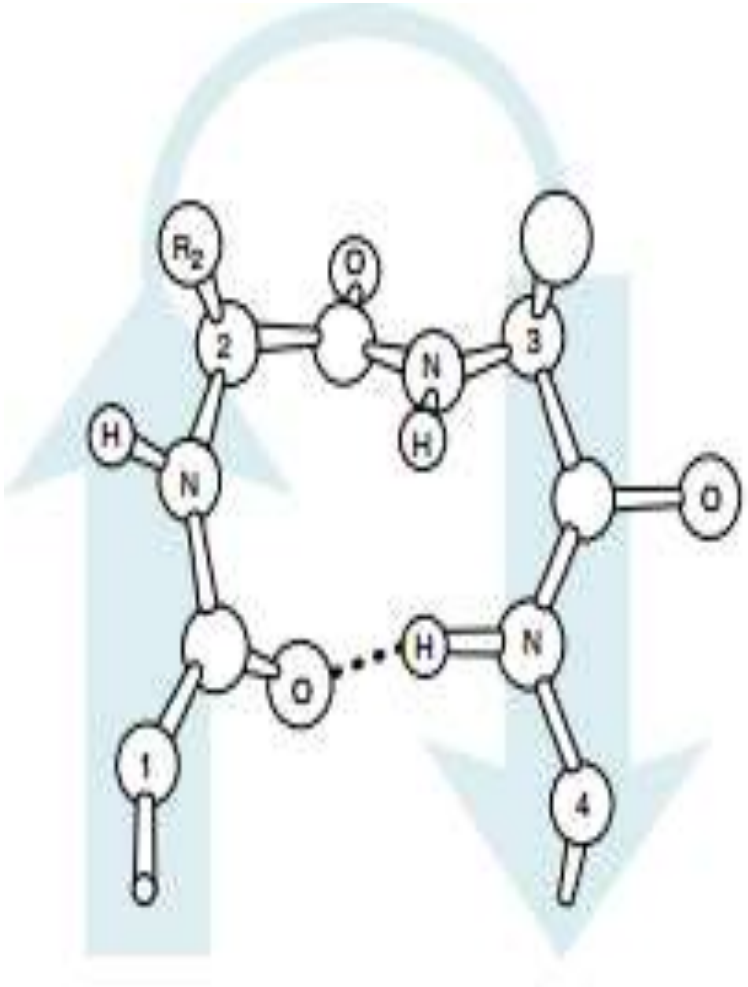
1CD8
Immunoglobulin-like β sandwich
Immunoglobulin
Antibody variable domain-like
CD8
Human (*Homo sapiens*)

END

The Secondary Structure

- In contrast to regular repeating element in the twist of an alpha Helix or a pleat of Beta sheet, **bends, loops,** and **turns are** non- regular secondary structures that do not have a repeating element.

The Secondary Structure

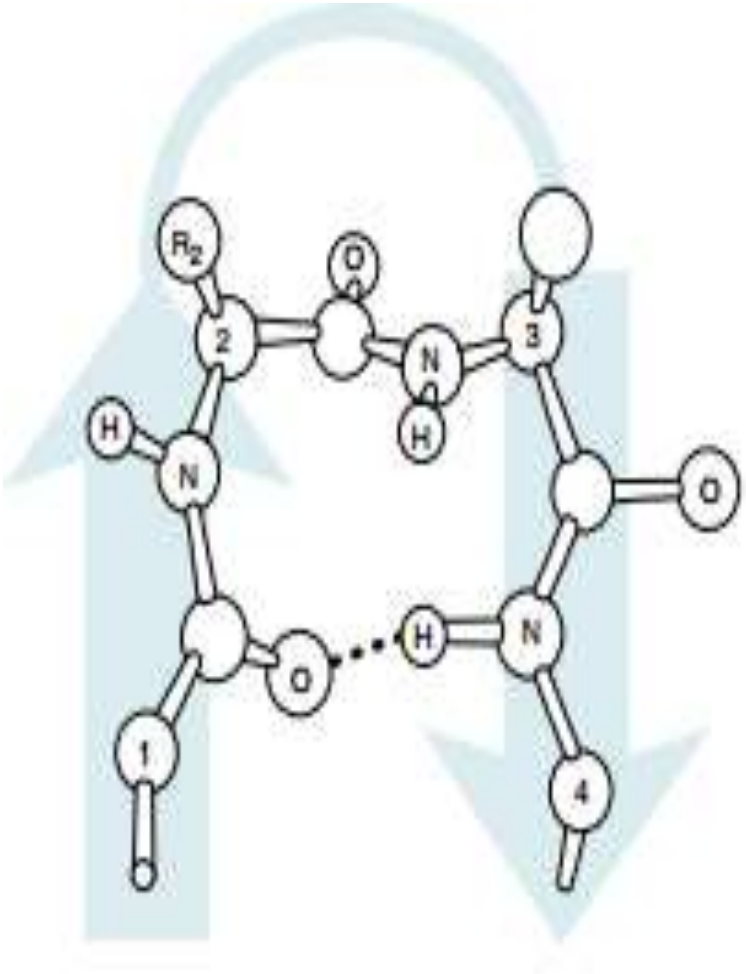


- **Beta-bends** are short regions usually involving four successive amino acid residues.
- They often connect strands of antiparallel beta-sheets.

The Secondary Structure

- β-Bends reverse the direction of a polypeptide chain, helping it form a compact, globular shape.
- The structure is a 180 degree turn involving four amino acid residues

The Secondary Structure

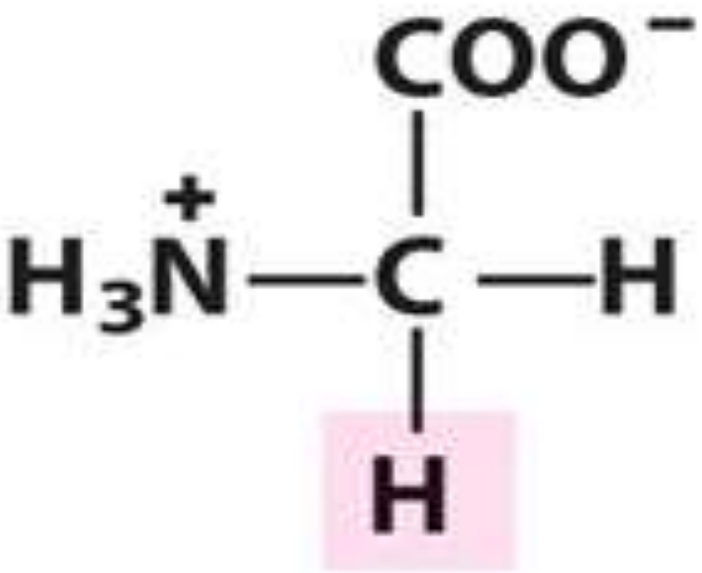


- The carbonyl oxygen of the first residue forms a hydrogen bond with the amino-group hydrogen of the fourth residue.

The Secondary Structure

- β-Bends were given this name because they often connect successive strands of antiparallel β-sheets

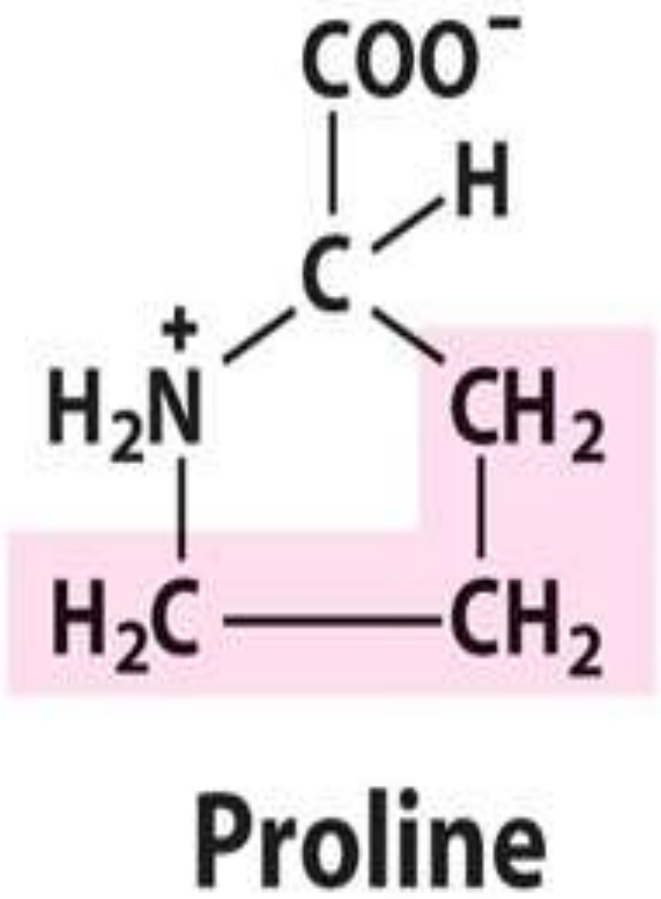
The Secondary Structure



Glycine

- Glycine and Proline residues often occur in β turns
- Glycine because it is small and flexible

The Secondary Structure



Proline

- Proline because peptide bonds involving the imino nitrogen readily assume the cis configuration, a form that is particularly amenable to a tight turn

END

The Secondary Structure

- A **motif** also called a **super secondary structure** is simply a recognizable folding pattern involving two or more elements of secondary structure and the connection(s) between them.

The Secondary Structure

- Structural motifs are intermediate between secondary and tertiary structures.

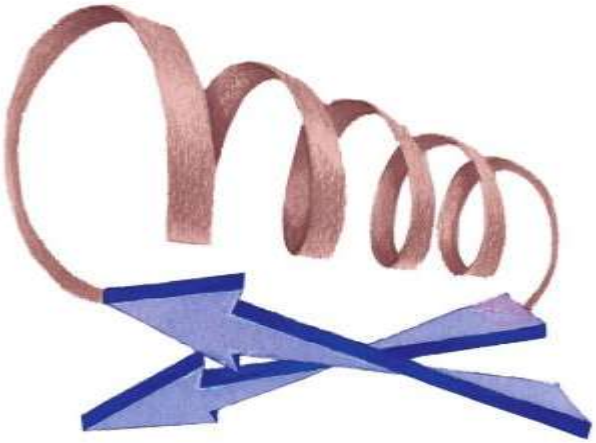
The Secondary Structure

- A motif can be very simple, such as two elements of secondary structure folded against each other, and represent only a small part of a protein
- An example is a **β-α-β loop**.

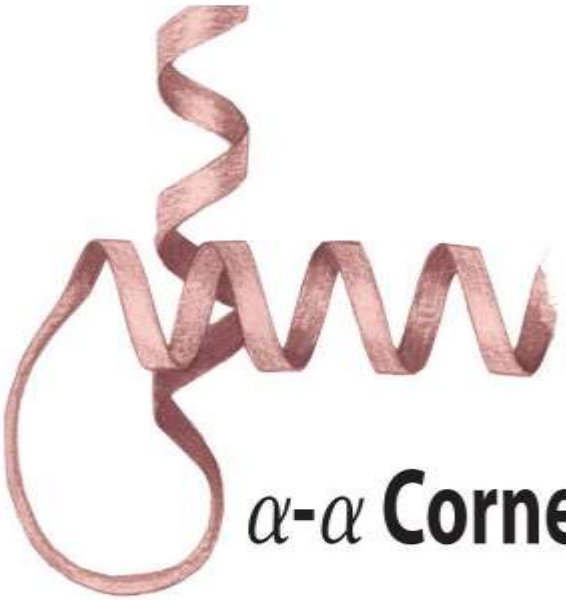
The Secondary Structure

- A motif can also be a very elaborate structure involving scores of protein segments folded together, such as the **β barrel**

The Secondary Structure



β-α-β Loop

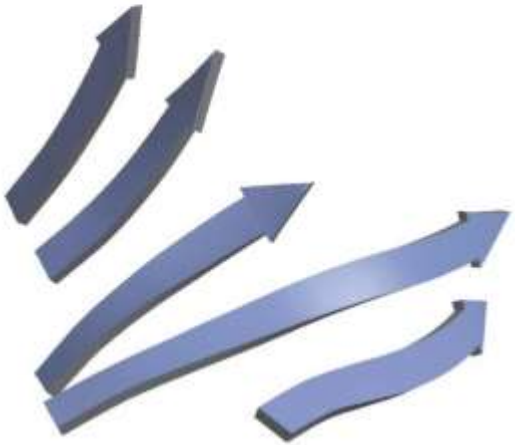


α-α Corner

The Secondary Structure



β Barrel



Twisted β sheet

END

The Tertiary Structure

The tertiary structure

- The tertiary structure of a protein is the folding pattern of the secondary structural elements into a three-dimensional conformation.

The Tertiary Structure

- It indicates, in three-dimensional space, how secondary structural features—helices, sheets, bends, turns, and loops—assemble to form domains and how these domains relate spatially to one another.

The Tertiary Structure

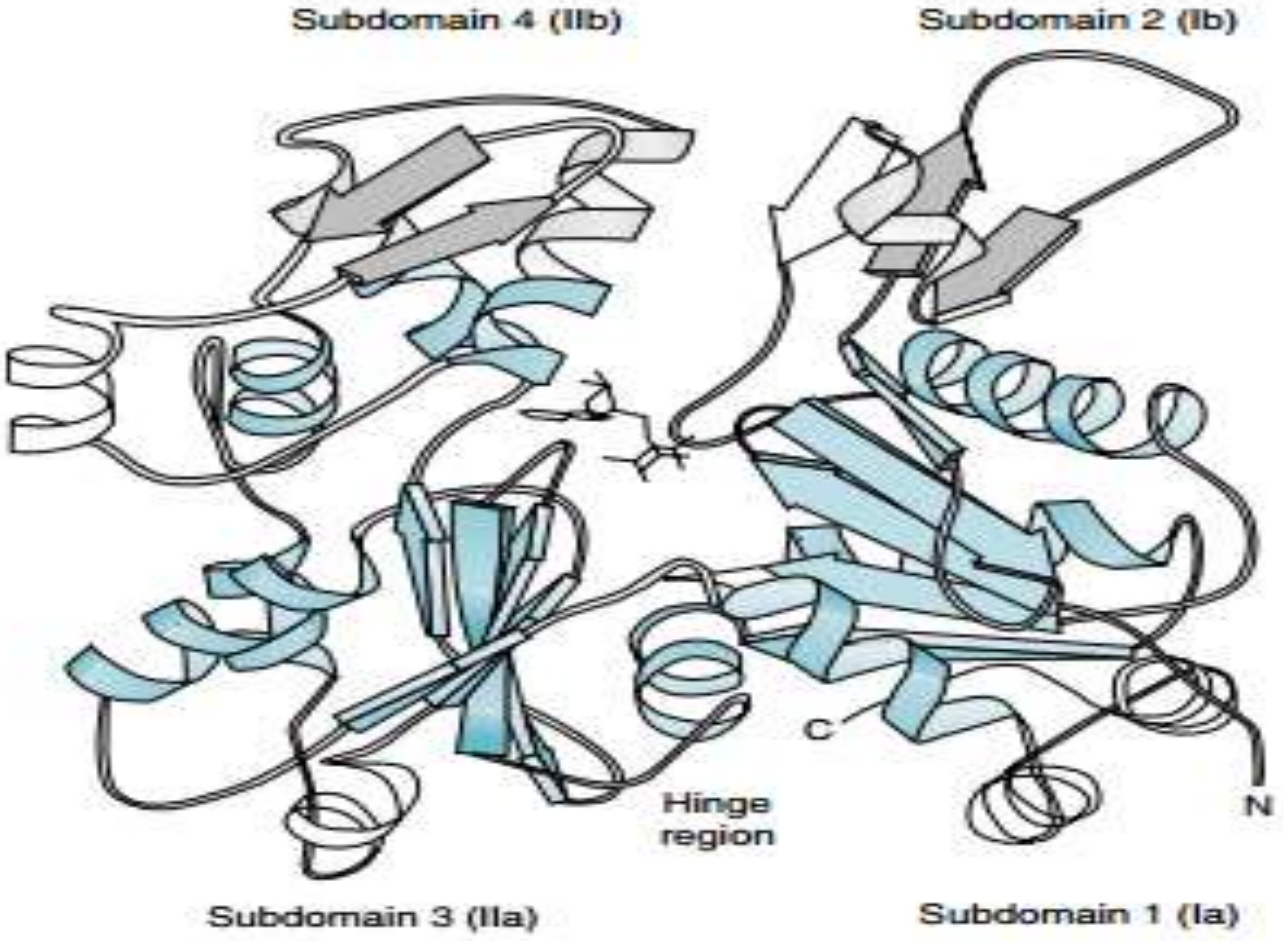
Domains in the Tertiary Structure

- The tertiary structure of large complex proteins is often described in terms of physically independent regions called structural domains.

The Tertiary Structure

- A domain is a section of protein structure sufficient to perform a particular chemical or physical task such as binding of a substrate or other ligand.

The Tertiary Structure



The Tertiary Structure

- Each domain is formed from a continuous sequence of amino acids in the polypeptide chain that are folded into a three-dimensional structure independently of the rest of the protein.

The Tertiary Structure

- Two domains are connected through a simpler structure like a loop.

END

The Tertiary Structure

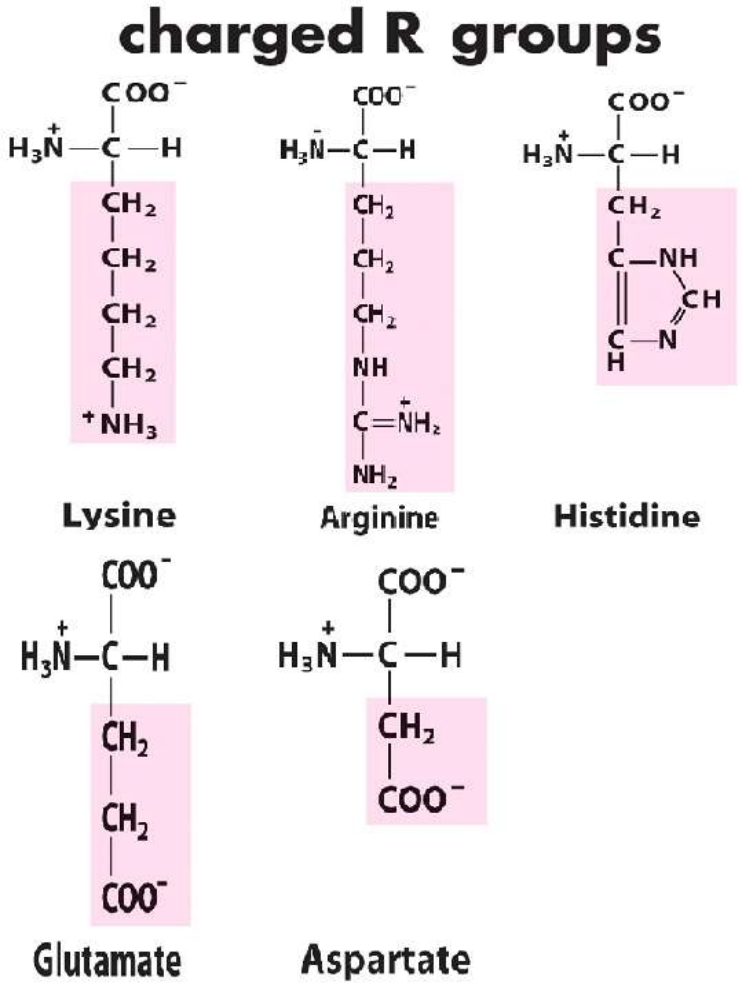
- Molecular forces that stabilise three dimensional tertiary structure include:
- Electrostatic Interactions
- Hydrophobic interactions
- Disulphide bridges
- Hydrogen bonding

The Tertiary Structure

Electrostatic Interactions

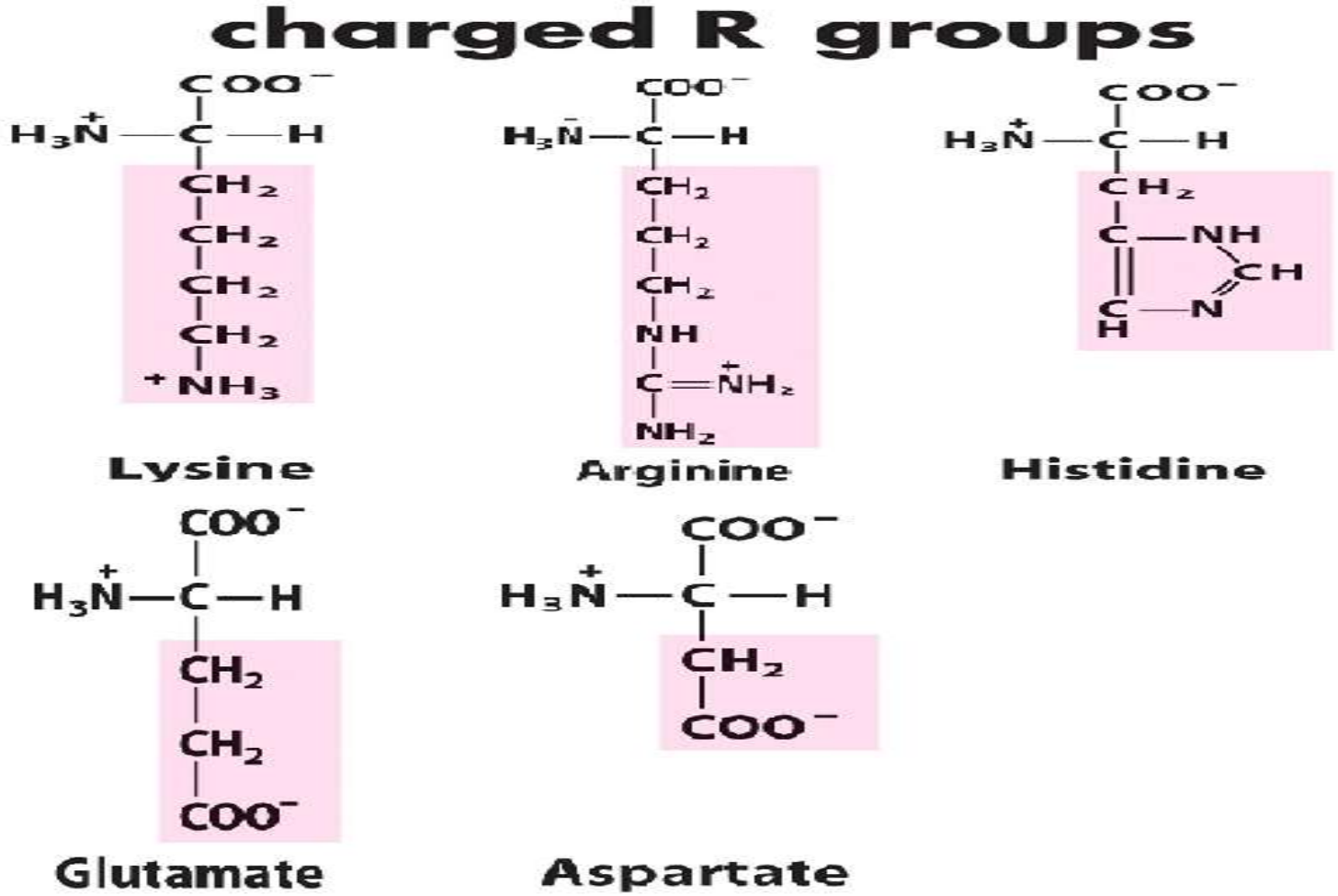
- The **positive** charges on the basic amino acids enables them to form ionic bonds (electrostatic bonds) with **negatively** charged groups, such as the side chains of acidic amino acids.

The Tertiary Structure

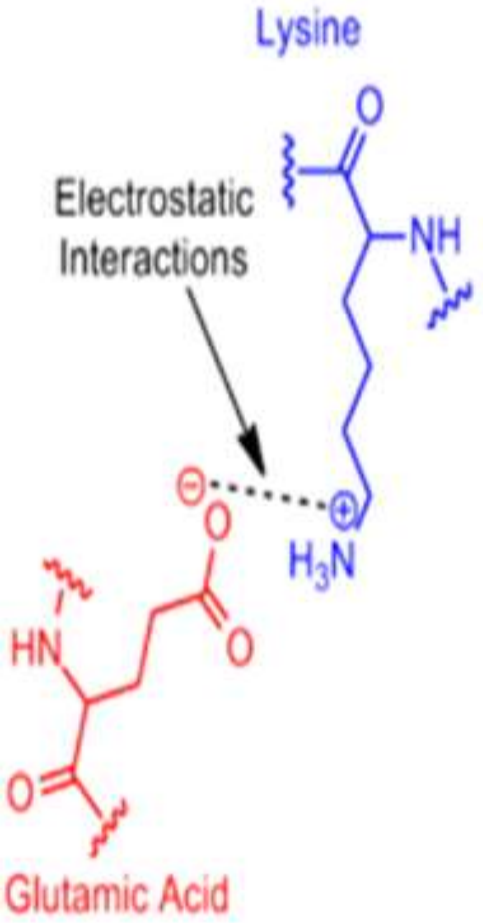


- **Electrostatic Interactions**
- The **positive** charges on the basic amino acids enables them to form ionic bonds (electrostatic bonds) with **negatively** charged groups, such as the side chains of acidic amino acids.

The Tertiary Structure

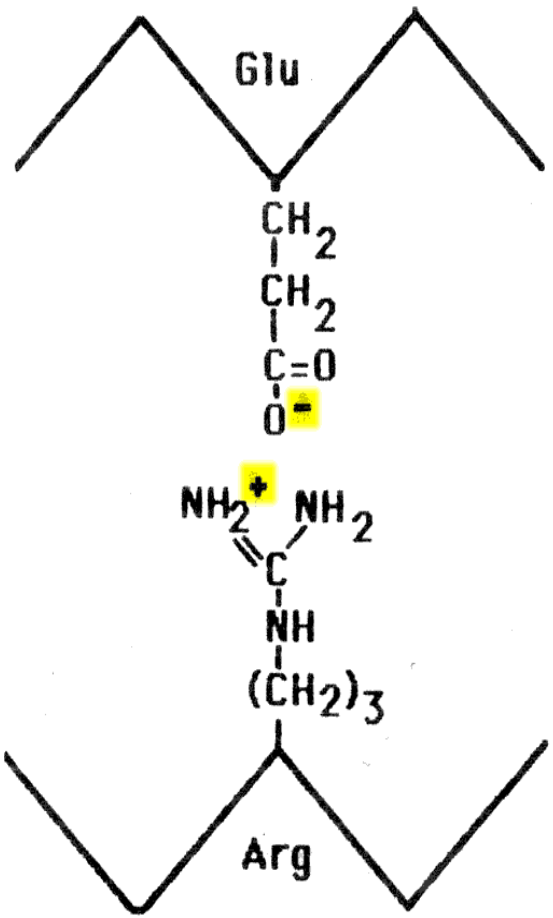


The Tertiary Structure



- The positive charges on the basic amino acids enables them to form ionic bonds (electrostatic bonds) with negatively charged groups, such as the side chains of acidic amino acids.

The Tertiary Structure



- The positive charges on the basic amino acids enables them to form ionic bonds (electrostatic bonds) with negatively charged groups, such as the side chains of acidic amino acids.

The Tertiary Structure

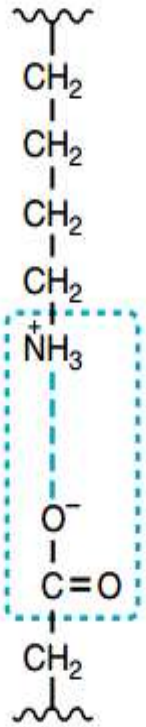


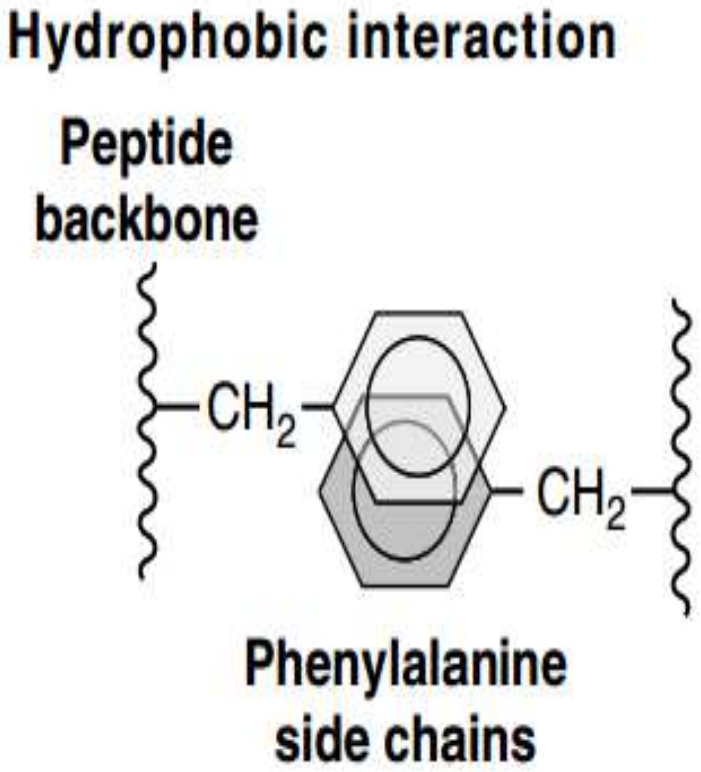
Fig. Electrostatic interaction between the positively charged side chain of lysine and the negatively charged side chain of aspartate.

- The positive charges on the basic amino acids enables them to form ionic bonds (electrostatic bonds) with negatively charged groups, such as the side chains of acidic amino acids.

The Tertiary Structure

- **Hydrophobic Interactions**
- Hydrophobic Interactions occur between non-polar uncharged side chains such as between aromatic groups of phenylalanine side chains.

The Tertiary Structure



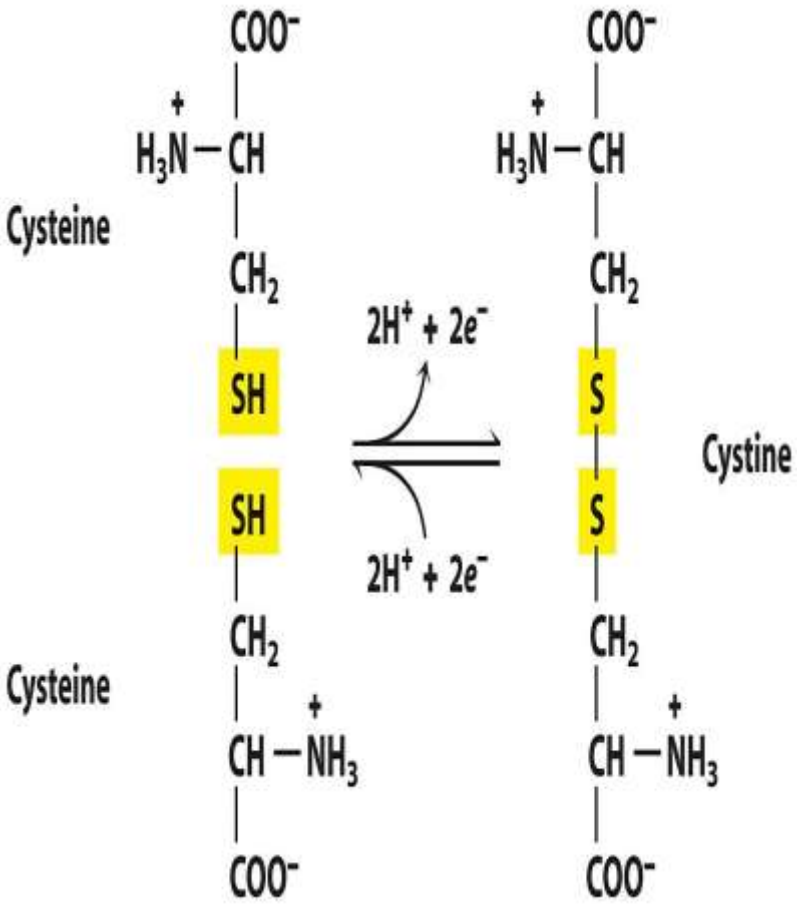
- Hydrophobic interactions occur between non-polar uncharged side chains such as between aromatic groups of phenylalanine side chains.

The Tertiary Structure

Disulfide Bridges

- The amino acid cysteine in a protein can form a covalent disulfide bond with another cysteine molecule through spontaneous (nonenzymatic) oxidation of their sulfhydryl groups.

The Tertiary Structure



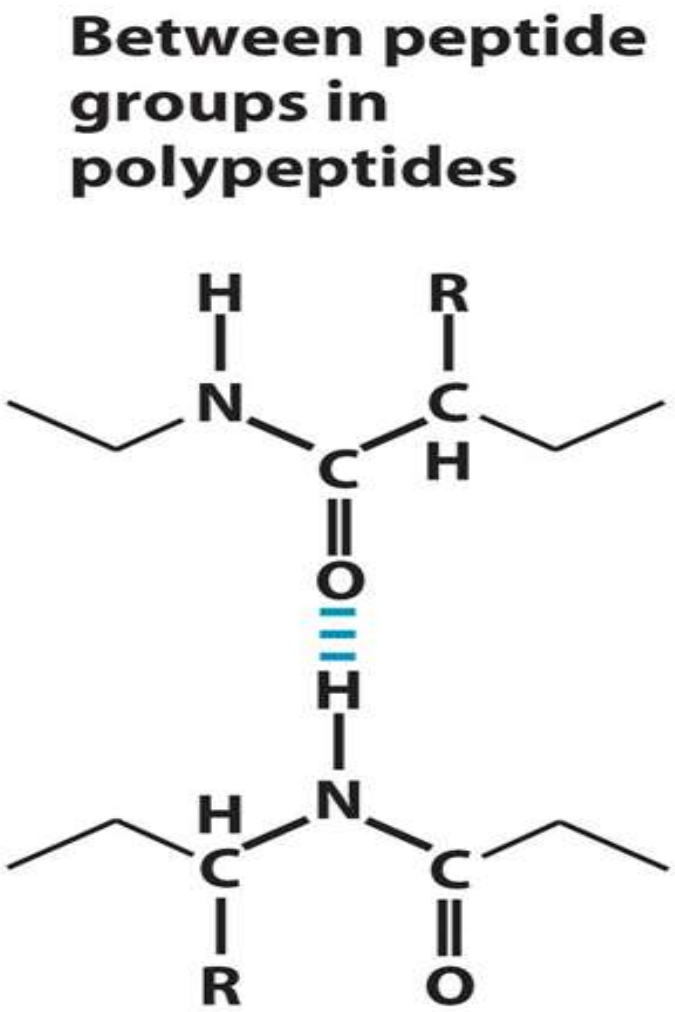
- The amino acid cysteine in a protein can form a covalent disulfide bond with another cysteine molecule through spontaneous (nonenzymatic) oxidation of their sulfhydryl groups.

The Tertiary Structure

Hydrogen bonds

- Hydrogen bonds in which a hydrogen atom is shared by a nitrogen (or oxygen) in the peptide backbone and an oxygen atom in an amino acid side chain.

The Tertiary Structure



- Hydrogen bonds in which a hydrogen atom is shared by a nitrogen (or oxygen) in the peptide backbone and an oxygen atom in an amino acid side chain.

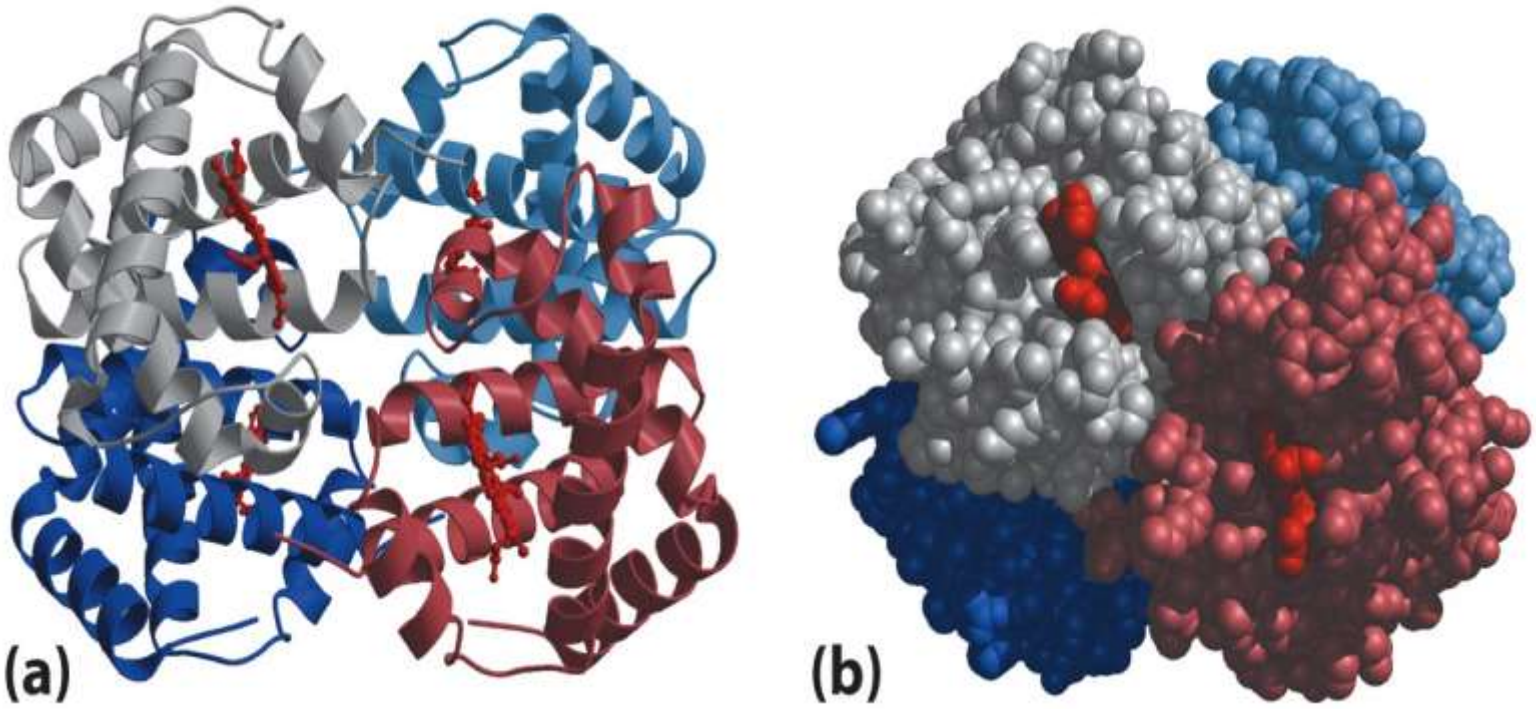
The Quaternary Structure

- Some proteins contain two or more separate polypeptide chains, or subunits, which may be identical or different.

The Quaternary Structure

- The arrangement of these protein subunits in three-dimensional complexes constitutes **quaternary structure.**

The Quaternary Structure



The Quaternary Structure

- Many proteins have multiple polypeptide subunits (from two to hundreds)
- The association of polypeptide chains can serve a variety of functions

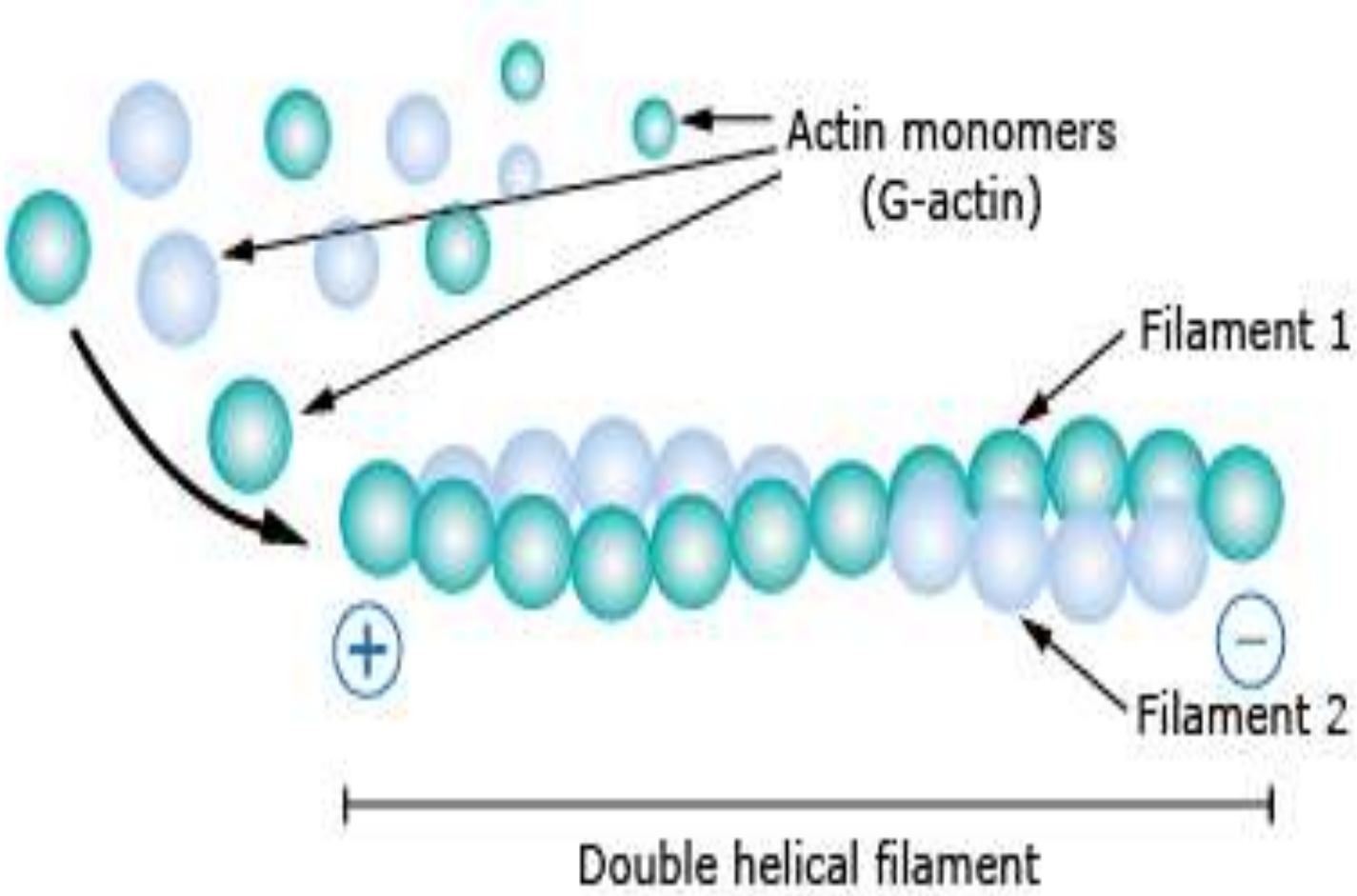
The Quaternary Structure

- A multisubunit protein is also referred to as a **multimer,** with the prefixes “homo” or “hetero” used to describe identical or different subunits, respectively.

The Quaternary Structure

- Most multimers have identical subunits
- or repeating groups of nonidentical subunits, usually in symmetric arrangements.

The Quaternary Structure



The Quaternary Structure

- The repeating structural unit in such a multimeric protein is **protomer**
- Assembly into a multisubunit structure increases the stability of a protein.

The Quaternary Structure

